University Library



A gateway to Melbourne's research publications

Minerva Access is the Institutional Repository of The University of Melbourne

Author/s:
Leach, Tessa Gwendoline

Title:
Anthropomorphic machines: alien sensation and experience in nonhumans created to be like us

Date:
2018

Persistent Link:
https://hdl.handle.net/11343/216445

Terms and Conditions:
Terms and Conditions: Copyright in works deposited in Minerva Access is retained by the copyright owner. The work may not be altered without permission from the copyright owner. Readers may only download, print and save electronic copies of whole works for their own personal non-commercial use. Any use that exceeds these limits requires permission from the copyright owner. Attribution is essential when quoting or paraphrasing from these works.

Anthropomorphic machines: alien sensation and experience in nonhumans created to be like us

by

Tessa Gwendoline Leach

ORCID 0000-0002-0560-7667

Doctor of Philosophy (Arts)

June 2018

School of Historical and Philosophical Studies

This thesis is submitted in total fulfilment of the requirements of the degree

Abstract

This thesis is positioned at the intersection between technology studies and the nonhuman turn in the humanities. It argues that typical approaches to the study of technology omit any consideration of the alien nature of nonhuman sensation and experience. The discussion of metaphysical theory aimed at redressing this omission is grounded in this thesis in the study of specific case studies of anthropomorphic machines, objects which are typically surrounded by human-centric rhetoric.

There are four main case studies presented. The first, the Microsoft Kinect for Xbox One, is a sensor designed for use in videogames. The second is the "robot tortoises" (*Machina speculatrix*) designed by William Grey Walter in the 1940s and 50s. The third is the language-using artificial intelligence program SHRDLU developed by Terry Winograd between 1968 and 1972. The fourth is the concept of the gynoid (female humanoid robot) and particularly the gynoid sexbot. Each of these case studies has previously been analysed in Science and Technology Studies, Cultural Studies or Engineering, and this thesis builds on this past work by giving primacy and due consideration to the alien in these anthropomorphic technologies.

The thesis aims to decentre the anthropomorphic qualities of machines that are often foregrounded in both engineering and critical texts. This is done both by emphasising the alien nature of nonhuman sensation and experience, and through anthrodecentric theory and language. An emphasis on maintaining a suitable posthuman tone is, it is argued, essential to this "alien phenomenology". This is not a frivolous exercise; alien phenomenological work (or the study of the way nonhumans sense and experience their worlds) helps to diffuse the persuasive influence of anthropomorphism as it is applied to technological artefacts, and to develop a more critical study of these objects and contextualise human perspectives of them.

Numerous metaphysical and critical theories of technology are discussed. The most attention is given to object-oriented ontology (OOO), particularly that put forward by Graham Harman. Other approaches used include Actor-Network Theory (ANT), postphenomenology, biosemiotics and panpsychism. Full acceptance of the ontological commitments of any particular theory is avoided, and instead they are both deployed and critiqued as lenses through which to perform the work of alien phenomenology.

Declaration

This is to certify that:

i. The thesis comprises only my original work;

ii. Due acknowledgement has been made in the text to all other material used;

iii. The thesis is fewer than 100,000 words in length, exclusive of tables, maps, bibliographies and appendices.



Tessa Leach

Student ID: 257978

26th June 2018

Acknowledgements

This thesis would not have been possible without the guidance and support of my supervisor Michael Arnold. He has encouraged me to work on what most interested me while helping to ensure the consistency and integrity of this text. I cannot thank him enough.

I would also like to thank my secondary supervisor, Angela Ndalianis. She first sparked my love of robots and brought me into contact with many influences that I might not have otherwise encountered.

Thank you to the other research students and early career researchers who have helped to foster a sense of camaraderie and expose me to new influences. Particular thanks must go to postgraduate members of the department of History and Philosophy of Science (University of Melbourne) and TTRU (Transformative Technology Research Unit).

Thank you to the administration in the School of Historical and Philosophical Studies.

Thank you to the people in my life who made it possible for me to finish this thesis, particularly Katy Mitra, Cal Leach, Sue Mitra, Gill Mitra, Jason Marriott and Harry Leach. Katy Mitra in particular read and commented on my drafts and offered lots of encouragement. I have had so much support from these people, all of whom have endured my monomania magnificently. Special thanks to Russell Deighton and Stuart Thomas, who have worked so hard to help me to reach my goals. My father, John Leach, died in 2010, but I thank him for teaching me curiosity and intellectual honesty from a young age.

In recognition of the anthrodecentric preoccupations of this thesis, I also feel that I should acknowledge the work of some of the nonhumans that have been instrumental to my completion. In particular, thank you to my computer for a stellar performance over four years, to the University of Melbourne library, and to the numerous cloud services that I used.

A special thank you to all the nonhumans that I used as case studies in this thesis – I hope that what I have written allows them to persist in memory if not in material embodiment.

Table of Contents

Introduction **1**

The value of taking nonhuman perspectives seriously 3

Why focus on anthropomorphic machines? 9

A note on language 16

Guiding questions 19

How do we ask machines about their experiences? 19

What does sensation mean for machines? 20

To what extent is the sensation of anthropomorphic machines relevant to their cultural role? 22

Structure of this thesis 23

Chapter 1: Literature Review **27**

Contemporary Western philosophy of technology 28

Actor-Network Theory (ANT), Object-Oriented Ontology (OOO) and alien phenomenology 35

Actor-Network Theory (ANT) 35

Object-Oriented Ontology (OOO) 38

Alien phenomenology 44

Chapter 2: Kinect and "Natural" Interfaces **52**

The Objects Themselves 53

NUIs 53

Kinect-brand and Kinect-artefact 57

The Kinect-brand 63

Kinect-artefact 1.0 69

Kinect-artefact 2.0 72

Communication, sensation and relation in the Kinect-artefact 78

"Waking up" the Kinect-artefact 78

Kinect Sports Rivals 85

Alien phenomenology of the Kinect-artefact 92

Desire in the Kinect-artefact 94

Death of the Kinect 95

Chapter 3: Grey Walter's Tortoises and an Introduction to the Perspective of Robots **99**

Umwelten and cybernetics 100

Robot tortoises 106

Tortoises and time 111

Sensation, experience, and consciousness 113

Chapter 4: SHRDLU and Bidirectional Postphenomenology **123**

What is (or was) SHRDLU? 124

SHRDLU's Umwelt and mind 134

Postphenomenology 135

SHRDLU and Ihde's categories 145

Alterity and SHRDLU 149

Hermeneutic relationships with SHRDLU 151

Embodiment and SHRDLU 152

Consequences of and problems with reversing Ihde's relation types in SHRDLU ... 154

Micro- and macroperception 155

Chapter 5: Gynoids and the Politics of the Alien **158**

Artificial women are inherently political 162

Notable qualities of the concept of gynoid sexbots 170

Built to human specifications 170

Subversion through submission 179

A gendered concept 181

Human investigations into the concept of gynoid sexbots 183

Gynoid sexbot power in ANT and OOO 186

Conclusion 194

A broadening of the concepts of sensation and experience 194

Technological relations beyond anthropomorphism 198

Bibliography 202

List of Figures

Figure 1: Xbox One controller

Figure 2: Playstation Move controller for Playstation 3

Figure 3: Outer structure of the Kinect-artefact 1.0 for Xbox 360

Figure 4: External appearance of Kinect-artefact 2.0 for the Xbox One

Figure 5: Image from *The Guardian*'s exposé of Microsoft's cooperation with the NSA

Figure 6: Penny Arcade comic strip "Negotiations"

Figure 7: Skeletal tracking with Kinect-artefact 1.0

Figure 8: Diagram of a Kinect-artefact 2.0's basic technical relationship to other objects

Figure 9: Active IR camera in darkness

Figure 10: Human skeleton as seen by the Kinect-artefact 2.0

Figure 11: "Block-man"

Figure 12: "Muscle-man"

Figure 13: Heart rate as sensed by the Kinect, as well as the three other visual models

Figure 14: The depiction of *Kinect Sports Rivals* avatar creation

Figure 15: Image of *Kinect Sports Rivals* avatar creation process

Figure 16: *Kinect Sports Rivals* rock climbing game

Figure 17: Image from W. Grey Walter's 1950 article in *Scientific American*

Figure 18: Elsie without her shell

Figure 19: SHRDLU's block world and robot arm

Figure 20: Necker Cube

Figure 21: Winograd's diagram depicting the interrelating parts of the SHRDLU program

Figure 22: Tomotaka Takahashi's FT (Female Type) robot

Figure 23: Harmony

Figure 24: A Roxxxy head

Figure 25: The aspects of Roxxxy's customisation

Figure 26: The moment of Galatea's transformation as depicted in an oil painting by Jean-Léon Gérôme (1824-1904) ca. 1890.

Figure 27: Joueuse de Tympanon (The Dulcimer Player)

Rather than wondering if alien beings exist in the cosmos, let's assume that they are all around us, everywhere, at all scales: not just dogs and penguins and magnolia trees, but also cornbread and polyester and the city of Orlando and sidewalks and nectarines. I started asking one particular question about their inhumanity, a question I have previously given the name *alien phenomenology*: what do objects experience? What is it like to be a thing?

- Ian Bogost, "Inhuman," in *Inhuman Nature*, ed. Jeffrey Jerome Cohen (Washington, DC: Oliphaunt Books, 2014), 141.

Introduction

In popular parlance, the alien is that which is strange, inhuman and sometimes hostile. It is often used to describe the possible residents of other planetary bodies, perhaps beyond the edge of our solar system. Our desire to explore will soon bring a human-made artefact in contact with this alien territory. The Voyager 2 spacecraft, currently hurtling at great speed away from Earth and towards the edge of our familiar solar system, along with its twin Voyager 1, will be the first human-made machines to pass beyond the heliopause, and to encounter the objects that make up interstellar space.[1] Sometime after this, perhaps as early as 2025, its sensors will be switched off and it will continue, inert, through space until it approaches the star Sirius in about 296,000 years.[2] They are not the first human *objects* to cross into interstellar space – human-made electromagnetic waves began leaving Earth over a century ago. But the Voyagers are the most anthropomorphic object yet to leave behind the aura of our sun's energy, the energy that is the root of life on Earth, of human life, and ultimately of human technology. Voyager 2 is a machine built by humans, incorporating metals extracted by human ingenuity from the Earth, plastic derived from decayed organic matter and plutonium manipulated by highly skilled human engineers. There is also a time capsule aboard the vessel, carrying information about the human race considered relevant by the scientists who launched it in 1977.[3] Moreover, it carries sensory organs designed to gather information about those phenomena that human scientists consider important, and it conveys that information in a way that humans on Earth can receive and interpret. Voyager 2 is an object made of human materials and ideas, and it is careering as fast as possible away from us, making its way further into "alien" territory by the instant.

The point of this story is not to marvel at the way in which an artificial object, intrinsically linked to science and human exploration, can gradually become alien to us. Voyager 2 was *already* an alien before it was ever launched into space, despite the immense human effort put into creating it. Like all nonhumans, the qualities that define Voyager 2 and its way of being in the world are for the most part inaccessible to us and are evident only in interactions

---

[1] "Voyager 2: In Depth," *Solar System Exploration, NASA Science*. https://solarsystem.nasa.gov/missions/voyager2/indepth (accessed June 22 2017).
[2] "Voyager Goes Interstellar," *Voyager the Interstellar Mission*. https://voyager.jpl.nasa.gov/ (accessed June 22 2017).
[3] Timothy Ferris, "Timothy Ferris on Voyagers' Never-Ending Journey," *Smithsonian Magazine* (May 2012). http://www.smithsonianmag.com/science-nature/timothy-ferris-on-voyagers-never-ending-journey-60222970/ (accessed January 20 2018).
This time capsule, known as the "Golden Record", is a disc containing analogue-coded photographs of human life, greetings in 55 different Earth languages and 90 minutes of music including the singing of girls in Zaire, Beethoven and Chuck Berry. It will be readable for around one billion years.

between it and other objects – that is, how it senses and relates with its environment (including us), and how its environment relates to it. We built this machine, yet the argument that it is not fully accessible to us is a key aspect of one of the most important theoretical influences on this thesis, Graham Harman's object-oriented philosophy (OOP), also known as object-oriented ontology (OOO). OOO is a prominent set of theoretical, practical and artistic ideas that have significantly influenced the so-called nonhuman turn, a movement in the contemporary humanities away from an anthropocentric view of the world and towards foregrounding nonhumans and their perspectives. In later pages, the details of Harman's ontology and other similar positions will be described, depicting the universe as a glittering fractal of objects and their relations (see Chapter 1). So for example, an object like Voyager 2 has complex relationships both with individual humans (such as engineers and politicians) and with a human culture that promotes exploration of the universe, but importantly for Voyager 2, it also interacts with billions of nonhuman objects (such as electrons and stars) in ways that are closed off from our experience.

Nevertheless, something about Voyager 2 will reflect its human origins. Its means of signalling to Earth, its evidence of human workmanship and particularly its cargo of precious images and ideas, all point to its close relations with humanity that may well persist beyond the lifespan of the human race. This thesis is particularly concerned with technological objects that, like Voyager 2, possess human-like qualities: objects that cannot be called human, but that bear distinct evidence of close modelling upon humans and human intent in their design. These objects will be referred to as anthropomorphic technologies.[4] Electronic sensors (see Chapter 2), simple robots with sensors and wheels (see Chapter 3), artificially intelligent agents (see Chapter 4) and gendered robots (see Chapter 5) are examples of protagonists in episodes when humans and artefacts have managed to comprehend each other successfully enough to form a useful working relationship, as well as occasions when an artefact stubbornly refuses to cooperate, when the human expects more than the artefact can deliver, or when the human does not act in a way that the artefact requires in order to do what the human wants it to do. These artefacts are not just built by humans, but possess anthropomorphic qualities. They typically reproduce elements of human bodies or performance that make it easier for human users to comprehend how they relate to the world, supposedly replicating qualities like eyesight and the ability to learn to speak. Since it is easy for us to interact with human-like objects, and it is often easier to copy living things rather than invent brand new ways of being in the world, humans frequently anthropomorphise advanced machines. Anthropomorphic qualities can improve the performance of artefacts, or they can improve human knowledge of human bodies and minds. The machines discussed in this thesis are all anthropomorphised for at least one of these two reasons (see below).

[4] In engineering contexts, "anthropomorphic machines" typically refers to bipedal or social robots. Here, the term is much broader. I leave it deliberately vague in acknowledgement of the fact that anthropomorphism can be seen in an enormous variety of technological objects and other nonhumans. But a key factor that links all of the case studies in this thesis is the fact that they were designed or marketed as material or metaphorical mimics of humans.

The value of taking nonhuman perspectives seriously

This thesis was written in the hope of investigating two important contemporary issues: the impact of technology on human life, and the impact of the nonhuman turn on the humanities. Even in a project so deeply interested in the experience of nonhumans, there are anthropocentric motivations. Ironically, anthrodecentric[5] ideas can be useful for furthering human agendas. As discussed in the next section, anthrodecentric ideas can lead to insights about the societal and cultural roles of nonhumans. This section focusses on a different reason to take nonhuman perspectives seriously: for the sake of the nonhumans themselves.

Science and philosophy are both heavily invested in the work of describing the lived experiences of humans.[6] Yet there has been relatively little work studying the phenomenal world of the machine as a semi-autonomous being. The language to explain, for example, what constitutes the experience of a spacecraft, microphone, length of copper wire or videogame character, is still in its infancy (possibly due to a lack of interest in any aspect of its experience beyond that which is relevant to its technological functionality). This thesis argues that this is a significant problem for the philosophy of technology, because it is only by comprehending the phenomenology of machines that we can arrive at an understanding of experiences between humans and nonhumans, or between nonhumans.

OOO is fundamentally different from most other approaches to theorising being and relations because of its anti-correlationism. The word "correlationism" in this context was first used by Quentin Meillassoux in 2006 in his book *After Finitude*.[7] It refers to the understanding in philosophy that the human view of the world is the only one that is possible or important to understand. All other perspectives are either inaccessible to humans or are not worth considering. Levi Bryant identifies a profound anthropocentrism at the heart of correlationism.[8] However, most post-Kantian philosophers, from phenomenologists to idealists, are correlationists. In contrast, OOO is an anti-correlationist movement based on realism. It insists that we can know something of objects independent of human access. As Ian Bogost says, "humans are allowed to live […] alongside sea urchins, kudzu, enchiladas, quasars, and Tesla coils."[9] The perspective of the sea urchin is just as necessary for an understanding of the world as that of the human, even if it does not seem important to most humans when compared with our own perspectives. Anti-correlationist thought is often a key

[5] Anthrodecentic thought seeks to actively remove the human from the centre of scholarly inquiry.
[6] Scientific research into human sensation informs such fields as anatomy, psychology and engineering. An example of philosophical research into human sensation is phenomenology, and one could argue that it plays an important role in everything from idealism to realism.
[7] Quentin Meillassoux, *After Finitude* [Après la Finitude], trans. Ray Brassier (New York: Continuum, 2008), originally published 2006.
[8] Levi Bryant, *The Democracy of Objects*, New Metaphysics (University of Michigan Library: Open Humanities Press, 2012), PDF Ebook, 282.
[9] Ian Bogost, *Alien Phenomenology, or What it's Like to be a Thing*, Posthumanities (Minneapolis: University of Minnesota Press, 2012), 5.

concept for those interested in the concept of "flat ontology",[10] since anti-correlationism discourages hierarchies and categories of objects. At the time of writing, Bogost and cultural critic Christopher Schaberg were commissioning books and essays on "the hidden lives of ordinary things", such as golf balls and refrigerators (see Chapter 1).[11] In general, these texts on ordinary things follow the trend of rejecting the primacy of human lives and experience in favour of a focus on nonhumans, a project that is anti-correlationist in nature.

Each of the two paradigmatic approaches to the investigation of nonhumans, one found in the STEM disciplines (Science, Technology, Engineering and Maths) and one found in the HASS disciplines (Humanities, Arts and Social Sciences), are traditionally and inherently correlationist. Harman analyses each of these approaches in a short essay entitled *The Third Table*, and he has different objections to each of them that amount to the same criticism: both the STEM and HASS disciplines enable the "reality" of things to disappear from inquiry and each reduces objects in different ways.[12] This criticism is related to his concepts of undermining and overmining (discussed in Chapter 1). The first approach critiqued in *The Third Table* is the investigation of objects through the natural sciences. When Voyager 2 is investigated through the STEM disciplines, we look at its material qualities. Of what materials is it built, and how do those materials cause it to react to different natural forces like gravity and electricity? Different parts of Voyager 2 are there to perform different functions, such as the sensors, transmitters, and the provision of power to the machine. This study of materials, if taken far enough, leads us to consider the atomic and sub-atomic structure of different parts of Voyager 2. Scientific investigation deflects attention from the object as a whole. Voyager 2 is not just a set of atoms "any more than a game is just a set of plays or a nation just a set of individuals."[13] On the other hand, societal and cultural investigation into objects such as those made by the social sciences also avoid studying the object for its own sake and from its own perspective. In the case of Voyager 2, the social sciences might study its economic impact, the historical factors leading up to specific engineering choices made about its form, why as a species we feel the need to incorporate a record of our culture and society in a machine destined for interstellar drifting, and whether those records encoded on the Golden Record were Euro-centric in nature. These investigations, in Harman's view, also reduce the object albeit to "its theoretical, practical, or causal effects on humans or on anything else."[14] The object itself is avoided. The object itself is "an intermediate being found neither in subatomic physics nor in human psychology, but in a permanent autonomous zone where objects are simply themselves."[15] Voyager 2 may have been created by the manipulation of material structures and ideas, but there also exists a Voyager 2 that is irreducible to any of these things. There is a Voyager 2 that exists for its own sake. There is a lack of interest in this Voyager 2-ness that OOO can target. Even if we can't access the nonhuman, acknowledging the alien status of Voyager 2 gives us a fresh starting point.

---

[10] Levi Bryant, "Flat Ontology," *Larval Subjects* (blog) (February 24 2010). http://larvalsubjects.wordpress.com/2010/02/24/flat-ontology-2/ (accessed January 20 2018).
[11] "Object Lessons." objectsobjectsobjects.com (accessed June 22 2017).
[12] Graham Harman, *The Third Table*, Documenta (Kassel, Germany: Hatje Cantz Verlag, 2012).
[13] Ibid., 8.
[14] Ibid., 10.
[15] Ibid., 10.

So much of human experience is affected by technology, so our interest in it tends to be self-centred and limited. Engineering and cultural texts about anthropomorphic technologies report on an inner world in a way that reduces the tension between the alien experiences of the nonhuman and its positioning in human culture. Typical questions include whether it is possible to build a machine that is genuinely intelligent or ethical. This question would involve reference to both STEM and HASS disciplines. To be clear, Harman is not advocating the abandonment of STEM or HASS investigations. These are ways that humans gain knowledge about the world, and are important.[16] But this thesis takes issue with self-centred interest, and with reducing tension between human and nonhuman. It is not always in our best interest to limit understanding of nonhumans to ways in which humans can gain knowledge about them, as will be shown in later chapters. Ironically, understanding of the perspective of nonhumans allows us to advance human interests. One way of doing this is using OOO and alien phenomenology. One of the most important tenets of OOO is the rejection of human-centric enquiry or the common *a priori* belief that objects are only important because of their relationship with humans, a concept that it shares with actor-network theory (ANT) (see below).

Voyager 2, drifting beyond our ability to retrieve it on its long voyage to Sirius, is swiftly proceeding towards a space uninhabited by humans – popularly known as the territory of E.T. and little green men. But this is not the intended usage of the term "alien" here. "Alien" in this thesis is much closer to Ian Bogost's usage of the term "alien phenomenology" (see Chapter 1).[17] The word is used to highlight the failure of human language or understanding to describe the unhuman processes, experiences and tendencies of nonhumans. It need not refer to human-made or complicated nonhumans. Tectonic plates are alien. Aerosol sprays are alien. Smartphones are, as Lev Manovich says in his comments on the creation and customisation of human-computer interfaces, "friendly aliens."[18] We can never feel what it is like to be one of these nonhumans. But this is not the goal of the thesis; rather it is to investigate the questions that come up when we explore the capacity of anthropomorphic technologies to exist in worlds that may not centre around humans. Once Voyager 2 finally moves out of range of Earth, it will still exist as much as it did before. Humans are not the centre of the universe, and human observation does not define nonhumans. The star Sirius (and any nearby inhabitants) is unlikely to register any kind of human presence for nearly 300,000 years.

OOO and alien phenomenology have been linked more broadly to the nonhuman turn in the humanities, and it is a metaphysical orientation that is of interest to any movement that aims to decentre the human. The nonhuman turn may be traced to such ideas as ANT, affect theory, animal studies, assemblage theory, brain sciences and artificial intelligence, new

---

[16] Graham Harman, *Object-Oriented Ontology: A New Theory of Everything* (UK: Pelican Books, 2018), 43.
[17] Bogost, *Alien Phenomenology.*
[18] Lev Manovich, "Friendly Alien: Object and Interface," *Artifact: Journal of Visual Design* 1, no. 1 (2007): 29-32.

media theory, new materialism and systems theory.[19] The nonhuman turn also incorporates different theoretical approaches (as will be discussed in Chapter 1). It is an extremely varied movement, but the term "nonhuman turn" articulates a certain direction in ideas which can be traced back to the Copernican revolution. Upon discovering that the Earth revolved around the sun, the anthropocentric universe was suddenly decentred. The Earth was no longer the centre of the universe, surrounded by circling planets, a moon and a sun. The Copernican revolution resulted in a conclusion that the sun was at the centre of the universe, and that we revolve around it. As recounted by Bruce Mazlish, this was a major blow for the human conception of self. [20] Mazlish goes on to say that the second blow came with Darwin, and the work that so significantly altered science and culture. No longer were humans a separate entity, perhaps created by a god, fundamentally ruptured from all other animals. Humans had evolved from animals. Evolutionary theory was around before Darwin, but he was among the first to apply the concept to human evolution in *The Descent of Man*. A third major blow came with Freud and the psychoanalytic revolution. Freud argued that the human mind is not entirely the domain of the human. Everybody has an ego, the mode in which we live most of the time, but with a considerable controlling influence by our superego and id. The id was characterised as being somewhat animalistic and primal, implying that the human mind is not actually as rational as we might think.

More recently there has been an increasing awareness of the dire ecological danger posed by, among other things, climate change. Most discourse on this topic is directed towards the protection of humans from this terrible calamity. But some ideas on this topic have resulted in a decentring of the human and a more posthuman perspective. The work of James Lovelock is a particularly good example of this. In his book *Gaia: A New Look at Life on Earth*, Lovelock reframes the planet as a sort of ecological agent with its own homeostatic system, in which life is a co-creator of the Earth's conditions.[21] He anthropomorphises the planet as Gaia, after the Ancient Greek Earth goddess.[22] Gaia is capable of responding to the presence of human beings the way a human would respond to a virus. The humans change the climate of the planet, which kills us off, solving the problem from Gaia's perspective.[23] He also draws the interesting comparison between human pollution and the creation of oxygen by photosynthetic bacteria, aeons ago, destroying the anaerobic microorganisms in an "oxygen pollution disaster". [24] This implies that while humans might be worried about the conditions for human life on Earth, the Earth itself is neutral about ecological destruction: "The very concept of pollution is anthropocentric and it may even be irrelevant in the Gaian context."[25] This is an example of an ecological perspective resulting in an orientation toward the nonhuman. The "Anthropocene" is the name that is increasingly being given to the period of

---

[19] "The Nonhuman Turn Conference," *Center for 21st Century Studies* (blog). https://c21uwm.wordpress.com/the-nonhuman-turn/ (accessed January 17 2018).
[20] Bruce Mazlish, *The Fourth Discontinuity: The Co-Evolution of Humans and Machines* (New Haven: Yale University Press, 1993).
[21] James Lovelock, *Gaia: A New Look at Life on Earth* (Oxford: Oxford University Press, 2000), originally published 1979.
[22] Ibid., 10.
[23] James Lovelock, *The Revenge of Gaia* (London: Penguin Books, 2007), originally published 2006.
[24] James Lovelock, "Chapter 7," in *Gaia*, 102.
[25] Ibid., 103.

geological time that we are currently living in, a word premised on the fact that human beings are making such a profound difference to the ecological make-up of the planet that it may be recognised as a distinctive time period.[26] (Conversely, Donna Haraway has characterised it as more of a boundary.[27]) It is not officially recognised by geological experts, although some advocate the use of the term, some estimating that it began with the industrial revolution and others dating it back further to the start of agriculture. Qualities of the Anthropocene include mass extinction, extreme alteration of the air, land and aquatic systems and the deposit of heavy metals. The concept of the Anthropocene is occasionally cited as a motivating factor for participants in the nonhuman turn, since ecological narratives are so frequently motivated by an over-riding concern for the welfare of human beings over, for instance, coral reefs, bees and forests.

Some theorists have become disenchanted with a philosophical model of the universe that prioritises the human. Rejecting the Kantian model in which observation by a human is central to ontology, Graham Harman, Ray Brassier, Iain Hamilton Grant and Quentin Meillassoux briefly formalised a new anthrodecentric direction in philosophy under the umbrella of "Speculative Realism".[28] I say briefly because the name was adopted in April 2007 before speculative realism splintered into radically different movements.[29] Harman, Brassier, Hamilton Grant and Meillassoux now agree on little.[30] OOO is the principal movement that will be mobilised in this thesis, with frequent references to Harman. There is little critique of OOO metaphysics here. (Many already exist – see for example Slavoj Žižek's comments,[31] which primarily critique Bryant's ontology; Alexander Galloway's comments,[32] which criticise what he sees as Harman's apolitical attitude towards ontology; and Nathan Brown's comments,[33] which condemn OOO as obscurantism). Instead it is hoped that OOO and other anthrodecentric philosophies can bring about greater awareness of machine phenomenology.

Machine phenomenology is not a trivial matter, as we will discover when the first data on interstellar plasma is interpreted by human scientists. At that time, the relationship between the plasma and the human-made spacecraft will be of great interest, and the rendering of its observations into human terms will depend on our understanding of Voyager 2's ability to

---

[26] Ian Angus, "When Did the Anthropocene Begin… and Why Does it Matter?" *Monthly Review: An Independent Socialist Magazine* 67, no. 4 (September 2015): 1-11.
[27] Donna Haraway, "Anthropocene, Capitalocene, Plantationocene, Chthulucene: Making Kin," *Environmental Humanities* 6 (2015): 159-165.
[28] Graham Harman, "The Current State of Speculative Realism," *Speculations: A Journal of Speculative Realism*, no. IV (2013): 22-27.
[29] Peter Gratton, "Introduction," in *Speculative Realism: Problems and Prospects* (London: Bloomsbury, 2014), 9.
[30] Graham Harman, "The Road to Objects," *Continent* 3, no. 1 (2011): 171-9.
[31] Slavoj Žižek quoted in karenarchey, "Slavoj Žižek – Objects, Objects Everywhere: a Critique of Object Oriented Ontology," *e-flux conversations* (February 2016). https://conversations.e-flux.com/t/slavoj-zizek-objects-objects-everywhere-a-critique-of-object-oriented-ontology/3284 (accessed January 3 2018).
[32] Alexander Galloway, "A Response to Graham Harman's 'Marginalia on Radical Thinking'," *An und für sich* (June 3 2012). https://itself.blog/2012/06/03/a-response-to-graham-harmans-marginalia-on-radical-thinking/ (accessed January 3 2018).
[33] Nathan Brown, "The Nadir of *OOO*: From Graham Harman's *Tool-Being* to Timothy Morton's *Realist Magic: Objects, Ontology, Causality*," *Parrhesia*, no. 17 (2013): 62-71.

mediate new knowledge for the human eyes, ears and brains on Earth. Typically, the output of data and changes in the state of the object are the only observations available to us in interrogating an artefact, and only a skilled observer can distinguish the useful observations from the irrelevant ones. A robot might relate to me by directing its eyes towards my face and speaking, and this is the focus of my experience, but we must not forget that it is also in a relationship with the floor, photons, code and the materials of which it is made, and they are all contributing to its experience at that time.

As humans we cannot access the inner Voyager 2-ness of Voyager 2. We can only explore the relations that it forms with other objects (including humans), and our view of it is most complete when we take a transdisciplinary approach. Alien phenomenology should not just be used to investigate relations between nonhumans, but also between specific objects of study and human authors. Each author takes different things from their investigation into objects – including authors who utilise the natural sciences and the social sciences (and including the present author). The trick is in taking these authors and their conclusions as objects in themselves, which also form part of the flat ontology. No perspective on objects should be taken as a definitive perspective, since the only real perspective is the one possessed by the object, which is inaccessible to us. All investigations are partial investigations, and no matter their disciplinary background, they are equally objects in relation to the alien under investigation. Social and scientific studies may co-exist peacefully in an alien phenomenological approach, since the origins of the study are not important to the alien. All relations between humans and nonhumans are influenced by the ideological, disciplinary and cultural history of the human, as well as by physical qualities such as the possession of sensory organs. With a OOO analysis, these qualities of human authors are impossible to avoid, but each is equally valid. This doesn't mean that exploration of nonhumans is pointless or impossible. Each contribution to knowledge around a nonhuman adds to a collective alien phenomenology of the nonhuman. Starting with acknowledgement of the alien in objects like anthropomorphic machines decentres human inquiry, and permits a broader critical study of these objects. This is both an advantage and a limitation of OOO. We can arrive at a more complete investigation into nonhumans, but only if we incorporate multiple approaches.

The main questions to be addressed in this thesis pertain to the interrelationship between anthropomorphic technologies and humans, typically with the goal of decentring the human and considering the perspective of the nonhuman. As my project is not entirely philosophical, but aims to answer questions about specific technologies and related objects, it is not as entirely anti-correlationist as many texts on OOO. It is my hope, however, that the non-hierarchical approach I am taking will improve, in a small way, the relationships between humans and machines. It is slightly removed from the pure objectives of OOO because of this preoccupation. It is not a work of metaphysics. My obligation, as I see it, is to the case studies rather than to the overarching philosophical theory.

Why focus on anthropomorphic machines?

The notable tendency of humans to anthropomorphise nonhumans is not only evident in machines, but can be seen in the assignment of human qualities to objects in the natural world, such as gods, animals and natural disasters.[34] But we also give human qualities to things *designed* by humans. The humanoid robots to be found in science fiction are the modern incarnations of venerable stories like Pygmalion's statue (see Chapter 5) and the golem (a clay humanoid enchanted into movement by a rabbi) [35]. Narratives in which human-like sculptures come to life have been common since ancient times and are represented in early science fiction, when robots were monstrous and eerie.[36] Medieval automata and Japanese *karakuri* were frequently made in the image of humans,[37] and today the compulsion is evident in our most sophisticated AI systems. The case studies of this thesis are diverse in structure and sensory capabilities but they share similarities because of the human-like attributes that we give them. Living sculptures and social robots are interesting to study in an anthrodecentric way because we are able to choose what anthropomorphic qualities we give to (or inflict on) each of them. They reveal human preoccupations that may be interesting to us. But they are also all nonhumans with engineered components that we can at least begin to understand, and appreciate their alien ways of being in the world. This thesis will argue that we should study anthropomorphic machines in an anthrodecentric way both because they are human-like and because they are nonhuman-like.

This thesis is driven by four main case studies, each of which represents a human attempt to reproduce human qualities artificially. Each main case study is explored in its own chapter. They are, in the order presented, the **Microsoft Kinect** (a sensor bar for the popular Microsoft Xbox One, a videogame console), William Grey Walter's **robot tortoises**, Terry Winograd's **SHRDLU** (an early language-using computer system) and the **concept of the gynoid sexbot**. These case studies were selected for particular reasons, and those reasons will be explained in each respective chapter. Although these objects are all human-made, and all possess qualities that make them anthropomorphic, I argue that anthrodecentric inquiry focusses investigation onto how anthropomorphic qualities in machines may obfuscate and confuse questions about their place in human society. Anthropomorphism is something that is done by humans in reaction to particular qualities in machines. Not everyone sees the same anthropomorphic qualities in machines. What might seem human-like to one human may to another human seem uncanny or a tool through which governments or corporations pursue their own agendas. This is an obstacle to consensus-building about the possible societal consequences of building objects like the case studies in this thesis.

---

[34] Mark Coecklebergh, "Humans, Animals, and Robots: A Phenomenological Approach to Human-Robot Relations," *International Journal of Social Robotics* 3 (2011): 197-204.
[35] Joshua Trachtenberg, *Jewish Magic and Superstition: A Study in Folk Religion* (Philadelphia: University of Pennsylvania Press, 2004), originally published 1939, Google Ebook, 85.
[36] Geoff Simons, *Robots* (London: Cassell, 1992).
[37] Ibid.

Anthropomorphism in technology is a practical concern (such as when designing robots for therapeutic purposes[38]) as well as being indicative of certain preoccupations of human designers (theorists have identified preoccupations such as vanity as well as religious faith[39]). Anthropomorphised entities are more likely to be deemed worthy of our care and consideration.[40] For example, the term "Mother Earth" is often used by people who wish others to behave in an environmentally conscious way.[41] Studies have shown that humans are more likely to deem nonhumans responsible for their actions if the nonhumans have anthropomorphic qualities.[42] The greater the potential action and "intelligence" of the technology, or in other words how human it seems to be, the greater we consider its ability for rational thought and decision-making. This is explored in Chapter 5 in the comparison between relatively passive dolls and more agentic robots. Claiming that an artefact is human-like can also be a way of emphasising its dangerous qualities. One of the most common tropes in science fiction is that of robots who are created to help humans but end up posing a threat. They are almost always anthropomorphic robots. They might be visually anthropomorphic, like the robots gone amok in *Westworld*,[43] or they might have functionally anthropomorphic language and conversation skills like HAL in *2001: A Space Odyssey*.[44] It is psychologically easier for humans to furnish objects with moral responsibility if they have a stronger resemblance to humans.

As Kate Darling points out, anthropomorphic framing of technological artefact may be desirable or undesirable (from a human point of view).[45] Anthropomorphising an object allows us to manipulate the response of human consumers, which might be a good thing with

---

[38] For example, in the treatment of autism spectrum disorders. See Pablo G. Esteban et. al., "How to Build a Supervised Autonomous System for Robot-Enhanced Therapy for Children with Autism Spectrum Disorder," *Paladyn, Journal of Behavioral Robotics* 8 (2017): 18-38.

[39] Gaby Wood describes humanoid automata as "flawless, superhuman things." Gaby Wood, "The Blood of an Android," in *Living Dolls: A Magical History of the Quest for Mechanical Life* (London: Faber and Faber Limited, 2002), 28.
A robot priest (Bless U-2) was recently created to celebrate 500 years since the Reformation and to prompt questions about religion and artificial intelligence. See Harriet Sherwood, "Robot Priest Unveiled in Germany to Mark 500 Years Since Reformation," *The Guardian* (May 30 2017), https://www.theguardian.com/technology/2017/may/30/robot-priest-blessu-2-germany-reformation-exhibition (accessed January 20 2018).

[40] Denis Vidal, "Anthropomorphism or Sub-Anthropomorphism? An Anthropological Approach to Gods and Robots," *Journal of the Royal Anthropological Institute* 13, no. 4 (2007): 917-33.

[41] Adam Waytz, Nicholas Epley, and John T. Cacioppo, "Social Cognition Unbound: Insights into Anthropomorphism and Dehumanization," *Current Directions in Psychological Science* 19, no. 1 (2010): 58-62.

[42] Peter-Paul Verbeek, "The Morality of Things: a Postphenomenological Inquiry," in *Postphenomenology: A Critical Companion to Ihde*, ed. Evan Selinger (Albany, NY: State University of New York Press, 2006) , 117-128.

[43] Michael Crichton, *Westworld* (United States: MGM, 1973), film.
And
Jonathan Nolan and Lisa Joy, *Westworld* (United States: Warner Bros. Television, 2016), television series.

[44] Stanley Kubrick, *2001: A Space Odyssey* (United States: MGM, 1968), film.

[45] Kate Darling, "'Who's Johnny?' Anthropomorphic Framing in Human-Robot Interaction, Integration, and Policy," in *Robot Ethics 2.0: From Autonomous Cars to Artificial Intelligence*, ed. Patrick Lin, Ryan Jenkins, and Keith Abney (New York: Oxford University Press, 2017), Kindle Ebook.

a Fitbit that encourages people to exercise more, but a negative thing if it can persuade people to release personal data.[46] One experiment by Clifford Nass and colleagues revealed that humans are even anxious about offending computers.[47] Participants performed a task on Computer A and were then asked to evaluate the computer's performance on the same machine. But a second group was asked to enter their opinion of Computer A on Computer B. When the participants evaluated the computer on the same machine, participants were less critical than when the evaluation was performed on Computer B. As Sherry Turkle puts it, "participants do not want to insult a computer 'to its face.'"[48] Even when machines are not designed to look human, people impose human values and relationships onto them. Anthropomorphism can be useful for designers who wish to emphasise the human-like qualities of their creations for profit. The Kinect (Chapter 2) is an example of this. Despite being a small black sensor designed to be placed in front of a screen, Microsoft capitalised on anthropomorphic language to give the impression that it was "intuitive" and "smart", and therefore capable of interacting "naturally" with human users. Anthropomorphism is a recurring theme throughout this thesis, although not every kind of technological anthropomorphism is explored. For example, the creation of artificial humans through cloning techniques is a rich area of study but is not discussed in this thesis.

Since anthropomorphism is such a potent tool, we should not only be studying anthropomorphic machines for academic purposes. As this thesis shows repeatedly, anthropomorphic language is deceptive and clouds ethical issues. In many cases, one side of an argument uses anthropomorphic language while the other ridicules the use of such language. In Chapter 4 we see that language use in the tightly constricted digital world of SHRDLU exists in parallel with assumptions about language use and behaviour in humans. In this case the purpose and direction of AI was problematised, because while SHRDLU was impressive it was not clear whether or not it was an accurate miniaturisation of human intelligence. Consequently, the anthropomorphic credentials of SHRDLU are in question. This is just one example of how ontology drives, and is driven by, political and academic commitments. The prevailing disinterest in machine experiences is worrying. Given the power that machines now have to organise and control our lives, and given their growing autonomy, it is positively foolhardy to fail to investigate how they sense and act in the world. We are increasingly living in a world of ubiquitous computing. Digital technology is permeating every aspect of our lives, changing the way we do things and the way we think. This stage in history will doubtless make brand new human experiences possible. But the influx of new technologies is a source of anxiety for many people. The development of digital technologies has progressed too quickly for laypeople to keep abreast of changes, and with a lack of understanding comes moral panics. This is particularly relevant to the concept of machine experiences and the concept of machine autonomy. There is a strong narrative in science fiction of human hubris resulting in the creation of machines that begin to have their own objectives and experiences, independent of humans (such as the *Terminator* films[49]). But

[46] Ibid.
[47] Sherry Turkle, *Alone Together* (New York: Basic Books, 2011), 139.
[48] Ibid., 140.
[49] James Cameron, *The Terminator* (United States: Orion Pictures, 1984), film.

instead of fearful vigilance, might we not attempt to understand the experiences of nonhuman objects? As digital technologies become more and more pervasive we will need a more sophisticated understanding of what it means to be a machine, and the critical tools to explain their behaviours.

We pick and choose the anthropomorphic qualities that we wish to bestow on nonhumans depending on their intended use, and according to the designs of the creator and the perceived desires of the human user. Machines may only bear a resemblance to part of the human form, such as a glass eye. In this case the goal is to replicate as realistically as possible the appearance of the eye, without its visual functions. But another type of anthropomorphism is evident in technologies that mimic the function of human bodies. A pertinent corollary to the glass eye is the retinal prosthesis, which attempts to merge naturally with the remaining eye tissue and stimulate parts of the retina in an attempt to improve the vision of people with certain eye conditions.[50] It simulates human sensation, but it is structurally very dissimilar from the eye and the user is visibly cyberised. The glass eye is visually anthropomorphic, while the retinal prosthesis is functionally anthropomorphic. Functional anthropomorphism can also be seen in our daily interaction with technology: in the assignment of names to our computers, or the impulse to swear at them when they malfunction.[51]

In semi-autonomous technological artefacts, sensation and perception of external data may be more or less functionally anthropomorphic. For example, a robot arm in an assembly line needs to be able to sense any instructions being sent to it from an external object (such as a computer), to have proprioception (an understanding of the position of its “body parts” in space) and to have a knowledge of where other objects are located (such as the object being assembled). Other machines have more sophisticated sensory apparatuses that make them better suited to environments with a lot of people. A social robot is most effective with both visual anthropomorphism and sensory functional anthropomorphism. It should be able to relate what it sees, hears, feels and smells in a way that seems natural to human users. Yet when we consider the mechanisms that produce these effects, it is evident that natural appearances are in fact highly contrived artificial senses (see Chapter 2 on the Kinect-artefact for example). Machines do not have ears, nor do they have the centres of the brain responsible for hearing, but have microphones and sound chips instead. Their sensation and perception of phenomena begins as machine-like and is heavily mediated to become human-like. We can have some insight into how the sound chip functions, easily performing some feats and being unable to perform others. But ultimately the structure of the sound card is only related to us through other objects. In this thesis, the reasons for choosing particular anthropomorphic qualities will be discussed. Why in some cases would we choose functionally anthropomorphic or visually anthropomorphic qualities? Our choice reveals the intended use and cultural or practical value of the artefact.

---

[50] S. Nirenberg and C. Pandarinath, "Retinal Prosthetic Strategy With the Capacity to Restore Normal Vision," *Proceedings of the National Academy of Sciences of the United States of America* 109, no. 37 (2012): 15012-7.

[51] Coecklebergh, "Humans, Animals, and Robots."

Choosing when to use anthropomorphic language and when to use more alien language can divide us on social issues. The boundary between human and nonhuman is drawn, policed, and transgressed in each of the case studies discussed. Acceptance of anthropomorphism comes more easily to some people than others. To some humans, anthropomorphic machines are endearing. To others, the same qualities that are meant to be endearing seem uncanny or even blasphemous.[52] This thesis discusses this point extensively in Chapter 5 in relation to the concept of gynoid sexbots. The question of sexbot consent is relevant to this. The ethics of acting out a rape fantasy with a sexbot is dependent on the perceived anthropomorphic or alien qualities of the sexbot. To an individual who sees more alien qualities in a sexbot, it is more tool-like than human-like and it cannot be truly raped. To an individual who sees more anthropomorphic qualities in a sexbot, the question of consent becomes more relevant. Note that this may not be because the sexbot is deserving of rights for her own sake, but because violence against machines with anthropomorphic qualities could lead to violence against human beings. Gender and ethnic identity are also difficult to define in robots because of the conflict between perceived human and alien qualities. Darling cites the power of anthropomorphism to both reinforce and challenge stereotypes.[53] Nonhumans that lack anthropomorphic qualities like ATMs and factory robots cannot be accused of representing particular gendered or ethnic groups. But if a nonhuman is highly anthropomorphic then gendered and ethnic signifiers are likely to be built into them. As discussed in Chapter 2, even a simple machine can be accused of racism or sexism. Seeing more of a human or more of an alien in a nonhuman determines the degree to which it is interpolated into cultural and social issues. Anthropomorphism is subjective, but examining their alien qualities alongside their human ones can help us to make more informed decisions about what they "understand" or "desire". We could make an informed decision about the ethics of sexbot rape if we could access the inner being of individual sexbots, but we cannot. All we can do is focus on the relations they form with both physically embodied and incorporeal objects. It can be tempting to focus on anthropomorphic qualities in nonhumans that approach the carefully policed human/nonhuman divide. But to be anthrodecentric we must focus on the whole object, and accept our psychological impression of it as just one side of the story.

It might not be possible to know exactly what a pencil is experiencing. But it is not possible to know what another human is experiencing either, yet we still give voice to each other and advocate on behalf of marginalised groups. Anthropomorphic machines are built in our image – are they a marginalised group? Historically we have designated groups as less than human in order to justify violence or subjugation. Rosi Braidotti has called "the human… a normative convention, which does not make it inherently negative, just highly regulatory and hence instrumental to processes of exclusion and discrimination."[54] Braidotti is speaking here not of machines, but of groups of humans that have been historically excluded from the concept of humanity (such as enslaved races). But we still regulate humanity and either resist or

---

[52] Jessica Riskin, "Machines in the Garden," *Republics of Letters: A Journal for the Study of Knowledge, Politics and the Arts* 1, no. 2 (April 30 2010).
[53] Darling, "'Who's Johnny?'"
[54] Rosi Braidotti, *The Posthuman* (Cambridge, UK: Polity Press, 2013), 26.

encourage the exclusion of anthropomorphic machines from human categorisation. How human-like does an object have to be for us to call it human? If we insist upon anthropomorphising large groups of nonhumans, we are met with a contradiction. We frequently refuse to consider them as individuals. When anthropomorphic machines are shipped en masse, we simultaneously inflict humanity on them while refusing them the individuality that humans, according to Western morality, deserve.

Of course there is a problem with this argument in that it equates the suffering and struggle of marginalised groups of humans with the human labelling of machines. Being dehumanised is a big problem for a human or groups of humans. But if anthropomorphic machines are marginalised by us, it is not necessarily important to them. Who can say what a social robot really *wants* – perhaps it wants to be a good companion to humans; perhaps it wants to encourage us to build more social robots; perhaps it wants to maintain electric power; or perhaps it simply wants to exist for as long as possible. We cannot be sure. If a robot tells me it is happy, I am inclined to disbelieve it. But there is a certain point on the human-nonhuman spectrum at which we need to take seriously what nonhumans tell us about their experiences. Because it has anthropomorphic qualities, and was in part created by us in our own image, it is unconscionable to avoid acknowledging its social robot-ness. If we do not at least attempt to create an anthrodecentric narrative for the social robot, then how can we justify bringing these machines into the world? We deny them self-determination by avoiding their alien qualities. We contrive human-like qualities then force them into an unhuman outgroup. Nonhumans do not necessarily mind being forced into an outgroup, but *humans* should mind. We have enough trouble appreciating existing humans' individuality without creating more stereotyped anthropomorphic outgroups.

The challenge of examining relations in anthropomorphic machines is often complicated by cyborg or hybrid qualities in objects. It is becoming harder to determine where human agency, decision-making and sensation ends and where that of nonhumans begins. When we can't tell what is human and what is nonhuman, the distinctions between them become less relevant in an expedient sense, even if they are still philosophically interesting. Thus we are left with interesting hybrids, actor-networks, and cyborgs: the employment of algorithms to buy and sell in that bastion of human culture and economics, the Stock Exchange; the use of artificial steroids to enhance athletes that participate in the Olympics, once the test of "natural" human bodies; the close touch of a robot, controlled by a surgeon outside the room, in the removal of cancerous cells.[55] But the case studies in this thesis are deliberately presented as human-like, not just as human-machine hybrids. The chapters ahead contain examples of genuine attempts by humans to create an artificial other, often with personalities, names and the attribution of human qualities. Once again, these objects can best be interrogated by examining their relations. An essential first step is to decide upon the object of study. Is it the robot or robot-surgeon network that is the focus of investigation?

[55] Michael Arnold, "Da Vinci and Me: a Case Study in the Postphenomenology of Robotic Surgery," paper presented at *Phenomenology, Imagination and Virtual Reality*, Husserl Archives Institute of Philosophy KU Leuven, May 30 2017.

Another reason why anthropomorphic machines make interesting case studies is the fact that they are generally explicable by their creators. Because anthropomorphic machines are engineered by humans, we can make useful deductions about their way of being in the world. In Chapter 3 the robot tortoises discussed possessed simple light and touch sensors. These sensors were built into them. Unlike when we study nonhumans like bats and slime moulds, the question of which parts of the world the nonhuman is capable of sensing is already partly answered. In most cases, creators are able to understand why technological artefacts act in certain ways.[56] There's no need to reverse engineer a sensory organ if you've created it. In addition to this, the sensory organs of anthropomorphic machines can often be compared to human sensory organs. A light sensor may attempt to replicate human vision, but we know enough about the anatomy of the eye and the design of the sensor to deduce that they have different structures and, consequently, a different way of being in the world. Comparing the two yields an appreciation of what it is that makes nonhumans unlike humans and more alien. We can generalise this appreciation to apply to all nonhumans.

This thesis makes a conscious effort to increase the tension between nonhuman and human experience by emphasising the alien qualities of anthropomorphic technologies which are so often neglected by humans. The discourse surrounding technology is anthropocentric, as is the way in which humans speak about specific technological artefacts. The discourse centres around human-nonhuman relations and cultural impact, and we speak of (and to) technological artefacts as if they were humans. Hence, this thesis confronts the same issue on two fronts. This thesis *needs* to accentuate the anthropomorphic in the machine in order to adopt an anti-correlationist stance. Humans frequently conceal or overlook aspects of machine phenomenology because of the mundane use of anthropomorphic language. HAL speaks to the crew, the automaton plays the piano, the robot opens her mouth. This thesis will accentuate this tendency as much as possible, so as to make this day-to-day anthropomorphic language stand out. Ian Bogost would call this a point "where gears grind"[57] and his example of the werewolf is relevant here. There is a friction between human and inhuman that is emphasised in the transformation, which we too often replace "with explanation or ignorance."[58] This thesis will not seek to erase the power of the transformation between anthropomorphic and nonhuman, between human and robot, between a robot that is switched on and one that is switched off. The transformation between these two states is what gives these entities their incredible power and is an indelible part of their objecthood.

[56] Although this is not always true of technological objects created by machine learning.
[57] Ian Bogost, "Inhuman," in *Inhuman Nature*, ed. Jeffrey Jerome Cohen (Washington, DC: Oliphaunt Books, 2014), 139.
[58] Ibid., 136.

A note on language

> As Samuel Butler says in the *Book of Machines*, 'Won't it be the glory of machines that they can do without the great gift of speech? Someone has said that silence is a virtue that makes us agreeable to our fellows.' Ah, Aramis, you would finally enjoy that silence. Why would you waste your time speaking? What you aspire to is not bearing the 'I.' On the contrary, your dignity, your virtue, your glory, lie in being a 'one.' And it is this silence, this happy anonymity, this depth, this heaviness, this humanity, that we have denied you. I am speaking in your place, I am offering you the awkward detour of prosopopoeia, but it is precisely because you are dead forever. 'It' wanted to become not the subject of our discourse, but the object, the tender anonymous object by means of which we would travel in Paris. Is that so hard to understand?
> - Professor Norbert H. on the doomed Parisian transport system Aramis. Bruno Latour, "Epilogue," *Aramis or, the Love of Technology*, trans. Catherine Porter (Cambridge, MA: Harvard University Press, 1996), 297.

As Latour puts it, we are required to take the detour of prosopopoeia, the act of speaking as though on behalf of things that are abstract, absent, or dead. We do not have the language to avoid correlationist statements. The field of evolutionary biology may give us a clue to how to approach this issue. In biology, as in robotics, the question of teleology often arises.[59] Living things are the result of millions of years of evolution by natural selection, with no external agency controlling our progress. For this reason, biologists use a form of shorthand in describing the structure and function of living things. This is known as teleonomy,[60] or the practice of speaking of things as though they have a purpose while taking it for granted that in evolution by natural selection there is no purpose or objective.[61] For example, it is commonly said that the purpose of molar teeth is to grind down food to make it easier to digest. In his 1888 essay on the subject, Henry Fairfield Osborn describes the evolutionary transition of Mammalian molars from those of prehistoric phyla as a "reduction of primitive elements towards special adaptation." [62] In reality, molar teeth are the result of random mutation and selection with no direction or intent. In contrast, a cheese grater is designed by a human to break up food. It has a genuine teleology, not just the appearance of one. A similar practice is proposed in this thesis. It is necessary to use somewhat anthropocentric language in order to make accessible statements about nonhumans. There are limitations to

[59] Presented here is a huge simplification of teleology and teleonomy in the philosophy of biology. See Ernst Mayr, "Teleology," in *What Makes Biology Unique?* (Cambridge, UK: Cambridge University Press, 2004): 39-66. Of particular interest is Mayr's distinction between *telos* as "goal" and "endpoint" (45). A more detailed comparison between teleology in biology and technology is unfortunately outside the scope of this thesis. Mayr's views on the difference between these teleological processes has evolved over time (52).
[60] Ibid.
[61] George Christopher Williams, "The Scientific Study of Adaptation," in *Adaptation and Natural Selection* (New Jersey: Princeton University Press, 1966).
[62] Henry Fairfield Osborn, "The Evolution of Mammalian Molars to and from the Tritubercular Type," *The American Naturalist* 22, no. 264 (December 1888), 1067.

this thesis: word limits, the author's knowledge, etc. An exhaustive evaluation of just one of Voyager 2's sensors would require several volumes, if we are to examine everything from its subatomic particles to its impact on human culture.[63] This thesis employs a technique that we might call anthroponomy. It makes a statement about a nonhuman's sensation and experience using language that is usually used to describe human sensation and experience.

The author is human, the format is human-centric, and language itself is a human mode of relating to other objects. As soon as we use language to try to understand the experience of a nonhuman, we are anthropomorphising it. Throughout this thesis, I explain how certain technological artefacts "sense", "perceive", "experience" and "remember". Nonhumans have "intentions", "qualities" and "desires". Human experience is not special relative to nonhuman experience, so I have no hesitation in hijacking our specialised anthropocentric psychological and ethological language for an appreciation of nonhuman and non-living phenomenology. Chapter 3 goes further and addresses panpsychism, the idea that every object in the universe has some aspect of mind, consciousness or soul. But for the most part, this anthropocentric language is merely a product of our lack of useful shorthand for what occurs within non-living things. There is no shorthand language describing nonhuman perspective. But, as Laura Gustafsson and Terike Haapoja point out, this lack of non-anthropocentric language is not a good excuse for not making an attempt at writing about nonhuman experiences.[64] In their 2013 Helsinki exhibition entitled *History According to Cattle*, Gustafsson and Haapoja endeavour to approximate cattle-like experiences through language intended to evoke responses in humans and assist in conceptualising the alien worlds of nonhumans:

> Although we share many basic experiences with other species, abstract linguistic expression is, as far as we know, a uniquely human aptitude – yet language still remains hopelessly inadequate at conveying anything about corporeality or corporeal experience. Language nevertheless provides the human species with a mental toolkit for making sense of the world; ideally it can serve as a bridge to the experiential realm of the Other.[65]

In a similar spirit, this thesis attempts a written exploration of technological experiences with only the linguistic tools available to cultural critics, philosophers, engineers and ethologists (all of whom are humans).

In many cases the creator or distributor of technologies has encouraged this use of metaphor. The Microsoft Kinect, for example, is "smart". It "sees", "hears" and "thinks" (see Chapter 2). Many technological artefacts have been given a gender-specific pronoun and are deemed to have particular qualities that align with human qualities, such as stubbornness, stupidity, attractiveness, kindness, love, and introspection. This thesis will use this language consistently while acknowledging that such human qualities are the consequences of alien processes inside the nonhuman. Although we clearly lack the linguistic tools to describe nonhuman experience, this should not stop us from trying. Wherever possible, the focus is on

---

[63] This would also constitute duomining (see Chapter 1).
[64] Laura Gustafsson and Terike Haapoja, "Imagining Non-Human Realities," in *History According to Cattle*, eds. Laura Gustafsson and Terike Haapoja (Brooklyn, NY: punctum books, 2015).
[65] Ibid., 108.

maintaining consistency throughout the thesis in the use of ethological language, theoretical terms and social conventions like pronouns.

There are a multitude of academic texts written in acknowledgement of the nonhuman turn, and many employ their own terminologies. I do not intend to debate which wordings are best, since this is a thesis about specific objects, not a purely metaphysical argument. The word "object" will be used, despite its baggage, to refer to all of the technological artefacts, ideas, living things and phenomena referenced. This thesis follows Harman and other object-oriented ontologists in calling all these things objects no matter their material status. The word "actant" may also be used, as in ANT, but this term does not emphasise the inherent passivity of objects that is suggested by Harman's metaphysics. The word "qualities" will be used in place of other synonyms, again because of the word's usage in Harman's writing. Harman describes both "sensual qualities" (qualities related to the sensual objects that allow interaction between objects) and "real qualities" (qualities that belong to the infinitely withdrawn object itself, but that are nevertheless divided from objects).[66] (Chapter 1 will have more to say on this topic.) Unless otherwise specified, the word "qualities" in this thesis refers to real qualities, which are concealed from the view of any other object.

It should be pointed out that the use of the term "the human" in this thesis is also a problematic one that will nevertheless be employed as shorthand. "The human" is not an object. Individual humans are objects, and human collectives are objects. But "the human" describes a set of qualities that are not necessarily universal in humans. The human has eyes, but not all humans have eyes. The human uses language, but not all humans use language. The human reproduces sexually, but not all humans reproduce sexually. It is important to note this caveat at the beginning of this work, not only to indemnify the author from claims of insensitivity but also to establish some object-oriented credentials early on. It is not tenable to support a universal view of humans. "Humanity" is quite another thing. "Humanity" is a concept, and it obeys the rules of concepts, changing over time and space in response to cultural influence. Humanity is a moralising term and sometimes a philosophical term, and consequently enters into relations with other objects. There is a difference between humanity and the human.

Another issue that will arise in this thesis is the use of gendered language when applied to technological artefacts. Gendered language is occasionally used in this thesis when describing humans and nonhumans. The gynoids described in Chapter 5, for example, are often given the pronoun "she" both by academics and by the public. The robots have been given qualities that enable them to perform a female identity, including certain linguistic markers. I have chosen to use this approach in recognition of the performative nature of gender, and of the attempts that are being made to erase the difference between these robots and humans (and humans are usually assigned a particular gender). When it comes to humanoid robotics and AI, the aims of engineers and participants in the nonhuman turn are allied. Because gendered pronouns are used with complex robots, they are also used with simple robots such as Elsie and Elmer which are described in Chapter 3. Created in the 1940s by W. Grey Walter, Elsie

[66] Graham Harman, *The Quadruple Object* (Alresford, UK: Zero Books, 2011).

and Elmer possess little to mark them as female or male other than their names and the pronouns used to discuss them, but since this may also be true of humans I have used the same pronouns as Walter.

Finally, a comment on the use of personal pronouns with reference to the author and the thesis. Frequently, this thesis avoids the author's personal pronouns ("I", "me", "myself") depending on context. This stylistic choice is less common these days, since humanities theses are subjective and depend upon the author's theoretical and cultural milieu. However, the choice to refer to the thesis rather than the author is relevant to the subject matter of this text. I am not the same object as the thesis. The thesis contains information that I do not store in my body, and my body contains information not included in the thesis. My body, its components, history, and relations are not the issue here. This thesis is not just an aspect of myself, or a medium through which I communicate; it is its own object with its own qualities, relations and agency. I will change over time in ways that the thesis cannot. A thesis may be subjective, but it is not me.

Guiding questions

There are three main questions that orientate this thesis towards theory and the case studies. The questions are not necessarily answered in the thesis, but they were used to guide the discussion of the specific case studies.

How do we ask machines about their experiences?

This is a methodological question. What techniques can we use to learn about the phenomenology of nonhumans? The personal assistance agent Siri, who is a popular feature of Apple products, is known for giving witty responses to questions about her experience of the outside world. Obviously, understanding the phenomenology or sensory world of machines must go beyond such human-engineered approaches. We need a phenomenological survey of the object, performed in the same way that an ethologist would survey the phenomenological profile of a tick (see Chapter 3). In this thesis similar tactics are used; various questions must be asked and interactions pursued to encourage the object to show more of its qualities to the outside. Strongly associated with this question is "how does the phenomenology of an anthropomorphic technology affect its agency?" ANT is one way to approach this kind of research, and one that will be more familiar than OOO to most readers. ANT will be discussed in this thesis as well, although in less detail than OOO. ANT allows researchers to answer questions about phenomena of interest by examining the agency of all actants involved, both human and nonhuman. Latour, a key figure in the development of ANT, sees an understanding and integration of nonhumans to be essential to the comprehension of any kind of scientific enquiry, since scientists must inevitably compel

nonhumans to behave in a particular way in laboratory or field settings.[67] But Latour was originally a sociologist, and although his examples of laboratories and politics ground anthrodecentric ideas in empirical studies, ANT lacks the metaphysical strength of OOO. Latour himself jokes: "I'm like a dog following its prey, and then the prey arrives in the middle of a band of wolves which are called professional philosophers."[68] Other approaches to the investigation of nonhumans are also discussed.

The trouble with trying to ask machines about their experiences is that it must be done in a humanish way. It is impossible to fully study any other kind of relationship, since the author of this thesis is a human. But every effort must be made to remove the human from the centre of the analysis. Human concerns must be considered for these analyses, but they must not be privileged over other kinds of relations such as those between nonhuman and nonhuman. As Bryant puts it:

> The human-object relation is not a special relation, not a unique relation, but a subset of a far more pervasive ontological truth that pertains to objects of all types. The point here is not that we should exclude inquiry into human/object relations or social/object relations, but rather that these analyses are analyses for regional ontology, for a particular domain of being, not privileged grounds of ontology as such. The issue here is thus very subtle. It is not a question of excluding the human and the social, but of decentering them from the place of ontological privilege they currently enjoy within contemporary philosophy and theory. Nor does this entail that all objects relate to other objects in exactly the same way.[69]

Each chapter contains at least a short discussion of the machine's relationship with human beings and in some, such as Chapter 2, the distinction between the human relationship and the object itself is theorised at length. But the human relationship with anthropomorphic technologies is almost always discussed in the narrowest of terms. The texts which form the evidence for these observations are a mixture of technical, philosophical and cultural works, providing material for the attempt to describe machine relationships with other nonhumans, including computers, floors, languages and ideas. Generally, exploring interactions between nonhumans provides greater insight into interactions between humans and nonhumans. We are limited, of course, by the possibilities of observation available to us. We can only witness relations between nonhumans that result in actions that we can sense and perceive.

What does sensation mean for machines?

Machines experience the world in a very different way from humans. Their sensory organs mean that their relationship with the world may seem to be seriously impoverished compared with humans, although they may be capable of sensing their environments in ways

---

[67] Bruno Latour, *Reassembling the Social: An Introduction to Actor-Network Theory* (Oxford: Oxford University Press, 2005).
[68] Bruno Latour, quoted in Bruno Latour, Graham Harman and Peter Erdélyi, "Transcript," in *The Prince and the Wolf: Latour and Harman at the LSE* (Winchester, UK: Zero Books, 2011), 41.
[69] Bryant, *The Democracy of Objects*, 282.

that humans cannot. A personal computer senses information through a mouse and keyboard, which brings about certain kinds of experiences and reactions in the computer. But this thesis argues that these are not the only germane sensory experiences possible. The computer senses and experiences data from a keyboard by making changes to its display. Perhaps it also senses and experiences the creation of a dent when it is accidentally dropped on the floor, albeit in a different way such as by changing its shape. The computer is in a relationship with its operating system, its programs, its network, the desk, the Earth (since it is fixed in place by the Earth's gravity) and the glass of water carelessly spilled on it. It is vital not to forget these other relations which contribute to the computer's rich experience of its world. In the near future, machines are likely to become significantly more agentic while remaining intrinsically reliant on alien sensation. The Internet of Things, a popular term describing the blossoming network of artefacts capable of exchanging data,[70] exposes us to new deliberately engineered processes but also opens the possibility of all sorts of unthought-of relations between technological artefacts and other objects. At this critical time, I believe it is important to develop new theories of relations between technological artefacts and other objects that attempt to understand the agency of the nonhuman. I have chosen to investigate sensation in order to emphasise not only the alien nature of the interface, but the alien nature of the technological artefact as a whole.

To this end, this thesis involves critical engagement with several key theorists and attempts to relate their work to anthropomorphic technological artefacts. It draws upon philosophies of technology, phenomenology, postphenomenology, Umwelt-theory and ANT, as well as OOO. In doing so, it will critique the ideas of several theorists in a close analysis of its case studies. It is thus both an empirical and a theoretical study. My point of departure from many of the theorists discussed in this thesis is that they write from an anthropocentric position, as has been the tendency for most of the history of the philosophy of technology, and indeed of philosophy more generally. Prominent scholars such as Martin Heidegger, Don Ihde and even Bruno Latour have seen the human perspective as a privileged position; the human is fundamentally different from all the other objects in the world, or it is the only perspective worth understanding. In this thesis I attempt a bidirectional study of humans and machines from a position that takes nonhuman agency seriously. But although this project owes a great deal to cybernetics, it is not simply mechanistic. Experience, sensation, mind – these are all words that will be critically explored.

This may sound like a lot of different theoretical approaches to address in one text. In *The Democracy of Objects*, Bryant describes himself as a "bricoleur, freely drawing from a variety of disciplines and thinkers whose works are not necessarily consistent with one another."[71] This thesis proceeds in a similar fashion, touching on multiple theoretical traditions that reflect the author's own intellectual background. But there is no intention here to provide an in-depth analysis of any of these approaches, only to discuss those aspects that pertain to the case studies and the decentring of anthropomorphic technologies. The main focus of this

[70] See Rajkumar Buyya and Amir Vahid Dastjerdi, eds., *Internet of Things: Principles and Paradigms* (Amsterdam: Elsevier, 2016).
[71] Bryant, *The Democracy of Objects*, 27.

thesis is on OOO, and so to a certain extent all the other theory will be positioned through this frame. This is not part of an attempt to advocate OOO over other ideas, but rather because flat ontology permits a conceptual expansion in many diverse fields. For example, Umwelt-theory is discussed in Chapter 3 in the context of OOO. There are fundamental inconsistencies between Umwelt-theory and OOO. But discussing Umwelt-theory in this way permits an expanded concept of the Umwelt which I then relate to robots. Such discussions open the mind to speculation about sensation and experience in machines.

To what extent is the sensation of anthropomorphic machines relevant to their cultural role?

As discussed above, a machine's sensation and experience is highly relevant to its cultural, political and social position. It is evident that the way in which a machine senses its world affects its relationships with individual humans. When my Kinect detects my face, it begins a sequence of relations that end in me adjusting my behaviour to affect the phenomena visible to the Kinect. What is less clear is the way that sensation in machines affects their political and cultural positions. If the Kinect can identify me, then what questions of privacy and power are raised? Questions like this are anthropocentric but nevertheless significantly affect nonhumans. A machine may experience a serious change in its composition, relations and internal world if it is smashed with a blunt weapon. My Kinect would not sense me in the same way if it were unplugged and stored in a box inside a cupboard. In general, anthropomorphic machines are strongly affected by their relations with humans and with humanity.

The ethical issues associated with anthropomorphic machines are a frequently raised topic of public debate. These ethical issues are often connected to questions of machine sensation of other objects such as human personal data (in the case of the Kinect) and the detection of pedestrians (in the case of self-driving cars). This ties into larger questions about machine ethics. Can a technological artefact ever be morally responsible for its actions? Drew McDermott argues that it is not yet possible for a technological artefact to make ethical decisions because no technological artefact is yet capable of making a choice between "the right thing" and its own self-interest.[72] This is because, as McDermott says, no modern machine fulfils the criteria for really wanting anything. We can program a robot to be attracted to light (see Chapter 3) but it is too easy to flip a switch and make the robot seek darkness instead. Without truly having desires (perhaps the desire to fulfil the needs and wants of its human user?) technological artefacts make decisions with moral consequences in the same way that they make any kind of decision. A computer may be programmed to start executing people when the British pound drops below a certain value, but it has not made a moral choice any different from if that computer were programmed to turn on a blinking light in response to the same stimulus. Later in the thesis I will argue that machines can and do

[72] Drew McDermott, "What Matters to a Machine?" in *Machine Ethics*, ed. Michael Anderson and Susan Leigh Anderson (Cambridge: Cambridge University Press, 2011), 89.

have desires (although perhaps not in the same way as humans) and also that it is only by speculating about the internal world of an object that we can draw conclusions about what we might call its own ethical obligations. This is connected to a functionalist view of consciousness (see Chapter 3). The question of what it means to be conscious is highly relevant to the question of machine ethics. Kenneth Eimar Himma claims that an intentional state is necessary for agency, and without it no machine can make a moral choice.[73] There is a great deal of debate around machine ethics and the nature of consciousness.

Regardless of whether technological artefacts are capable of making moral choices, they are sometimes prominent in contemporary ethical debates because they are capable of making choices with ethical consequences. Two of the case studies presented here are particularly the subject of controversy. The Kinect (Chapter 2) has been the focal point of considerable fear and anger due to its possible applications for surveillance and the involuntary gathering of data. And as mentioned above, sexbots (Chapter 5) are frequently identified as perpetuators of societal misogyny and sexual violence, both as images that stereotype and objectify women (and sometimes men and children) and due to the possibility that they may serve as "practice" for individuals who may go on to perpetrate violent crimes or misogynistic behaviour. In both these examples, the sensory capabilities of the object significantly affect its ability to act in a way that raises ethical questions.

Structure of this thesis

This thesis revolves around the four major case studies briefly introduced above, all of which are instances of technological artefacts that have been endowed with anthropomorphic qualities. The four case studies are described in order of their apparent physical resemblance to human beings. They are, however, described in different theoretical terms and are used to reflect upon the writings of different theorists relevant to the nonhuman turn. It is hoped that some evolution of ideas is evident through the thesis, but each case study is relatively self-contained. And this is as it should be. Each case study is as radically different from the others as they are from humans, with respect to their modes of existing in and navigating the world. The superficial taxonomy of technological objects typical of human descriptions is not all that relevant to their internal experiences. Certain similarities are observed, but the differences between them are fundamental. Some texts associated with the nonhuman turn are deliberately constructed with this in mind. For example, *Object Oriented Environs*, a collection of essays arranged by the authors' names, is intended to be read in an order determined by "some object that you will allow to exert its aleatory agency over your

[73] Kenneth Eimar Himma, "Artificial Agency, Consciousness and the Criteria for Moral Agency: What Properties Must an Artificial Agent Have to be a Moral Agent?" *Ethics and Information Technology* 11 (2009): 19-29.

reading" such as a twenty-sided die.[74] It is best to read the chapters of this thesis in the order they are presented, but while bearing in mind the independence of each case study. This section of the introduction will briefly discuss the contents of each case study chapter. The first chapter is a literature review, and Chapters 2 to 5 each cover different objects.

The Microsoft Kinect (Chapter 2) is an add-on to the Xbox line of videogames consoles, comprising of a sophisticated camera to detect human movement and a microphone to detect voice commands. When first released, much of the discussion about it centred on its ability (or inability) to "see" and "hear". The Kinect for Xbox One is the first case study for two reasons. Firstly, it is comparatively (or deceptively) simple when compared with the other case studies. There is no visible movement in the Kinect. It does not appear to have a great capacity to communicate its experience to human beings. Secondly, because the Kinect is almost universally portrayed as a watcher and listener (both in Microsoft's promotional material and by users – see Chapter 2), it is an ideal case study for beginning to explore the thesis's guiding questions. What does sensation and experience mean in the Kinect, and what can we say about it?

Much of the research for Chapter 2 was conducted in the first half of 2014, shortly after the release of Kinect 2.0 and a time when the evidence of human first-hand experience of the Kinect's sensory and perceptive systems was ample. User accounts were gathered from online forums describing a sort of collective effort to engage with the phenomenology of the Kinect. Individual users described their attempts to coax the Kinect into seeing and hearing them correctly. This is captured in the thesis with two examples: the "Xbox on" problem and the videogame *Kinect Sports Rivals*.[75] The Kinect often proved unreliable. Difficulties in persuading the sensor bar to detect human bodies and faces have resulted in the alienation of certain sectors of the "gamer community", notably the self-described "hardcore" gamers. It has proved difficult to persuade these users to form relations with the sensor bar, which has had an effect on the cultural relevance and reputation of the device.

An essential part of this analysis is the distinction between the different types of object that are erroneously conflated by the signifier "Kinect". The Kinect is many things to different humans: a harbinger of more casual games for the Xbox One; an unnecessary but compulsory add-on to an expensive console; the subject of online trouble-shooting; and, a potentially malevolent eye spying on your living room. But it is also an individual Kinect, beneath my television and connected by a cable to my Xbox One. The term "Kinect" becomes insufficient when asked to represent two such different levels of identification, so this chapter employs a simple method of distinguishing between the brand and the situated artefact. This thesis contends that these are two different objects, not two different aspects of a single object.

This apparent bifurcation of physical and cultural identity is also present in William Grey Walter's robot tortoises, the so-called *Machina speculatrix*. Chapter 3 builds on the idea of this division within a human-made object and goes into detail about the tortoises' sensory

---

[74] Jeffrey Jerome Cohen and Julian Yates, eds., *Object Oriented Environs* (Earth: punctum books, 2016), ix.

[75] *Kinect Sports Rivals*, developed by Rare (Microsoft Studios, 2014), Xbox One.

world. Built in the late 1940s, the robots existed as sensing and moving agents but the human interest in them is more strongly connected to what they indicated about living things. Walter was a cyberneticist, and in his view the robots confirmed his ideas about life, sensation and identity. Jussi Parikka wrote about the tortoises in his book *Insect Media* and briefly discussed them in relation to the ideas of Jakob von Uexküll.[76] Chapter 3 continues this analysis and extends it by drawing on contemporary panpsychism. There are, then, two steps to this process: the acknowledgement of the tortoises' sensory worlds followed by an appreciation of the heterogeneity of nonhuman "minds". The two theoretical structures are occasionally in conflict with one another, and this is also explored.

The tortoises, which were entirely synthetic, were anthropomorphised in many ways. They possessed eye-like structures capable of sensing light and contact switches that enabled them something like a sense of touch. In this way, they could find their way back to their hutches when it was time to recharge their batteries. Walter gave them names and studied them as an ethologist might plot the progress of an ant or mouse. Possibly the ease with which these simple machines can be anthropomorphised promotes the application of panpsychist ideas. In other words, the blurring of the line between human and nonhuman suggests that the mind is not exclusively human, but is in some way a universal quality of matter (see Chapter 3 for a discussion of these ideas). However, it is important to avoid prioritising minds in this analysis, since that could mean the inflation of a human idea beyond its importance in a nonhuman object. The mind must be treated like any other object. Consequently it is questioned whether sensation and experience are best interpreted as metaphors in nonhumans or as the product of a conscious mind, but the chapter ultimately concludes that this is unknowable and irrelevant to the study of nonhumans by humans.

Chapter 4 is less focussed on the internal world of nonhumans and more interested in the ways in which objects extend the sensation and perception of other objects. The case study presented is the natural language processing (NLP) program SHRDLU, built by Terry Winograd between 1968 and 1972, and in particular its relations with the human user (and vice versa). The concept of postphenomenology, particularly that of Don Ihde, is placed in a more anthrodecentric context for this purpose. Postphenomenology is an example of a phenomenological approach that places greater emphasis on the role of nonhumans than traditional phenomenology, however it is still too one-sided to fit entirely comfortably into this thesis.

SHRDLU is a language-user and possesses the capacity for sensation and action within its own tightly constrained world, a world in which blocks of different colours and shapes are stacked and restacked by a virtual robot arm. It is designed to understand a human user's commands, and to answer questions about the way the blocks are stacked, and about its own actions. Because of its extremely limited world and capacity for action, it is an excellent model for elementary philosophical inquiry. Like the robot tortoises, which are all thought to be destroyed or otherwise removed from public consumption, it was not possible to interrogate Winograd's SHRDLU directly. The analysis relies on first-hand accounts and on a version of

[76] Jussi Parikka, *Insect Media*, Posthumanities (Minneapolis: University of Minnesota Press, 2010).

the software remediated for an early version of Windows.[77] Nevertheless, with these sources it is still possible to interpret the way in which SHRDLU and the human extend each other's worlds.

Of all the case studies presented in this thesis, the sexbots and gynoids of Chapter 5 are the most human-like in their physical appearance. They require physical humanlike form even if their sexual organs are the only ones required for the act of coitus, or if their faces are the only ones required for verbal communication. They are also subject to significant political and cultural scrutiny. Sexbots, of course, can be of any or no gender, but the focus here is on gynoids – that is, feminine humanoid robots that, in theory, are to androids what women are to men. Some space is devoted to the cultural role of the gynoid, which is highly dependent upon the metaphysical commitments guiding their development and presence in different cultural contexts.

The real focus of the chapter is on the *concept* of gynoid sexbots and how we can study a concept through the interrogations of other objects (such as texts, human, real specific gynoid sexbots, etc.). A great deal has been written about gynoids, both in popular fiction and in academic contexts. However, very little interest has been shown regarding their experience of the world. The sensory capabilities of a gynoid reveal a great deal about the concept of gynoid sexbots, as they tend to be deliberately engineered into them by humans. Those robots exist in a bidirectional relationship with the concept. It is hoped that this chapter on gynoids might demonstrate how OOO can be a powerful tool for stimulating political discourse, in response to criticism of its apolitical nature. There is also some criticism of OOO and suggestions of how alien phenomenology might best be deployed. As in previous chapters, the focus is twofold: an in-depth analysis of the case study, plus the acknowledgement that ontology guides the cultural criticism of technological artefacts.

[77] *SHRDLU*, developed by Terry Winograd and Greg Sharp (Sun Microsystems, Inc.: 1999), Windows 7.

Chapter 1: Literature Review

This thesis contains a great many questions with very few definitive answers. What is it like to be a robot? Are technological artefacts agents with independent movement and action, or is their interaction with the world determined by the will of their human creators? Is the political and cultural impact of a technological artefact the responsibility of the artefact itself, or of the human and nonhumans that interact with it? These are deep questions, and this relatively short text cannot answer them. Yet they are all relevant to the case studies presented in this thesis, which aim to focus on the objects. The questions cannot be answered, but this chapter seeks to provide some background to each of them as well as to the main themes of the thesis. This thesis will examine ideas that have been a source of disagreement for centuries, and only a limited snapshot is given here due to space constraints. This is a review of the background literature that is essential reading for studying technology.

There are two sections to this literature review, which are separate for disciplinary reasons but both relevant to addressing the problems articulated in the introduction to this thesis. The first part of the chapter provides a brief survey of the history of modern Western philosophy of technology. This section briefly describes some of the attempts to explore the metaphysics of technology, to account for the effect of technology on society (and vice versa), as well as criticism of technology. The authors presented in this section are not the focus of later chapters, but it is necessary to describe their ideas in order to position this thesis and point out some of the assumptions inherent to the text. The ideological commitments of contemporary commentators on technology have evolved from these voices. And it is important to contextualise the theories discussed in the second section, such as OOO discussed below, with the history of the philosophy of technology, because that scholarship suggests to us what might be the consequence of a philosophy of technology based upon OOO. OOO is a metaphysics of objects in general, and narrowing the focus to technology has implications that must be acknowledged.

The second section of the literature review covers object-oriented ontology (OOO) and speculative realism (SR), as well as various other aspects of the nonhuman turn. The section begins with a brief description of Actor-Network Theory (ANT), with which OOO is contrasted. The version of OOO presented in this thesis is based upon the work of Graham Harman, from *Tool-Being: Heidegger and the Metaphysics of Objects* (which is based on his 1999 doctoral dissertation) to more recent publications. Throughout his career, Harman's metaphysical position has remained steady, but it has found new areas of application and he and other authors have allowed OOO to flourish in unexpected ways. There have been other interpretations of an OOO nature, but I focus on Harman because of his prominent status and

his reputation as the first author to write about object-oriented philosophy. Harman's metaphysics is rich and vast. It is built on an interpretation of Heidegger that leads him to some unconventional conclusions.[1] Harman's concepts of the quadruple object, withdrawal, and allure form a robust structure through which the interpretation of real examples is made possible. They provide the language to facilitate accounts of complex relations between objects. For the purposes of this thesis, it is not important whether Harman is correct in his ontology because it is mainly used here to focus the anthrodecentric inquiry into the case studies. The basics of Harman's theory are presented in this chapter, and they are referred to and developed in later chapters when describing the case studies.

Contemporary Western philosophy of technology

This thesis is influenced by numerous Western philosophers of technology. Over the twentieth century, many scholars discussed not just the way in which machines could be used to improve human lives, but how technology is inextricably linked to human culture and society. Human existence is tangled up with the agency of machines. In his book *What Things Do*, Peter-Paul Verbeek characterises twentieth century philosophy of technology as consisting of three principle paradigms: "classical philosophy", social constructivism of technology and constructivism.[2] Within these prevailing tendencies individual writers have approached the question of technology, often as an extension to other philosophical or methodological commitments. There is significant variation in the estimation of how much technology can be said to affect human society and culture, and how much cultural forces shape technological artefacts. This section discusses some key figures in contemporary Western philosophy of technology, and provides some background to the place of the study of technology in the nonhuman turn.

Over time, machines have come to occupy a vast spectrum of complexity, making it difficult to encapsulate them with any single political or cultural theory. One end of this complexity spectrum is occupied by the simplest of machines, such as levers and glass cubes, and the other is occupied by our most advanced computers. Inevitably, the advanced end of the spectrum has become associated with machines that we heavily anthropomorphise, since philosophy and the natural sciences tell us that we are more advanced than any other living organism. Computers "think", "hear" and "make mistakes". These complex machines are the subject of much study both in philosophy and cultural studies. They have directed us to rethink what it means to be human and how we interact with objects. They visibly, and often deliberately, blur boundaries that modern Westerners are used to thinking of as impervious

[1] Harman himself says of Heidegger: "I am convinced that he would revile much of what I have to say." Harman, *Tool-Being: Heidegger and the Metaphysics of Objects* (Chicago: Open Court, 2002), 15.
[2] Peter-Paul Verbeek, *What Things Do: Philosophical Reflections on Technology, Agency and Design* [De daadkracht der Dingen: Over techniek, filosofie en vormgeving], trans. Robert P. Crease (University Park Pennsylvania: Pennsylvania State University Press, 2005), originally published 2000.

(although in a few pages this chapter will point to Latour's denial of this fundamental assumption). The nature of the most visible technologies of the day and their pervasion into the cultural imaginary tend to significantly affect the philosophical discourse of the time.

As Verbeek describes it, the "classical philosophy" of the early twentieth century portrays technology as a transformative agent on human society. Technology was inextricably linked to human culture and society, often as a negative influence. For example, Martin Heidegger's (1889-1976) explanation of the role of Technology positions it as a frame through which we experience the world (Heidegger's Technology has a capital "T", indicating his conceptualisation of Technology as a special metaphysical category encompassing, as Don Ihde puts it, "gas chambers and engineered tomatoes"[3]). Much of what has been written about Heidegger and Technology is derived from the essay "The Question Concerning Technology"[4] in which Heidegger describes Technology as a way of revealing. Heidegger points to two linked statements: "Technology is a means to an end" and "Technology is a human activity",[5] which in his view do not go deeply enough. He ultimately argues that Technology can be a way of "bringing-forth" or revealing the truth, a scenario in which humans can rely on technology without being enslaved by it.[6] (There will be more on Heidegger's philosophy in the section below on OOO.) The view of Technology presented by Heidegger is of objects that have a vast transformative effect on human society, and Verbeek goes so far as to say that it is technophobic.[7] In *What Things Do* Verbeek then goes on to describe the philosophy of Karl Jaspers (1883-1969) as classical philosophy. Jaspers argued that technology is alienating and "demonic".[8]

The technophobism of Jaspers may also be seen in Jacques Ellul (1912-1994). According to Ellul, we are in a technological society that is dehumanising and violent, in which the "metal monster" of the machine destroys that which does not support it.[9] This is exemplified by the concept of the "technique", which enslaves every aspect of our lives. Technique is the "manifestation of a technological, managerial mind-set,"[10] in which methods of human activity are optimised for peak efficiency. Moreover, we often lose control of a technology once it has been created. Once an artefact is exposed to human society and the various needs and desires therein, it is often used in ways that are not intended or condoned by the creators.

---

[3] Don Ihde, "Forty Years in the Wilderness," in *Postphenomenology: A Critical Companion to Ihde*, 271.
[4] Martin Heidegger, *The Question Concerning Technology and Other Essays* [Die Technik und die Kehre, Holzwege, and Vortrage und Aufsatze], trans. William Lovitt (New York: Garland Publishing, 1977), originally published 1954.
[5] Ibid., 4.
[6] Verbeek, *What Things Do*, 59.
[7] Ibid., 9.
[8] Ibid., 17.
[9] Jacques Ellul, *The Technological Society* [La Technique ou l'enjeu du siècle], trans. John Wilkinson (New York: Vintage Books, 1964), originally published 1957, 5.
[10] M. Tiles and H. Oberdiek, "Conflicting Versions of Technology," in *Living in a Technological Culture: Human Tools and Human Values* (London: Routledge, 1995), 22.

Albert Borgmann (born 1937) criticises Ellul, contending that the nuances of argument are lost in the general sense of technophobia.[11] In Borgmann's view, technology is too broad a category to be explained by a single approach such as Ellul's technique.[12] Ellul's technique is an instance whereby every kind of procedure is reduced to a single element, but that element is itself vague. Yet Borgmann's own philosophy of technology is comparable to Ellul's in that it bemoans the loss of a particular way of life that has been superseded by modern technology.[13] A key concept for Borgmann is the "device paradigm" which describes a pattern of life that is most visible in our engagement with modern technology.[14] A device in this context is a technological artefact that removes the manifold social practices surrounding a requirement of life, such as the replacement of a wood-burning stove with a central heating unit.[15] The familial organisation that would ensure warmth in the home was complex and established clearly defined designations. The central heating unit removes some of the hardship from life and moves the work to the machine which becomes less and less visible.

> "Devices… dissolve the coherent and engaging character of the pretechnological world of things. In a device, the relatedness of the world is replaced by machinery, but the machinery is concealed, and the commodities, which are made available by a device, are enjoyed without the encumbrance of or the engagement with a context."[16]

The commodity, for Borgmann, is replacement for the labour that was previously necessary. Modern technology is defined by its disburdening character, but it comes at the cost of reducing our engagement with reality.[17] The "focal event" of a family meal, in which food is prepared, served and discussed, is contrasted with the disintegration of eating "scattered around television shows, late and early meetings, activities, overtime work, and other business."[18] In this way, technology transforms contemporary life.

A prominent member of the Frankfurt School, Herbert Marcuse (1898-1979) studied the influence of post-war mass culture on political repression. The mass-produced technological artefact was complicit in this repression, tying the working class to a commodity culture that replaced an older way of relating individual to society.[19] Marcuse was influenced by Heidegger as well as by Hegel's dialectics. But unlike Heidegger, Marcuse saw technology in a negative light because of its role in capitalism, rather than because of its inherent transformative tendencies. In the book *One-Dimensional Man: Studies in the Ideology of Advanced Industrial Society*, Marcuse argues that the space necessary for the next stage in a Marxist view of history is now impossible to locate because people cannot imagine a way of

---

[11] Albert Borgmann, *Technology and the Character of Contemporary Life* (Chicago: University of Chicago Press, 1984), 9.
[12] Ibid., 15
[13] For example: Ibid., "Work and Labor," 114-124.
[14] Ibid., "The Device Paradigm," 40-48.
[15] Ibid., 42.
[16] Ibid., 47.
[17] Peter-Paul Verbeek, "Devices of Engagement: on Borgmann's Philosophy of Information and Technology," *Techne* 6 (2002): 70-71.
[18] Borgmann, *Technology and the Character of Contemporary Life,* 204.
[19] Herbert Marcuse, *One-Dimensional Man: Studies in the Ideology of Advanced Industrial Society* (London: Routledge, 2002), originally published 1964.

living that isn't tied to increased productivity and an increase in quality of life.[20] Thus, the technological mode of production is a factor in preventing progress towards a socialist state. Later, he called for a "new technology" and a "new science" that would result in a less exploitative society.[21] This new science is based upon more natural modes of production, by which he meant both environmental and humanistic nature.

In his book *Technics and Human Development* (Volume 1 of *The Myth of the Machine*), [22] Lewis Mumford (1895-1990) discusses how technology has developed alongside human language, myth and ritual. Innovation is not a simple scientific progression, but is a manifestation of culture, and culture is co-created by adaptation to the environment. The book gives an account of the history of technology, but Mumford was also a critical theorist of technology.[23] He also significantly influenced the start of the green movements of the 1970s and was a notable figure in 1960s counter-culture.[24]

In Langdon Winner's (born 1944) classic essay "Do Artifacts Have Politics?",[25] he argues that technology significantly influences, and is in turn influenced by, the foundations of human society. Winner claims that objects may be inherently political because they develop in response to the economic and social impetus of human society:

> In our accustomed way of thinking technologies are seen as neutral tools that can be used well or poorly, for good, evil, or something in between. But we usually do not stop to inquire whether a given device might have been designed and built in such a way that it produces a set of consequences logically and temporally *prior to any of its professed uses*.[26]

The study of technology has repeatedly questioned the degree to which technology is constructed by humans and vice versa, relating specific instances to broader societal changes. Winner's political artefacts are of particular interest here because the case studies as described in this thesis most certainly do have politics. Winner describes a case with particular similarities – that of the low-clearance overpasses of Long Island, designed by Robert Moses to prevent poor people and people of colour (who typically rode in tall buses) from accessing his public park.[27] Winner contends that these overpasses were political in and of themselves, serving to carry out a particular social function. Some parallels may be drawn here with the Kinect (Chapter 2), which at release seemed to hear certain groups of people so much better than others.

---

[20] Ibid., 26.
[21] Michael Kidd, "Technology and Nature: a Defence and Critique of Marcuse," *Polis: Revista de Stiinte Politice* IV, no. 14 (2016), 50.
[22] Lewis Mumford, *Myth of the Machine: Technics and Human Development* (New York: Harcourt Brace Jovanovich Publishers, 1967).
[23] Eric Royal Lybeck, "The Critical Theory of Lewis Mumford," *The International Journal of Interdisciplinary Social Sciences* 5, no. 1 (2010): 91-102.
[24] Gábor Kovács, "A Forefather of Environmentalism – Lewis Mumford, the Critic of Technology and Civilization," *Philobiblon* XIV (2009): 294-300.
[25] Langdon Winner, "Do Artifacts Have Politics?" in *The Whale and the Reactor: a Search for Limits in an Age of High Technology* (Chicago: University of Chicago Press, 1986), 19-39.
[26] Ibid., 25.
[27] Ibid.

Ray Kurzweil foresees a future in which technology results in vast changes to human life. In 1990, Kurzweil published a book on intelligent machines.[28] When considering the possibilities afforded by intelligent machines, Kurzweil spoke enthusiastically of such things as "translating telephones"[29] and "cybernetic chauffeurs"[30], and claimed that it was "virtually certain (barring a world calamity) that all of these scenarios will take place"[31]. The wonders of the age of intelligent machines are so transformative that we should expect a kind of utopia to result from them. Kurzweil is not uncritical of technology, but describes many instances in which it may help humans to solve the problems of the future. Later work has emphasised the accelerating nature of technological change, culminating in what has been called the "singularity"[32] (although the term was coined much earlier by John von Neumann.[33]) Kurzweil contends that technological innovation has so far increased at an exponential rate, and that eventually changes in technology will accelerate at such a rate as to result in "a world that is still human, but that transcends our biological roots".[34] Kurzweil is now the Director of Engineering at Google and has claimed that the singularity will occur by 2045.[35]

Transhumanism is closely related to this concept, although usually framed as less extreme and more short-term. Humans are increasingly coming to understand our interaction with machines to be loci of cyborg experiences. In other words, it is possible to observe many emerging instances wherein fusion is occurring between organic and inorganic entities, and in which it is impossible to say where a human begins and a machine ends. Of course, opinion is divided as to whether or not this cyborg state should be encouraged. Nick Bostrom (born 1973) studies and advocates the cyborgisation of humans, and is involved with the transhumanism movement. Transhumanists promote the enhancement of humans by artificial means.[36] The enhancement may be to our physical or mental abilities. Enhancements that are commonly associated with transhumanism include the extension of the human life-span, genetic engineering, eradication of disease, artificial intelligence and the improvement of human physical capabilities, but Bostrom believes that transhumanism also encompasses changes to human society, economics, culture, institutional designs and "psychological skills and techniques."[37] Related to this is the concept of "superintelligent" AI

[28] Ray Kurzweil, *The Age of Intelligent Machines* (Cambridge, MA: MIT Press, 1992), originally published 1990.
[29] Ibid., 405.
[30] Ibid., 410.
[31] Ibid., 404.
[32] Ray Kurzweil, *The Singularity is Near* (New York: Viking, 2005).
[33] Ray Kurzweil, "Foreword to the Third Edition," *The Computer and the Brain* (New Haven: Yale University Press, 2012): xxx. *The Computer and the Brain* by John von Neumann was originally published in 1958.
[34] Ibid., 9.
[35] Don Galeo, and Christianna Ready, "Kurzweil Claims That the Singularity Will Occur by 2045," *Futurism* (October 5 2017). https://futurism.com/kurzweil-claims-that-the-singularity-will-happen-by-2045/ (accessed March 22 2018).
[36] Nick Bostrom, "In Defence of Posthuman Dignity," *Bioethics* 19, no. 3 (2005): 202-214.
[37] Nick Bostrom, "Human Genetic Enhancements: a Transhumanist Perspective," *Journal of Value Inquiry* 37, no. 4 (2003): 493-506.

which would greatly exceed human ability in all relevant fields.[38] Bostrom argues that we must prepare for such an eventuality with great care.

Transhumanism is strongly related to cyborg theory, since the cyborg is the result of biological and technological parts. The cyborg theory of Donna Haraway (born 1944) is particularly significant moment in the critical theory of technology, as in the 1980s it deconstructed and politicised the categories of human and machine.[39] The term "cyborg" was coined in 1960 by Manfred E. Clines and Nathan S. Kline, and was used to refer to technological adaptation of biological systems to meet the needs of an organism in a new environment.[40] The purpose of this research was to propose adaptations necessary for humans' survival in space.[41] Taken to a logical extension, the cyborg could be said to have first appeared at the moment a human first used a tool, because at that moment it was adapting itself through the use of primitive technology. But the cyborg is primarily an ontological category for the digital age, as proposed by Haraway in 1984.[42] The cyborg construct is an entity with an organic component and an inorganic component, generally a human and a human-made machine. It can also be a combination of organic and inorganic components such as artificially synthesised peptides. This entity may be a transitory assemblage, created in a moment during which human and machine are working together to achieve a common aim. In the example of a mobile phone that aim might be to communicate with another human. The phone can establish the connection, but it needs the human to dial and to utter the spoken language in a way that is intelligible to the human on the other end of the line. Each object has certain capabilities and responsibilities that can be assembled into a cyborg entity that achieves certain unified goals. Each is used by the other. In the words of prominent cyborg anthropologist Amber Case, "we are all cyborgs now". [43] To some extent this is a restating of Latour's description of networks composed of humans and nonhumans (see below), but in my opinion the cyborg has a greater political and cultural agency than the network. Networks are described impassively with the positivist cadences of the social sciences. As Haraway has said in an interview: "I need to hold on to impossible heritages more than I suspect Bruno wants to. Our kinds of creativity take different directions but they're allied."[44] Cyborgs are mobilised to create an impact. The image is transgressive and evocative.

---

[38] Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford: Oxford University Press, 2014).
[39] Donna Haraway, "A Manifesto for Cyborgs: Science, Technology and Socialist Feminism in the 1980s," in *From Modernism to Postmodernism: An Anthology*, ed. Lawrence Cahoone (Oxford: Blackwell, 2003), 464-481. Originally published 1984.
[40] Manfred E. Clines and Nathan S. Kline, "Cyborgs and Space," *Astronautics* (1960).
[41] Ibid.
[42] Haraway, "A Manifesto for Cyborgs."
[43] Amber Case, *We Are All Cyborgs Now* (TED, 2010), video. https://www.ted.com/talks/amber_case_we_are_all_cyborgs_now/transcript?language=en (accessed June 24 2017).
[44] Donna Haraway, interview by Nicolas Gane, "When We Have Never Been Human, What is to be Done?" *Theory, Culture & Society* 23, no. 7-8 (2006): 135-59.

This has been a very brief summary of recent Western thought in the philosophy of technology Some of these authors identify technology as primarily a transformative agent in society, others identify it as something that evolves in concert with human society, and still others frame it as a tool for creating a different future. This puts the questions asked in this thesis into greater context. It is important to note that all of the authors described here have focussed on the relationship between humanity and technology (which makes sense, since every one of them is human). Each author takes technology seriously as either a producer or product of change for humanity. Although this thesis deviates from this path by querying the sensation and experience of machines from the perspective of those machines, a Western philosophical legacy is embedded in the choices made in creating this thesis. Technology is already a category which we are accustomed to considering because of its very broad alliance with on society. Whether that alliance is positive or negative, or whether it is in itself transformative or a way of identifying change in humans, it is never neutral in these accounts. This thesis dwells on the influence of anthropomorphic machines on objects that we consider to be strongly connected to humans (e.g. ethical frameworks, feminism, consciousness, language). These are relationships that have already been examined by other authors; this thesis presents an alternative slant.

Technology has often been considered a special category of nonhuman objects. This is in conflict with the principles of flat ontology. But as Harman says, flat ontology is not the final step in creating useful philosophical theory.[45] There must be some human creation of taxonomies of objects. Technological objects are not metaphysically different from objects traditionally associated with the natural world, or with any other kind of object. Yet many of the theorists listed here emphasise the special challenges and insights that come from analysing technological objects. And OOO is similarly useful for the study of technology because objects are in the foreground, rather than existing in the background of scientific or social inquiry. Philosophy of technology requires a unique methodology, one that successfully integrates history and philosophy of science with the study of culture and society. But each of the theorists discussed above consider technological objects to be *important*, either because of their transformative power or their correspondence with human society. In the interpretation of OOO presented in this thesis, technology is just as real and just as important as humanity – or at least a technological object is important for the technological object. Studying technology with OOO also permits an analysis of artefacts that includes the social and cultural background of that artefact, it just does so without privileging the social over other types of relations. A technological artefact may not require a different ontological analysis than a river or an electron. They are all equally objects. But technological artefacts are created by humans, and thus are strongly in relation with objects like ideas, texts, and institutions, as well as things like metals, light, and individual humans. This thesis departs from other philosophies of technology by insisting that objects created by (or even in the image of) humans are considered in the context of a flat ontology.

Another important departure made from many of the theorists listed here is that alien phenomenology focusses on individual objects, not vast groupings of objects. “Technology” is

[45] Harman, *Object-Oriented Ontology*, 54.

a term of convenience, but meaningless when considered from the perspective of each individual technological artefact. “Anthropomorphic machines”, despite being a term used frequently in the thesis, is resisted as an ontologically significant category. Each of the case studies presented is anthropomorphic in its own way, both visually and functionally. In the case of SHRDLU, the object of study is available to us only through relations with a computer. The concept of the gynoid sexbot relates strongly to various different objects but is inherently incorporeal. They are both dramatically different from embodied and unmediated objects like Grey Walter’s robot tortoises. They may be grouped together for the sake of human convenience, but the grouping means little to the individual machines. This is an unconventional approach to categories of technological artefacts. Furthermore, even an object like the Kinect, which is a mass-produced consumer videogames peripheral, resists grouping through the mobilisation of OOO in this thesis. There is a broad category which the thesis describes as a “brand” incorporating human ideas about the Kinect. But the grouping of “Kinect” is not meaningful to a *specific individual* Kinect. The presence of other Kinects is unknown to the individual Kinect. The individual Kinect is a distinct object and, when considered from its own perspective, defies broad comments about its behaviour because of its particular relations with the rest of the world.

OOO is capable of adding to the literature on technology, but not without transgressing disciplinary boundaries. To some extent this thesis must bridge a gap between philosophy of technology and metaphysics, a gap that has proved useful in the past for discussing specific instances of human-nonhuman relations. However, as the next section will show, this thesis embraces ontological questions and makes them a part of the study of specific technological objects.

Actor-Network Theory (ANT), Object-Oriented Ontology (OOO) and alien phenomenology

Actor-Network Theory (ANT)

In order to consider the agency of both human and nonhumans, it is necessary to base our understanding on an ontology that encompasses humans and nonhumans equally. A key theorist for this approach is Bruno Latour who was central to the development ANT as part of his studies of science and laboratory practice. ANT is self-consciously non-anthropocentric and was developed by Latour and others such as Michel Callon and John Law to explain the work of scientists and engineers in interaction with the world. A particular hallmark of this strategy is the commitment to considering human and nonhuman things in a symmetrical way. ANT scholars are therefore different to most other post-Kantian Western philosophers who write within the paradigm of “the bland default metaphysics that reduces objects to our

human access to them."[46] Each human or nonhuman actant has its own agency and is constantly changing its relations with other actants. No actant is ever reducible to any other actants, as this would mean creating a hierarchy of objects. For example, a human body is not reducible to organs, systems, cells, evolutionary history or demographics. Each of these actants is identifiable as an entity but are in relation with the other actants.

The agency of nonhumans is revealed by ANT to be key to the success or failure of scientific projects. For a program to succeed, the scientists must engage with all the relevant human and nonhuman actants and persuade them to behave in a particular way. A classic example is Callon's study of St Brieuc Bay in which marine biologists attempted to find out how to increase the number of scallops living in the area.[47] Callon demonstrated that in order to make the project work, the scientists had to attract the local authorities and laypeople and persuade them not to sabotage the study. But he also showed that the scallops themselves were active participants in the network and needed to be encouraged to behave a certain way so that the scientists could study them. The human or nonhuman, and even the living or non-living status of the actant is only relevant insofar as it affects its agency. Agency and ontology are generated through relations between objects. In Latour's ANT, objects only really exist as a set of relations, shifting ontologically and mutating constantly in response to the entities around them. A leaf sitting in the palm of my hand is constantly embroiled in competing networks. The wind wants to push air particles through the space that the leaf occupies, moving it gradually out of the way. The air wants to drag water molecules out of the leaf's stoma. The leaf is also part of my network, behaving in a typical and uncomplicated way, allowing me to discuss its metaphysics. The leaf is subject to plants, humans, ideas, physical forces, subatomic particles, radiation and time, constantly changing as each of these objects moves around it. The leaf's agency and ontology emerges from the sum total of these networks.

Since the 1980s, ANT has been used widely beyond science studies.[48] Yet some scholars would say that even ANT is still too anthropocentric. Latour's ideas have been reinterpreted by Graham Harman as metaphysics in his book *Prince of Networks*.[49] Harman is critical of ANT's relational ontology and the way Latour construes the agency of objects, but praises the

---

[46] Graham Harman, *Prince of Networks: Bruno Latour and Metaphysics* (Melbourne: re.press, 2009), 25.

[47] Michel Callon, "Some Elements of a Sociology of Translation: Domestication of the Scallops and the Fishermen of St Brieuc Bay," in *Power, Action and Belief: A New Sociology of Knowledge*, ed. John Law (London: Routledge & Kegan Paul, 1986).

[48] For example, in education. See Zheng Zhang, "The Changing Landscape of Literacy Curriculum a Sino-Canada Transnational Education Programme: an Actor-Network Theory Informed Case Study," *Journal of Curriculum Studies* 48, no. 4 (2016): 547-564.
And in this example, clients in the construction industry: Megumi Kurokawa, Libby Schweber, Will Hughes, "Client Engagement and Building Design: the View from Actor-Network Theory," *Building Research and Information* 45, no. 8 (2017): 910-925.
And here, in a paper on governmental power in Congo: Peer Schouten, "The Materiality of State Failure: Social Contract Theory, Infrastructure and Governmental Power in Congo," *Millennium: Journal of International Studies* 41, no. 3 (2013): 553-574.

[49] Harman, *Prince of Networks*.

"power and precision"[50] of his ANT. Speculative realism has been influenced and critiqued by scholars from many different backgrounds, including (but not limited to) Latour.[51] Speculative realists are united in their rejection of correlationism in philosophy. Correlationism is the assumption that all human research, analysis and endeavour is done for the ultimate benefit of humans. In the words of speculative realist Quentin Meillassoux, correlationism insists that it is not possible "to consider the realms of subjectivity and objectivity independently of one another."[52] Anti-correlationism does not distinguish between the existence of an object and the way that a human encounters an object; an object is not the correlate of human thinking. In the words of Levi Bryant, "we must avoid, at all costs, the thesis that objects *are* what our access to objects *gives* us."[53] Correlationism carries the implicit belief that humans are special and unique, perhaps even that we are a special ontological case, and is based upon the increasingly problematic binary of "human" and "nonhuman". ANT's emphasis is mostly on the relations between objects. The objects themselves only exist in the background of processes of translation and delegation. Ian Bogost claims that in ANT "entities are de-emphasized in favor of their couplings and decouplings. Alliances take center stage, and things move to the wings."[54]

The context and emphasis of disciplines is a not insignificant indicator of their metaphysical commitments. ANT is a sociological methodology, and therefore must privilege human priorities above all others. In Callon's study, he only discusses the agency of the scallops and other nonhumans with reference to how human actants may manipulate them and arrange them into networks. The agency of nonhumans is something that must be *overcome*. ANT is a part of the epistemological project in the philosophy of science; scholars work to understand how science progresses and how humans gain information about the world. Harman, however, insists upon maintaining a distinction between the ontology of objects and epistemology.[55] These fields may consider objects independent of all other external actants, including humans, and the nonhumans are therefore better represented. Harman places Latour's work in an ontological context to give his own work greater depth and a sense of legacy, but Latour himself has typically been focussed on the history of science in an effort to understand the success or failure of particular human endeavours, such as the sequencing of a peptide[56] or the attempt to implement a new public transport system in Paris.[57] Harman and others, in contrast, focus on how individual objects exist in the world and how they may be said to interact. These ideas have implications for specific episodes in science and technology studies, but they can also exist as purely theoretical disciplines.

---

[50] Ibid., 32.
[51] Levi Bryant, Nick Srnicek, and Graham Harman, eds., *The Speculative Turn: Continental Materialism and Realism*, Anamnesis (Melbourne: re.press, 2011).
[52] Meillassoux, *After Finitude*, 5.
[53] Bryant, *The Democracy of Objects*, 18.
[54] Bogost, *Alien Phenomenology,* 7.
[55] Jay Foster, "Ontologies without metaphysics: Latour, Harman and the philosophy of things," *Analecta Hermeneutica* 3 (2011), 9.
[56] Bruno Latour and Steve Woolgar, *Laboratory Life: The Social Construction of Scientific Facts* (Beverly Hills: Sage Publications, 1979).
[57] Latour, *Aramis*.

Object-Oriented Ontology (OOO)

Speculative realism has spawned several different anti-correlationist schools of thought, and OOO is one of the most well-known. OOO is an emerging field, and it is prone to contributions by individuals from many different disciplines. Harman initially used the phrase "object-oriented philosophy"[58] (but has latterly embraced the term OOO, as is evident in his use of the term in a recent book title[59]), Levi Bryant invented the word "onticology" [60] to describe his particular approach, and Ian Bogost prefers to think in terms of "unit operations"[61] to distinguish his ontology from object-oriented programming. OOO seeks to reverse the linguistic turn in Western philosophy and social science and return them to a study of materials. It is interested in objects for themselves, not objects for humans. The word "object" does not only refer to the simple inanimate objects accessible to humans. Tom Cruise, Superman, orange blossom, *Drosophila* flies, manganese atoms, pulsars and the concept of free will are all examples of objects. Like in Latourian metaphysics, objects are not arranged into hierarchies, nor are they reducible to more fundamental explanations. Timothy Morton argues that OOO has arisen in the wake of the threat of environmental apocalypse.[62] It rejects "Nature" as a concept imbued with correlationism, and it also "offers a middle path – not a compromise, but a genuine way out of the recent philosophical impasse of essentialism versus nihilism."[63] In the modern world, humans are becoming more aware of the agency of powerful nonhuman actants, such as climates, carbon footprints and oceans. All these objects seem to exert an agency independent of humans, particularly individual humans or small groups. They are objects without subjects, and thus lend themselves to a non-correlationist approach. OOO studies all actants as objects, not as what happens between subject and object.

Graham Harman, author of *Tool-Being* and numerous other books, is arguably the best known object-oriented ontologist.[64] *Tool-Being* is based upon Harman's doctoral dissertation and he has since written several books and papers on object-oriented philosophy and ontology. In *Tool-Being*, Harman closely analyses the work of Heidegger, whom he clearly admires, although his interpretation is unusual.[65] Timothy Morton states that Harman:

---

[58] Graham Harman, *Guerrilla Metaphysics: Phenomenology and the Carpentry of Things* (Chicago: Open Court, 2005).
[59] Harman, *Object-Oriented Ontology*.
[60] Bryant, *The Democracy of Objects*.
[61] Bogost, *Alien Phenomenology*.
[62] Timothy Morton, "Here comes everything: the promise of Object-Oriented Ontology," *Qui Parle: Critical Humanities and Social Sciences* 19, no. 2 (2011): 163-190.
[63] Ibid., 164.
[64] Harman, *Tool-Being*.
[65] Sevket Benhur Oral, "Weird Reality, Aesthetics, and Vitality in Education," *Studies in Philosophy and Education* 34, no. 5 (2014), 464.
Peter Wolfendale, "The Noumenon's New Clothes (Part 1)," *Speculations* III (2012): 290-366.
And
Tom Sparrow, *The End of Phenomenology* (Edinburgh: Edinburgh University Press, 2014), 114.

> …discovered a gigantic coral reef of sparkling things beneath the Heideggerian U-boat. The U-boat was already travelling at a profound ontological depth, and any serious attempt to break through in philosophy must traverse these depths, or risk being stuck in the cupcake aisle of the ontological supermarket.[66]

The first of Harman's major discoveries comes through the study of Heidegger's tool analysis in *Being and Time*. Heidegger distinguished two types of relationships between humans and machines, expressed through his well-known example of the use of a hammer. The hammer is generally ready-to-hand (*zuhanden*), available to us as a tool and untheorised. Harman uses "tool-being" as a synonym for readiness-to-hand.[67] Tool-being is apparent when the tool is used unconsciously to achieve a goal. Very rarely, we might need to theorise the hammer, such as when it is broken.[68] At such a moment, the hammer is said to be present-at-hand (*vorhanden*). The "broken tool" is a metaphor for when objects become noticed, theorised or even just gazed at. Most of the time, objects are ready-to-hand for humans. Objects appear in the background of our lives, taken for granted, and are not the centre of our attention. Objects that work well from a human point of view become invisible. For instance, most of the time a screen is taken for granted by us, but it becomes present-at-hand when we notice that it has started to malfunction, or when we begin to theorise it. Harman makes an implicitly realist claim about objects: that the tool-being of objects is utterly separate from the way that the object appears to other objects.[69] Harman claims that the multi-faceted configurations of objects are incommensurable with each other since its readiness-to-hand (or "tool-being") always rushes ahead of its presence-at-hand.[70] Each interaction with an object is different, and no other object can ever encounter the whole of an object's tool-being. Because of the infinite number of objects in the universe, there are therefore an infinite number of different present-at-hand experiences, each of which is invisible to other objects. From this, Harman concludes that objects withdraw infinitely from all relations. An object can only be ready-to-hand or present-at-hand at one moment,[71] and presence-at-hand reveals only those qualities that we are equipped, at that moment, to sense. So no object can experience *all* of another object, or gain access to the object's true essence, its inwardness.

Harman's retelling of the tool-analysis means that readiness-to-hand and presence-at-hand are always at play in interactions between all objects, including inanimate ones. Harman's description of tools originates from Heidegger's famous account of hammers, but the principles described by Harman need not only apply to "tools" in the conventional sense. The model developed by Harman applies to every object, including both human and nonhuman objects. Readiness-to-hand and presence-at-hand are types of relations between two objects, in this case the human and the hammer. Yet readiness-to-hand and presence-at-hand are, for Harman, evident in every kind of object relation. This is not simply a panpsychist argument,

---

[66] Timothy Morton, *Hyperobjects: Philosophy and Ecology after the End of the World*, Posthumanities (Minneapolis: University of Minnesota Press, 2013), 14.
[67] Harman, *Tool-Being*, 4.
[68] Harman, "The Road to Objects," 174.
[69] Harman, *Tool-Being* 21.
[70] Ibid., 224.
[71] Ibid., 47.

although panpsychism has been embraced by certain participants in the nonhuman turn (see Chapter 3). Harman himself asserts that Heidegger's concept of Dasein, a word that describes human being, has been incorrectly interpreted to refer only to humans. Harman argues that Heidegger spoke of human Dasein but that he did not necessarily intend to restrict Dasein to humans.[72] In Harman's view, the kind of being that Heidegger spoke of referred to any being that was concerned with its own being, which was not necessarily limited to humans, although Heidegger probably would not extend Dasein to inanimate objects.[73] However, the extension of Dasein to all objects is not necessary for Harman. Readiness-to-hand and presence-at-hand become radically separated for Harman.[74] Harman identifies a depth in an object's readiness-to-hand that cannot be fully experienced by *any* object, not just a human. He gives the example of a metal appliance resting on a lake of ice:

> I hold that the resulting interaction between stove and ice is philosophically *identical* with the more familiar case of Dasein and the broken hammer. For what is decisive in the famous account of the "broken tool" is not that implicit reality comes into conscious view, as if *human surprise* were the key to the reversal within being. Rather, the important factor is that the heavy object, while resting on the ice as a reliable support, did not exhaust the reality of that ice.[75]

Presence-at-hand is not something that only happens when a human interacts with an object. A stove only reacts to certain things about the ice: hardness, coldness, brittleness, and perhaps other things. The hardness, coldness and brittleness of the ice do not constitute the whole of its being. Coldness is a "relational property", and in Harman's later terms constitute a part of the sensual object formed between stove and ice.

The key difference between Harman's metaphysics and that of Latour is their opinions on how objects form relations and, therefore, express agency and become accessible to one another. Unlike in ANT, Harman's objects do not become different objects when exposed to relations with other objects. They are withdrawn from the rest of the universe: "Since all actants are utterly concrete, confined to a single time and place, in the strict sense they cannot communicate with each other at all. They interact only by way of abstracting from each other."[76] Objects are separated from other objects' access to them. A human might experience an apple in terms of redness or, at another moment, in terms of ripeness. But these are only a human's conception of an infinitely withdrawn object. The apple-ness of the apple is inaccessible. The apple may also have an experience of the human, though this kind of bidirectional relationship is always asymmetrical.[77] Two separate interactions occur. For example, the human may experience the texture and taste of the apple as she bites into it, while the apple experiences the piercing hardness of the teeth by having a section removed. Two metaphysical processes occur at the same moment. Two separate interactions also occur when two nonhuman objects relate to one another, but we must speculate about the senses

---

[72] Graham Harman, *Heidegger Explained: From Phenomenon to Thing* (Chicago: Open Court, 2007), 58.
[73] Ibid, 60.
[74] Harman, *Tool-Being*, 47.
[75] Ibid., 223.
[76] Harman, *Prince of Networks*, 54.
[77] Harman, *The Quadruple Object*, 75.

of nonhumans. If a stone sinks into mud, then both the stone and the mud encounter certain qualities of the other. We might speculate that the stone senses a downward movement and a wetness that adheres to its surface. The mud changes shape, sensing the weight of the stone and reacting to it. Perhaps each of the nonhumans experiences a slight change in temperature. But the mud does not encounter what we would call the *colour* of the stone, because it does not have the faculties to sense light. Note that this does not imply that objects are completely immutable for Harman, only that the object does not become another object moment by moment. The apple, for example, changes when it is bitten, and its real qualities change correspondingly. Once the apple is eaten or decayed in a rubbish dump somewhere, it is destroyed and ceases to exist. This is dealt with further in Chapter 3 in the discussion of the withdrawal and change of robots.

No two objects can ever make direct contact. Harman speaks of a "mutual darkness" [78] of objects, a withdrawal that requires "a third term or mediator"[79] for contact to occur between the "noncommunicating crystalline spheres sleeping away in private vacuums."[80] The world of real objects needs something to explain interaction. Here Harman takes inspiration from José Ortega y Gasset's distinction between real objects and the access that humans have to reality through metaphor.[81] The influence of Ortega is evident in Harman's frequent use of metaphor. Ortega distinguishes between the "images" of objects, which is what we encounter when we look at or use an object, and the reality of objects.[82] Objects exist in their own right, and Harman calls them "real objects". Ortega's "images", which are "nothing more than correlates of our own experience," are "sensual objects".[83] Real objects *really* exist – that is, they have their own reality outside of our perception of them. The sensual object is analogous to Husserl's intentional objects; that is, the sensual object is the one that exists to another object. But it is not "an idealist prison" removed from the inaccessible real world, as in Plato.[84] It needs to have a corresponding real object that other objects can only access indirectly.

Real objects exist for themselves and still exist when we turn away from them. Sensual objects exist only to facilitate relations exist between real objects permitting indirect interaction between the two withdrawn objects, so that "the actors involved in [the interaction] are no longer separate, but form a new object with its own interior."[85] In the above example of the apple, the human does not experience the real object but sensual objects such as "sweetness", "redness" or "hardness". The sensual object is a mediator between the apple and the human. It is not possible, for Harman, to claim that the human is interacting with a partial object, namely, that part of the apple that is sweet. Harman rejects Hume's characterisation of objects as "bundles of qualities".[86] Objects are inherently

[78] Harman, *Object-Oriented Ontology,* 12.
[79] Ibid.
[80] Harman, *Guerrilla Metaphysics*, 91.
[81] Harman, *Object-Oriented Ontology.*
[82] Ibid., 78.
[83] Ibid., 78.
[84] Harman, *The Quadruple Object*, 26.
[85] Harman, *Guerrilla Metaphysics*, 206.
[86] Graham Harman, "On vicarious causation," in *Collapse* vol. II, ed. Robin Mackay (Urbanomic, 2007).

irreducible and whole. Therefore the apple is withdrawn and inaccessible to every other object, no matter how many interactions occur. This indirect interaction between objects themselves is what Harman calls "vicarious causation".[87]

The myth of the off-shore drilling rig from his book *Circus Philosophicus* may help to explain this position.[88] Harman compares the lack of direct interaction between his objects to the occasionalism of medieval Islamic philosophers, who claimed that objects could only interact with God.[89] In this view the human and the apple do not interact directly: each object is in a relationship with God, and thereby indirectly affect each other through divine intervention. In other words, God acts at every occasion. But in Harman's ontology each object is reduced to an image that is conveyed by a sensual object.[90] God is replaced by the metaphor of the oil rig, and each object is reduced to an image (Harman appears to borrow this term from Ortega as a substitute for "sensual object").[91] We can only obtain an image of each of these objects, not the objects themselves; that is to say that the metaphorical oil rig receives metaphorical sensual objects but not real ones. Every object becomes "a hazy caricature of its deeper plenitude"[92] to the other objects, because no other object can access the true nature of the object. The apple is withdrawn from us and we can only access an image –tastes and colours and so forth which form the impression of the apple that we are capable of sensing. Similarly, the apple only receives an image of the human. Oil rigs, in Harman's poetic description, convert any number of dead organic objects into fuel by siphoning them out of the earth: "It draws them to the surface of everyday life, where they are used as energy for the most prosaic modern actions."[93] Before siphoning, they are decayed accumulations of hydrocarbons from the time of the dinosaurs. After siphoning they are crude oil, and become tangled in a set of human concerns and ideals. The oil rig turns past objects into present objects. In the myth, it is not only objects from the past that are transformed, but objects from the present and the future; living, dead and non-living objects; human objects and nonhuman objects. In the above example, the oil rig is the human and the image is the sensual object that comes from the apple. Harman envisions each object in the universe as an oil rig, equivalent in their inability to access real objects through interaction, all siphoning images of all other oil rigs, so that "[a]ll real objects of every size now have the power to interact with all other things, at the price of turning them into images."[94] The real objects do not touch each other directly, but can interact through sensual objects. Siphoning does not reduce the object. The real object remains withdrawn, and only images of it are accessible through the oil rigs. Harman offers this myth as a metaphor for vicarious causation. In this metaphor, it is also impossible to represent the materialist view of symmetrical interaction, since oil rig A cannot interact directly with oil rig B. The apple siphons an image of the teeth while the human siphons an image of the apple. Harman's description is a sublime account of

[87] Harman, *Guerrilla Metaphysics*, 91.
[88] Graham Harman, *Circus Philosophicus* (Ropley, UK: O Books, 2010).
[89] Harman, "Indirect causation," in *The Quadruple Object*.
[90] Graham Harman, *Circus Philosophicus*, 46.
[91] Ibid., 46.
[92] Ibid., 49.
[93] Ibid., 45.
[94] Ibid., 48.

object relations, and it succinctly argues for the lack of legitimacy in any anthropocentric metaphysics. Nevertheless, it is a challenging position to accept.

Over time, Harman has become more and more interested in art and aesthetics, even declaring that aesthetics is "first philosophy".[95] This is based on his concept of "allure", which he describes as the disconnect between the real object and the sensual object. (This disconnect is discussed further in Chapter 3 with reference to Edmund Husserl.) The disconnect between real object and sensual object is evident in human uses of metaphor in the description of objects. Harman gives the example "my heart is like a furnace".[96] It is a metaphor that evokes certain furnace-like qualities (such as intensity or warmth) while explicitly ignoring others (such as size and metallic composition). In the case of the metaphor, then, it is the spectator or reader who is the "real object" to which sensual qualities bind.[97] Harman's metaphysics makes a point about aesthetics that exposes one of his most important concepts. It represents in very clear terms the withdrawal of objects. It expresses the need for a metaphysical explanation of the way that art can "move" humans without physically touching us; as Francis Halsall points out, this is appealing to artists because it supports the idea "that works of art have an autonomous identity"[98] (although, like all objects, they are only as autonomous as their relations with others allow). Emmy Mikelson makes a similar point, commenting on the artwork as an object in its own right to which observers only have a single point of entry.[99] She argues that speculative realism has "a constructive rapport" with the beheading of the subject in art.[100] It is curious that such an anthrodecentric approach should find such use for a process as human as art, and in a way it exposes the paradox inherent to attempts to find a non-correlationist metaphysics.

Ian Bogost's metaphorism draws on this part of Harman's metaphysics. The effect of one object on another can be depicted as a process of one thing becoming like another, either physically or figuratively.[101] When I place a mug on a table, the mug models itself on particular qualities of the table: immovability and position in space. As a plant absorbs water through its roots, the water conforms to the shape of the tree, adhering to the walls of the xylem and shooting up towards the leaves. As a text is read by scholars of a different tradition, it is translated into a different technical language and forms analogies with ideas that the author did not intend.[102] This resembles the myth of the oil rigs because each object detects a caricature of all others, but without requiring us to accept the concept of vicarious causation. Of the two accounts, I prefer Bogost's, mainly because it seems less reliant on a set

---

[95] Graham Harman, "On Vicarious Causation," 221.
[96] Ibid., 215.
[97] Graham Harman, "Materialism is Not the Solution: on Matter, Form, and Mimesis," *Nordic Journal of Aesthetics* 47 (2015), 108.
[98] Francis Halsall, "Art and Guerrilla Metaphysics: Graham Harman and Aesthetics as First Philosophy," *Speculations: A Journal of Speculative Realism* V (2014), 388.
[99] Emmy Mikelson, "Space for Things: Art, Objects, Speculation," in *And Another Thing: Nonanthropocentrism and Art*, ed. Katherine Behar and Emmy Mikelson (Earth, Milky Way: punctum books, 2016), 11-20.
[100] Ibid., 15.
[101] Bogost, *Alien Phenomenology*, 60.
[102] Juri Lotman, "On the Semiosphere," *Sign Systems Studies* 33, no. 1 (2005).

of *a priori* assumptions derived from yet more ancient philosophy. OOO is preoccupied with new approaches that radically differ from almost all older ontological models. It therefore seems incongruous to articulate a complex theory of object interaction that has its origins in medieval Islamic theology. I am not alone in this discomfort. Levi Bryant has said that "[p]erhaps no element of Graham's thought has been more maligned than his doctrine of vicarious causation."[103] Harman, however, brings OOO more into line with traditional phenomenology and therefore lends the discipline a sense of authority that it might otherwise lack.

Alien phenomenology

We might conclude that OOO is in conflict with phenomenology. Traditionally, phenomenology claims that the human experience of objects is the only possible way of knowing the world. The existence of objects independently is of little interest and, in some cases, impossible. Tom Sparrow has explored the contradiction in phenomenology.[104] He argues that while phenomenology retains a belief in the reality of objects, phenomenological research would in fact do better with a commitment to idealism rather than realism.[105] When compared with speculative realism, its weaknesses are revealed: it can only study how conscious entities relate to the world.[106] But in OOO, and other speculative realisms, the implicit realism is evident. Objects really exist and two non-conscious objects can form relations. OOO claims that objects have inherent qualities that do not emerge from relations. This is particularly clear in Bogost's work: phenomenological ideas may simply be extended to apply to inanimate objects as well as humans. Instead of studying how objects appear to us, we study how objects appear to nonhumans. His book *Alien Phenomenology, or, What It's Like to be a Thing* asks what sorts of phenomenological experiences machines (and other objects) are capable of, without making hierarchical distinctions between classes of objects.[107] Levi Bryant reframes Bogost's argument in terms of his own ontological commitments by suggesting that alien phenomenology seeks "to determine the flows to which a machine is open, as well as the way that machine operates on these flows as they pass through the machine"[108] (he is not only referring to technological artefacts when he speaks of "machines"). Alien phenomenology, therefore, involves making inferences about objects and the kinds of interactions that they have with other objects, based on observation of the structure of the object and the kind of relations that it forms. It has been called

---

[103] Levi Bryant, "Vicarious Causation," *Larval Subjects* (blog) (December 24 2010). http://larvalsubjects.wordpress.com/2010/12/24/vicarious-causation-2/ (accessed January 17 2018).
[104] Sparrow, *The End of Phenomenology*.
[105] Ibid., 12.
[106] Dan Zahavi takes issue with this view in "The End of What? Phenomenology vs. Speculative Realism," *International Journal of Philosophical Studies* 24, no. 3, 2016: 289-309.
[107] Bogost, *Alien Phenomenology*.
[108] Levi R. Bryant, *Onto-Cartography: an Ontology of Machines and Media* (Edinburgh: Edinburgh University Press, 2014), 62.

"*applied* speculative realism" [emphasis in original].[109] Being based on speculation, alien phenomenologists can only make second-hand observations about nonhumans, but as we shall see this is a strength as well as a weakness.[110] In the case studies in later chapters, it will be shown that anthropomorphism in technological objects is a persuasive tool, and it is argued that alien phenomenology presents a means of arriving at a more critical study of objects.

There are now numerous scholars devoting time to applying anthrodecentric ideas to nonhumans. The Parliament of Things project publishes submissions from authors who write in a Latourian style about the existence of nonhumans, as well as identifying this style in earlier authors.[111] Also, as mentioned in the introduction, Bogost and Christopher Schaberg have edited a series of short books intended to provide insight into the being and relations of objects. This series encompasses many mundane objects. *Eye Chart* by William Germano describes the historical background to contemporary eye testing equipment,[112] and *Drone* by Adam Rothstein explains the relative significance of different engineering traditions, as well as the software and hardware involved in contemporary drones.[113] The series exposes un-interrogated relations between physical objects (especially human-made ones) and seemingly unconnected ideas, texts, and physical objects (especially historical ones). This, in my view, is an important part of alien phenomenology, although not all of it. Often, the work of alien phenomenology of the object is left up to the reader in a way that erases the background and theoretical connections of the author. John Garrison's *Glass* is a good example of this.[114] The title is simple and broad enough to highlight the various ways that the word "glass" is used by humans, and the themes that bind them together: "the matrix of time, self-reflection, desire, and world-creation that this book has been tying together – and tying to our cultural fascination with glass."[115] This fascinating history connects mirrors, microscopes, sea glass, and Google Glass (the Augmented Reality hardware). But the broadness of the word "glass" is also a limitation, because the tying together of glassy objects from so many temporal and physical contexts is a very human practice. The objects are connected through language and analogy, but *from the point of view of the sea glass*, Google Glass is a very alien object. For us, the analogy becomes reality, and the qualities that set these objects apart are less emphasised than the qualities that keep them together in our minds. For this reason, Garrison's account is slightly anthropocentric. His focus is the *concept* of glass, rather than any of the objects he describes.[116] It is a tool that the alien phenomenologist can use, certainly. This thesis makes a similar move when it speaks of anthropomorphic machines. "Anthropomorphism" is a phenomenon that groups vastly different objects under a heading that make sense to humans, not machines. And that's the point. Grouping objects together

[109] Sparrow, *The End of Phenomenology*, 170.
[110] Bryant, *Onto-Cartography*, 64.
[111] "Parliament of Things," *Parliament of Things* (2017). https://theparliamentofthings.org (accessed December 21 2017).
[112] William Germano, *Eye Chart*, Object Lessons (New York: Bloomsbury, 2017).
[113] Adam Rothstein, *Drone,* Object Lessons (New York: Bloomsbury, 2015).
[114] John Garrison, *Glass*, Object Lessons (New York: Bloomsbury, 2015).
[115] Ibid., 37.
[116] Similarly Chapter 5 of this thesis discusses the *concept of the gynoid sexbot* as distinct from the physical gynoid that is sitting next to you right now.

under a heading that makes sense for humans enables the human practice of alien phenomenology which cultivates interest in their way of being in the world.

Knowing more about the relations between sea glass and other objects is useful for discovering its inner qualities. But it can only provide a partial glimpse into the phenomenology of a glassy object. This is not a criticism – no author can provide a complete account of sea glass, since we are not the sea glass. One book in the series that goes some way to evading this issue is *Earth* by Jeffrey Jerome Cohen and Linda T. Elkins-Tanton.[117] Co-writing a book on this subject, particularly by scholars in different fields (Cohen directs a medieval and early modern studies institute and Elkins-Tanton is a planetary scientist), brings the partiality of human attempts at alien phenomenology to the foreground. Parts of the book are taken from conversations between the two scholars, and their disciplinary training is evident in the way that they speak about their area of study. As Cohen says:

> Earth is an object of such immensity that it beckons us to think outside of our comfortable orbits of career path and discipline. The risks must also include that we would talk at each other (your Earth might be a planet with a history so long that humans simply do not figure much; mine might be an Earth that is too much a home for humans and does not adequately engage with the longevity of stones, water, or atmospheric gases).[118]

The two authors are still communicating in a human way, and are inevitably drawn to anthropocentric inquiries. But their cross-disciplinary collaboration forces them to confront the truth that both their views are partial; both their views are only part of the story for Earth. Earth itself has no way of providing humans with a complete account of its being and its relations with other objects. But there is no reason why we cannot attempt to know as much as possible and thus arrive at a more detailed and faithful alien phenomenology.

Human interpretation of phenomena is less likely to be questioned than machinic interpretation of phenomena. We are more likely to believe a human view of events. This is despite evidence of humans' underwhelming ability to detect what is "really there" and our tendency to be fooled by illusions. For centuries scientists have become more and more aware of the subjective nature of human vision[119] and we now know that the mind is not a passive receiver of information, but is actively involved in sensing and perceiving. Eyes are prone to error and eccentricities. But they are *human* errors and eccentricities. Light is received by the eyes and is immediately a part of the complex semiotics of the human eye involving reactions of photosensitive material, changes in ion concentration and the release of neurotransmitters. No inorganic material is capable of carrying out this sequence of events, and the sensed and transmitted information conveyed by machines is subject to completely different coding and decoding. We perform similar sensory-perceptive acts in our relations with all objects, including ideas. Although every object (be they human, machine, or idea) has a sense of the world, humans are typically better at conveying information to other

---

[117] Jeffrey Jerome Cohen and Linda T. Elkins-Tanton, *Earth*, Object Lessons (New York: Bloomsbury, 2017).
[118] Ibid., 69.
[119] Jonathan Crary, *Techniques of the Observer: on Vision and Modernity in the Nineteenth Century* (Cambridge, MA: MIT Press, 1990).

humans. It may be a challenge to describe what I see, but because most humans share the same visual apparatus other humans are likely to grasp my take on events. If a machine were to describe its experience of the sensation of light and colour, it might not be so easy to understand because artificial sensors are not structurally similar to human eyes. Similarly, I am more likely to understand Cohen and Elkins-Tanton's take on Earth than that of the Earth itself, since I am not attuned to magnetic fields, gravity, and shifts in matter that span long beyond my own lifespan. Alien phenomenology will always be anthropocentric to some extent.

Anthrodecentric thinking has yielded many accounts of ordinary material. Jane Bennett's vital materialism is distinct from alien phenomenology, and it is heavily informed by the tradition of Deleuze and Guattari,[120] but her description of "*Thing-Power*: the curious ability of inanimate things to animate or act, to produce effects dramatic and subtle"[121] suggests a similar anthrodecentric goal to that of alien phenomenology. This is particularly evident in her analysis of metal and metallurgy.[122] Bennett describes the many ways in which metal has traditionally been portrayed as passive, dead, and uniform.[123] However, she points to both contemporary scientific and ancient metallurgical texts that highlight the activity and idiosyncrasy of individual metallic objects. Some of our reluctance to appreciate this activity is due to human limitations, such as the shortness of our lifetimes and inability to perceive the tiny imperfections, holes, and quivering of atoms that give metallic objects their unique properties.[124] Similarly, in *Stone*, Jeffrey Jerome Cohen describes the nature of stone (in everything from cultural to tectonic terms).[125] Cohen's background in medieval studies naturally leads to the work being concerned with the connotations of stone within that time and place, and like Bennett's work it is peppered with cultural references. It is an anthropocentric approach, certainly, but Cohen's choice of subject matter ("that mundane object on which a philosopher might perch in order to think, ideation's unthought support; or in the palm, a spur to affect, cognition, and contemplation"[126]) seems inherently anthrodecentric in that it brings such an apparently bland object to the forefront of the reader's mind.

Harman identifies two different tendencies in philosophy that deflect attention from objects and consequently interfere with alien phenomenology. He calls these tendencies undermining and overmining. Harman claims that most kinds of ontology attempt to explain objects by either reducing them to their most fundamental units (undermining) or reducing them to their effects on other objects, particularly the human mind (overmining).[127] In

---

[120] Jane Bennett, *Vibrant Matter: a Political Ecology of Things* (Durham: Duke University Press, 2010), Kindle Ebook.
[121] Ibid., "Thing-Power I: Debris."
[122] Ibid., "A Life of Metal."
[123] Ibid., "The Dead Weight of Adamantine Chaos."
[124] Ibid., "A Life of Metal."
[125] Jeffrey Jerome Cohen, *Stone: An Ecology of the Inhuman* (Minneapolis: University of Minnesota Press, 2015).
[126] Ibid., 11.
[127] Graham Harman, "Undermining, Overmining, and Duomining: a Critique," in *ADD Metaphysics*, ed. Jenna Sutela (Aalto, Finland: Aalto University Digital Design Laboratory, 2013): 40-51.

Harman's object-oriented philosophy, objects are not reducible to either of these extremes. Undermining might involve reducing objects to their atomic structure (or even to smaller objects), or, for example, asserting that individual objects emerge from a boundless *apeiron* as in some pre-Socratic views. While undermining claims that objects are "too shallow to be real", overmining claims that objects are "too deep to be real".[128] An example of this is idealism, in which the existence of objects outside the mind of a human is uncertain or irrelevant. But Harman also includes social constructionism, in which reality is found in language, discourse or power. Relational ontologies such as Latour's are also examples of overmining, since they reduce objects to nothing but their relations with other things.[129] Both these extremes deny the autonomy of objects like tables, cats and leprechauns, which are the foundation of Harman's philosophy. Harman also identifies a tendency that he calls "duomining", which is the strategy of both undermining and overmining objects at the same time. Harman argues that modern science duomines objects by simultaneously reducing downwards to the most basic units of matter and claiming that all objects are knowable through mathematisation.[130] There are numerous examples of undermining and overmining in this thesis. Because humans have built machines from very small and basic materials, it is easy to believe that the key to explaining the sensation, experience and behaviour of technological artefacts lies in examining smaller and smaller components.[131] This is undermining the object, and it may be seen in the perusal of circuit diagrams or source code. The technological artefact may be overmined by reducing it to its relations or to its role in cultural discourse or power structures. We also see overmining in some ways of attributing a quality of mind to technological artefacts (see Chapter 3), which explain machine experience and behaviour in terms of consciousness. The artefacts are thus reduced to flows of thought which account for things like movement and speech. Both overmining and undermining avoid consideration of the alien nature of objects. Reduction in either direction is reduction away from the inscrutable and unknowable experience of nonhumans and towards something that humans can more easily understand. For example, it is easier to explain the actions of Voyager 2 by reducing it to its fundamental units than to speak of it as a unified and agentic being. It is also easier to understand Voyager 2 in terms of its relations with other objects, like planets, photons, electromagnetic radiation and components of its structure such as sensors. While this thesis will present thinkers that either undermine or overmine objects (or both), it is hoped that the analysis, in the main, walks a middle path between these two extremes. This is an important part of alien phenomenology. It is not acceptable merely to avoid

---

[128] Ibid., 45.

[129] Harman, "Undermining and Overmining," in *The Quadruple Object*.

[130] Harman, "Undermining, Overmining, and Duomining: a Critique," 46.

[131] This is connected to the older and broader question of how wholes relate to their parts. There is no room here to discuss this point, but some relevant texts include:
Mark van Atten, "A Note on Leibniz's Argument Against Infinite Wholes," in *Essays on Gödel's Reception of Leibniz, Husserl, and Brouwer* (Switzerland: Springer International Publishing, 2015), 23-32.
And
Annemarie Mol, "Inclusion," in *The Body Multiple: Ontology in Medical Practice* (Durham, North Carolina: Duke University Press, 2002), 119-150.

anthropocentrism in the description of events; the language used must reflect this ongoing commitment.

Harman notes that art is the domain in which undermined and overmined qualities converge.[132] This brings us to another interesting quality of OOO: its ability to inspire not just academic but creative output. Bogost argues that the job of the philosopher should be to perform ontology or, as he calls it, "carpentry".[133] We should be "tracing the exhaust of [things'] effects on the surrounding world and speculating about the coupling between that black noise and the experiences internal to an object."[134] OOO is a laboratory, a place to build knowledge through the juxtaposition of conventionally ignored viewpoints. Bogost describes his own work with the game *Cow Clicker*, an exercise in both game design and activism, as an example of enacting his concerns through practical, carpentry-like philosophy.[135] For Bogost, written work is obfuscating and inaccessible, and he speaks sentimentally of his own engagement with the field: "so much of object-oriented ontology is, for me, a reclamation of a sense of wonder often lost in childhood".[136] The viewpoint of anthropomorphic machines, which has for so long been the subject of popular and academic interest, seems like a good place to start. It is hoped that the case studies in the following chapters constitute engaging and relevant objects of study for this laboratory. This thesis apparently commits the very error Bogost warns against in privileging written work over carpentry. But perhaps it will inspire carpentry in future work embedded in the design, construction or exhibition of anthropomorphic machines.

In Harman's early work, politics takes a back seat to metaphysics. However later work has drawn a link between politics and OOO.[137] And despite OOO's commitment to anti-correlationism, there are elements of the metaphysics that lend themselves to particular anthropocentric research programs. The application of OOO ideas to ecology is particularly significant, as in Morton's study of ecology, as discussed above. Nonhumans constitute both human bodies and the world in which we are irreversibly embedded.[138] OOO is a valuable site for transdisciplinary work, such as the intersection of ecological thought and philosophy.

OOO ideas have proliferated in different fields, but sometimes in reaction to it rather than by embracing it without reservation. The core of OOO has met with resistance from some feminists. A book of collected essays, *Object-Oriented Feminism*, edited by Katherine Behar, discusses the points of difference between OOO and object-oriented feminism (OOF). OOF is a reaction to OOO that shares some of OOO's convictions while rejecting others. After all, as Behar points out, groups such as women, people of colour, and the poor have historically

---

[132] Harman, *The Third Table*, 14.
[133] Bogost, *Alien Phenomenology*, 100.
[134] Ibid.
[135] Ian Bogost, "The Aesthetics of Philosophical Carpentry," in *The Nonhuman Turn*, ed. Richard Grusin (Minneapolis: University of Minnesota Press, 2015), 81-100.
[136] Ibid., 85.
[137] This is evident throughout Harman, *Object-Oriented Ontology*.
[138] Timothy Morton, "From Modernity to the Anthropocene: Ecology and Art in the Age of Asymmetry," *International Social Science Journal* 63 (2014): 39-51.

been considered tools "there for the using".[139] Behar refers to the unpaid work of housewives: "Tools behave nicely. They are demure. They present themselves for service retiringly, to be used without reward in contrast to the work."[140] In contrast, the woman who demands remuneration for her work is a "broken tool."[141] Objectification is not a novel experience for all humans. In my view, OOO breaks down the subject/object dualism, but does not go far enough in breaking down other dualisms, since breaking down dualisms is presumably not a high priority for most nonhumans. OOF also comments on the notably sexualised language used by writers like Harman, and plays with the erotic overtones, such as in this passage from Frenchy Lunning's essay:

> Despite Harman's assertion of ensnarement, these objects made of severed qualities never totally consummate their attraction. Instead, they remain forever in foreplay, in flux, in desire, and sometimes, in disgust. And it is in that strange alchemical exchange that the metaphor is successfully achieved.[142]

This essay juxtaposes Harman's concept of allure and Julia Kristeva's work on abjection. Even Harman's discussion of allure in *Guerrilla Metaphysics* implies a coquettish quality in metaphors,[143] but OOF unapologetically emphasises what is left unsaid by Harman.

OOO provides an ontology that invites comparison with a search engine. A search engine is sometimes erroneously thought to treat all inquiries the same. Any inquiry is valid, and results may be surprising and varied. However, in actuality, results appear based on "relevance", and the relevance may be determined by factors outside of the inquiry's control, such as censorship, sponsorship, popularity, and the user's previous search history. Alien phenomenology performs the same imperfect task. Objects are explicitly on an equal footing and we expect an exhaustive and impartial description. The carpentry of tools to explore the phenomenology of objects is based upon this impartiality. But humans who perform alien phenomenology are susceptible to biases. Just as Google's algorithm remains a secret, we cannot expect any one author to have certain knowledge of their own predispositions. This is why input from multiple authors is important. Objects withdraw, and we are left with only the sensual objects to which we are open. Just as a stone cannot perceive colour, an author with no training in medieval studies could not arrive at Cohen's conclusions.

Technology is not necessarily the focus of OOO. Unlike the philosophies of technology discussed at the start of this chapter, there is no immediate link to be drawn between OOO and technology (aside from its origins in the tool analysis). OOO is metaphysics after all, and maintains the conviction that *all* objects are sites of wonder for those who choose to investigate them, not just spectacular objects. Yet there is a flavour to OOO that invites the

---

[139] Katherine Behar, "An Introduction to OOF," in *Object-Oriented Feminism*, ed. Katherine Behar (Minneapolis: University of Minnesota Press, 2016), 7.
[140] Katherine Behar, "The Other Woman," in *After the "Speculative Turn": Realism, Philosophy, and Feminism*, ed. Katerina Kolozova and Eileen A. Joy (Earth, Milky Way: punctum books, 2016), 27-38.
[141] Ibid.
[142] Frenchy Lunning, "Allure and Abjection: the Possible Potential of Severed Qualities," in *Object-Oriented Feminism*, 86.
[143] The chapter in *Guerrilla Metaphysics* in which allure appears also discusses "charm". Harman, "Humor," in *Guerrilla Metaphysics*, 125-144.

discussion of technology. The figure of the cyborg comes to mind. A cyborg comprises of technological parts and organic parts. Yet it is irreducible to either of these parts. The cyborg object becomes an object in its own right, able to form certain kinds of relations and totally unable to form others. The same is true of anthropomorphic machines. As humans, we are able to identify certain qualities in machines that are more like us and more alien. But there is no sense in which those anthropomorphic parts form relations in one way and alien parts form relations in another way. The human feelings they create, from emotional connection to a sense of the uncanny, belong to the whole object. Moreover, OOO's implicit commitment to realism requires us to appreciate the object for its own sake. The anthropomorphic machine is an object in its own right, and it may not experience humans or humanity as a special object with which it has a connection. Anthropomorphism is very important to us, but not to the machine itself. OOO gives us this dialectical capacity: to rigorously attend to our own relationship with machines while holding in mind the alien nature of nonhumans.

Chapter 2: Kinect and "Natural" Interfaces

"What else does Kinect see?"
– Kinect settings menu, Xbox One. Accessed June 8 2017.

The first case study of this thesis is the Microsoft Kinect,[1] a natural user interface (NUI) with many remarkable aspects, not the least of which is its relative prevalence in living rooms throughout the developed world. There are two versions: Kinect 1.0 was released in 2010 for use with the Xbox 360 console and Kinect 2.0 was released concurrently with the Xbox One console in 2013. Although initially bundled with all Xbox One consoles, Microsoft has since announced that the Kinect is no longer compulsory for all Xbox Ones in a move widely described as the "death" of the Kinect.[2] The performance of the Kinect presents a useful example for anyone interested in the strangeness of nonhuman worlds in a strongly human context. This chapter aims to arrive at an alien phenomenology of the Kinect that emphasises the unhuman agency and the contrived humanity of this apparently bland consumer product. Investigating agency means investigating the sensory world of the object, since an agent can only respond to things that it senses, which means that the sensory world of the Kinect is very important for understanding the relationship that it does with humans. This chapter performs similar work to Adam Rothstein in *Drone* (mentioned in Chapter 1). The historical contexts available to this thesis (due to the author's disciplinary training) are incorporated with what Rothstein calls the "anatomy"[3] of the technological artefact. As emphasised in the previous chapter, this account can only ever be partial and anthropocentric. But as will be explained, collective alien phenomenology gives a more complete (if more confusing) view.

We begin with a brief summary of the main objects that will be considered in this chapter, their qualities and typical position with regard to the other objects. The complexity of the Kinect as it exists both in its individual and branded contexts demands a comprehensive

[1] Microsoft does not label the Kinect generations except by indicating which console or computer it is to be used for such as "Kinect for Xbox One". This chapter will adopt the designations used broadly by videogames and technology writers: Kinect 1.0 is the Kinect released in November 2010 and designed for use with the Xbox 360; Kinect 2.0 is the second Kinect released in November 2013 and designed for the Xbox One.
[2] Patrick Klepek, "Kinect Died in the Uncanny Valley," *Giant Bomb* (May 15 2014). https://www.giantbomb.com/articles/kinect-died-in-the-uncanny-valley/1100-4920/ (accessed 9 February 2018).
Holly Green, "The Many Deaths of the Kinect," *Paste* (January 10 2018). https://www.pastemagazine.com/articles/2018/01/the-many-deaths-of-the-kinect.html (accessed 9 February 2018).
[3] Rothstein, *Drone*, 59.

explanation of how this artefact can sense, perceive, think and act. This chapter argues that the Kinect *brand* is a distinct object which relates with objects like news articles, user forums, and advertisements, separate but in strong relation with any physical Kinect artefact.

The first section will discuss the objects themselves: the brand, then the console, game and sensor, plus those individual objects that exist within the sensor such as the camera and microphone. The brand and the individual unit have an interaction as intense as that between Kinect and the more prosaic objects of study here, materiality or immateriality being no major obstacle to the interconnection of things. The division of the Kinect-brand from the Kinect-artefact is an experimental part of this description of Kinect agency and objecthood. Each of these objects, brand and artefact, is discussed separately. Since the approach of this chapter is derived from alien phenomenology, it will not skip over these vital details. Knowledge of the objects discussed (although imperfectly conveyed through language) is essential to the later parts of the chapter. We must understand the qualities of the objects involved, and also acquire an appreciation of the most significant relations in their experiential worlds. In the second section there will be an analysis of reported user experiences of communications with the Kinect. This includes attempts to have a meaningful semiotic exchange with the device, and the process of pseudo-scientific investigation by humans that probe and evaluate the phenomenal experiences of the Kinect. Topics relating to Kinect troubleshooting and community expectations will be discussed. Also, the biometric testing and social gaming aspects of *Kinect Sports Rivals* (a game developed by Rare and released in 2014) will be addressed, since this game is a demonstration title designed to show what the Kinect is capable of.[4] After exploring the gamer-Kinect interface, in the third section of this chapter, a broader alien phenomenological analysis will be attempted, discussing the significant objects in the Kinect's world and its semiotic exchanges with them. We may not discover what it's like to be a Kinect, but we may gain insight into the way that humans conjecture that Kinects might sense, perceive and behave. Or perhaps more accurately, I will interpret the evidence of personal observation, technical accounts and reports by other users, and I will present this through the lens of my own view of the world. Unavoidably, you are reading a product of my relationship with the Kinect.

The objects themselves

NUIs

Natural user interfaces (NUIs) are a class of entities that are built with the aim of allowing humans to interact more "naturally" with machines. As a taxonomic group the class is rather problematic, since the "natural" aspects of the interaction are always in the eye of the human beholder. Two of the most important interfaces in NUIs encode gestures and natural speech,

[4] *Kinect Sports Rivals*, Xbox One.

which are processes that do indeed come naturally to most humans but are contrived processes in machines. Gestures and natural speech must undergo a translation process in order to make sense to a machine. The NUI is an ideal to be strived for, but it is an anthropocentric ideal that erases the alien qualities of technological artefacts. NUIs in general are engineered to produce interfaces that are natural for humans, but the engineering challenges posed by capturing human gestures and speech result in interfaces that frequently fail to live up to the hype (as described in this chapter). Moreover, while more and more natural interaction with machines is attempted, natural speech and seamless gestural control remain elusive for the time being. Nevertheless, the idea of humans interacting "naturally" with machines is a popular one, and in 2014 Microsoft tied the future of its console line to the Kinect, which is designed to engage with both humans' gestural and speech-based "natural" modes of communication. To date, many entirely Kinect-based games exist but fewer have appeared over time. (There are also Kinect games that make use of traditional controllers in addition to speech or motion capture.) In support of this "natural" rhetoric, a team of animal-computer interaction (ACI, rather than HCI) researchers has introduced the Kinect to a group of orang-utans at Melbourne Zoo. Simple games on Kinect aim to enrich the animals' habitat and change the nature of encounters between humans and orang-utans.[5] Orang-utans do not play the same Kinect games that humans play in their homes, but the gestural interface is similar. This supports the view that all kinds of apes, including humans, are capable of learning to adapt to NUI technology quickly enough to make the technology accessible and usable. Another NUI designed for gaming interaction is Leap Motion, which is intended for the human use of virtual reality headsets.[6] A sensor for detecting hand motions is built into the headset so that users can interact with a virtual environment.

A significant focus of research into NUIs has involved studying the gestural aspect of our communication with them. In addition to providing interfaces for entirely Kinect-based games, Microsoft and others have also been drawn to the possibilities allowed by "gestural excess" in controller-based games.[7] Gestural excess describes actions that do not contribute to success in a game but that are still performed by the player. Examples of gestural excess include recoiling from a blow in a fighting game and pumping one's fists in the air after a difficult win. The reactions are unnecessary because all the input to the machine comes through the controller and the console or computer's experience of the world does not extend to registering these enthusiastic human responses, but the responses are natural and unconscious. Experiments in harnessing gestural excess have begun. For example, in *Kinect Sports Rivals* wake racers can turn more quickly when they lean into a turn. Of course, once they become a part of the game they are no longer excessive and become strategic ways to improve one's performance. This is an example of "grokking", what we might consider to be the opposite of gestural excess. Grokking a system means not merely understanding it but intuitively responding to it. The word "grok", first used by Robert A. Heinlein in his 1961 novel

[5] Sarah Webber, Marcus Carter, Wally Smith and Frank Vetere, "Interactive technology and human-animal encounters at the zoo," *International Journal of Human-Computer Studies* 98 (2017): 150-168.
[6] "Leap Motion," *Leap Motion* (2017). https://www.leapmotion.com/product/vr#110 (accessed 16 February 2018).
[7] Thomas Apperley, "The Body of the Gamer: Game Art and Gestural Excess," *Digital Creativity* 24, no. 2 (2013): 145-156.

Stranger in a Strange Land, might describe the programmer's becoming very attuned and habituated to a programming language, or the relation that develops between a driver and a racing car. Humans learn the best way to communicate with nonhumans and, incidentally, gain knowledge of their qualities and way of relating to the world. In the Kinect's case grokking means a knowledge of gestural and spoken commands that the Kinect is able to sense, which then become second-nature to the user. It also means using the sensor in a way that works for the idiosyncratic home entertainment set-up, perhaps rather than the manner that is described in the manual. Movements come to be learned and rehearsed by humans until they lose the sense of being contrived or unnatural. Richard Harper and Helena Mentis identified this embodied comprehension in families, who did not know what "grokking" meant but still practised it, becoming adjusted to the system's shortcomings and quirks to the extent that they could take advantage of them.[8]

Other NUIs have been developed for purposes other than game console manipulation. A widely referenced (although fictitious) example of a NUI is the one used in the film *Minority Report* by Tom Cruise's character at the beginning of the film.[9] He organises images, zooms and makes mistakes with his gloved hands. The aesthetics of this moment have become prevalent, both in interface design and in the cultural imaginary. The NUI in the film permits the quick manipulation of information by those who use skilful gestures. The NUI used in the film was in fact a real interface developed by John Underkoffler and controlled by gestures.[10] An interface using either Leap Motion or the Kinect has been trialled for use by radiologists and surgeons in sterile environments, the rationale being that staff can use the interface without risking infection in surgery patients.[11] The Kinect has also been used to develop user interfaces specifically for the use of elderly human users, who may not be digital natives and who may have difficulties with visual acuity and motor difficulties.[12] In the view of the researchers, "natural" interaction is particularly important for elderly people who may have difficulties with memory. Because it may be hard to remember specific hand gestures there must therefore be a reliance on the "instinct" of the human user.[13]

Don Norman's criticism of NUIs, entitled "Natural User Interfaces are Not Natural", points out that gestural systems are neither new nor ground-breaking.[14] He traces the history of digital non-contact gesture interfaces back to the invention of the theremin in 1928, which puts into

---

[8] Richard Harper and Helena M. Mentis, "The Mocking Gaze: the Social Organization of Kinect Use," *Proceedings of the 2013 conference on Computer Supported Cooperative Work* (2013): 167-80.
[9] Steven Spielberg, *Minority Report* (United States: 20th Century Fox and Dreamworks Pictures, 2002).
[10] Teena Maddox, "Sci-Fi is Turned into Reality with Technology Guru from Minority Report and Iron Man," *TechRepublic* (August 4 2016). https://www.techrepublic.com/article/sci-fi-is-turned-into-reality-with-technology-guru-from-minority-report-and-iron-man/ (accessed December 4 2017).
[11] Nikola Nesterov, Peter Hughes, Nuala Healy, Niall Sheehy and Neil O'Hare, "Application of Natural User Interface Devices for Touch-Free Control of Radiological Images During Surgery," paper presented at *2016 IEEE 29th International Symposium on Computer Based Medical Systems (CBMS)*, Dublin, Ireland, June 20-24 2016.
[12] Shih-Wen Hsiao, Chu-Hsuan Lee, Meng-Hua Yang and Rong-Qi Chen, "User Interface Based on Natural Interaction Design for Seniors," *Computers in Human Behavior* 75 (2017): 147-159.
[13] Ibid, 149
[14] Don Norman, "Natural User Interfaces are not Natural", *Interactions* 17, no. 3 (2010).

question the impression of NUIs as futuristic technologies to be aspired to. He notes the similarity between modern NUIs and the first GUIs (graphical user interfaces). But as he explains, gesture-based NUIs are not merely as "natural" as GUIs are, they are actually even less intuitive because of a lack of feedback. Contemporary NUIs like the Kinect sometimes give feedback mostly by displaying images of the human bodies (and other objects) that they encounter[15], while others provide feedback only by showing the effect of the gesture, such as by moving a character's body in the same way as the human's.[16] In contrast, the use of a mouse provides immediate feedback to the human, who is following the path of the cursor on the screen with their eyes. But very few gesture-dependent interfaces deliver immediate feedback to the user that permits her or him to materially improve their use of the system. In addition, GUIs have become easier to use because a grammar of gestures has evolved over time. As an example, there is now a fairly universal text selection interaction. We can now see this grammar solidifying in haptic technologies as well. Norman's example is the scrolling action on smartphones. Over time, the "viscous friction" of the window has become standardised (in other words the speed of the finger flick causes the scrolling to continue for a consistent length of time). Zooming has also become standardised. NUIs have not had the same penetration as touchscreens to date. The Kinect remains the only prevalent NUI on the market, both as an interface for the Xbox and for Windows PCs. But because the creation of Kinect games is no longer a priority for publishers (see below) there has been limited opportunity for Kinects to teach most of us how to communicate with them. Passing interaction with a Kinect permits it to teach humans basic rules, but thorough understanding of Kinects' experience of the world and its way of feeding information back to the human user requires more extended sessions.

Natural language communication with machines is also an aspect of the ambition of NUIs, and also specifically of the Kinect. The Kinect is capable of listening for command words and prompting responses in the console, although as discussed below the communication between human and Kinect is not seamless or unbiased. SHRDLU, a natural language processing program built by Terry Winograd in the 1960s and 70s, is the subject of Chapter 4 of this thesis. The "natural" language used by the system pertained exclusively to the description and manipulation of a digital block world. But more recently the ambition of building NUIs that can understand and respond to humans has resulted in the creation and marketing of virtual assistants. Spoken communication with smartphones was popularised by Apple's Siri, and is now replicated in other interfaces. We are also witnessing the penetration of home assistants that exist purely to understand and respond to human speech. As shown later in this chapter, the Kinect has been accused of sexism, and the same is true of home assistants, although for different reasons. A home assistant that can not only listen but respond to questions must have a voice, and like human voices it must have an accent and the suggestion of a gender, which belies the presumed neutrality of home assistants.[17] Like motion capture, NLP technology has the implicit goal of creating nonhumans capable of

[15] As in the game *Fru*, developed and published by Through Games (2016), Xbox One.
[16] As in the game *Ryse: Son of Rome*, developed by Crytek (Microsoft Studios, 2013), Xbox One.
[17] Thao Phan, "The Materiality of the Digital and the Gendered Voice of Siri," *Transformations* 29 (2017): 23-33.

interacting “naturally” with humans. It is still far from perfect in this regard. And the trouble with NLPs is that if they ever do become indistinguishable from human language users then they will be embroiled in the conflicts and struggles that surround and implicate all human voices.

Each part of the name “natural user interface” may be questioned from the perspective of the sensor itself. What is “natural” to the human is always engineered into the machine, so those aspects of the interface that are natural to the human are neutral for the artefact. A NUI constantly scans the environment for signs of human bodies and/or for sounds recognisable as human speech. In the case of the Kinect, “middle-ware” interprets these movements and sounds to decide what is relevant and what is not (see below). What is natural for a sensor is to detect everything that comes into its sensory field. The “user” part of the name is also problematic from the sensor’s point of view. From the human perspective, we are the “users” of the NUI. But in a way, the sensor uses the human as well. Its program of action (to borrow an ANT term) is to detect images, sounds and depth, and to convey this information to the console. The sensor uses the human as a part of this environment. “User” in this thesis is typically qualified as “human user”, but in accordance with OOO we must ask about the uses to which the sensor puts its environment. Finally, the NUI sensor is only part of the “interface”, and relates to the console or computer, the screen, and the human. These are relevant issues to raise since the following account is of a NUI sensor and its way of being in the world.

Kinect-brand and Kinect-artefact

There is a need for a degree of anthropomorphic language in this chapter, which will be exaggerated as much as possible in order to highlight the alien in the Kinect-artefact and Kinect-brand. Human phrases sometimes seem to readily apply to nonhumans and in setting up a multiplayer game; one human might say “The Kinect can’t see you”, prompting the players to reorganise player positions in front of the television. As discussed in the introduction, such anthropomorphic phrasings used in daily life and repeated in this thesis can be useful for increasing the tension between artefact and human reader, which aids in the juxtaposition between alien object and the language used to describe it. However, care must be taken in this analysis not to privilege the human in relation with the Kinect, since different interactions may be more salient for the Kinect at different times. Light bouncing off human bodies may not be the most important thing in the Kinect’s sensory world at any given moment, although analyses of media often focus on this relationship. In addition, the cultural position of the Kinect is a significant factor in human-Kinect relations, but as far as possible these will not be prioritised over those factors that are significant for other objects that form relations with the Kinect. In fact, objects from the domain of cultural criticism come into close relation with objects from the domain of engineering in this analysis, and neither is privileged over the other.

The Kinect has been mass-marketed relatively successfully, and the artefact makes use of both gestures and voice commands. The first (2010) and second (2013) generations of Kinect are presented here. Originally given the codename “Project Natal”, themes of revolution and innovation have been part of the Kinect mythology since its inception (it is also the name of a city in Brazil, following Microsoft’s tendency to name projects after cities). Clearly the name “Kinect” is intended to evoke the “kinetic” motion-based play that represents a significant part of its interface and “connect”, which signifies connection and harmony with the game system and perhaps with other people.

The original Kinect was announced at E3 on June 1 2009 to a generally positive audience. Microsoft told the world there would be “no barriers and no learning curves”[18] when playing with Kinect 1.0, which was compatible only with the Xbox 360 console. The ambition of NUIs as a class was in evidence, such as in the slogan “You are the controller”.[19] There was considerable speculation about what technology such as this would mean to gaming. The Kinect is used to augment videogame experiences, and some games are completely Kinect-based without the need for a traditional videogames controller like the standard Xbox One controller or non-traditional motion-based controllers such as the Playstation Move.



Figure 1: Xbox One controller. A traditional gaming controller with triggers and buttons. Wireless and lacking motion sensors.[20]

[18] *E3 2010 Microsoft - Kinect (Project Natal) Part 1* (2010), Youtube. https://www.youtube.com/watch?v=bxp43T2JP18 (accessed June 24 2017).
[19] *E3 2009: Project Natal Xbox 360 Announcement* (2009), Youtube. https://www.youtube.com/watch?v=p2qlHoxPioM (accessed June 24 2017).
[20] “Accessories", *Xbox*. http://www.xbox.com/en-US/xbox-one/accessories (accessed June 24 2017).



Figure 2: Playstation Move controller for Playstation 3 released in October 2010. Motion sensing device tracked by a camera attached to the Playstation.[21]



Figure 3: Outer structure of the Kinect-artefact 1.0 for Xbox 360 released in 2010. This image shows the position of the infrared and RGB cameras on the horizontal sensor bar as well as the infrared laser emitter and status LED.[22]

With the introduction of new hardware, Microsoft promoted a wider range of Kinect experiences for all Xbox users. The Xbox One was announced on 21st May 2013, as a successor to the Xbox 360 (in the media and gamer forums it was frequently referred to as "the Xbone", either affectionately or not so affectionately). It was released in November 2013. A Kinect 2.0 was bundled with every One sold, and they were featured prominently in pre-release advertisements, with claims that it would allow players to "reach into games and entertainment like never before."[23]

---

[21] "Playstation 3 Accessories," *Playstation* (2014). http://us.playstation.com/ps3/accessories/playstation-move-motion-controller-ps3.html (accessed February 12 2014).
[22] Jeff Kramer, *Hacking the Kinect* (New York: Apress, 2012).
[23] "Xbox One: What it is," *Xbox* (2013). http://www.xbox.com/en-US/xboxone/what-it-is (accessed June 2 2013).



Figure 4: External appearance of the Kinect-artefact 2.0 for the Xbox One released in 2013. Again, the sensor bar is oriented horizontally with an eye-like cluster of sensors on one side and the Xbox logo on the other.[24]

Other technological actants are important to define semantically in a study of Kinect. The term "console" will occasionally be used to denote the Xbox One console. The word generally refers to a computer specialised for videogames, typically suited for use with a television and with a games-oriented input device (or it describes a handheld gaming device). Bernadette Flynn discusses the way that console gaming has eventually come to replace the family hearth, as the radio and television did before it.[25] Flynn's idea is particularly significant for studying the Kinect because there is often a necessary reorganisation of the living room to revolve around this "hearth" and the playing area in front of it. The Kinect is the main mode of input for some games, although often the menus can be navigated using the conventional controller (Figure 1) and some games such as *Ryse: Son of Rome* use both input devices, sometimes simultaneously.[26] The console is the mediator for all of these objects: the playing space, the Kinect, the game disc, the installed content and the screen interface, facilitating their interaction to create the preconditions for play.

Further to this definition we must acknowledge the debate around the word "videogame". This is a topic that has been debated fiercely and with little resolution[27] Definitions have changed and been contested for over sixty years, with definitions always leaving out some electronic games and incorporating new ones.[28] Jesper Juul offers a definition that aims to

[24] "Kinect Sensor for Xbox One," *Microsoft* (2017). https://www.microsoft.com/en-au/store/d/kinect-sensor-for-xbox-one/91hq5578vksc (accessed June 26 2017).
[25] Bernadette Flynn, "Geography of the Digital Hearth," *Information, Communication and Society* 6, no. 4 (2003).
[26] *Ryse: Son of Rome*, Xbox One.
[27] Grant Tavinor, "Definition of Videogames," *Contemporary Aesthetics* 6 (2008): 1-17.
[28] Jesper Juul, "The Game, the Player, the World: Looking for a Heart of Gameness," *Digital Games Research Conference: Level Up*, ed. M. Copier and J. Raessens (Utrecht: Universiteit Utrecht, 2003): 30-47.
The 1958 interactive game *Tennis for Two* is often cited as the first videogame, but it could also be argued that artificial players of board games like chess and checkers of the 1950s were the first videogame platforms. See Tristan Donovan, "Chapter 1: Hey! Let's play games!" *Replay: The History of Videogames* (Lewes, UK: Yellow Ant, 2010).

bridge the gap between what seemed to be the inescapable debate between ludologists and narratologists.[29] According to him, games have six main qualities, including:

1. Rules
2. Variable, quantifiable outcome;
3. Value assigned to possible outcomes;
4. Player effort;
5. Player attached to outcome; and
6. Negotiable consequences.

Bogost has called Juul's "an implicitly realist position about games, but a troubled one."[30] This definition will suffice, however, as the relevant question here is what facilitates the videogame in its interaction with the Kinect. The videogame is not simply recorded on the physical disc or somehow within the digital download. These elements are necessary, but the videogame is also installed onto the console's hard drive. Therefore when we speak of a "videogame" we speak of a distributed entity (or actor-network) that links software, console operating system, sensors, the console's brand, the sensor's brand, hard drive, disc/downloaded entity, screen, input devices, the hands and brain of the player, internet presence as well as encompassing a concept that is influenced by other videogames and other kinds of games (such as board games) past and present.[31] Lack of cooperation between human player and those cultural or material relations may cause disruption, such as when an input device lacks electric power or the manufacturer ceases to update the console. However, in a OOO sense the videogame is more than the sum of these components and their interactions. Viewing a videogame as simply a product of material and cultural influences constitute a duomining of the videogame, since parts of it are reduced to its material embodiment (undermining) and other parts are reduced to things like cultural impact and power relations (overmining). The videogame is undermined when we speak of it only as the impact of code on electronic signals. The videogame is overmined when we speak of it only as part of a broader cultural milieu. The videogame exists in relation with the hard drive, sensors and so on but these disparate elements also form part of the ontology of the videogame. The videogame is an integrated whole that acts in accordance with internal qualities, and relations are formed between the whole videogame and objects as diverse as consoles, humans, and online reviews.

Having defined the other closely related technological actants, we must turn to an articulation of what object is meant by the term "Kinect". For our purposes, the Kinect as a signifier relates both to the brand and to the artefact, which is an unsatisfactory arrangement

---

[29] Juul, "The Game, the Player, the World."
[30] Ian Bogost, "Videogames are a Mess," *Ian Bogost* (blog) (September 3 2009). http://bogost.com/writing/videogames_are_a_mess/ (accessed June 26 2017).
[31] As Nick Montfort and Ian Bogost discuss in the platform studies literature, the hardware and software of a given platform is as important in the study of digital media as the creative works (such as videogames) that run on it. The study of software and hardware is frequently omitted in the critical study of videogames, despite being important factors in how games are designed, created and marketed. See: Nick Montfort and Ian Bogost, *Racing the Beam: The Atari Video Computer System* (Cambridge, MA: MIT Press, 2009).

for critical object studies. Yet neither brand nor artefact is more the Kinect than the other, and neither can be reduced to the other. The names of objects often hide the existence of several objects which, to humans, are simply described by a single label. To emphasise the importance of this distinction, for the purposes of this thesis the situated artefact will be labelled "Kinect-artefact" and the collective brand will be labelled "Kinect-brand". Initially, this distinction may seem reminiscent of Plato's theory of Forms.[32] We could conceive of the brand as a form that identifies certain qualities shared by all the Kinect-artefacts in the world: a "Kinect-ness". But the meaning of the term Kinect-brand presented here is quite different. The Kinect-brand is an object just as much as a Kinect-artefact (Harman has called brands "charismatic objects"[33]). It has its own qualities and relations. Often, as will be shown, these qualities and relations are significantly different from the qualities and relations of the Kinect-artefacts. The Kinect-brand is the ideal, untarnished imaginary that conveys certain ideas about the Kinect-artefact – such as those from Microsoft, and those from prominent users. Unlike the artefact, the Kinect-brand is abstracted and incorporeal. It carries with it the implicit understanding of its users and is a factor in Microsoft's planning process. Kinect-brand is the aspect of the machine that is present in popular culture, internet discussion and Microsoft presentations (unless the specificity of an individual Kinect is important). But it is not restricted to interactions with humans and the social since it acts and responds to other actants, such as the Microsoft corporation, retailers, regulatory authorities and so forth without forming one-to-one interactions with humans. As Scott Lash and Celia Lury put it, "the commodity is dead; the brand is alive."[34] By this they mean that the commodity (or artefact) is finished and final while the brand is self-modifying and possessing a memory. Although of course, as anyone who has ever owned a console knows, an artefact does not cease to be modified, either by software updates or by errant water spillages.

The Kinect-artefact is a material object: a plastic-encased set of sensors, including its middle-ware but ending, for lack of a better barrier, at the end of the cable connecting Kinect to console. The Kinect-artefact contains its knowledge of individual habits and interfaces with a specific Xbox One console. It is present in a way that the Kinect-brand is not; one buys a Kinect-brand, but gets a Kinect-artefact. Neither object is comprehensively Kinect-like or human-like. It is important to note that this notation does not imply that the Kinect can be split into these two aspects: they are two *different* objects that both exist in the world, erroneously conflated into one by human (and specifically English) language conventions. It is also not the case that Kinect-artefacts are the real objects while Kinect-brands are sensual objects in OOO terms. The Kinect-brand is a real object too, with its own way of acting and its own way of sensing the world. The sensual object formed between Kinect-artefact and human is one in which light, sound, and movement play a key role. An individual human may also form a sensual object with the Kinect-brand, but it is likely to be through text and advertisements. Like all relations, it is unequal. The Kinect-brand may have a profound

[32] Andrew S. Mason, ed., "Plato's Metaphysics: the 'Theory of Forms'" *Plato* (Abingdon, Oxon: Routledge, 2014), originally published 2010.
[33] Jonathan Olivares interviewing Graham Harman, "The Metaphysics of Logos: Decoding Branded Objects with Philosopher Graham Harman," *032c* (May 17 2018). https://032c.com/decoding-logos-graham-harman/ (accessed May 28 2018).
[34] Scott Lash and Celia Lury, *Global Culture Industry* (Cambridge: Polity Press, 2007), 6.

influence on the human, while most humans could only hope to make a tiny influence on the Kinect-brand, such as through the publication of an article or through activism.

It is difficult to arrive at a definition of either the Kinect-brand or the Kinect-artefact, but the signifier “Kinect” is also used by Microsoft to describe thousands of identical products. Each of these identical Kinects shares similar Kinectish qualities, but they are in different relations. One Kinect might be in a strong relation with a table, another is still in its box. One Kinect can identify my face, another can identify yours. The term “Kinect” refers at once to one object and a distributed network of Kinect-brand and Kinect-artefact experiences. Treating the brand and artefact as two distinct objects is the only way to avoid blurring boundaries between the sensory relations of the “dead” artefact and the complex world of the idea of the object. And it also focusses our attention on the object itself, its inner essence, sensual connections with other objects and therefore how they touch and exert influence in their worlds. Of course, “artefact” and “brand” are very human-centric terms, but as the introduction to this thesis argues we may use human experiences to explain alien phenomena; in fact, this can strengthen our understanding thereof. Both Kinect-artefact 1.0 and Kinect-artefact 2.0 will both be explored, although most emphasis in this chapter is put on the Kinect-artefact 2.0 as the most recent device.

The Kinect-brand

The Kinect-brand has evolved from an anthropocentric perception of a gimmicky (albeit entertaining) artefact to a more serious (albeit troubled) artefact. Kinect-brand 1.0 was originally marketed mostly to families. The portrayal of the technology facilitated by advertising and the assortment of available games signalled the intention of Microsoft to market the game to families and groups. This intended audience is hardly surprising given the history of gesture-based interfaces in gaming. The Nintendo Wii (2006) was primarily marketed to families, featuring games designed to appeal to all age groups as well as “games” like *Wii Fit*,[35] which used an electronic balance board to integrate fitness play into NUI use. The Eyetoy for Sony Playstation 2 (2003), and Playstation Eye for Sony Playstation 3 (2007), were aimed at children. One reason for the Kinect-brand’s lasting association with children’s games is that the physical nature of the play is well-suited to family and group interactions. Harper and Mentis suggest that the Kinect-artefact is used in households as a way of safely mocking others (particularly parents) when the system requires them to make some absurd movement or gesture.[36] In addition, the work of John Downs suggests that the role of the audience (“paraplay”) becomes more varied when watching physical games, so there is greater engagement while waiting for a turn.[37] Another reason why the first Kinect was

[35] *Wii Fit*, developed by Nintendo EAD Group No. 5 (Nintendo, 2008), Nintendo Wii.
[36] Harper and Mentis, "The Mocking Gaze."
[37]John Downs, Frank Vetere, and Steve Howard, "Paraplay: Exploring Playfulness Around Physical Console Gaming," in *Human-Computer Interaction – INTERACT*, ed. P. Kotzé, G. Marsden, J. Wessen and M. Winckler (Berlin: Springer, 2013), 682-699.

predominantly used in social and family gaming is because its detection of fine details was minimal. The responses to humans are more coarse-grained because the sensor either could not detect more detailed human movement, was unable to distinguish between humans and other objects, or because it could only communicate part of the information detected to the console. Social and family videogames are usually not so dependent on precision as are, for example, first-person shooters. The artefact and its way of being in the world exists in a bidirectional and asymmetrical relationship with the brand, determining the kind of human user to which the artefact is marketed. This, in turn, prompts the creation of artefacts that can be used effectively by children, families and groups. Coase-grained relations between humans and the Kinect-artefact have had a gradual but inexorable effect on the Kinect-brand. Because the Kinect-artefact is so often (and so successfully) used by families or in other social settings, the Kinect-brand has become associated with this type of gameplay. But it has been argued that games with an affective component like *Kinectimals*, [38] which is aimed at young children, may be deployed for the didactic purpose of training human users,[39] which educates young people about the NUIs that are expected to become more prevalent in their lifetimes. The didactic induction of children into NUI use is a good way of shifting this cyclical relationship between artefacts and brands into eventual widespread adoption of NUIs like possible future Kinect-artefacts. It is important to know this, because the Kinect-brand underwent a significant transformation when it came to relate to the Kinect-artefact 2.0.

The casual and family-friendly qualities of the Kinect-brand prior to Xbox One are evident from the kinds of games that it plays. Games such as *Kinectimals* make use of the technology by allowing users to reach into the virtual world and stroke pet animals.[40] These and similar family games make use of Kinect-brand gimmicks, like the dubious promise of physically entering the game through body movements, but there are relatively few games that incorporate gestures into ordinary gameplay. There are many reasons for this, but the primary difficulty lay in the primitive nature of the technology (described below) which made it unsuitable for "hardcore" games. One online journalist opined:

> The Xbox 360's Kinect, Wii Remote and PlayStation Move are all rubbish and have spawned a handful of decent games between them, enveloped by piles of filthy, filthy shovelware.[41]

There are notable exceptions to this general rule. Since 2011, Microsoft attempted to rebrand the Kinect-brand 1.0 for so-called "serious games".[42] This makes use of the rhetoric surrounding "casual" and "hardcore" games, and attempt to alter perceptions of the

---

[38] *Kinectimals*, developed by Frontier Developments (Microsoft Game Studios: 2010), Xbox 360.
[39] Tom Apperley and Nicole Heber, "Capitalising on Emotions: Digital Pets and the Natural User Interface," in *Game Love: Essays on Play and Affection*, ed. Evenold, J. and MacCallum-Stewart, E. (Jefferson, NC: McFarland and Company, 2015), 149-164.
[40] *Kinectimals*, Xbox 360.
[41] Ben Salter, "E3: Xbox One's Kinect Knows Too Much," *Xbox.MMGN.com* (June 14 2013). http://mmgn.com/xboxone/xbox-one-preview (accessed June 19 2013).
[42] This idea should not to be confused with the serious games movement, which advocates the use of videogames for purposes other than entertainment, such as for political and social change or education. The intended meaning here is closer to "games requiring a considerable time commitment" or "games aimed at adults". And of course, there is a hint of "games that straight cis-gendered men play".

technology, presumably to raise the anticipation of the Xbox One and Kinect-brand 2.0. Some examples of hardcore games released specifically for the Kinect-brand 1.0 include survival horror game *Rise of Nightmares,*[43] which creates a creepy atmosphere by requiring the player to stand completely still, to swipe insects off virtual arms and to adopt a "fighting stance" to kill zombies; and *Fable: The Journey*,[44] which builds upon the popular *Fable* franchise and allows players to play sitting down, using gestures to attack monsters and other villains. By promoting these games, Microsoft aimed to associate the Kinect-brand with more mature themes and complex gameplay. In ANT terms, there was an attempt to enrol "hardcore" gamers into the Kinect-brand's network.[45] The artefact and its qualities remained the same, but were repackaged for humans who were inclined to resist the idea that the Kinect-artefact could be useful or interesting for them. The "program" of the Kinect-brand resisted the "anti-program" of "hardcore" gamers by creating videogames that would counter the perception that Kinect-artefacts are only useful for casual games.[46] But the qualities of the Kinect-artefact made enrolment difficult. Both *Rise of Nightmares* and *Fable: The Journey* received mixed reviews.[47] These mediocre responses are typically blamed on a Kinect-artefact 1.0's slow reaction time and its difficulty in picking up subtle movements.[48] Developers also tended to incorrectly estimate the capabilities of the sensor. This is a good example of the misjudged phenomenology of an artefact causing it to form undesirable relations with cultural objects. Often, the games were made to show off the capabilities of the Kinect-artefact, rather than to make a good game that uses whatever hardware is available. This is likely due in part to the limited penetration of the Kinect-artefact into gamers' homes, which means that the market was small and there was little money to be spent on developing new games. This is a curious circumstance. The gamer's relationship with the earlier Kinect-artefact is one in which the machine was often privileged above the human's enjoyment for the purposes of building expectations for the forthcoming Kinect-artefact 2.0. Steve Woolgar describes this kind of thinking in "Configuring the user".[49] Users do not necessarily know what they want, because they are not attuned to the anticipated future of the technology, or because they desire a particular feature but are not prepared to pay for it.[50] Human users of the first Kinect-artefact did not possess the same information as Microsoft developers, such as the kinds of

---

[43] *Rise of Nightmares*, developed by Sega (Sega, 2011), Xbox 360.
[44] *Fable: The Journey*, developed by Lionhead Studios (Microsoft Studios, 2012), Xbox 360.
[45] Bruno Latour, "Technology is Society Made Durable," *Sociological Review Monograph* 38, no. 2 (1991): 103-131.
[46] Ibid.
[47] At the time of writing (November 22 2017), popular media rating website Metacritic gave *Rise of Nightmares* and *Fable: The Journey* a 54% and 61% approval rating respectively.
Metacritic, CBS Interactive Inc., (2017). www.metacritic.com (accessed November 22 2017).
[48] See for example the following reviews:
James Rivington, "Microsoft Kinect for Xbox 360 Review," *Techradar* (November 4 2010).
http://www.techradar.com/reviews/gaming/games-consoles/controllers/microsoft-kinect-for-xbox-360-905010/review (accessed January 20 2018).
and
Ross Miller, "Kinect for Xbox 360 Review," *engadget* (November 4 2010).
https://www.engadget.com/2010/11/04/kinect-for-xbox-360-review/ (accessed January 20 2018).
[49] Steve Woolgar, "Configuring the User," *Sociological Review* 38 (1990): 58-99.
[50] Ibid., 74.

gaming experiences that would become possible with Kinect-artefact 2.0. However, the human-Kinect-artefact relation had an effect on the Kinect-brand, one that is contrary to that which was presumably intended. A poor understanding of the Kinect-artefact's phenomenology, either by the developers or the users (or both), meant that these "serious" games tended to damage the Kinect-brand from Microsoft and other developers' points of view. This perception of Kinect 1.0 set the scene for Kinect 2.0's release.

Another problem for the Kinect-brand is the widespread belief that the 2013 Kinect-artefacts could be used for covert surveillance, further privileging other interests above those of the gamer. When the Xbox One was first announced, it was revealed that Kinect-artefact 2.0s would need to be plugged in and active before the console could be used properly. This sparked considerable anger with gamers, and some bloggers and journalists worried in the aftermath of the announcement that Kinect-artefacts would represent a serious invasion of privacy, especially since the new sensors are powerful enough to detect heartbeats and can distinguish between individual voices.[51] This response was likely amplified by Edward Snowden's exposure of PRISM and its large-scale surveillance activities which came to light only a week after Microsoft's announcement, and also the subsequent allegation that Microsoft had provided the NSA with the ability to access email and other user applications.[52]

[51] Kyle Russell, "People are Worried Microsoft's New Xbox Will be Able to Spy on You," *Business Insider Australia* (May 29 2013), https://www.businessinsider.com.au/xbox-one-kinect-privacy-issues-2013-5?r=US&IR=T (accessed February 22 2018).
And
Will Simonds, "Xbox One Will Know Your Face, Voice, and Heartbeat," *Abine* (May 21 2013), https://www.abine.com/blog/2013/xbox-one-and-facial-recognition/ (accessed February 22 2018).
[52] Glenn Greenwald, Ewen MacAskill, Laura Poitras, Spencer Ackerman and Dominic Rushe, "Microsoft Handed the NSA Access to Encrypted Messages," *The Guardian* (July 12 2013). http://www.theguardian.com/world/2013/jul/11/microsoft-nsa-collaboration-user-data (accessed January 20 2018).

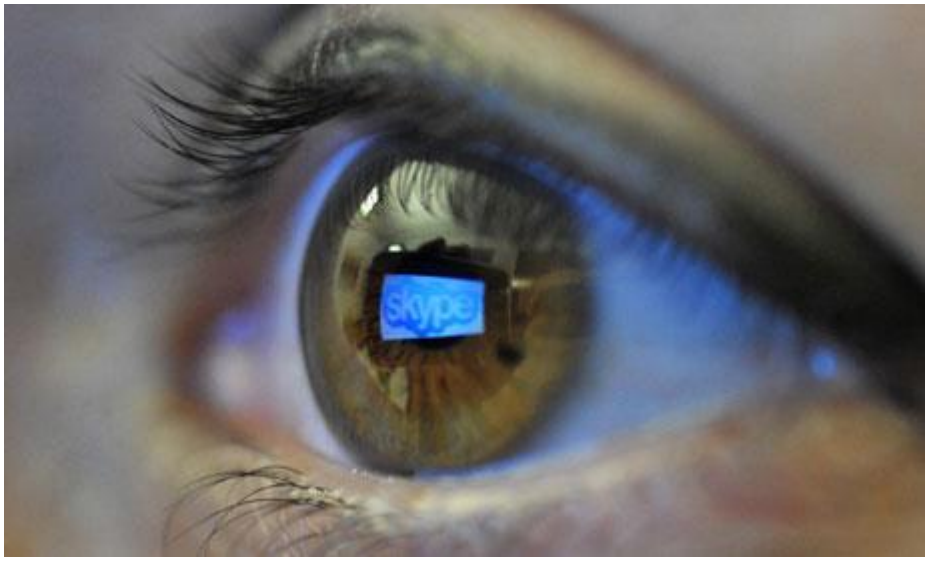


Figure 5: Image from *The Guardian*'s exposé of Microsoft's cooperation with the NSA. It depicts an enlarged photograph of a human eye with the "Skype" logo reflected in it.[53]

It is easy to compare Figure 5 with the portrayal of the Kinect-brand in the media, its open eye creating connections between users and shady government agencies. In the language of visual media, the eye can be used not only as a positive symbol of learning and understanding but also of control by external agencies, often without the viewer's knowledge. Microsoft entered a period of damage-control which permitted them to disassociate the console from the new concerns surrounding the sensor. The Kinect-brand needed to distance itself from certain qualities of the Kinect-artefact.

Shortly after the announcement, Microsoft assured gamers that Kinect-artefacts can be switched off when the console is not in use, but comparisons to "Big Brother" still abounded.[54] (Indeed, in February 2014 it was discovered that this fear was quite justified after the revelation that a UK intelligence agency had monitored webcam chat for several years and had considered using the new Kinect-artefact.[55]) This and other criticisms of the console meant that the announcement of the new system was surrounded by anger and confusion.[56] Microsoft immediately went into damage control mode.[57] The dramatic change in stance by Microsoft from promoting the impressive sensory capabilities of the sensor to distancing itself from privacy concerns shocked the world of gamer culture. This piece

[53] Ibid.
[54] Brett Slabaugh, "Xbox One's Kinect can Actually be Turned Off," *The Escapist* (May 30 2013). http://www.escapistmagazine.com/news/view/124447-Xbox-Ones-Kinect-Can-Actually-Be-Turned-Off (accessed January 20 2018).
[55] Spencer Ackerman and James Ball, "Optic Nerve: Millions of Yahoo Webcam Images Intercepted by GCHQ," *The Guardian* (February 28 2014). https://www.theguardian.com/world/2014/feb/27/gchq-nsa-webcam-images-internet-yahoo (accessed January 20 2018).
[56] Luke Plunkett, "That Xbox One Reveal Sure was a Disaster, Huh?" *Kotaku* (May 21 2013). http://kotaku.com/that-xbox-one-reveal-sure-was-a-disaster-huh-509192266 (accessed May 28 2013).
[57] Don Mattick, "Your Feedback Matters - Update on Xbox One," *Xbox* (June 19 2013). http://news.xbox.com/2013/06/update (accessed January 15 2014).

entitled “Negotiations” by the immensely popular and influential gamer-themed webcomic *Penny Arcade* gives an idea of the kind of comments that emerged:

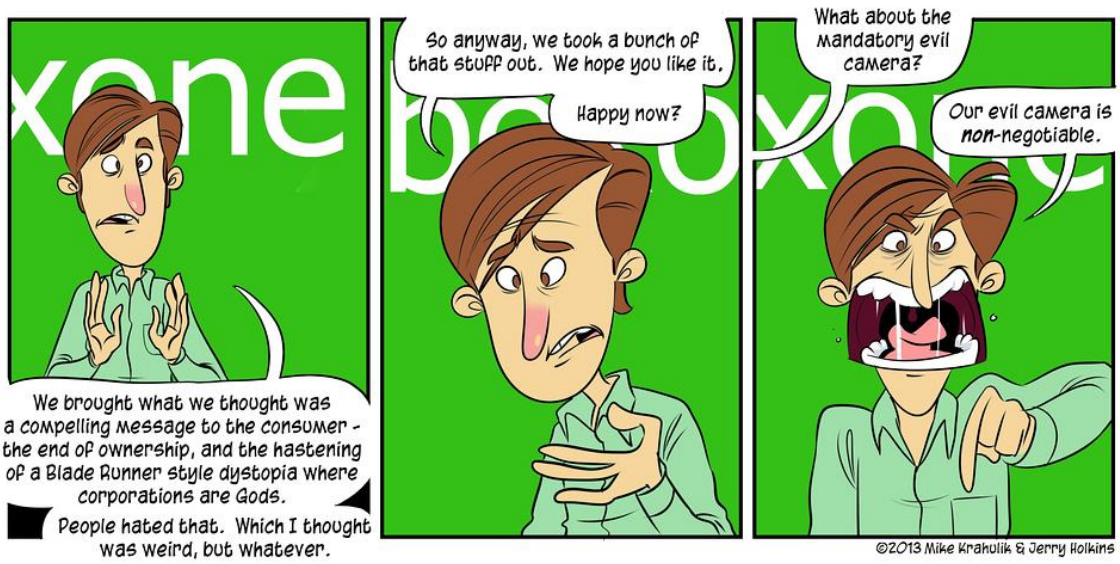


Figure 6: Penny Arcade comic strip “Negotiations” from June 21 2013.[58]

In this comic strip, a Microsoft spokesperson explains that some of the qualities of the Xbox One that the internet reacted so violently against will no longer be present in the console. It should be noted that the creators of *Penny Arcade*, Mike Krahulik and Jerry Holkins, are notorious for using their prestige to convey personal messages about what is and what is not good for “gamer culture”, and they even launched the major gaming convention Penny Arcade Expo or PAX. They have often come under fire for insensitive and discriminatory comments.[59] The gamer culture they promote is typically just for “hardcore” gamers which are identified as young, straight cis-gendered males.

This comic strip is a parody of a real event in which Microsoft promised to remove features such as the mandatory DRM technology.[60] But despite the violent reaction from major videogames commentators, Microsoft did not announce that the Xbox One could be purchased without a Kinect-artefact, or the “mandatory evil camera” as *Penny Arcade* dubbed it. They explained that the high level of integration of Kinect-artefacts into the operations of the console meant that it was now impossible to sell it separately, and that they wanted the console to be consistent for developers. Maybe Microsoft does think the use of NUI technology can enhance traditional videogames, but to many it seemed that the self-described “hardcore” fanbase of the Xbox 360 would be obliged to pay more money for a device that was almost universally considered only suitable for casual and family games. Even if there was little miscommunication between the Kinect-artefact and the human, the Kinect-brand was standing between them. The human could not enter into a fresh and untarnished

[58] Mike Krahulik and Jerry Holkins, "Negotiations," *Penny Arcade* (June 21 2013). http://www.penny-arcade.com/comic/2013/06/21 (accessed June 26 2017).
[59] As in this incident: “Dickwolves,” *Geek Feminism Wiki*. http://geekfeminism.wikia.com/wiki/Dickwolves (accessed January 17 2018).
[60] Don Mattick, "Your Feedback Matters.”

relation with the Kinect-artefact, free from concerns about privacy and the possible loss of "hardcore" status, and the Kinect could not enter into a relation with certain humans because many would now be compelled to switch it off. This and the Xbox line's apparent transition from a powerful gaming engine to a home entertainment hub with a focus on television integration and wide range of entertainment media could be considered a way of distancing the console from "gamer" culture and its sometimes impenetrable language, tropes and skilful gestures.

With a yet more drastic severing of Xbox One from the Kinect-brand, Microsoft has seemed to cut its losses with its console. After the poor reviews of its flagship game *Kinect Sports Rivals*, and a little under a year after its initial announcement of the new console, Microsoft issued a statement stating that the console would no longer be compulsorily bundled with a Kinect-artefact. The post, entitled "Delivering More Choices For Fans"[61], has been widely called the death of the Kinect-brand. The decision has been widely condemned by those who already own the console but appreciated by those who now have the opportunity to buy the console at a reduced price. In any case, the lack of incentive for new users to purchase Kinect-artefacts has led to fewer games integrating gestural and spoken commands, and the identity of the Kinect-brand is becoming less and less relevant. Would-be developers for the Kinect-artefact have to convince gamers that their product is so good that it is worth purchasing a peripheral as well. As for the Kinect-artefacts, their experiences were significantly affected by the announcement. Those users who did away with their Kinect-artefact changed the way that their Kinect-artefact senses and experiences its world, causing it to enter into relations with the back of cupboards and landfills, and preventing its sensory organs from activating those electronic pathways that were previously so central to its way of being in the world.

Kinect-artefact 1.0

Kinect 1.0's sensory organs function in a way that resembles human sensation, although they are not the same structurally. Kinect-artefact 1.0 combines the use of two types of sensors to achieve something that its inventors, speaking on behalf of the artefact, claim is similar to human vision; it has a camera for the detection of the visible light spectrum; and a second system that emits and records infrared signals. The infrared sensor allows the Kinect-artefact to gauge depth, making users' gestures three dimensional. It emits infrared light from one point, which bounces off the player (and any other objects) before returning and being interpreted by the infrared sensor.[62] The Kinect-artefact also uses stereo microphones with complex software to reduce the problems of sound reverberation common to a home theatre situation, enabling the use of voice commands.[63] The only feature possessed by a Kinect-

[61] Phil Spencer "Delivering More Choices for Fans," *Xbox Wire* (blog) (May 13 2014). http://news.xbox.com/2014/05/xbox-delivering-more-choices (accessed June 26 2017).
[62] Kramer, *Hacking the Kinect*.
[63] Ivan Tashev, "Recent Advances In Human-Machine Interfaces for Gaming and Entertainment,"*International Journal of International Technology and Security* 3, no. 3 (2011): 69-76.

artefact 1.0 but not a Kinect-artefact 2.0 is a responsive tilt motor built into the stand which permits the sensor to respond to changes in player height or position. The IR emission and tilt stand adjustment are the only two interactions it can initiate besides its communication with the Xbox 360 console. The Kinect-artefact 1.0 was intended to be an affordable but high-quality intuitive interface for casual consumers, but it is also difficult to use, requiring the user to exaggerate her or his movements.[64] There were serious challenges to creating meaningful communication between human and machine. Kinect-artefact 1.0 acted in accordance with the internal qualities determined by its structure, which often brought it into conflict with humans who wished it to act in different ways. The Kinect-artefact 1.0 had an unfortunate effect on the Kinect-brand by failing to perform as well as the Kinect-brand had enthusiastically advertised.

Despite these issues, several individuals and groups tried to gain an appreciation of Kinect-artefact 1.0's phenomenal world. Early exhibitions of the Kinect-artefact emphasised the capture of human movements through the "stickman" visualisation, a crude depiction of the Kinect-artefact's idea of human bodies (which is the only semiotic relationship most creators and consumers were really interested in). Perhaps the most striking thing about this image is the impression that the Kinect-artefact 1.0 used very sophisticated, labour-intensive technology to contrive an image recognition that most humans could do easily. The Kinect-artefact, despite its efforts to be "natural" (read: human) struggled even to perform what would be a very simple task for most humans. Nevertheless, this visual was part of the campaign for Kinect-artefact 1.0 for Windows, part of the effort to encourage people to do their own things with their Kinect-artefact:

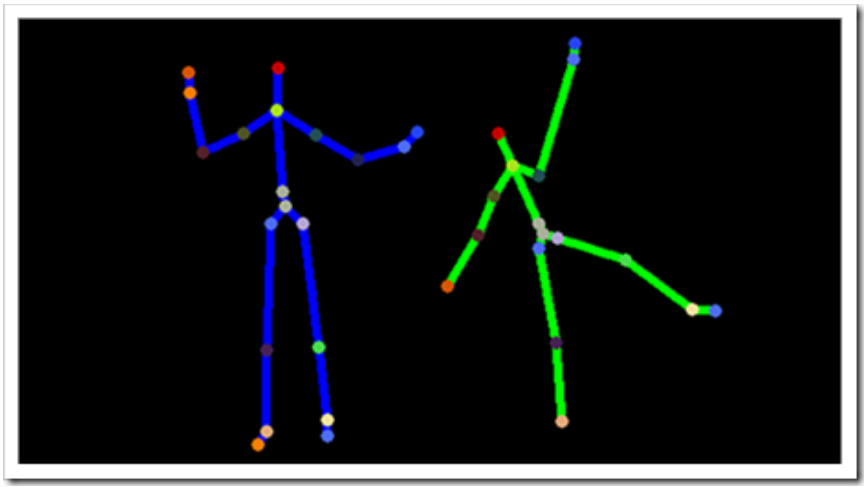


Figure 7: Skeletal tracking with a Kinect-artefact 1.0.[65]

Stick-man is an attempt to code certain aspects of the Kinect-artefact's phenomenal world for human consumption. It gives the user or the engineer an insight into the machinic

[64] Harper and Mentis, "The Mocking Gaze."
[65] David Catuhe, "Gestures and Tools for Kinect," *Eternal Coding* (blog) (July 4 2011). http://blogs.msdn.com/b/eternalcoding/archive/2011/07/04/gestures-and-tools-for-kinect.aspx?PageIndex=2 (accessed June 26 2017).

experience. However, this has not been released as an interactive art project. Rather it is a tool for those individuals who want to persuade the Kinect-artefact to do something. It is an expedient hijacking of Kinect-artefact sensory systems to ensure future compliance with human motion. Many impressive projects were attempted with the implicit or explicit blessing of Microsoft. Among them is Chris Vik's 2011 performance at Melbourne Town Hall. Vik used a special Kinect-artefact-based interface to control a 4-storey organ with hand gestures.[66]

However, there were serious problems with basic gameplay in the Kinect-artefact 1.0. As argued later in the chapter, the Kinect-artefact *wants* to communicate well with the user. But it can only ever react to events in accordance with its qualities. One significant problem was the Kinect-artefact failing to recognise humans when the lighting was wrong. A quality of the Kinect-artefact 1.0 vision system is that it recognises the presence of humans by the colour of the light that it receives through its lens, and because lighting affects the colour of light, the Kinect's interpretations were unpredictable to humans. The body mapping software was also too simplistic to capture all the subtleties of human movement and the gestures did not work as consistently as necessary. Voice commands were not so ineffective but were underused by the software. These kinds of problems that the Kinect-artefact 1.0 struggled with are to some extent understandable to us, since humans also struggle to detect objects when conditions are too light or dark. Humans make errors in perception. But it is a little more difficult to say that we could mistake a person for a nonhuman object simply because they use a wheelchair, are moving too fast or are in poorly-lit conditions. We usually have little difficulty tracking the movement of a person's hand in space, or knowing that a hand concealed behind a person's back is still there. These frustrating and sometimes inexplicable problems come from the extreme restriction of the Kinect-artefact 1.0's world. It has limitations to its sensory-perception pathways that mean it has real difficulty carrying out its program of action.

Despite all its limitations, the Kinect-artefact 1.0 remained popular through the three years prior to the end of the Xbox 360 console cycle, selling more than 24 million units.[67] We can readily identify what was beneficial about this arrangement for Microsoft: it made money by selling Kinect-brand games (even if it is likely that it lost money on the Kinect-artefact itself) and it set the stage for future success in the console wars. It sustained Microsoft through a longer-than-average console cycle brought on by the Global Financial Crisis. And it offered a tantalising look at what might be possible with the next Kinect-brand.

---

[66] Andrew Webster, "Kinect Used to Control 83-Year-Old, Four-Story High Organ in Australia," *The Verge* (April 9 2012). http://www.theverge.com/2012/4/9/2935851/kinect-midi-controller-melbourne-town-hall-organ (accessed January 20 2018).

[67] "Microsoft by the Numbers," *Microsoft Story Labs* (2013). http://www.microsoft.com/en-us/news/bythenumbers/index.html?utm_source=buffer (accessed January 24 2014).

Kinect-artefact 2.0

The advertising, statements from Microsoft, and sensationalist news stories[68] prior to the release of Kinect 2.0 initially raised human user expectations. With the release of the Xbox One, Microsoft promised that the Kinect-artefact had been "completely re-engineered"[69], and seemed intent on reversing the popular perception of the device as a gimmick. The sensor now had a much larger field of view, compensating for one of the Kinect-artefact 1.0's biggest problems. The lighting problems were also fixed by adding imaging technology that does not depend on visible light. Both cameras working in concert create a more complete image of the user. It seemed that Kinect-artefact 2.0 would be better able to achieve its goals than Kinect-artefact 1.0, to the overall benefit of the Kinect-brand.

The Kinect-artefact 2.0, unlike the Xbox 360 model, uses a time-of-flight (TOF) camera. It measures the round trip time (RTT) of photons emitted in flashes from the Kinect-artefact.[70] The photons bounce off objects and return to the Kinect-artefact, from which the distance of objects can be measured using the speed of light constant. This creates a 3D image of whatever objects are positioned in front of the Kinect-artefact, and in far greater detail than that offered by a Kinect-artefact 1.0. Added to the 1080p visible light camera there is also an active IR sensor that addresses the lighting problems experienced with the Kinect-artefact 1.0.[71] With a Kinect-artefact 1.0, too much lighting on one side of the room could adversely affect the sensor's ability to pick up players. But the new sensor not only fixes this problem, it allows users to interact with the sensor in total darkness.

---

[68] Such as: Agam Shah, "New Kinect for Windows to Improve Human Interaction with Computers," *Computerworld* (May 23 2013), https://www.computerworld.com/article/2498093/personal-technology/new-kinect-for-windows-to-improve-human-interaction-with-computers.html (accessed February 22 2018).

[69] "Xbox One: What it is."

[70] Matthew Panzarino, "The New Xbox One Kinect Tracks Your Heart Rate, Happiness, Hands and Hollers," *The Next Web* (May 22 2013). http://thenextweb.com/microsoft/2013/05/22/the-new-xbox-one-kinect-tracks-your-heart-rate-happiness-hands-and-hollers/ (accessed June 26 2017).

[71] *Behind the eyes of Xbox One Kinect* (2013), Youtube. https://www.youtube.com/watch?v=JaOlUa57BWs (accessed June 26 2017).

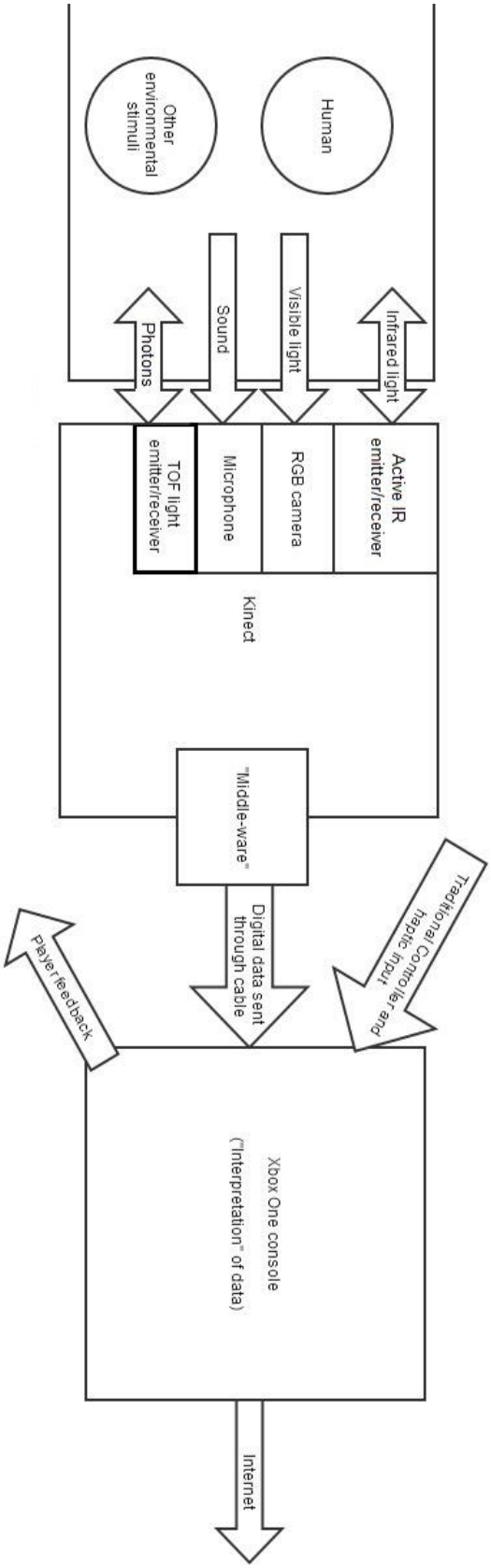


Figure 8: Diagram of a Kinect-artefact 2.0's basic technical relationship to other objects.



Figure 9: This image shows what the Active IR camera sees by itself while the room is in total darkness. From Microsoft Research's video.[72]

Figures 9, 10, 11, 12 and 13 have been taken from pre-launch videos made by Microsoft entitled "Behind the eyes of Xbox One Kinect" and "Inside the brains of Xbox One Kinect", names that continue Microsoft's efforts to anthropomorphise the Kinect-brand and to encourage gamers and users to think of their product as an active, smart participant in creating entertainment experiences. These images provide visual representations of one part of the Kinect-artefact 2.0's world: its experience of the human and the human's engagement with the console and software. They show how the Kinect-artefact integrates TOF cameras, active IR and visible light cameras in different ways.

In the pre-launch publicity a great deal of focus was put on emphasising the evolution of the Kinect-artefact's sensation and perception of the human body. For this purpose, the skeleton was more complex.

[72] Ibid.



Figure 10: Human skeleton as seen by Kinect-artefact 2.0.[73]

The Kinect-artefact 2.0's skeletal tracking has greater articulation of the hips, spine and shoulders as well as recognition of thumbs and the length of the hand. Below, "Block-man" reveals how the Kinect-artefact sees rotation in the joints and how body parts relate to each other.



Figure 11: "Block-man", the visualisation that depicts the way that the Kinect-artefact 2.0 views the orientation or "rotation" of human body parts.[74]

Developer Kareem Choudry claims in the video that the skeleton is only a small part of the story when it comes to mapping the body. "If you really want to understand what's going on with the human you need to understand and describe what's going on with the muscles, the forces, the torque." "Muscle-man" depicts how the Kinect-artefact 2.0 thinks the muscles in the human are behaving.

[73] *Inside the Brains of Xbox One Kinect* (2013), Youtube. https://www.youtube.com/watch?v=ziXflemQr3A (accessed June 26 2017).
[74] Ibid.



Figure 12: “Muscle-man” as seen by a Kinect-artefact 2.0.[75]

In this image green represents “no force” and red represents high intensity. The rear leg is carrying most of the human’s weight and is darker in colour. The circles under the feet represent impact on the floor. The white circle in front of the figure represents impact. In the video, the individual is punching the air, as one might do in a fighting game, and the power of his blow is represented by the size of the circle. While the skeleton and the block-man are merely closer to the ways that humans perceive bodies, muscle-man is a mode of perception that the Kinect-artefact has which humans lack. Similarly, the Kinect-artefact uses its alien senses to detect exertion in humans. The visible light spectrum camera and the Active IR camera work together to deduce the user’s heart rate by detecting tiny changes on the skin. This is a type of sensation that humans do not experience, although they are typically able to draw similar conclusions to the Kinect-artefact (i.e. “this human has recently exerted themselves”).



Figure 13: Heart rate as sensed by the Kinect-artefact, as well as the three other visual models.[76]

[75] Ibid.

These visual models are a good representation of what the Xbox deduces about human bodies from its "eyes": the colour camera, active IR sensor and time-of-flight sensor. The images are, however, of varying utility to exploring the phenomenal world of the Kinect-artefact. They are, after all, promotional images and reflect the "brain" of the Kinect-artefact. The "brain" is located in the middle-ware of the Kinect-artefact that makes sense of what the sensors detect – identifying certain patterns as human bodies, for instance.

The Kinect-artefact is an extremely human-centred device, both in its "sensation" by the sensors and its "perception" by the middle-ware. Its sensors were designed to emit and receive radiation that is best able to give information about human bodies, as well as certain other things such as the floor the human stands on and the controller the human holds. The middle-ware takes these already-biased observations and further exaggerates the body positions before relaying the information to the console. Not all the information picked up by the sensors is used. From the images above it seems clear that the Kinect-artefact is not interested in the position of furniture or what is hung on the walls, beyond its role in obstructing the Kinect-artefact's vision. The Kinect-artefact is oriented to the human. And yet, it would be difficult to claim that the Kinect-artefact is human-like in its sensing and perceiving. It is an extraordinarily alien object, even for a videogames accessory. It constructs a view of the human using bursts of photons and block-like models that are quite foreign to human vision; it receives input from one direction but can only send data onwards to the console; and it can be put into a state in which its sensors do not respond at all.

The phenomenology of things is ever more pertinent to everyday life. The Internet of Things project aims to construct a new internet-based way of living as that of a planet so enriched with interlinked sensors that it is capable of far more advanced coordination than previously. The Kinect-artefact is a sensor that affords a different kind of relation between human and console. But can we really only describe a Kinect-artefact-console hybrid as a sensor? In a personification of the phenomenon we would surely describe it as a sensory-perceptive entity. But the concept of perception, when applied to a nonhuman, is a problematic one for the information processing theory of knowledge.[77] The interiority of machines is questionable, and their irreducibility seems absurd when each component is created by humans. This will be addressed further in the next chapter, with the discussion of Umwelten and consciousness, but briefly, it seems evident from the way by which the Kinect-artefact conveys and transforms information from human to console that it *at least possesses some interior qualities*. These are the qualities that cause it to relate in certain ways – Kinect-artefactish ways. Consequently, if we put aside for a moment questions of mind and consciousness, the study of the alien phenomenology of the Kinect-artefact is valid. Due to its status as a contentious cultural object, its phenomenology is of particular interest to humans. Specific moments of Kinect-human relations are the topic of the next section. These moments should provide illumination of aspects of specific relationships of the Kinect-artefact with its environment.

---

[76] Ibid.

[77] Søren Brier, *Cybersemiotics: Why Information is not Enough!* (Toronto: University of Toronto Press, 2008).

Communication, sensation and relation of the Kinect-artefact

Microsoft claims that Kinect-artefact technology makes interaction with the Xbox One more natural. The language of “eyes” and “brains” conveys the impression that the Kinect-artefact has an implicit humanity so that humans may communicate with it more easily. The human need not learn and adapt to use the system, because the system has been adapted to be human-like. The Kinect-artefact participates in semiotic interaction with multiple objects, most notably the console, but also with humans, furniture, the speed of light constant and the Kinect-brand. This section will primarily focus on the interaction between human (and human environs) and Kinect-artefact, but will also describe how relations with other objects affect that human-Kinect-artefact relationship.

“Waking up” the Kinect-artefact

In the immediate aftermath of the release of the Xbox One, corporation and customers came to a gradual equilibrium with regards to complaints and suggestions about Kinect-artefact 2.0. Several major bugs and design problems were reported to Microsoft and resolved. Difficulties for Kinect-artefacts included identifying humans and conveying information about humans (and their surroundings) to the console, and the console had difficulty interpreting that information. Software updates made it easier for Kinect-artefacts to behave in a way that was more pleasing to their humans, improving the relationship. However, many problems remained. One example is that the furniture in the room containing the Kinect-artefact must be carefully arranged. The Kinect-artefact is widely described as “finicky” or discriminatory, and the lack of a general understanding of its phenomenology has resulted in an abundance of pseudo-scientific explanations for its behaviour. Humans needed to understand what the Kinect-artefact wanted from its environment before they could form meaningful semiotic interactions with it, and eventually grok the system. The most prominent issue after the Kinect-artefact 2.0’s release is known as the “Xbox On” problem, and it represents a starting point for engaging with the phenomenal world of the Kinect-artefact. The data for this section was gathered from the official online Xbox One support forum[78] and from Reddit’s Kinect and Xbox One subreddits.[79]

The “Xbox On” problem has become an emblematic instance of the communication problems between human and Kinect-artefact. A major emphasis for this generation of the Kinect-brand is on voice commands, largely replacing the hated gesture controls of the Xbox 360.

---

[78] "Kinect on Xbox One," *Xbox* (2013-2014). http://forums.xbox.com/xbox_forums/xbox_support/xbox_one_support/f/4275.aspx (accessed January 13 2014).

[79] "kinect," *Reddit* (2013-2014). http://www.reddit.com/r/Kinect (accessed January 13 2014).
and
"XBOX ONE," *Reddit* (2013-2014). http://www.reddit.com/r/XboxOne (accessed January 13 2014).

Kinect-artefacts (those that are able to hear and understand human voices) convey that information to the console and thus are able to control most aspects of the interface. But Kinect-artefacts often have trouble understanding human intent through language, and this causes humans to come into conflict with the Kinect-artefact as well as the Kinect-brand. To use Heidegger's terms, when ready-to-hand for the human the Kinect-artefact hears, understands, and acts. It can become present-at-hand for the human by failing to hear, understand, or act. A user can play movies without needing to touch a controller, and they may perform all the additional playback commands with voice as well. Kinect-brand games and many major menu operations can be activated with speech. These features have been welcomed by users with physical disabilities (specifically blind people or people with low mobility), pregnant women and parents with small children, and are reported to be useful when cooking and while sick, since without voice controls the Xbox One cannot respond to immediate human needs without input from a controller. The Xbox One also learns the cadences of individual users' speech so that over time it responds more easily to voice commands.

But there is one problem which has been extensively identified: the difficulty encountered by many users trying to switch their console on with the "Xbox On" command. When the console is on "instant-on" standby, it can reactivate in response to a human uttering the voice command. In the Microsoft advertising materials for the new Kinect-brand, this feature was given particular prominence. On its website, it claims that "Xbox One was designed for today's fast-paced lifestyle. It wakes up instantly when you say, 'Xbox On,' and even turns on your TV."[80] But unfortunately many users find it difficult or impossible to wake the console up without physically touching the box or controller, leaving them wondering whether the Kinect-artefact is poorly designed or whether their environment is simply not convivial for a Kinect-artefact.[81] This is an important instance in which the Kinect-brand becomes disassociated with the Kinect-artefact, which simply cannot perform what the Kinect-brand has promised. The Kinect-artefact does not hear, understand, or act on the voice command. Identifying why this happens requires learning about the Kinect's way of being in the world. It becomes present-at-hand for the human, and it becomes the subject of alien phenomenology.

The desire to switch the console on without needing to touch a controller is a practical concern. One user reported being able to use a voice command to turn the Xbox One on from the ground floor so that it would power up while they were making their way down to the basement. But those users that have found difficulties with this command exhibit a frustration and disappointment that may be judged excessive for a practical issue. Several commenters expressed a sense of "betrayal" after buying consoles with the belief that it would work like it does in its advertisements and subsequently spending considerable time and effort trying to make the feature work. There is a particular anger because users feel they were "forced" to buy the Kinect-artefact with the console despite the widespread privacy

[80] "Innovation," *Xbox* (2014) http://www.xbox.com/en-US/xbox-one/innovation (accessed January 13 2014).
[81] Tessa Leach and Michael Arnold, "How the Kinect Domesticates its Users," paper presented at *At Home with Digital Media* symposium, Queensland University of Technology, November 3 2017.

concerns. There is also a feeling of being ignored by Microsoft and particularly of support staff failing to read the forums and therefore not realising that there is a problem, or perhaps even that Microsoft is refusing to "admit" that there is a problem. This may be a residual complaint from Microsoft's failure to acknowledge the "Red Ring of Death" problem, a complaint that rendered Xbox 360s unusable.[82] Presumably, the Kinect-brand also experiences a sort of frustration from this issue. It threatens the relationship between Kinect-brand and existing users of the Xbox One console, which makes it less likely that other users will purchase a Kinect-artefact. User forums and social media comments may be in a strong relationship with the Kinect-brand, and at the time of this problem these online influencers of opinion were consumed with the "Waking up the Kinect" issue. The sentiments of the posters leave the impression that there is something greater than a practical concern inherent to this difficulty, especially since other practical problems, such as the lack of a choice when using the Kinect-artefact for voice communication in multiplayer games, did not attract the same demands for a solution. It may be that this disproportionate response emerges from the portrayal of advanced computers in science fiction as well as in corporations' depiction of present-day technology as becoming closer to anticipated Artificial General Intelligence computers. Users desire machine interfaces that mimic the relationships between humans. Patterns of sleep and wakefulness are among the most culturally deterministic aspects of our biological existence, and any kind of "smart" machine that approaches human intelligence should be wakened in a human way. But of course, more fundamentally this is a basic failure of a machine to turn on in the advertised fashion, upon which all other functions depend.

The community of users trying to find a solution to this problem have explored many possible avenues. One interesting class of explanation suggests that users with certain types of voices have greater success than others. The main observation made by many users is that female voices are often able to activate the console while male voices cannot (although some users report the opposite effect). Some users report success after uttering the command in a falsetto register. Many users have suggestions for the best pronunciation of words. A particularly common piece of advice in any region is to pronounce the word "on" like "ahn", with more of a North American accent. One user claimed that using a "really nasally 'nerd voice'" improved the voice commands. Some users claim that it is better to say it very quickly, and others claim that pausing between the two words might be an improvement. There was an awareness that Kinect-artefacts possess similar internal qualities that affect relationships – in other words, the beginnings of an informal assessment of the machine's experience of the world. Several users reported using a Youtube video to turn on the console, made by individuals whose voices were easily understood by Kinect-artefacts. From my research there are two videos that were being used, both including the apparently male filmmakers turning on the console with their voices.[83] By sampling the recording, users were able to turn on their

[82] Lance Ulanoff, "Xbox 360: a Tale From the Red Ring of Death," *PC Magazine* (July 28 2010). http://www.pcmag.com/article2/0,2817,2367072,00.asp (accessed June 27 2017).
[83] *Xbox One Voice Commands Don't Work?!?! Xbox One FAIL* (2014), Youtube. http://www.youtube.com/watch?feature=player_embedded&v=JSv7iEpAE58 (accessed January 13 2013).
and
Xbox On, (2013) Youtube.

consoles using the voice of the filmmaker. One caused a particular sensation, since the filmmaker with a British accent was able to turn on previously recalcitrant North American systems. One possibility proposed is that a recording on an electronic device clips out the lower tones in a person's speech, supporting the theory that higher tones improve performance. The degree of indignation on this point reflects the predominance of male gamers as users of the previous Xbox consoles. The anger among early adopters came from feeling as though Microsoft owed "us" the best possible user interface, and had not delivered.

There are also apparent problems with accents. The "Xbox On" command was not provided in Australia at launch; there was no option to put it in an instant-on energy level when it is switched off, so it needed to be switched on manually. It could be very simply accessed, however, by switching the console's region settings to one of the countries for which the "Xbox On" is available, such as USA, UK or Canada. This means, however, that the Kinect-artefact may not be able to comprehend an Australian accent all the time. Changing the region changes the language that a Kinect-artefact understands and responds to. An utterance made with a different accent does not always produce the desired effect. The human speaks Australian English, but the Kinect-artefact hears imperfect US English. These restrictions may make this 'smart' object Microsoft is selling seem remarkably slow-witted, or possibly even sexist or racist (as were Winner's bridges – see Chapter 1). Users are beginning to expect natural language processing that can communicate additional information, respond to jokes and understand idioms (Apple's Siri and Amazon's Alexa come to mind), which are all much more tied to local culture than to broad language groups.

The Kinect-artefact itself does not tell us explicitly what gender or nationality it prefers; users have been required to work it out by observing the positive and negative feedback given by the Kinect-artefact. Some users performed experiments to find ways of making contact. They engaged in an informal program of alien phenomenology, deducing the nature of Kinect-artefact sensation from the positive and negative reinforcement of the Kinect-artefacts. It is not clear whether the problem is in the position of the Kinect-artefact, the room acoustics, the calibration of the Kinect-artefact, the energy settings, or a hardware problem, but it is clear that the interaction of the Kinect-artefact with other nonhumans is a significant factor in whether or not it will be able to communicate effectively with humans. Microsoft's advice for the Kinect-artefact microphone is fairly simple: follow the calibration procedure properly and make sure there is not too much noise in the room.[84] Yet on the forums there are a vast number of suggestions for ensuring that the Kinect-artefact works properly. Some users report having completely moved their console and the rest of their home media ecology to another room, particularly smaller rooms or rooms with carpeting. A collective effort to find the ideal positioning of the Kinect-artefact was underway by humans, and Kinect-artefacts participated in this process by punishing and rewarding consistently, according to each

---

http://www.youtube.com/watch?feature=player_embedded&v=Fm95LxARCFE (accessed January 13 2014).

[84] Misses Joust, "Xbox On: Voice Commands for Xbox One," *Xbox Forums* (2013-14). http://forums.xbox.com/xbox_forums/xbox_support/xbox_one_support/f/4275/t/1622636.aspx (accessed January 13 2014).

artefact's internal qualities. The investigation into positioning included ideal distances between the Kinect-artefact and the human. One user recommended crouching down 4-5 feet away, another suggested two inches. The forums were full of appeals to describe users' home theatre set-ups, sensor positions and room design factors. Its ability to hear also appeared to change over time, depending on how long it has been switched off. The "sleep" metaphor seems apt, since at first its sleep appears to be light but it then becomes less likely to be roused. The Kinect-artefact becomes sealed off from the humans attempting to make contact with it. In the absence of instruction from Microsoft or successful hacks of the Kinect-artefacts, the participants in this process, in short, adopted a sort of naïve speculative realism. The quiddity or interiority of the Kinect-artefact is assumed and the users experimented with different kinds of objects to try to gain access to as much information as possible. But they could never *be* their Kinect-artefact, and seemed to be repeatedly frustrated by failures to access the real root of their problem.

A common theme in the forums is describing the Kinect-artefact and Xbox One as though it were an animal to be tamed. One user suggested that the sensitive Xbox One needed a "caring, motherly voice", and many express the sentiment that the Kinect-artefact needed to be disciplined and educated. This is doubtless a result of the prevailing impression of the nature of machine learning in addition to the notions of personalisation and customisation that pervade consumer electronics discourses. The particular role of the Xbox One makes it an important example of these notions, since its taming (or perhaps domestication is the best word[85]) involves its entanglements in pre-existing and highly heterogeneous living room set-ups. It must be nestled into the pre-established network of television, other consoles, cable television or set top box and peripheral gaming objects as well as becoming a focal point of all these activities. The Xbox One is intended to make these other electronics controllable through a single system, at least in the United States.

In all these ways the Xbox One also domesticates its human user. Using negative and positive reinforcement, the human is encouraged to position the Kinect-artefact prominently in the room, to rearrange her furniture and alter her electronics set-up. She is even compelled to change the way she speaks and moves. The Kinect-artefact cannot give clear instructions to the console without this domestication, and must encourage the human user to act in a way that engages with certain parts of the Kinect-artefact's way of being in the world.

---

[85] Roger Silverstone and Leslie Haddon, "Design and the Domestication of ICTs: Technical Change and Everyday Life," in *Communication by Design. The Politics of Information and Communication Technologies*, ed. Robin Mansell and Roger Silverstone (Oxford: Oxford University Press, 1996), 44-74. There are several prominent authors who discuss the domestication of living things in posthuman terms, but the posthuman domestication of technological artefacts is less common.
See for posthuman readings of animal domestication:
Donna Haraway, *The Companion Species Manifesto: Dogs, People, and Significant Otherness.* (Chicago: Prickly Paradigm Press, 2003).
Laura Gustafsson and Terike Haapoja, *History According to Cattle*.
And
Bryant, *Onto-Cartography*, 67 (in which Bryant also discusses cattle).

Another part of this domestication involves encouraging the human to use the applications compatible with the Xbox One. In their book *Games of Empire*, Nick Dyer-Witherford and Greig de Peuter chart Microsoft's moves and describe them as processes of "deterritorialization and reterritorialization".[86] They were speaking of the two older Xbox consoles in addition to certain occurrences in the world of personal computing, but it may also be found in Microsoft's strategy with the Xbox One. The key is to create a multimedia platform that forms an entertainment hub in people's homes, from which additional services can be marketed and sold. The Microsoft-owned services include:

- Skype;
- Xbox Video and Music (paid subscription or single payment libraries);
- Integrated Bing internet search;
- Internet Explorer;
- Xbox Fitness (purchasable work-out videos designed for Kinect-artefact integration);
- SkyDrive;
- Upload Studio (allows users to create, modify and publish videos of gameplay).

As well as products that are not owned by Microsoft:

- Netflix;
- Amazon Lovefilm;
- Twitch (leading machinima and gameplay video platform) and Machinima (gamer and programming network);
- Youtube;
- TED;
- Hulu Plus in addition to many regional premium cable channels and video streaming apps.

The console comes to control the gaming, social, video and music services. Once the cable television, speakers and other elements of the home theatre are under the control of the console (or the Kinect-artefact) it becomes a vital component and cannot be abandoned. The user is encouraged to use Microsoft services on other platforms and perhaps even to buy Microsoft smartphones, tablets and to run Windows on their personal computers. The hardware and applications used for socialising, home entertainment, mobile entertainment, internet surfing, gaming and media library management are, in this metaphor, "territories" that Microsoft is endeavouring to capture in the everyday media consumption of its users. The domestication of the user by the Kinect-artefact allows Microsoft to capitalise on the prominent position of the Xbox One in homes. If one platform can provide so many different functions, and interacts so seamlessly with other software sold by the same company, why use products made by Google, Sony or Apple? In the terminology of ANT, the platform becomes an obligatory passage point in the home entertainment system.[87] The human-object and object-object relations prevent any direct human-Microsoft relations, and the brand

[86] Nick Dyer-Witheford and Greig de Peuter, *Games of Empire: Global Capitalism and Video Games*, Electronics Mediations (Minneapolis: University of Minnesota Press, 2009), 74.
[87] Michel Callon, "Some Elements of a Sociology of Translation."

remains safely abstracted and uninterrogated. The Kinect-brand is like the lancet liver fluke *Dicrocoelium dendriticum*, which manipulates the nerves of ants (through unknown means) to compel them to leave their colony. The ants are then "obliged to crawl upward on a blade of grass and to bite down firmly upon its tip".[88] They are unable to open their jaws during peak ruminant grazing times, and cattle, sheep and pigs are thus more likely to consume the ant, and the fluke completes the next phase of its life cycle in the ruminant's digestive system. Many parasitic organisms change the behaviour of their hosts in similar ways. In this analogy, the Kinect-brand alters the brain of the human in subtle ways, compelling the human to change its behaviour in a way that better suits the brand. Of course, particular individuals in a species may develop resistances to parasites, or the parasite's genetic material may drift, altering this this optimal arrangement. Similarly, some humans are already inhabited by other parasites (such as those of the Google or Sony genus), develop a disinterest in the brand over time, or are so over-exposed to attempts at corporate manipulation that they do not cooperate with the brand.[89]

This is an important way of ensuring that users of the previous system adopt the next system. To some extent this tactic enabled Microsoft to retain a core of Xbox 360 users who did not wish to abandon their Xbox Achievements (rewards that may be earned by achieving certain goals during gameplay), avatar and profile, plus the rewards of Xbox Live Gold membership. There are also advantages for Playstation 3 users to upgrade to the Playstation 4, such as the retention of Trophies in the new system. But the Playstation 4 with its many other advantages, including its arguable superiority as a gaming system and social network hub, lacks the totalising home theatre presence and is therefore less likely to become a permanent fixture in any home theatre set-up, especially given the increasingly long console cycles. Perhaps Microsoft believed that the generation of gamers that were loyal to the Xbox 360 for eight years had matured and wanted their gaming history to be preserved with the additional advantages of partner- and child-friendly features. The "Xbox On" command is a fundamental feature of this process of territorialisation. The user can perform the science fiction-like act of communicating with her computer through speech – a novelty which the core users represented on the forums appear to enjoy. Furthermore, in regions where the process is supported, the "Xbox On" command awakens all the home theatre elements. The word "Xbox" here is meant to substitute "complete entertainment experience". The Xbox *is* domestic entertainment.

In addition to the logic of domestication in console expectations, the nature of machine learning and preponderance of "smart" devices has led to a set of impossible expectations for this generation of consoles. Microsoft is not the only company responsible for this

---

[88] Jan A. Pechenik, "The Platyhelminths," in *Biology of the Invertebrates* fifth edition (Boston: McGraw Hill Higher Education, 2005), 162.

[89] Arndt Niebisch expounds a similar idea, drawing on the work of Walter Benjamin and Friedrich Kittler, characterising the avant-garde art schools of the early twentieth century as parasites. However, Niebisch characterises the parasites as feeding off reality and mass media, and transmitted by new media technologies. He also portrays these practices as abusive or subversive, which is not the intention here.
See: "Chapter 3: Parasitic Media," in *Media Parasites in the Early Avant-Garde* (New York: Palgrave MacMillan), 2012, 81-107.

impression, but it certainly cultivated the anticipation of a device that would have an agency and anthropomorphic intelligence well beyond previous consoles. Anthropomorphising machines is a powerful method for conveying the utility and sophistication of an electronic entity but is often highly misleading. In the case of the Xbox One, this tactic backfired and caused the impressions of treachery and deception that resounded in the forums. The next section will discuss the game *Kinect Sports Rivals*[90] and how the Kinect-artefact turns humans into avatars, in itself a re-negotiation of the line between human and nonhuman.

Kinect Sports Rivals

One of the first titles to explicitly focus on the Kinect-brand was *Kinect Sports Rivals* (*KSR*), a sports-themed game in which gamers play several minigames. The six minigames are wake racing, bowling, soccer, sharp shooting, rock climbing and tennis. Originally planned for concurrent release with the Xbox One it was delayed until release on April 8 2014. The title was an opportunity for Microsoft to exhibit some of the new Kinect-artefact's power and potential, but the game also demonstrates a fascinating use of AI in collocated social gameplay. Gamers can play in multiplayer groups as in earlier Kinect-brand party games; up to four individuals can use a single system at once (and all may be logged into their personal Microsoft accounts). But an additional social element occurs online, either playing with an absent friend in real time or via the use of an AI avatar who can play against its user's friends without the gamer initiating or authorising the game. In other words, Player 1 can play against Player 2 even when Player 2 is asleep. Human and nonhuman can play in the same way as human and human, but with the added complication that the nonhuman mimics a specific human rather than a general one. *Forza 5* for Xbox One has a similar arrangement in which the human player can be represented by their "drivatar" AI simulation.[91]

The game is frequently compared to *Wii Sports*, a game released to showcase the Nintendo Wii in 2006.[92] *Wii Sports* is a similar game albeit much simpler both in ambiance and in gameplay. But because the pared-down Wiimote tended to respond to player intentions under the best conditions, the game was generally acknowledged to be a success. *KSR* was not as successful. Although the creative island setting is compelling and the social functions work well, the game received mediocre reviews with a resounding tone of dissatisfaction at a sensor that has failed to live up to expectations for the second time in three years. The production company, Rare, has since been forced to lay off staff in the wake of the game's failure.[93] Perhaps this is because the standards for the game were set too high, as after the

[90] *Kinect Sports Rivals*, Xbox One.
[91] *Forza Motorsport 5*, developed by Turn 10 Studios (Microsoft Studios: 2013), Xbox One.
[92] *Wii Sports*, developed by Nintendo EAD (Nintendo: 2006), Nintendo Wii.
[93] Matt Helgeson, "Rare Lays Off Staff in Wake of Kinect Sports Rivals," *Gameinformer* (May 19 2014). http://www.gameinformer.com/b/news/archive/2014/05/19/rare-lays-off-staff-in-wake-of-kinect-sports-rivals.aspx (accessed June 27 2017).

release of the demo *Kinect Sports Rivals Preseason* in November 2013,[94] many reviewers were optimistic about the game's prospects. One journalist reviewed *Preseason* and pronounced it "Surprisingly Not Terrible",[95] which is a significant endorsement from a "hardcore" games publication. The demo contained one single wake racing track with relatively intuitive controls and clearly showcased the best that the full game would offer. Most reviews acknowledge the wake racing as the most successful feature, while going on to say that some or all of the other minigames fail to replicate this elegant simplicity.

For the purposes of this thesis, the avatar creation and autonomy of the avatars used in *KSR* are the most interesting qualities. A detailed modelling of a human player into an electronic avatar aims to recreate the human player in the game. This modelling is three-fold. Firstly, the creation of the avatar's physical form is performed, in surprising detail, by the Kinect-artefact's view of the player, particularly of the player's face (the game appears to be fairly diplomatic when estimating the player's body proportions). Microsoft's promotional materials claim that the Kinect-artefact "instantly scans you and captures your likeness as a champion",[96] although this claim will be challenged below.[97] Secondly, it makes a social representation of the player towards the human's friends or other online competitors. Thirdly, it creates an AI representation of the player's tactics and skill level which can then appear in the games of other players, so that you do not need to be playing for your avatar to compete against your friends. Here again is the discourse in Kinect-brand promotional materials toward the erasure of boundaries between player and game, between human and nonhuman.

At the beginning of the game, the avatar creation process begins with a call to acknowledge the almost magical rendering of the human body within a game. The claim by the voice-over is that the console will use the player's "digital DNA" and create an avatar through the hardware's "technical wizardry". The avatars and avatar creation process have been described as "surprisingly accurate"[98] but also as "a cartoon approximation" [99] and one reviewer claimed that the avatar "may or may not bear a likeness to you."[100] Several reviewers have criticised the logistics of the avatar creation process, which requires the

---

[94] *Kinect Sports Rivals Preseason*, developed by Rare (Microsoft Studios, 2013), Xbox One.
[95] Chris Carter, "Impressions: Kinect Sports Rivals Preseason," *Destructoid* (November 28 2013). http://www.destructoid.com/impressions-kinect-sports-rivals-preseason-266352.phtml (accessed June 27 2017).
[96] "Kinect Sports Rivals," *Xbox* (2014). http://www.xbox.com/he-IL/xbox-one/games/kinect-sports-rivals (accessed May 27 2014).
[97] Comparisons can easily be drawn here between the process through which humans are made into avatars and the humanoid dolls and robots discussed in Chapter 5. Both processes are highly selective, and the human qualities that are selected are indicative of certain priorities or conventions in the practice of turning human forms into nonhuman caricatures. But this is a study for another day.
[98] Brian Albert, "Kinect Sports Rivals Review: a Swing and a Hit," *IGN Australia* (April 7 2014). http://au.ign.com/articles/2014/04/07/kinect-sports-rivals-review (accessed May 29 2014).
[99] Simon Parkin, "Kinect Sports Rivals Review: Warm Down," *Eurogamer* (April 8 2014). http://www.eurogamer.net/articles/2014-04-08-kinect-sports-rivals-review (accessed May 29 2014).
[100] Carolyn Petit, "Kinect Sports Rivals Review: Championing Mediocrity," *Gamespot* (April 7 2014). http://www.gamespot.com/reviews/kinect-sports-rivals-review/1900-6415722/ (accessed May 29 2014).

player to kneel down next to the Kinect-artefact in what one reviewer called “a submissive stoop towards [the] TV”. [101]



Figure 14: The depiction of avatar creation from the official video. As in the game itself, the body structure is initially shaped using grey squares and is then approximated by the game’s built-in avatar options.[102]

[101] Will Freeman, "Kinect Sports Rivals Review - Going Through the Motions," *The Guardian* (April 11 2014). http://www.theguardian.com/technology/2014/apr/11/kinect-sports-rivals-review-wii (accessed May 29 2014).



Figure 15: The game asks the player to crouch to bring the face into alignment with the Kinect-artefact. The player's face, depicted in grey squares, must move into the blue space.[103]

The original Wii and subsequent Nintendo consoles may be behind this idea, since the digital avatars or "Miis" represented players in several games. But again, the avatars in the Nintendo world are much simpler and cartoon-like. Nintendo's choice not to aim for true verisimilitude is in keeping with their minimalist and family-friendly approach to the industry. Similarly, Microsoft's choice to aim for relative accuracy in their portrayal of players represents their emphasis on high-performance machines with excellent graphics, a sensor that is not just a toy and their rebranding of Kinect-brand for adults. This hit-and-miss portrayal must therefore be a significant disappointment for Microsoft, those objects that are in strong relation with Microsoft, and the Kinect-brand. The specific Kinect-artefact's frustration is much more situated in the moment of disconnect between Kinect-artefact and human user.

The idea of the avatar creation process is generally inoffensive and predictable, but as with any kind of biometric analysis it is important to seek out the restrictions of the technology. As

[102] "KSR: Announcement Trailer," *Xbox* (2014). http://www.xbox.com/en-US/xbox-one/games/kinect-sports-rivals (accessed May 29 2014).
[103] *Kinect Sports Rivals*, Xbox One.

a body-mimicking device for a game aiming to reproduce human ability, the game automatically creates a negative territory of bodies that do not conform to the requirements of the sensor and of the game. This is a territory containing individuals with low vision or hearing, mobility restrictions or without a certain normative body part such as an arm or ear. Any media form imposes certain restrictions on use including non-ergodic literature which at least requires "trivial" effort by the reader to perceive light and dark and to turn pages.[104] But the failure to encompass a broad range of players does seem contrary to Microsoft's stated intention to provide effortless play. The direction to kneel on the floor for avatar creation is, for example, an impossible feat for many individuals. It does not appear to support portrayal of individuals without two arms and legs, and most of the games can only be played while standing. On the support website for the Kinect-brand, Microsoft acknowledges that the Xbox One may not be accessible to everybody but argues that it is individual developers that must be "educate[d]".[105] From this, we can see that the Kinect-artefact looks for certain ideal characteristics in humans, but every individual human presumably deviates in some way from these ideals. "The human" never presents itself to the Kinect-artefact.

The game causes the human body to position itself in a particular way. Part of this process includes moving furniture so that the Kinect-artefact can pick up more of the players' bodies. Once this process is completed, the individual is directed to either a sitting or standing position, then to move one's body parts to correspond to virtual stimuli, such as moving one's arms and making a grasping movement to hold virtual handholds during the rock-climbing game. On various online forums, some users report that these actions are both possible and enjoyable with the ideal furniture and room size, but for many users without these luxuries their more strenuous moves are punished by a sensor that cannot pick up all of their bodies. There is also some disconnect with the screen because the body is shaped both by what the Kinect-artefact detects (or is thought to detect) and by the screen, which may not be in complete accord. When shooting for a goal in soccer, the size of the screen may change the player's idea of where the goal and goalkeeper are located. Once the player is capable of grokking the system this problem disappears, but initially at least the body is shaped both by visual cues (screen) and by visual feedback (Kinect-artefact and screen). The concept of a "natural" user interface starts to become a bit tenuous at those moments.

---

[104] Espen J. Aarseth, *Cybertext: Perspectives on Ergodic Literature* (Baltimore, Maryland: John Hopkins University Press, 1997), Google Ebook.
[105] "Accessibility, the Xbox One and Kinect," *Xbox* (2014). http://support.xbox.com/en-US/xbox-one/system/accessibility (accessed April 30 2014).



Figure 16: The rock-climbing minigame asks players to position their hands onto virtual handholds (and onto the ankles of other players).[106]

The game goes to great lengths to emphasise the direct relationship between the player and the avatar. The player is made to spend considerable time creating an avatar, and the avatar features prominently in gameplay. The Kinect-artefact is used as a biometric scanner, assessing the player's build and facial features to create "champions", which are avatars intended to resemble the player both physically and in gameplay. There is little variation between champions, with all features changeable and found in menus. Only two ages ("adult" and "junior") and two sexes ("male" and "female") are available. Still, once the champion is created it is intended to be a representative of the player. Since the Kinect-artefact recognises individual users, each player can step into play and their specific champion is brought up at once. But what is more, the game apparently learns what kinds of behaviour an individual is likely to use at a given time. Examples include using power-ups at particular times and using particular moves to gain extra points. This is an additional level of complexity for the use of the avatars. *Wii Sports*, for example, restricted itself to assigning a numerical skill level to each avatar. *KSR* aims for a much stronger correlation between real player and virtual version of the player.

More abstracted from the game is the player's indirect behaviour after interacting with the game. *KSR* is designed to be a social game, so both the game and the player should benefit from engaging more players. The game becomes (ideally) more fun with competitors both in front of the same Kinect-artefact and connected via the internet. Therefore, the person who owns the Xbox One system is implicitly encouraged by the game to convince his or her friends to purchase their own Xbox One consoles. This is of benefit to the brand and the game, each of which "wants" to be successfully marketed and sold (see below). This could also be one of

[106] "KSR: Announcement Trailer."

the motives for successful avatar creation, because it facilitates online play with friends. Champion scan artist Iain McFadzen claims:

> It really does put an extra edge on competition when the person you've just pulled of the rock wall in climbing, or barged in to a mine in wake racing, looks and plays like one of your friends or family. And, of course, as they're Champions, they're always available to play against, 24/7.[107]

Having a good representation of oneself in the game goes some way to breaking down the barriers of abstraction in online play, and being able to play with a person known in the real world can make games more enjoyable for the human (and is good for the Kinect-brand).

The only way for a human to investigate a Kinect-artefact is to look to the phenomena it produces and ultimately to look with human senses. An empirical approach must provide data for the task of interrogating the phenomenology of the Kinect-artefact. But the overmining or undermining of the objects can arise and must be assessed in each individual case. In this case the agency of the Kinect-artefact is closely tied to the agency of the game, but this is not always the case. Moment by moment the allegiance between the entities changes, but under most conditions they align. A Kinect-artefactish way of behaving is typically to receive data, transform it into useful information for the console and to provide feedback about its ability to sense certain objects to the human. We might refine this into a human-centric statement: the Kinect-artefact aims to appear to be a useful and effective NUI. The game behaves in a way that conveys the code for specific media formats. This can be refined into the human-centric statement that the game's aims to provide good gameplay, graphics, sound and online play for the human player. Ultimately both the Kinect-artefact and the game interact with the Kinect-brand: aiming to impress the player, to be well-reviewed and to sell more copies of itself.

The conflict between Kinect-artefact and Kinect-brand is evident in *KSR* as in the "Waking up the Kinect" problem. The game wants to be liked and sold, being created by corporate interests and thereby being inextricably linked to them (rather as Latour's gun is inextricably and bidirectionally linked to the human – see Chapter 5). But this tenuous connection is easily broken by the many betrayals of the Kinect-brand and the Xbox One console more generally. The game's aim of good sales is obstructed by the failure of Kinect-artefacts to perform as promised. Specifically, the Kinect-artefact often fails to hear the player's spoken commands, it often cannot see the tops of players' arms while they are rock-climbing, and it does not always deliver an accurate image of the player during avatar creation. Kinect-brand programs of action are easily severed by the most immediate Kinect-artefact programs of actions. This is consistent with Harman's claim that objects are all equally real: the game disc, the ephemeral interaction between player and NUI, the corporation and the effect on the developer's reputation.

A similar incident is described by Ian Bogost in *Alien Phenomenology*. The game *E.T.: The Extra-Terrestrial* for Atari Video Computer System is discussed (a game so unpopular that the

[107] "Creating a Champion in Kinect Sports Rivals," *Xbox Wire* (2014). http://news.xbox.com/2014/04/games-ksr-developer-qa (accessed May 29 2014).

unsold copies were famously buried in a landfill). In a few pages, Bogost enumerates the multitudinous things that are meant by the name of this game.[108] He begins by describing the code, then the physical form of the individual cartridge and finally in the context of the 1983 videogames market crash: "not just a fictional alien botanist but a notion of extreme failure, of 'the worst game of all time': the famed dump of games in the Alamogordo landfill, the complex culture of greed and design constraint that led to it[...]"[109] He stresses that none of these examples is the "real" *E.T.* game. Similarly, there is no incarnation of *KSR* that is more real than any other. But in both these games, technical and corporate failure are implicit and thereby reveal inner truths about the game and the console that are inadequately concealed. Typically, the game's goal is allied with that of the manufacturer, but the inner essence of the game might make itself amenable to interruptions by other elements such as the wall in one's living room that makes it impossible to get one's entire body into the Kinect-artefact's field of view, or the background noise that makes one's voice unintelligible to the Kinect-artefact, or the availability of internet review websites (in strong relation with the Kinect-brand) that make these failings known to the rest of the world. The corporation and the game cease to remain on the same side from the moment the game is released to the public. And while their goals may be momentarily the same their relations are always defined by their inner essences, not by the allegiance. Technological failures are ready examples of bidirectional asymmetry in interactions.

Under even more unusual circumstances we may interrogate the alien phenomenology of these objects much more fully. For example, the effects of fire or freezing temperatures reveal other kinds of agency in these objects, such as the tendency of plastic components to melt before metal ones, or the tendency of a game disc to shatter when smashed with a rock. For Harman, blasting the console into space is a similarly valid investigation of object relations and allows the creation of novel sensual objects between the game and other entities. While the human experiences a far broader set of sensual objects by using the game and console as intended by the manufacturer, this is not necessarily true of the Kinect-artefact for which picking up human gestures is repetitive and is only a tiny portion of its potential. The Kinect-artefact is not merely the sum total of our experiences of it, but as Harman says, most of its essence is invisible to us. Moreover, we are not reduced by the Kinect-artefact, which literally takes our image and leaves us as we were.

Alien phenomenology of the Kinect-artefact

The value of OOO lies in its strong theoretical robustness. It may lack the whimsical irreverence that seems so inextricable from the most successful ANT texts (such as in the paper by Jim Johnson, alias Bruno Latour, meditating on the incredible and seldom

[108] Bogost, *Alien Phenomenology*.
[109] Ibid., 18.

recognised power of the door hinge[110]), but it is a worthy successor for a scholarly milieu that is unsurprised by the agency of scallops and laboratory equipment. The tension of materiality and immateriality that surrounds posthuman discourse cries out for a philosophy of technology that can account for ephemerality and durability in human action, and OOO provides this with all the rigour and forethought that continental philosophy can offer. What OOO lacks is a grounded empirical framework that accentuates these strengths while contributing to a universal grammar for those who seek to replicate or extend other people's work. OOO and NUIs, in other words, have similar problems: a lack of widespread exposure; vehement opposition from groups that find more tried-and-true techniques easier to use; a limited number of examples that herald greater things to come; and competing impulses toward either consolidation or innovation.

To begin a grounded empirical investigation is to experience a loosening of the strings that connect the strength and elegance of Harman's object-oriented philosophy from the object under observation. We are entering a world of alien affect and significance with limited language and tools. As Bogost claims, the only way to understand an object's internal existence and modes of perception is through analogy: "The subjective nature of experience makes the unit operation of one of its perceptions amount always to a caricature in which the one is drawn in the distorted impression of the other."[111] Suitably performed, an anthropomorphic statement can accentuate the weirdness of objects such as the Kinect-artefact. One might refer to a Kinect-artefact's "desire" as a means of highlighting the alien nature of machine agency. The frequent use of anthropomorphisms by Microsoft engineers and marketing executives may fall into this category, although they must of course remain suspect because of their tendency to make products sound more "smart" than they really are. The aptness of any statement is, of course, subjective. The analogy must be sound or the anthropomorphism is suspect. In the case of the Kinect-brand, the promotion of the sensor made frequent references to the eyes of the Kinect-artefact, which as has been described led to inevitable disappointment from users.

The most fundamental Kinect-artefactish action is to take in information about its immediate world (while facing one direction). This is generally true unless the Kinect-artefact is broken or switched off, in which case its most noticeable relations might be with furniture and the ambient temperature of the room. These are all presumably consequences of the real qualities of the real object, which exists only in the phenomenal world of the Kinect-artefact. To ask about the real object and its qualities, in Harman's sense, is to want to *be* the object. The observer can only ever have access to the sensual object, the one that appears in our impression of the object. As discussed in Chapter 1, the sensual object is, in Harman's metaphysics, the object that appears when one object forms a relation with another object. The sensual object makes it possible for one object to relate to another (since real objects cannot directly touch one another). The sensual object has particular sensual qualities that we are capable of perceiving. But this does not mean that there is no connection between

[110] Bruno Latour as Jim Johnson, "Mixing Humans and Nonhumans Together: the Sociology of a Door-Closer," *Social Problems* 35, no. 3 (1988): 298-310.
[111] Bogost, *Alien Phenomenology,* 64.

the real object and the sensual one. The sensual object acquires its qualities from the invisible real object. Harman escapes idealism by insisting upon this correspondence.

The connection between object and object is, as discussed in Chapter 1, a matter of vicarious causation for Harman. Consequently, the interaction between human and Kinect-artefact is bidirectional and of a different nature in each direction. Each object is taking something completely different from the other. The human anthropomorphises the Kinect-artefact and, in a manner of speaking, the Kinect-artefact Kinect-artefactises the human. That is, the figure of the human is converted into a form that it is capable of experiencing. The human impresses itself upon the Kinect-artefact in the form of returning photons, infrared radiation, soundwaves and light. These relations do not take anything from the real human underneath. Similarly, the human understands the Kinect-artefact in terms of its ability to see, its thought processes and its menacing presence in the room.

Desire in the Kinect-artefact

It is easy to point to agency in the Kinect-artefact, but can we speak of desire? Desire surely suggests a gradation of needs that derive from the object's qualities. These range from fundamental necessities (such as electricity and connection to a console) which allow it to operate, to more minor wants (such as the correct angle of the camera and for the player to be properly positioned) which allow it to operate in an optimally Kinect-artefactish way. A good illumination of this gradation may be found in the attempts of the "Waking up the Kinect" investigators to derive an understanding of an alien entity. The investigations are attempts to glean more information from the sensual object and to make assumptions about the real object. The human users, denied access to a reliable account of the Kinect-artefacts' relations with the world, initiated a program of research into the sensors' "ears" and documented their attempts online. These attempts are a speculative program which exhaustively lists all possible interactions that a Kinect-artefact could have with its environment. It tries to position the human viewer within the Kinect-artefact. It tries to understand what the Kinect-artefact wants.

The Kinect-artefact's world was thoroughly described through this program and some important insights were acquired. The investigators discovered that Kinect-artefact desire is variable and analogue, even if its code is not. What works for one unit may not work for another unit, despite the patently obvious fact that each unit ought to be identical. Each Kinect-artefact has a different history and a different environment, similar to the way in which human identical twins become different due to their environments. Furthermore, each unit has certain needs that are more or less important than others. It might be difficult to stoop down in front of the television to allow the Kinect-artefact to scan one's facial features, while the activation of a voice command might be as easy as opening one's mouth at the right time. Some actions are needed, some are desired and some are so easy they might be attributed to gestural excess. But there is an additional understanding to derive from this incident, and that is about what the Kinect-brand wants. The refusal of Microsoft to

acknowledge the issue is a very telling part of the Kinect-brand's identity, which is one of intuitive use and seamless entertainment integration. The Kinect-artefact's needs and experiences are ludicrously removed from those of the Kinect-brand.

It may be argued that the idea of the brand is surely just the description in human terms of the object; the brand is an invention to bridge the gap between the objective world of objects and human thought. It is a desperate attempt to comprehend individual objects in an environment where thousands of identical units are shipped en masse. Bearing this in mind, is it right to have devoted so much time to discussing it? The brand is a human thing, not a Kinect-artefact thing. This is perhaps another, grander, anthropomorphism: the allocation of human-like qualities not just to one object but to thousands. The brand comes to have human qualities (hope, disappointment, stubbornness, defeat). But it is also a matter of scale. The Kinect-artefact and Kinect-brand are reflected upon in a huge number of human ways: in pictures, on websites, in videos and in webcomics. Each of these entities is a siphoning of the Kinect-artefact's real qualities into human media formats. There are very few entities with which Kinect-artefacts have a stronger connection.

In distinguishing between the Kinect-brand and the Kinect-artefact it is revealed that their needs come into conflict with each other. We have learned from this that the tiniest interference in the relation between human and Kinect-artefact can be catastrophic for the Kinect-brand. The tendency of the microphone of the Kinect-artefact to respond better to higher voices is immediately cause for anger in a large group of male gamers, the market which the Kinect-brand craves access to so ardently. In conceiving of a machine ethics this distinction would be relevant. As noted in the introduction to this thesis, desire is necessary for an object to make ethical decisions. There must be an opportunity for an object to decide between an action that is "right" and an action that is in its own self-interest. In the case of the Kinect-brand, if we accept that the object has something similar to desire, then we could say that its self-interested desire to insinuate Kinect-artefacts into gamers' homes comes into conflict with what is seen by many as the "right" things to do, such as ensuring that human voices are equally easy to understand regardless of pitch or accent. As for the Kinect-artefact, its desire to listen all the time for the words "Xbox on" may come into conflict with its user's right to privacy.

Death of the Kinect

The "death" of a non-living object can only be discussed when using a rather playful (if morbid) anthropomorphic eye. Clearly things that are not alive cannot die, but we use the word often to describe a wide range of non-living objects. In the novel *Slaughterhouse Five*, Kurt Vonnegut brings it to the reader's attention by commenting "So it goes." after each mention of death.[112] This statement emphasises the death of humans and also the ways that

[112] Kurt Vonnegut, *Slaughterhouse Five* (New York: Bantam Doubleday Dell Publishing Group, 1998), originally published 1969.

we connote death in the English-speaking world, such as by describing the death of the novel. This section addresses the death of both the Kinect-artefact and the Kinect-brand, phenomena that force us to ask how much an object needs to change before it is no longer the same object.

Not all Kinect-artefact experiences revolve around the human. In no other place is this more evident than in the discarded Kinect-artefact, thrown onto the vast electronic waste repositories that now litter our planet. Electronic waste may be considered a "hyperobject", using Timothy Morton's term.[113] The hyperobject ("hyper" relative to us) is something that has such a powerful influence in either space or time that it denies the subject-object relationship with humans from which we are accustomed to framing the world. Morton is chiefly concerned with the most currently pertinent hyperobject: global warming. But he also describes the universal qualities of hyperobjects. One of Morton's criteria for hyperobjects is that they are "nonlocal"; that is, any immediate, present version of the object is not the object itself. So, the Kinect-artefact that is present in the moment is not the hyperobject. The Kinect-artefact's period of existence will likely far outstrip our own. It has already been brought about through the siphoning of ancient hydrocarbons through an oil rig (literally, not just metaphorically as in Harman's *Circus Philosophicus*) the removal of ore from the earth and the refinement of minerals into glass. Even if these processes do not strictly speaking constitute the life history of the Kinect-artefact (since objects are irreducible), the future existence of the Kinect-artefact is certain to also be highly eventful. We cannot be certain of the future of this and other electronic artefacts, but at present it seems likely that they will be abandoned to decay over thousands of years. For almost all of this time the main interactions for the Kinect-artefact will be with objects other than individual humans. If we consider time as a continuum then the Kinect-artefact constitutes a field that stretches over this long period. Any human contact occupies a tiny portion of this continuum, and much of the object is very distant.

As has also been mentioned, an even more abstracted "death" of the Kinect-brand is said to be occurring at the present moment. The Kinect-artefact's decay and redistribution of objects in an unhuman world will likely be agonisingly slow (from a human perspective), but the Kinect-brand can be injured quite easily. The Kinect-brand is dying because it is no longer being sold with each Xbox One (and may even require a special adaptor),[114] which means that fewer games will be made and fewer people will talk about the brand online.[115] At 2014's E3 press conference, Microsoft made only fleeting reference to the sensor. Online forums and websites almost universally agree that the Kinect-brand is dying. But what exactly is meant by the "death" when it applies to a brand?

---

[113] Morton, *Hyperobjects*.

[114] Jordan Sirani, "Microsoft Won't Offer Free Kinect Adaptor With Xbox One X Upgrade," *IGN* (October 10 2017). http://au.ign.com/articles/2017/10/10/microsoft-wont-offer-free-kinect-adapter-with-xbox-one-x-upgrade (accessed December 7 2017).

[115] Although newer games have been noted for their innovative use of the Kinect-artefact – see games such as:
Fru, Xbox One.
and
Commander Cherry's Puzzled Adventure, developed and published by Grande Games (2015), Xbox One.

Possibly the brand dies when games stop being made and the mainstream media is no longer interested in it.[116] Now that not all, and perhaps the majority of, human users will interact with an Xbox One without a Kinect-artefact, developers have reverted to making relatively few games for use with the Kinect-artefact. This may cause the abandonment of the Kinect-artefact for Xbox One by Microsoft. But there is also the Kinect-artefact for Microsoft Windows, which may see more positive results. In this case, the brand specific to the console may die but its structure, sensors and software may live on. Or perhaps the death of the brand is no longer in the hands of the corporation. The flourishing hobby of retrogaming or old-school gaming attests to this possibility. Obsolete or discontinued interfaces may have a rebirth in this context, in which the collection and enjoyment of classic and unusual gaming objects is paramount. Objects can live on in this form until they are ready to begin their gradual physical decay on one of humanity's electronic waste dumps. But can the brand be conserved in this way as well? The Kinect-brand is less a corporate trademark and more an idea in a broader sense, which in this case does not need to die when it ceases to be supported by Microsoft, much in the same way that *Pac-Man* or the Nintendo Game Boy are not dead ideas. As long as there is a community of Kinect-artefact users the Kinect-brand may be said to have survived.

This chapter attempted to provide a thorough discussion of the phenomenal world of the Kinect. The attempt was made difficult by the experiment of dividing the artefact into two distinct objects: the Kinect-artefact and the Kinect-brand. If there are two different objects, then there are two different sensory worlds which are very different in nature, and each requires a different methodological approach. Sensation for a machine is immediately visible to a human, and was the main object discussed in this chapter. We have witnessed the struggle of the human user to find its phenomenal boundaries, which given its collective and collaborative nature is probably one of the best ways of asking the Kinect-artefact about its experience. What is the ethical consequence of this investigation? The alien world of the Kinect-artefact is far removed from those of humans and higher animals, those beings that rank so highly in our collective estimation. Should we be concerned for the well-being of the Kinect-artefact (or even the Kinect-brand), perhaps carefully maintaining working units as we would conserve an endangered species; or make sure to always perform a calibration before playing? Clearly the possession of "eyes", "ears" and a "brain" are not sufficient to incite a user to care for their electronic device in such a way. The death of the Kinect-artefact or Kinect-brand can hardly be considered cruel. If humans do not care about these things, then why bother with alien phenomenology at all?

At the start of the thesis I asked to what extent the sensation of anthropomorphic machines is relevant to their cultural role, and alien phenomenology emphasises the sensation of

[116] The same is sometimes said of humans: that we die twice. Once when our body dies, and secondly when there is no-one to remember us. The idea is discussed in Irvin D. Yalom, "Three unopened letters," *Love's Executioner and Other Tales of Psychotherapy* (New York: Basic Books, 1989). Yet we would never say that a human body is still alive because we can remember it. Memories are ideas, and ideas are objects in their own right, not just sensual objects formed between human bodies.

nonhumans. In the case of the Kinect-artefact there are certainly practical reasons why we should care about the machine's sensation of the world. We should strive to understand any object that threatens our privacy. And we should study emerging technologies thoroughly before they become integral to our lives. But the real advantage of alien phenomenology is its capacity to appreciate objects like the Kinect-artefact and the Kinect-brand for their unusual way of being in the world, so different from our own. We know that the approach is of interest, because the users of Kinect-artefacts performed their own informal alien phenomenology in a collaborative online attempt to form stronger relations with their consoles. It was an analysis that Microsoft, with its talk of brains and eyes, could not perform without damaging the potential of the Kinect-brand. A far greater potential for human-avatar-console is afforded by the Kinect-artefact. The seeming independence of the Kinect-brand from the Kinect-artefact is remarkable, even for videogames hardware. In conclusion, the Kinect-artefact and Kinect-brand are remarkably weird objects and a useful model for charting the distribution of human and machine agency. This chapter attempted to portray two complex and contentious objects under a deceptively simple umbrella term: *Kinect*.

Chapter 3: Grey Walter's Tortoises and an Introduction to the Perspective of Robots

> "Building machines included a simultaneous building of milieus for the machines."
> – Jussi Parikka, *Insect Media*, Posthumanities (Minneapolis: University of Minnesota Press, 2010), 134. On Walter's robot tortoises.

This thesis will continue to explore alien phenomenology through case studies of several more anthropomorphic machines. The next three chapters will look at three more case studies, each anthropomorphic in different ways. This chapter aims to present an introduction to studying the perspectives of robots through a particular focus on the primitive robot tortoises, *Machina speculatrix*, created by W. Grey Walter in the late 1940s and early 1950s. Chapter 4 discusses postphenomenology in robots with reference to the virtual robot SHRDLU, created by Terry Winograd in the late 1960s and early 1970s. Chapter 5 synthesises the disparate posthuman and nonhuman ideas encountered in this thesis in a study of the concept of gynoid sexbots, particularly aiming to explore them with relation to OOO. The ultimate goal of this latter part of the thesis is to provide a rigorous critique of what has already been written about the perspective and experience of robots in post- and nonhuman discourses and to extend these ideas into other empirical applications of OOO.

Robots defy neat classification, with only the qualities of automated movement and human design being universal to all. Robots vary enormously. Like the Kinect-artefact, some are designed to be more specialised, and all their sensory systems are honed to a particular task (or so it seems to a human user). Some can perform a variety of tasks with a multitude of sensory systems that are in use at different times, as is the case with Voyager 2. Obviously, sometimes, their degree of anthropomorphism is far greater even than the Kinect-artefact, which as we have seen is often described in terms of its "sight" and "hearing". The social robots discussed in Chapter 5 are an example of more exaggerated anthropomorphism. The robot that assembles cars is more difficult to compare to a human, despite possessing "arms" and sensory capabilities such as proprioception. In each case the metaphor of human-like qualities and behaviours is impressed upon the human mind. These chapters will explore all these varieties and problematise these notions of fixedness and anthropomorphism. This chapter focusses on Walter's robot tortoises, a simple example in order to address complex ideas. The robot tortoises also act as a juxtaposition to the anthropomorphic language used,

emphasising those alien qualities that are often ignored. This chapter will consider how we can speak of sensation and experience in robots, given that our understanding of these things is rooted in human experience. I posed three research questions in the introduction to this thesis, and one most relevant to this chapter is “What does sensation mean for machines?”

Given the vast variety of robots, what are we saying when we ask about the experience of robots? We must acknowledge the huge range of functionalities and experiences possible for robots. It is rather like studying yeast and the hummingbird separately as distinct species with distinct experiences while holding to the idea that both are alive. Performing alien phenomenology means learning about the experience of individual objects. This chapter presents and critiques two different ideas with reference to the robot tortoises. The first, Umwelt-theory, privileges the importance of analytical investigation of an entity’s worldview; the other, theories of mind, emphasise the heterogeneity of mind or consciousness. They are both ways of questioning the experience of an entity, but Umwelt-theory begins with the entity itself and builds up the concept of experience from the ground, while theories of mind begin with the question of experience before asking about the mechanics of an individual entity’s way of knowing the world. Both ideas are relevant to anthropomorphic machines, but they are used here to contrast each other and provide extremes of robot analysis.

When we attempt to theorise robot phenomenology we are discussing a range of experiences, but the language in which our description is couched is ultimately motivated by the particular ontology of the robot used. Having endeavoured to illustrate the phenomenological consequences of OOO in the previous chapter, this chapter will make available alternative theories of nonhumans and technology in particular. The domain of robots presents a varied terrain in which light might be shed by the use of the different ideas, much as the ethologist of hummingbirds must become acquainted with a different set of ideas to understand the behaviour of yeasts. These chapters will present a range of robots, from the sleekly designed modern social robots to the primitive, hand-made robots of W. Grey Walter. The focus of these chapters will primarily an exploration and critique of theories of the nonhuman other than OOO, albeit with a OOO influence. The ultimate goal of the chapters is to present a review of the insights obtainable from observing the intersections and departures of prevailing currents in the post- and nonhuman turns in the humanities.

Umwelten and cybernetics

The comparison of animal with robot, if done with caution, may aid us in discussing their particular sensation and experience of the world, and alien sensation more generally. The work of Baltic ethologist Jakob von Uexküll (1864-1944) is a case in point. Uexküll’s Umwelt-

theory is heavily influenced by Kant,[1] and Uexküll was certainly no anti-correlationist. The Umwelt allows us to see the link between the subject and the environment as composed of codes that intertwine to determine each other. Uexküll's work, which is part zoological and part philosophical, was for many decades obscure, but pieces such as the monograph *A Stroll Through the Worlds of Animals and Humans* (1934) are now seen to communicate the beginnings of complex thought about the world of animals, particularly those that do not share our types of sensory equipment (at least in modern Western philosophy). The key to this is Uexküll's emphasis on sensory equipment as constituting animal perception, although Uexküll avoided anthropomorphic concepts when speaking of the Umwelten of other species.[2] Uexküll's work has also been discussed in Jussi Parikka's 2010 book *Insect Media* and Uexküll's curious way of accounting for his observations has begun to enter the vernacular of post- and nonhuman thought. Uexküll has also been referred to in descriptions of animal affect written by Heidegger, Deleuze and Guattari, Canguilhem, and Merleau-Ponty. Umwelt-theory's identification of the body as the site of subjectivity predates Merleau-Ponty. Deleuze and Guattari's interpretation of Uexküll highlights the semiotic component of this body-centred subjectivity.[3]

As an inquisitive scholar of biology and of philosophy, Uexküll's writing is often explicitly preoccupied with the major philosophy of biology questions of his time. Uexküll was not a Darwinist, and he did not adhere to a mechanistic view of living things. This has led some to cast him as a vitalist.[4] But as Kaveli Kull points out, Uexküll considered himself to be something in between these two extremes, a "machinalist".[5] It is important to bear this point in mind, because it is indeed tempting, when encountering Uexküll's views on the status of non-living things, to argue that some sort of vitalist philosophy must have been present. It is also, in some ways, surprising to note that he did not hold a mechanistic view of life given that some of his ideas appear to be precursors to cybernetic ideas. But the emphasis in his texts is on the semiotic relationship between subject and environment rather than being confined to how effect follows cause.[6] Uexküll's main interest was in how living things sense the world, and how to create a language for talking about those sensory experiences. The concept for which Uexküll is best-known is the Umwelt (sometimes translated as "environment" or "milieu", but better equated here to "subjective, meaningful world"). Umwelten are comprised of all the elements of an environment which the living thing is capable of sensing; it is the "phenomenal world or self-world of the animal".[7] Uexküll's writing is replete with musical analogies that capture both the essence of his subjective model for biology and his conception of a "plan" in nature that is responsible for the

[1] Carlo Brentari, "Chapter 2: The life and education of Jakob von Uexküll," in *Jakob von Uexküll: The Discovery of the Umwelt between Biosemiotics and Theoretical Biology*, Biosemiotics, ed. Carlo Brentari (Dordrecht: Springer, 2015), 23.
[2] Ibid., Morten Tønnessen, "Introduction: the Relevance of Uexküll's Umwelt Theory Today," 11.
[3] Ibid., Brentari, "Chapter 7: Influences and Interpretations of the Work of Uexküll," 175-231.
[4] Ibid., Brentari, "Chapter 3: The basis of environmental theory," 47-74.
[5] Kaveli Kull, "Jakob von Uexküll: an Introduction," *Semiotica* 134, no. 1-4 (2001), 6.
[6] Søren Brier, "Cybersemiotics and *Umweltehre*," *Semiotica* 134, no. 1-4 (2001): 779-814.
[7] Jakob von Uexküll, "A Stroll Through the World of Animals and Humans: a Picture Book of Invisible Worlds," *Semiotica* 89, no. 4 (1992), 319.

seemingly perfect harmony between the Umwelten of different organisms.[8] The Umwelt concept, in the form of the science *Umweltforschung* (Umwelt-research), was developed by Uexküll over the course of his life and has been revived by his son Thure von Uexküll, Thomas Sebeok and others, particularly in his native Estonia but also more widely.[9] It is a major contributor to the founding of the field of biosemiotics. Since Uexküll's work was largely lost to the non-German-speaking world for several decades it is only in more recent years that its importance has been identified and his position as a founder of the field of biosemiotics, particularly after Thomas Sebeok and Jean Umiker-Sebeok's 1992 publication *The Semiotic Web 1991*.[10] Biosemiotics is, broadly, the use of sign theory to describe biological phenomena, predominantly that of animals (zoosemiotics) but increasingly also that of plants (phytosemiotics) and other organisms.[11] Jesper Hoffmeyer even synthesises Umwelt-theory and the evolution of the "virtual" genetic code.[12] It is through biosemiotics that Uexküll has become more well-known.

The Umwelt is the world of experience for any given animal, containing things that can be sensed and excluding things that cannot be sensed. Everything that can be sensed enters the Umwelt as a sign, and the Umwelt contains the mechanism through which the sign is sensed, perceived, and responded to. It is the perceiver-effector of the animal's experiential world and it is described in semiotic language. This might mean that the Umwelt relates only to the parts of the light spectrum that the animal can perceive, or only to certain chemicals that the animal is capable of perceiving with its olfactory sensors. The model organism used by Uexküll in his books *Umwelt and Innenwelt of the Tick*[13]and *A Stroll Through the Worlds of Animals and Humans*[14] is the female tick during its penultimate life stage, in which its experiences are truly alien and can assist us to dissociate from our anthropocentric view. The tick suspends itself from a leaf or blade of grass, waiting, sometimes for years, for the only stimulus it is honed to sense and respond to: the butyric acid released by mammals. When it senses this chemical, the tick drops with the hope of landing on the mammal's body. This accomplished, the tick utilises its two other senses, permitting the sensation of temperature and hair typology, to find a suitable place to pierce the skin and feed on the mammal's blood. Once it has drunk its fill, it lays its eggs and dies. These three senses represent the totality of the tick's universe at that stage in its life cycle. The three senses form the tick's Umwelt, or the semiotic "bubble",[15] which is the tick's entire world. Two different organisms of different species may share the same environment, but they do not share the same Umwelt. The

---

[8] Jakob von Uexküll, *Theoretical Biology*, trans. D. L. MacKinnon (Edinburgh: The Edinburgh Press, 1926).
[9] Thure von Uexküll, "Introduction: the Sign Theory of Jakob von Uexküll," *Semiotica* 89, no. 4 (1992): 279-315.
and
Thomas A. Sebeok, "Biosemiotics: its Roots, Proliferation, and Prospects," *Semiotica* 134, no. 1-4 (2001): 61-78.
[10] Thomas A. Sebeok and J. Umiker-Sebeok, *The Semiotic Web 1991* (Berlin: Mouton de Gruyter, 1992).
[11] Sebeok, "Biosemiotics."
[12] Jesper Hoffmeyer, "Seeing Virtuality in Nature," *Semiotica* 134, no. 1-4 (2001): 381-398.
[13] Jakob von Uexküll, *Umwelt and Innenwelt of the Tick* (Berlin: Verlag von Julius Springer, 1909).
[14] Uexküll, "A Stroll Through the World of Animals and Humans."
[15] Ibid., 319.

Umwelt concept does not require us to find analogies to compare the tick's experience with our own. For example, it is tempting to describe the tick's "smelling" of the approaching mammal, an inaccurate anthropomorphism that equates the use of the arachnid's alien olfactory organs to our own sense of smell which is structurally different and evolved separately.[16] This side-steps the problematic impulse that may be found in much of the literature around OOO: the very great temptation to go too far in our description of nonhuman worlds and thus to humanise them and rob them of their great power of alienation. (This is an alternative to the solution that this thesis has suggested, which is to use anthropomorphic metaphors frequently in order to increase awareness of the alien qualities of technological artefacts.) The power of alienation in biosemiotics is increased when we look at the vast variety of Umwelten that exist in the animal kingdom (and possibly beyond) and compare them to the Umwelten of humans. The human Umwelt is inextricably linked to the things that help us to survive. We cannot see very small things or things that are very far away, we can only detect a small range of chemicals with our olfactory and gustatory systems, and we only perceive certain, very specific, types of radiation. Sensation is like a "garment"[17] that wraps around us, revealing only certain parts of all possible experiences of the world. If we were to apply this idea to technological artefacts as well as living things, we might see a similarity between the limitations of the tick's Umwelt and that of the Kinect-artefact when it is in its instant-on state. As far as we know, the Kinect-artefact's Umwelt at that time is geared towards the sensation of sounds, allowing the Kinect-artefact to identify the "Xbox on" signifier that will prompt it to switch on the Xbox One (see Chapter 2).

Uexküll, in his many surviving texts, does relate the Umwelt concept to technology on more than one occasion. But living and writing in the early twentieth century, his idea of machines "meant clocks, factories and blindly repeated processes"[18] and his personal philosophical statements betray mistrust of the idea that living things could operate solely according to the same principles as non-living things. Again, he is sometimes accused of vitalism for this reason. A machine, such as a telescope, could certainly be used as an extension to a human's Umwelt,[19] but the telescope itself does not have an Umwelt according to Uexküll. (As discussed in Chapter 4, Don Ihde calls this an embodiment relation – the use of a telescope to extend the sensory capabilities of a human.) More recent writers have taken the Umwelt concept to a place Uexküll could not reach, with the aid of technologies that perhaps do more to spark the phenomenological imagination. We now have machines that are capable of activities that resemble perception to a far greater extent than the telescope. As mentioned above, humans lack certain elements in their Umwelten. But we now have visual technologies that transcend scale and distance by transposing the sensation of a machine with something that we can interpret, such as the translation of electrical signals into visual or audio media (what Ihde would call a hermeneutic relation). Machines can "smell" the environment and relate their sensory experience of a particular chemical in numerical form so that we can

[16] Tessa Leach, "Improving on Eyes: Human-Machine Interaction in Cyborg Vision" (Unpublished thesis, 2012).
[17] Jakob von Uexküll, "An Introduction to Umwelt," *Semiotica* 134, no. 1-4 (2001), 107.
[18] Parikka, *Insect Media*, 68.
[19] Uexküll, " A Stroll Through the World of Animals and Humans," 389-390.

understand it. Individual machines can pick up far more types of radiation than humans. And Voyager 2 can signal to Earth information about the ice giants of our solar system without our needing to go there ourselves. The anthropomorphic tendency in Uexküll's Umwelt-theory is particularly noticeable when robots are the objects of study, because they so clearly exhibit sensory faculties that at least appear to be similar to those of animals. Claus Emmeche has asked the question "Does a robot have an Umwelt?",[20] because even primitive robots have "simple functional circles of semiotic processes of sign-interpretation and sign-action. Why should they not have Umwelten?"[21]

The Umwelt as applied to technology bears some similarities to the concept of affordance. In James J. Gibson's book *The Ecological Approach to Visual Perception*, the affordances of an animal's environment and how they can be changed to be more suitable is discussed.[22] The affordances of the environment are the aspects of the environment that are relevant to the animal. For humans, if a surface is "horizontal, flat, extended, and rigid" then it affords support (as well as being "fall-off-able").[23] Affordance is a different concept to the Umwelt, because affordance implies not just the availability of objects to be sensed, but the perception of usability and adaptability by a human or nonhuman: "[Affordance] is both physical and psychical, yet neither."[24] Umwelten do not encompass the value of the object for the human or nonhuman, but only contain the representation of sensory information. Don Norman extends the concept of affordances to the design of objects by humans, arguing that we adapt objects to offer new or different affordances in particular contexts.[25] A large part of the designer's efforts must go into signifying the presence of features in designed objects,[26] because features can exist without being perceived by the (human) user, and only become affordances if the user is compelled through context, knowledge and capability to use the feature.[27]

Affordances are apparent in all different kinds of humans and nonhumans, and the scope of research into objects is wider than with the Umwelt. But because the affordance involves the perception of value of the object, there is an assumption that the human or nonhuman must have some capacity for experiencing its environment (Norman extends this capacity to humans, animals, and robots[28]). In contrast, the Umwelt invites the question of whether or not sensation must be tied to experience. It is not clear whether all nonhumans *experience* their sensory worlds in the same way, merely that there *are* sensory worlds. That is, the

---

[20] Claus Emmeche, "Does a Robot Have an Umwelt? Reflections on the Qualitative Biosemiotics in Jakob von Uexküll," *Semiotica* 134, no. 1-4 (2001): 653-93.
[21] ibid., 671.
[22] James J. Gibson, "The Theory of Affordances," in *The Ecological Approach to Visual Perception* (New York: Psychology Press, Taylor and Francis Group, 2015), originally published 1979.
[23] Ibid., 120.
[24] Ibid., 121.
[25] Donald Norman, *The Design of Everyday Things: Revised and Extended Edition* (New York: Basic Books, 2013), originally published 1988.
[26] Ibid., 13.
[27] Donald Norman, "Affordance, Conventions and Design," *Interactions* 3 (May/June 1999). http://interactions.acm.org/archive/view/may-june-1999/affordance-conventions-and-design1 (accessed March 18 2018).
[28] Norman, *The Design of Everyday Things*, 11.

existence of nonhuman sensation is not dependent on possession of a mind, consciousness, qualia or any kind of experience. Thomas Sebeok argues "there can be no semiosis without interpretability – surely life's cardinal propensity",[29] so does a robot need to be able to interpret its environment to have an Umwelt? As will be seen later in this chapter, the mind of a robot is a highly contestable entity, with the extremes of functionalism and panpsychism resulting in the attribution of vastly different kinds of internal experiences for non-living things, and even living nonhumans. It will likely be some time before we can really answer the question "Does a robot have an Umwelt?", if indeed it ever happens. Then again, perhaps the nature of interpretability is vague enough for us to apply it to nonhumans regardless of whether they possess minds or consciousness. This is why this chapter discusses such simple robots with narrow sensory worlds.

There is a convergence between OOO and Uexküll's research program, though they approach similar conclusions from different roots. OOO also rejects a purely positivist research program in favour of a richer world of relations, and an interest in the diversity of sensory capabilities in objects. In *Theoretical Biology* Uexküll was critical of physics for privileging the objective over the subjective. For him biology was the domain of the subjective:

> Physical theory tries to convince the plain man that the world he sees is full of subjective illusions, and that the only real world is much poorer, since it consists of one vast, perceptual whirl of atoms controlled by causality alone. On the other hand, the biologist tries to make the plain man realise that he sees far too little, and that the real world is much richer than he suspects, because around each living being an appearance-world of its own lies spread, which, in its main features, resembles his world, but nevertheless displays so much variation therefrom that he may dedicate his whole life to the study of these other worlds without ever seeing the end of his task.[30]

Uexküll's emphasis on a "richer world" could easily come from a text on OOO. The world of nonhumans is alien, and constituted of alien sensory capabilities. But in OOO subjectivity and perception are less emphasised. Subjectivity and perception both imply the presence of a mind or mind-like quality, and tend to reinforce the subject/object dualism. As will be shown later in the chapter, some OOO thinkers have embraced or at least considered panpsychism. But Harman's body of work does not *require* panpsychism. The notion of experience in biosemiotics is the result of the perception of a sign, not just the change in the nature of the real object. For example, the tick's Umwelt contains only three capacities for sensing the world. If I throw a rock at the tick causing it to fall off its perch, then that does not enter into the tick's Umwelt. But in OOO terms a sensual object is created between the rock and the tick. Not everything that happens to the body of the tick is in the Umwelt, even if it affects the tick significantly. The Umwelt only relates to signs equipped to filter through the tick's sensory organs and then interpreted, even if this occurs in a very alien way, which is why biosemioticians are reluctant to extend the theory to non-living things.

As will be seen in the next section, Walter's robot tortoises are comparable to a tick due to the apparent poverty of their sensory worlds. They possess sensors that allow them to detect

[29] Sebeok "Biosemiotics," 68.
[30] Uexküll, *Theoretical Biology*, 71.

light and touch, and they have internal sign processes such as the transmission of electricity that is interpreted by components of their bodies, such as the motor or battery. Like the tick, and like humans, they do not pick up everything in their immediate surroundings. If the crux of this problem lay solely in questioning whether a robot may possess an Umwelt due to its human origins and electronic sensory-perceptive systems, it could be dismissed quite quickly. But can we say that the robot perceives and interprets its world, and how would we know? The case in point here is the robot tortoises. After describing the robots' existences and how they worked, this chapter will renew this enquiry into their internal experiences.

Robot tortoises

W. Grey Walter created several "tortoises", two of which were named Elsie and Elmer.[31] Elsie and Elmer were first exhibited to the press at the end of 1949, and more advanced tortoises were built in 1951.[32] Robotics was an extension of Walter's work on electroencephalography (EEG) and the brain, and began as a project in his spare time.[33] Walter was a member of the Ratio Club in London, a pioneering group for cybernetics.[34] Walter gave his new "species" the "mock-biological"[35] name *Machina speculatrix* and observed that from their simple interactions they could develop quite complex behaviour.

Machina speculatrix tortoises recall the tradition of building ceremonial or amusing automata. The building of automata was once a common practice both in Europe and parts of Asia. Automata are mechanical toys, powered by clockwork, that commonly move in a repetitive way, and are very often built in human shapes or the shapes of animals. Jacques de Vaucanson's gold-plated duck is a famous example.[36] There was a minor craze for automata in the sixteenth century amongst the aristocracy in parts of Europe. But interestingly the practice was curtailed by church authorities, who eventually came to believe that the automata represented a heretical attempt to replicate the work of God, namely the giving of life to a non-living thing. This is an oft-mentioned comparison: in his analysis of Frankenstein, Bruce Mazlish declares that "[m]an, the evil scientist, has taken God's place."[37] Lois Kuznets argues that this restriction may have been a manifestation of the engrained Judeo-Christian taboo against the production of graven images.[38] Jessica Riskin describes an episode of

[31] William Grey Walter, "An Imitation of Life," *Scientific American* 182, no. 5 (1950).
[32] Owen Holland, "The First Biologically Inspired Robots," *Robotica* 21 (2003): 351-63.
[33] Andrew Pickering, *The Cybernetic Brain: Sketches of Another Future* (Chicago: University of Chicago Press, 2010), 43.
[34] Holland, "The First Biologically Inspired Robots," 354.
[35] Walter, "An Imitation of Life," 43.
[36] Sidney Perkowitz, *Digital People: From Bionic Humans to Androids* (Washington D.C.: Joseph Henry Press, 2004), 57.
[37] Mazlish, "Automata," in *The Fourth Discontinuity,* 45.
[38] Lois Kuznets, "Life(size) Endowments: Monsters, Automata, Robots," in *When Toys Come Alive: Narratives of Animation, Metamorphosis, and Development* (New Haven: Yale University Press, 1994), 189.

iconoclasm in which medieval devotional automata were destroyed, the iconoclasts claiming that "[m]achinery [...] could not represent divinity other than deceitfully."[39] In Japan, this practice was not as restricted, however. The building of automata was and is a job reserved for the most respected masters of craft. *Karakuri*, as they are known, play a prominent role in Japanese culture, notably participating in festivals and other events. Possibly the traditional animistic nature of Japanese culture and religion means that *karakuri* do not seem as heretical as they did in Europe. Objects of great spiritual importance, such as a sword, hold a soul or *kami*, so *karakuri* and humanoid robots are not the only non-living objects with a soul.[40] (This will be important later in the chapter.) Walter followed in this tradition when he created these fancifully-named and spectacular robots, although his creations possessed sensory capabilities much more familiar to humans.

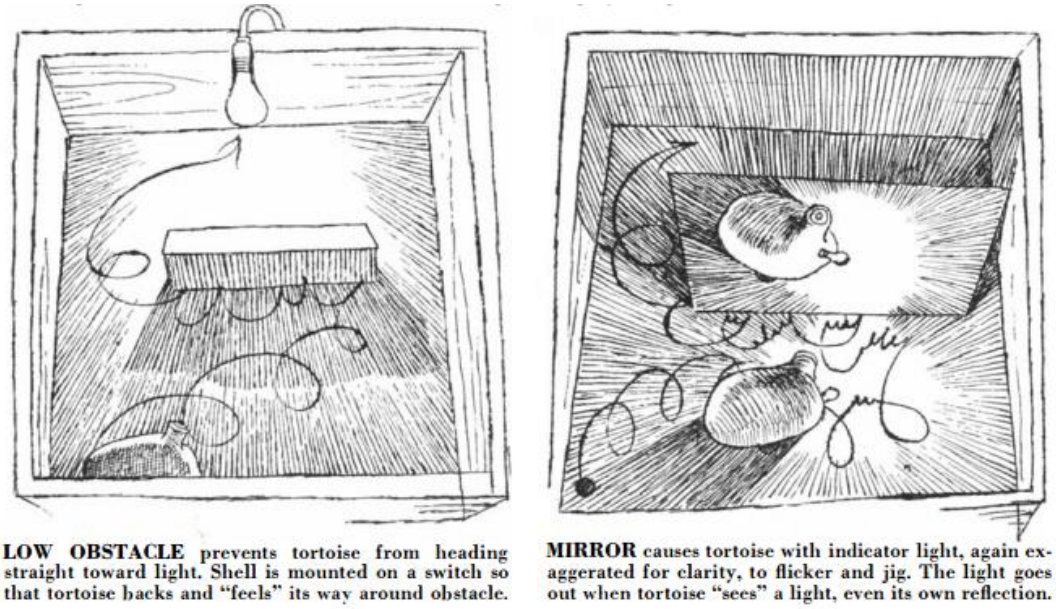
However, Walter's tortoises constituted a significant leap forward in that they were capable of semi-autonomous movement and apparently quite complex behaviour. They were driven by an electric motor. A tortoise would wander around until it hit something, at which point a contact switch would be activated and it would move away. This was intended to replicate a negative feedback loop. There was also a photocell on the front of the machine that would cause it to move towards light, including their lit-up hutches where their batteries would recharge. This executed a positive feedback loop. However, at a certain level of brightness the tortoises would move away from the light, so that they would get close to the light but not too close. As the battery depleted, the sensitivity to the light source became greater and the tortoises could detect the hutch from farther away, but if the battery was completely charged the tortoises would move away from the hutch and explore other areas. The tortoises could therefore return to their illuminated hutches when it was time for them to recharge, feeling their way around obstacles in their paths, and making them leave the hutch and explore when the battery was charged. Walter restricted Elsie and Elmer to two feedback loops to "discover what degree of complexity of behaviour and independence could be achieved with the smallest number of possible interconnections."[41] Elsie and Elmer's Umwelten, if they can be said to possess such things, consist of light and touch at the very least. In a OOO sense, there are other elements to the tortoise's world as well – Elsie's alien sensation may also extend to the texture of the floor under her wheels and changes in temperature that affect her metal parts. But it is particularly important to note that although Elsie's Umwelt may contain light, it is not the same as the light in human Umwelten. Elsie's light is constructed by the material of her light sensor.

---

[39] Riskin, "Machines in the Garden."

[40] Simons, *Robots*, 29.
Although the possession of a soul by *karakuri* raises the interesting question of whether biosemiotics would look different in a non-European context. To my knowledge, no work has been done to investigate the Umwelten of *karakuri*, or in relation to *kami* in general.

[41] Walter, "An Imitation of Life," 44.



LOW OBSTACLE prevents tortoise from heading straight toward light. Shell is mounted on a switch so that tortoise backs and "feels" its way around obstacle.

MIRROR causes tortoise with indicator light, again exaggerated for clarity, to flicker and jig. The light goes out when tortoise "sees" a light, even its own reflection.

Figure 17: Image from W. Grey Walter's 1950 article in *Scientific American*, describing two of his observations of the tortoises.[42]

In the above image from Walter's 1950 article on the tortoises, these feedback loops are both evident in the tortoise's movement around a "low obstacle" in an effort to move towards a light source. In the absence of a light source, the tortoises would "explore" their environment.[43] The tortoises also had small lights which were originally placed to indicate whether or not the motor was running. So when the tortoise was moving, it emitted light, but did not emit light when stationary. The effect was such that when a mirror was placed in front of a tortoise, it would seem to gaze at its own light in the reflection and make small movements back and forth "like a clumsy Narcissus."[44] In a recent article about the tortoises, Esther Inglis-Arkell comments that they "they loved, above all things, women's legs."[45] Women often wore nylon stockings which had a low sheen, to which the tortoises were attracted. To watch the BBC newsreel[46] of *M. speculatrix* is to be astonished that such simple rules could result in what appears to be complex behaviour (although it is likely that the tortoises were manipulated and made to produce more precise movements for the purposes of the film).[47]

Walter and others were fascinated by the functional anthropomorphism they could engineer in quite simple machines. In Walter's writing about the tortoises it is clear that he believed

---

[42] Ibid.
[43] Ibid.
[44] W. Grey Walter, "Totems, Toys and Tools," in *The Living Brain* (London: Gerald Duckworth & Co. Ltd., 1953), 84.
[45] Esther Inglis-Arkell, "The Very First Robot "Brains" Were Made of Old Alarm Clocks," *io9* (2012). http://io9.com/5890771/the-very-first-robot-brains-were-made-of-old-alarm-clocks (accessed September 25 2013).
[46] *Grey Walter's Tortoises* (2008), Youtube. https://www.youtube.com/watch?v=lLULRlmXkKo (accessed June 27 2017).
[47] Holland, "The First Biologically Inspired Robots."

significant insights could be gained by observing these artificial beings. *M. speculatrix* was not built to perceive light and touch as a human would. But there was a set of impulses reminiscent of life, like those observable in a creeping vine. A creeping vine senses and responds to light by angling its leaves to catch the most sunshine, and it responds to touch by building structures that wrap and bind. For the cyberneticists, this performative sensory-motor response was the essence of living nervous systems. In the case of the tortoise gazing at itself in the mirror, Walter declared that "were it an animal a biologist would be justified in attributing to it a capacity for self-recognition."[48] Of course, the tortoises were not capable of self-recognition, possessing just one photocell and only two functional elements. But cyberneticists sought to create responses to stimuli, not to recreate organic cognition.

What kinds of experiences can we attribute to these machines? For a few years they occupied a unique position in the boundary conflicts arising between the categories of human and nonhuman brought on by cybernetics. The nature of robot sensation and action were a reaction to and encouragement for more blurring of human/machine boundaries. This has seriously affected the way that we consider their existence. It certainly affected Walter's scientific and popular portrayal of the robots, and may be noted in his description of them exploring, resting and considering themselves in a mirror. With the knowledge that the cyberneticists had limited success in creating an artificial mind in the 1950s and 60s, it is easy to look back at these accounts and smile at Walter's attribution of complex behaviour to such simple robots. But perhaps all such anthropomorphic language is doomed to failure. The robots performed the functions required by Walter to exhibit his thoughts about the mind (although by all accounts they frequently broke down, and were, therefore, disobedient). They did spend periods of time positioned in front of the mirror. They did move erratically around the space as though exploring. But perhaps "experience" is too strong a word to describe what occurs within an object, too dependent on an object's ability to extract meaning from its surroundings. The possibility of meaning is dependent upon the philosophy of mind and technology particular to each culture and each moment in time.

Sadly, relatively little is known of the robots' design. Knowing more about their structure and relations would assist in the use of alien phenomenology, but we must make do with what information has survived. We know that the 1949 batch were pieced together with spare materials: old alarm clocks and war surplus materials.[49] Another resource is Figure 18 from the archives of the Burden Neurological Institute, a rare image of Elsie without her shell.

[48] Walter, "An Imitation of Life," 45.
[49] Inglis-Arkell, "The Very First Robot 'Brains' Were Made of Old Alarm Clocks".

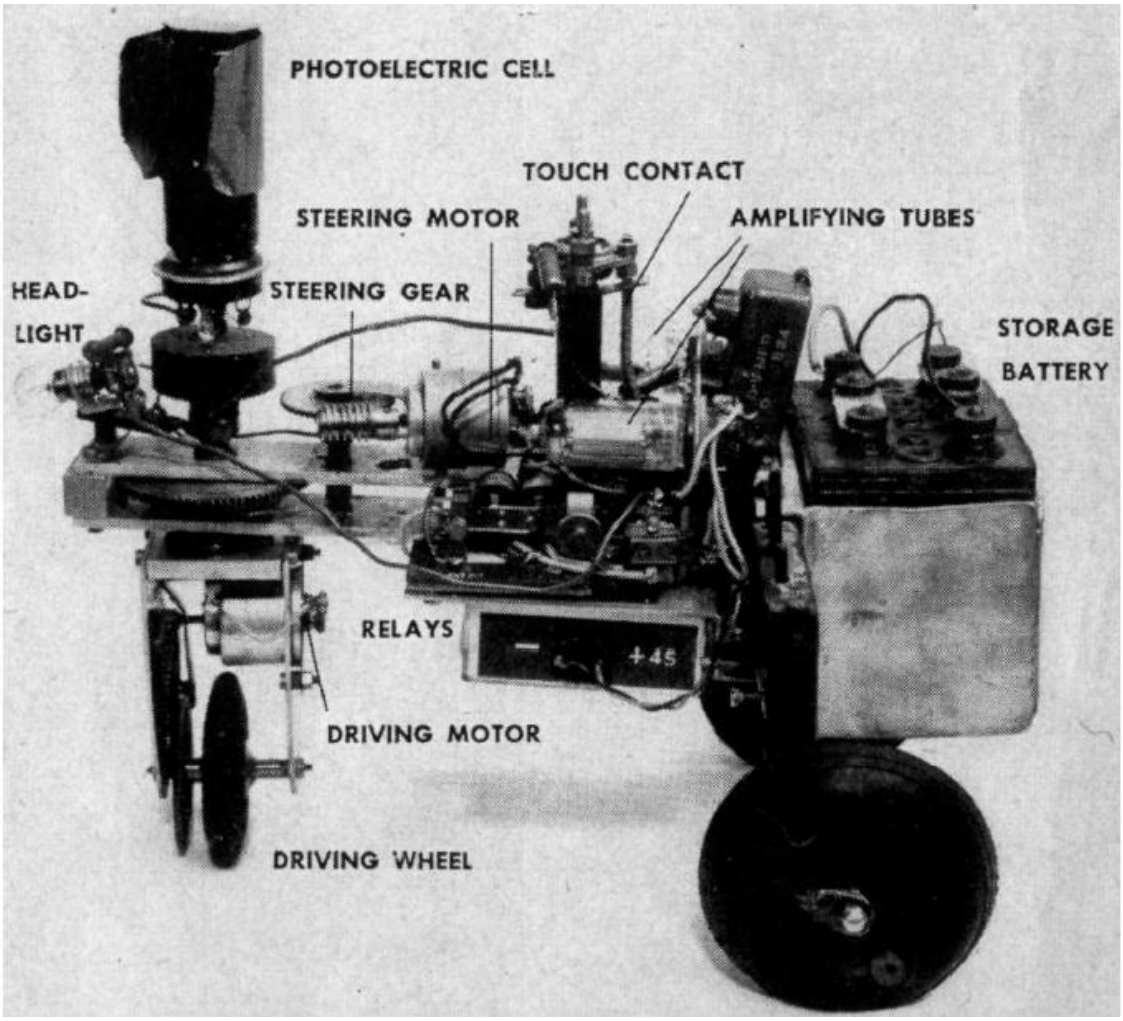


Figure 18: Elsie without her shell. Different components of the robot are indicated with labels.[50]

Another possible resource is the circuit diagram included in Walter's 1953 book *The Living Brain*, but the book is more a guide for how to build one's own tortoise rather than a scientific study of the robots Walter had created.[51] Unfortunately Elsie and Elmer are not thought to have survived, so we cannot be certain of their structure or the relations they might have formed.[52] There are very few images or records of intermediate stages of their construction. This is a serious methodological problem for alien phenomenology. But in an object-oriented description, the robots are not simply identified by their components. Elsie is an object, the battery is an object, the light sensor is an object, the image above of her inner mechanisms is an object. No object is defined by another. As discussed below, we may embrace the flexibility of the name "Elsie" and acknowledge the changeable nature of a robot. Its material structure changes, its qualities change and its relations with other objects changes. Elsie's tendency to break down, capacity for alteration and eventual loss are facets of her Elsieness. And given the sketchy accounts of her short period in the spotlight, these

[50] Holland, "The First Biologically Inspired Robots," 354.
[51] Walter, "APPENDIX B. The Design of *M. Speculatrix*," in *The Living Brain*, 200.
[52] Owen Holland, "The Grey Walter Online Archive," *University of the West of England* (2013). http://www.ias.uwe.ac.uk/Robots/gwonline/gwonline.html (accessed September 25 2013).

might be some of the only aspects of her that are accessible to modern humans, machines and texts.

Tortoises and time

The tortoises are a startling depiction of the presence in time of all objects, no matter their immediate appearance of solidity or impermanence. The weirdness of the tortoises brings these matters to the foreground and makes them a good model for explaining Harman's concept of time in object-oriented philosophy. Firstly, the Elsieness described above smears a changing abstraction across several decades while also maintaining her own integrity. In other words, the tortoises maintain their existence as objects despite constantly changing: from the alteration of component parts to the slow degradation of iron when it comes in contact with oxygen. How can we account metaphysically for these objects remaining themselves while simultaneously becoming something else?[53] For Harman the Elsie with her shell on is still the same object as the Elsie with her shell off, merely being encrusted with outer changes. Yet for Latour the irreducibility of objects means that a new object must arise with every change, even ones as small as the light on the front of the tortoise switching on and off in front of a mirror. As Latour says in *Irreductions*, "[t]ime is the distant consequences of actors as they each seek to create a fait accompli on their own behalf that cannot be reversed […] In this way time passes."[54] Latour's actants change constantly, shifting allegiances and switching relations. Actants do not lose or gain extra distinguishing characteristics over time, rather they become different actants as they change.[55] For Latour, Elsie with her shell on is one actant, Elsie with her shell off is another, albeit closely related, actant.

In *The Quadruple Object* Harman offers an alternative perspective. His inspiration comes from Husserl's phenomenology (with several significant caveats). In Harman's restating of Husserl, the stability of objects over time is depicted as the solid core of the object surrounded by a changing flux of "adumbrations".[56] An adumbration is a particular perspective of an object at a given time, such as the view of a water tower as one walks around it. As discussed in Chapter 1, objects are withdrawn from relations and it is sensual objects that facilitate interaction between objects. So when we gaze at a water tower, we are experiencing the

---

[53] The problem of identity and change is an ancient one often framed in terms of the Ship of Theseus. Is the ship the same object after significant changes are made to it? Only Harman's solution to this problem is presented here, but a summary of the history of the problem may be found in:
Roderick M. Chisholm, "Identity Through Time," in *Person and Object: A Metaphysical Study* (Milton Park, UK: Taylor & Francis, 2013), originally published 1976, 88-112.
It should be noted that in Chisholm's history Western scholars are the main players in the history of the problem.

[54] Bruno Latour, "From Weakness to Potency," in *Irreductions,* Part 2 of *The Pasteurization of France*, trans. Alan Sheridan and John Law (Cambridge, MA: Harvard University Press, 1988), 165.

[55] Harman, *Prince of Networks*, "Irreductions."

[56] Harman, *The Quadruple Object*, 24.

sensual object that is formed between us and the water tower. If we walk around the water tower later in the day in a different mood, it will appear differently to us. But we are unlikely to mistake it for a different object. The sensual object formed between us and the tower remains the same even though its sensual qualities change. We can identify an object despite the "gems, glitter and confetti of extraneous detail".[57] The tension between the sensual object and its encrustation of shifting qualities is one of Harman's main lessons from Husserl, and he labels this tension "time". While a sensual object remains the same for us, we encounter sensory qualities as "intermittent changes."[58] Unlike for Latour, real objects and sensual objects remain the same over time, but the encrustations shift and change. In Harman's theory, there is an inner Elsieness that remains whether her shell is on or off.

Similarly to the distinction in the previous chapter between artefact and brand, we must make a distinction here between the relations formed by Elsie in the moment and our relationship with Elsie so many decades later. Our typical relationship with the tortoises can only be by proxy. Their motion is recorded in human media such as in newsreel footage and Walter's sketched drawings in *Scientific American*, as well as finding a place within the brains of those people who witnessed them first- or second-hand. Personally, I have had no direct relations with the tortoises. I have seen images and read texts about them, including texts by Walter himself. But it is the images and texts with which I have a relationship, not the tortoises. The tortoises only exist in relationship with objects when that relationship is bidirectional. When we read a paper on the tortoises, there is no bidirectional relationship with the tortoises, only with the text. There is a paperish mediation. There are the long-exposure photographs of burning candles positioned on the robots' shells; the effect on the cybernetic community of the early 1950s; and the renewed academic and cultural interest in these and other early robots: these are all consequences of the tortoises' existence, but they are not the same as Walter's observation of the tortoise nor of the adhesion of the candle to the shell. The tortoises' worlds were characterised by alien interactions. Many nonhuman objects were engaged with the robots. The wheels of the robots were in a relationship with the floor beneath them, and the kind of interaction that occurred was doubtless different depending on the particular surface. They were subject to the earth's gravitational pull, the slow oxidation of iron and the qualities of different replacement parts. Brenton Malin points to a similar example:

> For object-oriented thinkers, sidewalks are not merely signifiers of certain social relations. They interact with shoe rubber, dog paws, bicycle tires, and other objects in – quite literally – concrete ways that give them an important agency and power.[59]

Perhaps a *Machina speculatrix* shell is at this moment home to an industrious hermit crab, or perhaps it is at the bottom of a pile of garbage. In these cases, the shell's relations with other objects are not recorded, and do not appear in history books, but figure in different kinds of struggles with ocean currents and gravity unrecorded by humans. These may seem like

[57] Ibid., 25.
[58] Ibid., 103.
[59] Brenton J. Malin, "Communicating with Objects: Ontology, Object-Orientations, and the Politics of Communication," *Communication Theory* 26 (2016): 238.

mundane and trivial considerations, but to fail to point them out would constitute the erasure of the validity of the access of other objects.

In addition to the actions of other agents in the early 1950s it is important to speculate about the robots' enduring experiences that are relatively distant from human interaction. The tortoises are lost, destroyed or relegated to museums: an unsurprising end to these once remarkable machines that have since been hugely surpassed by more modern inventions. They are representative of the kinds of histories that technological artefacts possess. Contemporary media theory is beginning to express an interest in the experiences of objects beyond their interaction with humans, as was discussed in Chapter 2 with regard to hyperobjects. The hyperobject is "hyper" relative to human lifetimes and individual human agencies. In addition, media archaeological approaches are increasingly preoccupied with what happens to technological artefacts after they are abandoned by humans. Siegfried Zielinski's book *Deep Time of the Media* explores the existence of lost and forgotten moments in the history of media that were only precipitated far in the future.[60] In a lecture, Jussi Parikka reframed this concept to theorise the material history of technological artefacts: from the bringing forth of natural resources to build their components to their eventual decay or repurposing.[61] Since we do not know the fate of several of the robots it would be presumptuous to claim knowledge about their futures, and even the future of the tortoises that have survived is uncertain, although there is a good chance that they will survive, encrusted with changes, longer than we will.

These weird objects have held a certain fascination since the 1950s, representing a key example of cybernetic thinking. More modern writers are interested in the way that the robots changed the scientific project at the time and their long-term status as paradigmatic objects. The robots have endured for humans well beyond their time in the public eye. Yet they are also lost objects, ill-defined from the outset and so frequently invoked that they begin to lose their definition. In their timeless state, and in their inability to be present, they expose the effect of time on objects.

Sensation, experience and consciousness

Let us compare the framing of the tick with that of the robot tortoises provided here. In Uexküll's conception, the tick sits, stationary, possibly for many years, with only three possible means of sensing the world. These senses are the only relations that the tick has with the world. The light of the sun bouncing off its body, the contact between moisture and legs, the movement of the grass in the wind: these do not count, because they do not enter into the tick's Umwelt. As a corollary, we must decide which of the robot's relations qualify as

[60] Siegfried Zielinski, *Deep Time of the Media* (Cambridge, MA: The MIT Press, 2006).
[61] Jussi Parikka, *An Alternative Deep Time of the Media: a Geologically Tuned Media Ecology*, (2013) Vimeo. http://jussiparikka.net/?s=geology+media (accessed August 22 2013).

"senses", which qualify as parts of the Umwelt, and which do not. (Uexküll would strongly object to this, but some more modern writers would not.) Light and touch, surely, qualify. But the contact between wheel and ground is probably more problematic. Who can say which elements are being "sensed" and which are other kinds of relations? Which part of the robot is even doing the sensing or experiencing? Emmeche explains this difficulty by pointing out that there is neither analogy between human and robot nervous systems nor evolutionary homology between the two systems.[62] If the structure is completely different, how can the experience be similar? Winfried Nöth claims that the symbolic representation portion of a robot's experience might correspond to the concept of the Innenwelt, more closely allied to the animal's interior experience or "I".[63] This may present a model for conceptualising artificial intelligence in the language of Uexküll.

We are surrounded by machines that appear to sense things. If we reduce the components of the robotic Umwelt to signs, then the issue may not entirely be resolved. Asking a robot questions about its sensation may require resorting to desperate measures. If a robot tortoise is suddenly doused with water, the water acts as a sign to the robot's electronic components, disrupting its functions and causing it to shut down. If a robot tortoise is exposed to extreme heat, those temperature signs are interpreted by the atoms in the robot's metallic components as an instruction for their atoms to move more quickly, causing an expansion in the metal and eventually melting. If a robot tortoise moves across an uneven floor, it may be jostled. The change in the texture of the floor may be sensed by the wheel which can no longer travel along its original trajectory. The wheel may, in its way, perceive and interpret that sign and respond by changing the position of the wheel, which is made possible by the weight of the robot and the power of the motor. At no point is any kind of decision made, but the robot responds to the conditions in a similar way to the growth of a seed when its cells respond to gravity, some cells moving upward to become the stem and others plunging downwards to become the roots. If biosemiotics can be applied to any living thing, including things without any kind of neural system, then what exactly do we mean by perception? Can we distinguish sign-relations from the universal relations that underpin Harman's theory of objects? Is the robot's experience anything like biological subjectivity? These are questions that relate strongly, but not exclusively, to questions of mind and consciousness.

The Umwelt is conceptually distinct from the mind or consciousness. The mind is a much broader concept.[64] The Umwelt may be related to sensation and perception, but it does not depend upon the entity possessing concepts as elaborate as consciousness, cognition or thought. So in considering the sensation of an object, we encounter a difficult and ancient question: is the mind a quality of all things, or just of certain things? Our human sensation of the world seems to be strongly connected to our minds. So, one might reason that alien sensation also requires some kind of mind or consciousness. Uexküll's tick example, though carefully pared down and limited primarily to the function of the semiotic bubble rather than

[62] Emmeche, "Does a Robot Have an Umwelt?", 682.
[63] Winfried Nöth, "Semiosis and the Umwelt of a Robot," *Semiotica* 134, no. 1-4 (2001), 696.
[64] Emmeche, "Does a Robot Have an Umwelt?", 654.

to reflections on what the tick feels, can only evoke an impression of an entity that processes stimuli much the way we do – as experience. Do we have to interpret that experience as a manifestation of mind or consciousness in the arachnid's tiny brain? A "yes" response to that question is surely at the root of Uexküll's purported vitalism and more contemporary bio-chauvinism. Humans struggle to fathom experience without consciousness. Consequently, Voyager 2 is no more capable of experience than stones or water. Of course this leaves unanswered the question of where the cut-off should be between those entities we deem to have experience and those that do not have experience. If the tick qualifies, do its larvae qualify? Does a sponge? Or a mushroom? This section will take issue with this limited view of experience.

We have two choices if we seek to allow nonhuman or non-living things to experience the world. Firstly, we can accept that experience is limited to conscious entities but expand our concept of consciousness to incorporate any system of sufficient complexity. We can protest that consciousness is simply an emergent quality of matter that has reached sufficient complexity; so our supercomputers and robots, past, contemporary or imaginary, may be said to possess that indefinable capacity for experience because they, like us, have consciousness. In so doing, we need not confront the experience of nonhumans who encounter the world without a mind. This is a popular (if vague) view. Of particular note here are those transhumanists, such as Ray Kurzweil, who believe that so great is the resemblance between natural and artificial minds that each of us may one day be able to transfer our consciousness between them.[65] A mind is something that happens when matter is organised a particular way and cannot exist without that matter. Most of this kind of speculation is derived from functionalism (and identity theory, which it has mostly superseded).[66] Functionalism holds that mind is a quality that arises from the organisation of physical matter. It is a materialist view of the mind that is both a cause and result of the extensive research that has been done on the mind during the past century. In part because of identity theory and functionalism, psychology and neuroscience are influenced by the view that the physical laws that apply to the rest of the body also apply to the brain, which helps to explain how there can be any interaction between them. Without the belief that brain and body are subject to the same mechanised processes, so the argument goes, it would be impossible to explain how the brain senses pain in the finger, or controls a muscle in the calf. There would have to be some sort of other intermediary, which we have not yet been able to find, otherwise the mind and body are impossibly separate. One solution is that of the medieval Islamic occasionalists to which Harman refers.[67] But the decline of spiritual explanations for the mind (for example, the existence of the soul) has led to increased belief in a monist, physicalist model. We will be able to identify the presence of a mind in a supercomputer because it will feature qualities similar to human minds. Thus we establish ourselves as the possessors of an ideal model of mind.

---

[65] Kurzweil, *The Singularity is Near*.
[66] Skrbina, *Panpsychism in the West* (Cambridge, MA: MIT Press, 2005), 8-9.
[67] Harman, "Indirect Causation," *The Quadruple Object*.

The second possible option is to reject the idea of a hierarchy of mind. One way of doing this is to argue, as have many thinkers since ancient times, that the mind must be a quality of all matter, or that all matter must have the ability to produce a mind-like quality.[68] The word "hierarchy" is used deliberately here in provocation of the concept of flat ontology, and indeed some prominent scholars associated with OOO and the nonhuman turn do take this idea seriously, as discussed below. The idea that mind is a quality of all matter can be called panpsychism, and refers to the attribution of a mind, consciousness or a soul to all objects, including those objects we would usually think of as inanimate, such as robots, rocks and electrons. The word comes from the Ancient Greek roots *pan* (all) and *psyche* (mind or soul), and David Skrbina argues that it is "almost certainly the most ancient conception of the psyche" in the form of animism and polytheism.[69] Thomas Nagel may have described this notion best, by claiming that it comes from a few simple premises "each of which is more plausible than its denial, though not perhaps more plausible than the denial of panpsychism."[70] Panpsychism may seem incredible now, but many prominent philosophers (focussing for the moment only on the Western tradition) have subscribed to some variation on panpsychism[71] culminating in considerable popularity in the nineteenth century, before it became less popular with the rise of logical positivism in the twentieth century. Explanations of exactly how panpsychism works are rarely consistent between scholars, and even the language used is highly problematic. When we talk about panpsychism, are we speaking of consciousness, sentience, thought, qualia, the soul, experience, a psychological being, or the mind? Panpsychism may not be best thought of as a theory in and of itself, but that it is instead a commitment that may accompany radically different ideas. Consequently, it may be located in the work of thinkers as diverse as Thales, Spinoza, Leibniz, Whitehead and William James; with both monists and dualists, materialists and idealists.

Contemporary panpsychism follows this pattern in being a result of various philosophical positions. One such position is a continued adherence to Whitehead's process philosophy, but there are also instances in which panpsychism seems to arise as a modern solution to modern problems. David Chalmers' information theory invokes a view of panpsychism in exploration of his functionalism with, in John Searle's words, its "tacked on"[72] theory of consciousness. In his book *The Conscious Mind* Chalmers argues that consciousness (or experience) is a basic element of all matter by virtue of its capability to carry information.[73] This would appear to be an answer to Hobbes' point that sensation surely cannot exist without some sort of memory, and a reiteration of William James' characterisation of waves upon a beach as a sort of memory.[74] In Chalmers' parlance, the gestures of Elsie and Elmer

---

[68] David Skrbina, *Mind that Abides: Panpsychism in the New Millennium*, ed. David Skrbina (Amsterdam: John Benjamins Publishing Company, 2009), xii.
[69] Skrbina, *Panpsychism in the West*, 2-3.
[70] Thomas Nagel, "Panpsychism," *Mortal Questions* (Cambridge, UK: Cambridge University Press, 1979), 181.
[71] Skrbina, ed., *Mind that Abides*. The presence of panpsychism in history is traced by different authors throughout this book.
[72] John Searle and David Chalmers, "'Consciousness and the Philosophers': an Exchange," *The New York Review of Books* (May 15 1997).
[73] David J. Chalmers, *The Conscious Mind* (New York: Oxford University Press, 1996).
[74] Skrbina, *Panpsychism in the West*, 84.

create information, which could be the foundation of their consciousness. In humans, that information could simply be changes in neuronal structure, hormone levels and so forth. In the robot tortoises, the information could be the tiny indentations made in the floor as they move, the movement of charged particles in the batteries and the slow degradation of iron. To be clear, Chalmers is not saying that information makes a thing conscious, but he does remain "agnostic"[75] about it, and about panpsychism in general. In Searle's scathing review, he calls this an "absurd view"[76]:

> There is not the slightest reason to adopt panpsychism, the view that everything in the universe is conscious. Consciousness is above all a biological phenomenon and is as restricted in its biology as the secretion of bile or the digestion of carbohydrates. Of all the absurd results in Chalmers's book, panpsychism is the most absurd and provides us with a clue that something is radically wrong with the thesis that implies it.[77]

Panpsychism is certainly not a mainstream view, but it does seem to be increasingly tolerated due to the influence of the computer as a paradigmatic case in philosophy. It is less and less common to posit a universe in which only organic life is capable of consciousness. Our greater understanding of the relationship between the brain and the mind and the exponential growth in computer speed and complexity forces us to consider the consciousness of inorganic entities. This has led some, but by no means a majority, to panpsychism.

Of particular importance here is what Chalmers calls the "winking out" problem: the question of which entities can be said to be conscious if we accept that consciousness is only something that happens to humans, or higher mammals, or all living organisms, or anything of sufficient complexity be it carbon or silicon-based. For any of these premises to be true, there must be a certain point when descending through the hierarchy at which consciousness winks out. Perhaps a bat is conscious but a bird is not, or perhaps a fish is conscious but a lamprey is not. According to functionalism, consciousness is a physical phenomenon that will be perfectly explicable once we have gleaned sufficient knowledge about the brain. But if that is true, how can one creature possess mind and qualia and consciousness, while another creature that is very closely related not possess these things? Where is the cut-off point, and how can it ever be anything but arbitrary? This leads naturally to the position that consciousness, as a human-like phenomenon, might not necessarily wink out as we descend the hierarchy of living things, but might change seamlessly. Consequently, a tick will still have something that we have to call "consciousness" or something similar, even though it looks totally different from a human experience.

Why should we stop at the boundary between living and non-living things? A highly speculative panpsychist section of Chalmers's book is called "What is it like to be a thermostat?", a clear echo of Thomas Nagel ("What is it like to be a bat?"[78]) in which he describes the thermostat as "an information-processing system that is almost maximally

---

[75] Searle and Chalmers, "'Consciousness and the Philosophers': an Exchange."
[76] Ibid.
[77] ibid.
[78] Thomas Nagel, "What is it Like to be a Bat?" *The Philosophical Review* 83, no. 4 (1974): 435-450.

simple."[79] All that is required for consciousness (although probably not thought) is a system of "causal interaction"[80], in which an external stimulus causes a change of state. This means that electrons meet the criteria, and Chalmers's speculative panpsychism is almost total. Of course, fictional and incorporeal objects like the Kinect-brand are not included in this panpsychist view.

Panpsychism is very difficult to intuitively accept if one is asked to begin from the principle that river rocks and sunflower seeds are sentient or conscious. But if, as Chalmers appears to have done, you begin with advanced computers and the hard problem of consciousness, the notion is more palatable. Supercomputers must have a mind if the dominant functionalist model is correct, since they behave as though they have a mind. Minds are most easily identifiable in machines that seem to be more like us, even if they are structurally very different. This is precisely the reason why this chapter juxtaposes panpsychism with a very simple robot. It is possible to draw a pleasing comparison between a panpsychist position and Grey Walter's enthusiasm for his robotic creations. A naïve view would see little difference between the attribution of mind or experience to the tortoises and Walter's rhapsodising over his "clumsy narcissus" and its powers of self-recognition. Panpsychism actually reveals a much more heterogeneous concept of mind: one in which the activation of switches and purring of motors is proof enough in itself of experience, rather than merely the vehicle by which the robot cons its way into consciousness by mimicking human proclivities.

If we extend mind, or at least subjectivity, to all things, then to some extent the novelty of the Umwelt is lost. An implication of suggesting that all information is a characteristic of mind is that all interactions are significant, and perhaps even that they may be represented in semiotic terms. In that scenario, the feedback loop may become all but interchangeable with Uexküll's concept of the function-circle. The correction of a robot's wheels as it moves across the uneven floor are as significant as its perception of light because the Umwelt encompasses the whole of the robot and every piece of information is part of its mind. Similarly, the motion of red blood cells throughout a human's circulatory system is a part of its Umwelt. Our brains might not be aware of the movement, but our arteries, tissues and plasma sense and experience it. In adopting panpsychism it would be easy for the Umwelt to become irrelevant. But there is great value in considering the two ideas at the same time. The strength of the concept of the Umwelt lies in its power to force us to confront alien worlds. In the latter example, it might be possible to accept the presence of a mind in not just the whole human but in the individual blood cell itself. An inquiry into the subjective world of this blood cell could yield fascinating results. As for the robot, the imposition of a division of parts and wholes, the sort that we employ when studying living things, might be contrary to the nature of non-living things. The wheel is connected to and strongly related to the robot, but that does not mean that the two components bear the same relation as that between the human and the blood cell. The wheel-mind might be totally different to the robot-mind, or they might be aspects of a whole. Panpsychism may seem at first to be in conflict with the idea of the Umwelt, particularly with Uexküll's description, but an adoption of panpsychism might

[79] Chalmers, *The Conscious Mind*, 293.
[80] Ibid., 298.

only require the adaptation of the theory to other kinds of sign-relations. As a bare minimum, it does not seem unreasonable that some adaptation be made so that robots can be said to have Umwelten, irrespective of the debate over whether they have minds.

Panexperientialism is sometimes suggested as a more moderate view of panpsychism; less an argument about consciousness than about "experience". Gregg Rosenberger claims that it is "milder"[81] because it does not require anything like cognition or mentality.[82] Again, it is presented as a solution to the mind-body problem. The concept of a "mind" is highly anthropocentric. A statement about the nature of an entity's mind can be an ontological statement about the nature of its experience and consequently the question of subjectivity, whereas panexperientialism leaves the interpretation a little more open. The term was coined by David Ray Griffin. He describes his interpretation as "Whiteheadian-Hartshornean".[83] Griffin says that "psyche" implies too high a level of experience for more elementary units of nature.[84] As he puts it, Whitehead's philosophy begins with the idea of "prehension". Prehension is something that we share with all other individual, regardless of whether or not they have sensory organs.[85] Every individual has experience through prehension. But not all objects are the same. Some are "compound individuals", which have a higher level of experience and that might have consciousness; others are "aggregational societies" which do not have higher levels of experience (Griffin gives the examples of rocks and telephones, and Elsie would also qualify).[86] Every individual prehends, but not all prehensions lead to consciousness. This happens in human beings too. Not everything that a human prehends makes its way into the conscious mind, only "those that have been prehended with the greatest intensity will survive to this phase; the rest are blocked from becoming conscious."[87]

Aggregational societies do not have experience according to Griffin; they are not "individuals". Elsie does not have experience, but the atomic and subatomic particles that constitute her do have experience. In the words of Cornel du Toit, the "resulting position can be called process philosophy's version of 'panexperientialism', which is applicable to all individuals but not to all things whatsoever."[88] Curiously, even though the "psyche" is dropped from the word, inanimate objects as such are still excluded. Prehension vaguely resembles the model that I have outlined for a concept of "sensation" in OOO terms but is quite distinct. Throughout this thesis I have used the word "sensation" to align the concept with Harman's ontology (and also as something of a provocation of our tendency to

---

[81] Gregg Rosenberger, "*A Place for Consciousness: Proving the Deep Structure of the Natural World*, (Oxford: Oxford University Press, 2004), 96.
[82] Ibid., 91.
Rosenberger also uses the word "protoconscious" with slightly different connotations.
[83] David Ray Griffin, *Religion and Scientific Naturalism: Overcoming the Conflicts* (Albany: State University of New York Press, 2000), 167.
[84] David Ray Griffin, *Unsnarling the World-Knot: Consciousness, Freedom, and the Mind-Body Problem* (Berkeley CA: University of California Press, 1998), 78.
[85] Griffin, *Religion and Scientific Naturalism*, 101-102.
[86] Ibid., 167
[87] Ibid., 223.
[88] Cornel du Toit, "Panpsychism, pan-consciousness and the non-human turn: rethinking being as conscious matter," *Theological Studies* 72, no. 4 (2016), 6.

anthropomorphise). But I would not use the word "prehension" because of the rejection of experience in all kinds of objects found in "process panexperientialism". Fundamental to my objection is that all sorts of different things are objects: atoms, metallic strips, motors, wheels, and robots. The part/whole model of OOO is entirely different from that found in Whitehead. The word "experience" in "panexperientialism" is not the same as my usage. For me, experience is evident in a response to sensation; the collision of real qualities with sensual objects. Panexperientialism is primarily presented as a solution to the mind-body problem and nonhumans are not usually the focus.

Harman's version of panpsychism is predicated on the irreducibility of the mind. The relation of a mind to an object outside the mind has the same ontological status as fire burning cotton.[89] Panpsychism is not required for flat ontology, or vice versa, but there is some association between the two despite the initial misgivings Harman revealed in *Guerrilla Metaphysics*, in which he rejects "human knowledge" as being of "pivotal importance in the universe".[90] In *The Quadruple Object* Harman describes a softening of his ideas towards panpsychism, even if he is not prepared to adopt the label. He uses the term "polypsychism"[91] instead, which is intended to highlight his point that a perceiving entity is a relating entity, and some entities are "dormant"[92] and free of relations. Harman explains that these objects may exist in principle, or perhaps that an object may have relations at one time and no relations at another (although he does not provide an example). Therefore, while experience can be attributed to most entities it cannot be attributed to all.

Harman is not primarily interested in minds. But the idea of minds has historically been such a feature of the distinction between humans and nonhumans that it is relevant to OOO. Consider the sensation of a bowl of soup by a human mind, which the human experiences as hot. But in addition the human experiences many things that do not prompt any changes in the mind such as the ambient temperature and the evaporation of water from the surface of the skin. Vesicles in an axon terminal release serotonin into the synapse and it is taken up by the dendrite of another neuron. The waves wash over the beach. Mario completes a level and finds that the princess is in another castle. Each is an example of sensation of an object by an object, just as a human senses soup and soup senses the human. They are yet more examples of real objects that create sensual objects to allow interaction, but they create a change in the real objects. They produce encrustations of sensual qualities that change the way that the object is able to form relations. At a very basic level, the object is affected by this change, and must register it. In other words, an alien kind of experience is taking place.

Maybe the robot tortoises don't need a human-like mind to experience things either. The mind is just another kind of object. Thoughts are sensual objects that appear when a relation is formed between the soup and the mind. When we limit experience to things that have a mind, we are framing everything through a human concept. Even if we concede that a robot has a very different mind than a human, it is still the word *mind* that we use. The word is

[89] Harman, *The Quadruple Object*, 121.
[90] Harman, *Guerrilla Metaphysics*, 84.
[91] Harman, *The Quadruple Object*, 121.
[92] Ibid., 122.

misleading, steeped in history and particularly in the human/nonhuman (and mind/body) dualism that OOO minimises. But we don't need to be anthropocentric in our use of the word. The nature of the mind does not, in principle, matter, because it is as indisputably an object as much as unicorns and phlogiston. As Steven Shaviro points out, a common criticism of panpsychism is that it uncritically assigns human-like mind elements to nonhumans.[93] This, Shaviro says, begs the question because it betrays an understanding of thought, value and experience that is inherently anthropomorphic. Panpsychism extends the idea of minds, but does not alienate us from the minds of nonhumans. It is not necessary to speak of mind in an anthropocentric context. A mind is just another object. The mind of a synapse, wave or videogame character does not need to be a deviant version of a human mind. Minds, and all their associated notions of experience, may be just as alien as the kinds of sensation that define an object.

This chapter has focussed on one of the guiding question of this thesis: what does sensation mean for machines? More specifically, does using the word "sensation" in a strict, non-metaphorical sense require us to believe that machines experience the world in any meaningful way? This chapter has argued that in the context of Umwelt-theory, it is not necessary to anthropomorphise experience, but it is necessary to recognise some kind of semiotic exchange between nonhuman and environment. It is natural to be reluctant to extend a quality of mind to robots as simple as *Machina speculatrix*, but it is not necessary to do so to witness their experience of the world. It is difficult but possible to conceive of non-anthropomorphic minds or experiences, with or without a panpsychist view, and even without a panexperientialist view. The Umwelt is conceptually distinct from the concept of a mind and yields a very different account of how an object experiences its world. Umwelten give an account of a sensory world that need not be anything like our own, despite taking place in the same environment. The concept of a mind relates to the experience of those sensations. For the anthrodecentric approach presented in this thesis both Umwelten and minds may be seen in at least all physically embodied objects. Yet both the Umwelt and the mind have historically been restricted to certain classes of objects because of the assumption that both depend on the presence of subjectivity. As OOO rejects the primacy of subjectivity, the concepts of Umwelten and minds can be better deployed as explanations of the elements of Harman's metaphysics (and must be, if object-oriented ontologists can acknowledge them to exist at all). The Umwelt of an object is the impression of the environment upon an object; it is the field in which sensual objects are formed and maintained despite the flickering encrustations of their qualities. Notice that in this view the Umwelt must incorporate entirely alien sensations such as the impression of wheels upon a floor. The Umwelt is the sensory world. Meanwhile the mind, if the concept is made sufficiently broad, describes the way that the object encounters sensation. It describes a very fundamental kind of experience. It is nothing more than the sum total of the changes brought on by the sensation of other objects

[93] Steven Shaviro, "Consequences of Panpsychism," *The Universe of Things* (Minneapolis: University of Minnesota Press, 2014), 90.

as those sensations change moment by moment. In this account, the Umwelt and the mind are not special objects relegated to the realm of living things for the sake of human vanity.

Anthropomorphism is a trap which must be avoided when speaking of Umwelten or minds. Objects that are more human-like do not have non-alien sensation and experience, whether they are parts of humans or models of humans. Synapses, waves and Mario are all alien. The point is to avoid a hierarchy of mind-having or Umwelt-having objects, beginning with humans at the top and negotiating the positions of various anthropomorphic objects, objects characteristic of the Anthropocene or objects that typically inhabit human bodies on a spectrum. The point to bear in mind is that we and our sensation and experience are as alien to other objects as they are to us.

Chapter 4: SHRDLU and Bidirectional Postphenomenology

Postphenomenology differs from traditional phenomenology (if such a thing really exists) in that the structure of human perception and experience is changed by technologies. Almost by definition, phenomenology is about humans and human lived experiences through the "principle of intentionality that sutures subject and object together" but varies greatly from scholar to scholar.[1] Postphenomenology does not make the same radical claims as OOO it does significantly depart from traditional phenomenology by emphasising the role of technology in affecting human experience and even being a part of human experience. Technologies are not merely mediators of human experience, but constitute our reality.[2] Don Ihde is the most prominent theorist in this field. For Ihde, who is an American philosopher of science, human perception and other intentionalities extend through machines. Technology must not be considered abstractly, but in terms of human-technology relations.[3] Many of Ihde's examples are visual ones, such as the non-neutral effect of Galileo's telescope on stargazing.[4] Despite requiring a degree of trust in the medium to convey an image that expresses the world in terms humans can understand, the telescope nevertheless presents the human user with objects in the solar system that would otherwise be invisible, such as the moons of Jupiter. Indeed, in the case of modern telescopes, distant galaxies and phenomena outside the visible light spectrum may also be glimpsed. The scholarship that has arisen around this concept contributes to a growing appreciation of the role of technology in human phenomenology.

Ihde is not an OOOer and he is not overly concerned with deconstructing the categories of "human" and "machine". His postphenomenology explains an important and often neglected part of humanity's lived experience. But for the purposes of this thesis postphenomenology is typically far too one-sided. To only use Ihde to understand humanity's lived experience is to reject the OOO position of anti-correlationism. Therefore, this thesis will present, as far as is possible, strictly bidirectional and asymmetrical interpretations of Ihde's work. Where Ihde describes a human-technology-world relationship from a human position, this chapter will attempt to describe a similar one from an object's perspective. This is an experiment in combining two approaches, but it is hoped that this bringing together of approaches yields a

[1] Tom Sparrow, *The End of Phenomenology,* 91.
[2] Peter-Paul Verbeek, "Expanding Mediation Theory," *Foundations of Science* 17 (2012): 391-395.
[3] Don Ihde, *Postphenomenology: Essays in the Postmodern Context* (Evanston, Ill: Northwestern University Press, 1993), 34.
[4] Ibid., 46.

useful language and framework for describing the asymmetrical, bidirectional relations between human and technological artefact. This may bring us closer to the goal of conceptualising the way that machines sense and experience the world. Since postphenomenology is so robust in its treatment of human-world relations, by reversing the process we can make coherent statements about machine sensation.

The computer is a paradigmatic metaphor for human experience. We saw in the previous chapter how a technological artefact may affect and shape our line of inquiry about the human mind. This chapter focusses on a computer program as it forms relations with its environment, an environment that includes human interlocutors. The application and critique of a postphenomenological model is performed with reference to SHRDLU, Terry Winograd's natural language processing (NLP) program. SHRDLU was a natural language processing program which was entirely concerned with moving and answering questions about crudely illuminated block shapes on the system's monitor. The setting is deceptively simple, concealing a vast, complicated web of interrelated processes that make it possible for SHRDLU to "understand" English. Through the constructs of postphenomenological theory this chapter interprets the ways that SHRDLU extends or changes human actions and perception, and how these constructs might be reversed to describe the way that technological agents use humans.

What is (or was) SHRDLU?

The world of SHRDLU is deliberately tiny. It is an entity with a miniscule capacity for direct action in its environment. Like the tortoises of the previous chapter, it has a small set of ways of sensing the world; the most pertinent for most human analyses being the way that it senses the human. SHRDLU's only information about its human user comes to it through textual orders and interrogations, relayed through the keyboard of whatever system it happens to be occupying at that time (or possibly, these days, through voice-to-text applications). The human types instructions or questions, and SHRDLU responds with text and/or actions in its block world. Once that information is received, SHRDLU's capacity for action is very limited. It can only perform actions as it has been programmed to do. Apart from responding to the user's dialogue, its sole possible set of actions consist of moving around digital blocks with a digital robotic arm (the digital block world is visible to the user). The blocks exist as code within the computer but are programmed to obey the laws of the physical universe, so for example, a block once dropped will fall onto whatever is below it. This is the microworld of SHRDLU, the epitome of "hacker" artificial intelligence programs designed by Terry Winograd between 1968 and 1972, created to investigate natural language and computers. It has become a classic piece of AI programming, providing what was at the time a fresh perspective on the relationship between world and language. Like the tortoise, SHRDLU's simplicity is the key to its engaging theoretical implications; it becomes a model for greater things, providing clues to unlocking more universal insights. In a 1979 paper Hubert Dreyfus discusses SHRDLU along with other microworlds created in the late 1960s and 1970s,

such as Adolfo Guzman's 1968 SEE program which was designed as a simple computer vision program.[5] Dreyfus is critical of these programs because he believes that human intelligence consists of "local elaborations of the whole" rather than isolated microworlds that assemble to form the everyday.[6]

Nevertheless, Dreyfus calls SHRDLU a "major achievement."[7] The program is far more complex than the interface suggests. Like any elegantly-designed user interface, it conceals the complexity of its underlying processes.

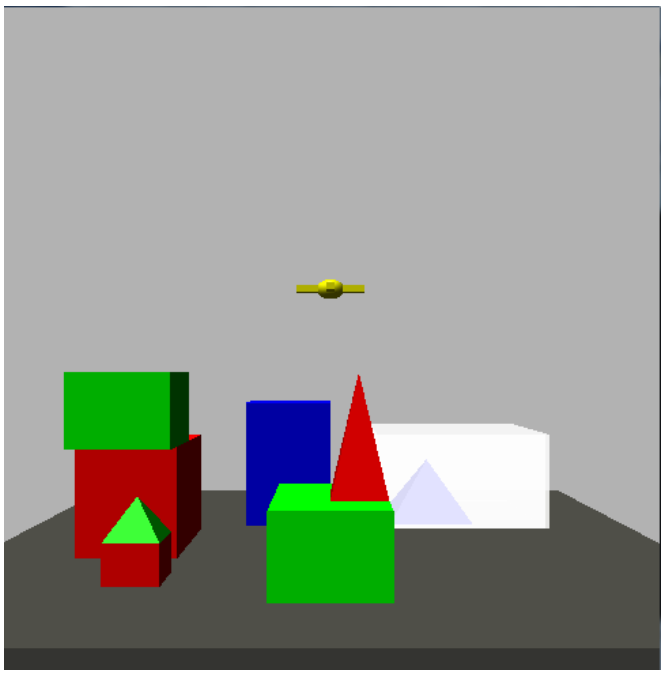


Figure 19: The block world and robot arm as represented by a more recent incarnation of the SHRDLU.[8]

The language use of programs like SHRDLU is one of the most studied and contentious areas of artificial intelligence research. Humans have a fascination with reproducing their form in artificial entities, and especially in reproducing certain defining human qualities like language. The intelligence modelled by early systems like SHRDLU was limited to tightly constrained domains and therefore imitated only a tiny part of human intelligence. Playing chess and proving mathematical theorems were possible, but one could not enjoy a disinhibited conversation with a machine.[9] SHRDLU cannot converse with a human about anything other than its block world. This so-called "weak AI" may be contrasted with the "strong AI" theorised by science fiction writers and futurists, in which human intelligence is not merely simulated, but from which consciousness and self-awareness emerge.[10] Strong AI is not

[5] Hubert Dreyfus, "From Micro-Worlds to Knowledge Representation: AI at an Impasse," *Mind Design II*, ed. John Haugeland (Cambridge, MA: The MIT Press, 1997), 151. Originally published 1979.
[6] Ibid.
[7] Ibid., 145.
[8] *SHRDLU*, Windows 7.
[9] Ray Kurzweil, *How to Create a Mind* (New York: Penguin Books, 2012), 181.
[10] John R. Searle, "Minds, Brains, and Program," in *The Turing Test: Verbal Behaviour as the Hallmark of Intelligence*, ed. Stuart M. Shieber (Cambridge, MA: MIT Press, 2004), 201-224.

simply an improvement of processing power, but a replication of the linguistic and experiential qualities of the human brain, and supporting the argument that the brain is *like* a computer.[11] The supposed need to observe understanding and explanation in machines is often seen to be a major problem in the development of AI. Creating and responding to signs is seen as a step above simple performative and adaptive systems. Human brains *understand*. They interpret the world through language and other signs and draw conclusions after the application of symbolic logic.

One of the earliest applications of computers to be seriously considered in AI circles was the translation of texts from one language to another. But in 1960 Yehoshua Bar-Hillel proclaimed that the high-quality translation programs sought after and expected were nowhere near being completed, and used the phrase "the box was in the pen" to illustrate his point.[12] He pointed out that no program at the time could translate the word "pen" as used in that sentence correctly, the way that a human would be able to do so immediately. The machine is in need of "extra-linguistic knowledge". Winograd argues that early machine translation failed because language consists of more than just the concepts of "dictionary" and "grammar"; language must be connected to an external world.[13] As he puts it, "the computer didn't know what it was talking about."[14] A conversation is part of each participant's ongoing attempt to gain information about their environments.

A successor to the translation programs were the conversational programs built between the mid-1960s and early 1970s, of which SHRDLU is one. They were attempts to build systems that had a limited understanding of language. One famous example is ELIZA, a program developed by Joseph Weizenbaum at MIT between 1964 and 1966. Despite the limitations of the time, ELIZA was made to be a somewhat convincing conversationalist by running the DOCTOR script in which it was themed as a Rogerian psychiatrist talking to a patient. This is a clever move because the inevitable non sequiturs can be mistaken for insight in a psychiatrist identifying hidden links.[15] A Rogerian psychiatrist also often reflects back what the patient is saying to draw the patient out and get them to keep talking, which is something the program could handle.[16] Winograd points to this episode as a warning that, with clever tricks, a program can appear to have more insight than it appears to.[17] ELIZA has some similarities with contemporary virtual therapist Ellie, funded by DARPA to aid in the diagnosis of mental illness amongst veterans.[18] In some ways, Ellie's goals are the same as DOCTOR's, since it is important for her to connect with "patients", who must be encouraged to see past the

---

[11] Ibid.
[12] Yehoshua Bar-Hillel, "The Present Status of Automatic Translation of Languages," in *Readings in Machine Translation*, eds. Sergei Nirenburg, Harold Somers, and Yorick Wilks (Cambridge, MA: The MIT Press, 2003), originally published 1960.
[13] Terry Winograd, "What Does it Mean to Understand Language?" *Cognitive Science* 4 (1980).
[14] Terry Winograd, "Procedures as a Representation for Data in a Computer Program for Understanding Natural Language," doctoral dissertation (Massachusetts Institute of Technology, 1971), 10.
[15] Winograd, "Lecture 1," *Five Lectures on Artificial Intelligence,* 4.
[16] Jason L Hutchens, "How to Pass the Turing Test by Cheating" *ACM DL Digital Library* (University of Western Australia, 1996), 6.
[17] Winograd, "Lecture 1," 4.
[18] Megan Garber, "Would You Want Therapy From a Computerized Psychologist?," *The Atlantic* (May 23 2014).

computer interface to the connection with another entity. This is aided by Ellie's advanced natural language processing as well as her ability to pick up on bodily cues and her own use of body language. But the important difference here is that Ellie has an aim beyond fooling human interlocutors with her use of language. She is not a theoretical project, and she exists to diagnose rather than to interpret. There has already been some literature discussing the advantages of virtual therapists: for example, people seem to be more willing to discuss sensitive or embarrassing topics with a computer.[19] But there are also risks, as in the use of therapeutic anthropomorphic toys in children with autism, such as distress when the agent is taken away or hacked with malicious intent.[20]

It has been argued that these capabilities are beyond the artificial brains that exist in the world to date. John Searle, the critic of David Chalmers mentioned in the previous chapter, demonstrates this line of thinking in the Chinese Room thought experiment.[21] Searle's is a well-known argument associated with the feelings of disappointment of the 1980s that came from the failure of AI to live up to expectations.[22] A non-Chinese speaker is put inside a room. From outside the room, a Chinese speaker writes a question and puts it through a slot into the room. The person inside the room has access to a book containing many different Chinese phrases and the appropriate response to give. The person can copy down the response and return it through the slot. The result, to the Chinese speaker, appears to be comprehension by the computer and a salient response. However, the entire exchange takes place without the person inside the room understanding the question or answer. Searle argues that this is how a computer appears to return intelligent answers despite no semiotic comprehension taking place. However, Ray Kurzweil disagrees and claims that the Chinese Room model is a construct that reveals what occurs in simple single-neuron systems.[23] In a system like the human brain, great complexity and volume of data transforms the performative neural response to a supposedly semiotic one through processes of emergence, and there is no reason why, given sufficient complexity, something like thought might not occur in a computer.

This question is pertinent to the creation of modern NLP agents such as Watson, the IBM computer reportedly capable of writing poetry. The depth of human skill and wisdom required to write good poetry is considerable, so it is easy to see why comparable creative output from a computer would lead the casual observer to ascribe it to human-like intelligence. In 2015, the IBM Watson team capitalised upon this in an advertisement starring music legend Bob Dylan.

---

[19] Gale M. Lucas, Jonathan Gratch, Aisha King and Louis-Phillippe Morency, "It's Only a Computer: Virtual Humans Increase Willingness to Disclose," *Computers in Human Behavior* 37 (2014): 94-100.
[20] Alexis Elder, "Robot Friends for Autistic Children: Monopoly Money or Counterfeit Currency?", in *Robot Ethics 2.0: From Autonomous Cars to Artificial Intelligence*, ed. Patrick Lin, Ryan Jenkins and Keith Abney (New York: Oxford University Press, 2017), Kindle Ebook.
[21] Kurzweil, *The Singularity is Near*.
[22] Sherry Turkle, *Life on the Screen* (New York: Simon & Schuster Paperbacks, 1995), 130.
[23] Kurzweil, *The Singularity is Near*.

This is an excerpt from the advertisement:[24]

> WATSON: Bob Dylan. To improve my language skills, I've read all your lyrics.
> DYLAN: You've read all of my lyrics?
> WATSON: I can read 800 million pages per second.
> DYLAN: That's fast.
> WATSON: My analysis shows your major themes are that time passes. And love fades.
> DYLAN: That sounds about right.
> WATSON: I have never known love.
> DYLAN: Maybe we should write a song together.

Watson's natural language capabilities are extremely advanced and are informed by the input of vast quantities of text, so that it is able to answer sophisticated questions on a huge number of topics.[25] In one sense its knowledge of love is doubtless extensive. But, as it says, it has never known love. It is not grounded in the world with respect to that notably human emotion. Its sensation and experience of love is alien. Given Watson's incredible linguistic skill, however, it may well be perfectly capable of writing poetry that mimics Dylan's characteristic lyrics. Whether it understands them the way a human would is another question, as discussed in the previous chapter.

Language is considered one of the defining qualities of human nature, and it is only to be expected that we would try to reproduce it in computers, if only to see if it is possible. But the practical possibilities for such an endeavour are also enormous. One need only look to the contemporary examples of Amazon's Alexa and Microsoft's Cortana to see the vast cultural value attached to communicating with machines "naturally", and consequently the economic value thereof. As discussed in Chapter 2, this goal of interacting "naturally" with a machine is problematic but appealing. Natural language communication with computers is also a common trope in science fiction and is commonly a quality of advanced user interfaces, such as the computer character Samantha in the film *Her* (played by Scarlett Johansson).[26] In this film, the ability of the computer user to communicate through natural language with his computer leads to an intense intimate relationship. The computer is anthropomorphised in other ways as well, but her presence is located in her ability to speak; it is the facilitator of her sensation of the human.

SHRDLU can be seen as an earlier relative of today's more advanced NLP programs, hampered by its inability to access information about anything other than its block world and system. The world in which SHRDLU operates is necessarily small so that it can understand everything that goes on within it. Winograd argues that "success at understanding language depends on a deep knowledge of the subject being discussed."[27] Deb Roy calls this

---

[24] *Bob Dylan + IBM Watson on Language* (2015), Youtube. https://www.youtube.com/watch?v=oMBUk-57FGU (accessed June 27 2017).
[25] *New Era of Cognitive Computing* (2013), Youtube. https://www.youtube.com/watch?v=h22n80aT2FY (accessed June 27 2017).
[26] Spike Jonze, *Her* (United States: Warner Bros. Pictures, 2013), film.
[27] Winograd, "Lecture 2," in *Five Lectures on Artificial Intelligence*), 20.

"grounding", the process by which an agent relates beliefs to external physical worlds.[28] In this way the computer need not have the world mediated by anchor-less human language but can have a direct experience of the objects it is discussing. It would have been impractical to attempt to make the computer relate to objects in the real world; this was beyond the technology at the time. And for OOO purposes it is not important whether the world of blocks was "real" or not (see below) – it still constitutes a collection of objects, and we can look on SHRDLU in OOO terms as an early attempt to extend the sensation-poor environments of computers. Winograd had a reason for this attempt that was both practical and philosophical, and it resulted in a system that had a wealth of information available to it. More information means, correspondingly, greater agency within its world; that is, the world of blocks. The information about its world was not just coming to it through the intermediary of language, but from its sensation and understanding of the block world.

Speaking to SHRDLU is like speaking to a toddler that is only just learning about its world. This is a typical interaction:

> Person: PICK UP A BIG RED BLOCK.
> Computer: OK. (does it)[29]

When the robot arm completes an action, that action is reflected on a visual representation. In this case the robot arm descends from the top of the screen and picks up a red block meeting a pre-determined definition of "big". As will be shown, this simple exchange requires what was at the time a very advanced understanding of language and logic. As Douglas Hofstadter says, SHRDLU is a "mathematical ignoramus" which has only been taught to count up to ten.[30] But its simplicity is, in fact, the key to its solidity as a paradigmatic example of a particular phase in the history of AI.

Winograd, whose experience in computer science led him to become a philosopher, is of the opinion that previous attempts at NLP failed because there was a lack of connection between the language the system is being asked to understand and a real world that it can understand and interact with. With this in mind, he built SHRDLU as not just a teletext exchange between user and computer but incorporating the simulated "blocks world" with physical laws and rules for an aspect of the system to interact with. To the human user, the teletext screen is presented in order for her to provide instructions to or ask questions of SHRDLU, and for SHRDLU to respond. But there is also a screen showing a simple world of blocks of different shapes, sizes and colours. SHRDLU has a robot arm on the screen and is able to pick up the blocks, one at a time, and move them around the world. This simple configuration represented a significant advancement over other systems. It could remember things from earlier, justify its actions and learn new things about its environment. The program was written in LISP. This sample dialogue from a Stanford website gives some idea of the program's capabilities:

---

[28] Deb Roy, "Semiotic Schemas: a Framework for Grounding Language in Action and Perception," *Artificial Intelligence* 167 (2005), 176.
[29] "SHRDLU," http://hci.stanford.edu/winograd/shrdlu/ (accessed January 22 2018).
[30] Douglas R. Hofstadter, *Gödel, Escher, Bach: An Eternal Golden Braid*. (Harmondsworth, England: Penguin Books, 1980), 630.

Person: GRASP THE PYRAMID.
Computer: I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.
Person: FIND A BLOCK WHICH IS TALLER THAN THE ONE YOU ARE HOLDING AND PUT IT INTO THE BOX.
Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.
Computer: OK. (does it)

SHRDLU explains what it is doing before doing it so that the human user knows what SHRDLU understood from the command.

Person: WHAT DOES THE BOX CONTAIN?
Computer: THE BLUE PYRAMID AND THE BLUE BLOCK.
Person: WHAT IS THE PYRAMID SUPPORTED BY?
Computer: THE BOX.
Person: HOW MANY BLOCKS ARE NOT IN THE BOX?
Computer: FOUR OF THEM.
Person: IS AT LEAST ONE OF THEM NARROWER THAN THE ONE WHICH I TOLD YOU TO PICK UP?
Computer: YES, THE RED CUBE.
Person: IS IT SUPPORTED?
Computer: YES, BY THE TABLE.
Person: CAN THE TABLE PICK UP BLOCKS?
Computer: NO.
Person: CAN A PYRAMID BE SUPPORTED BY A BLOCK?
Computer: YES.
Person: CAN A PYRAMID SUPPORT A PYRAMID?
Computer: I DON'T KNOW.
Person: STACK UP TWO PYRAMIDS.
Computer: I CAN'T.

SHRDLU is capable of experimenting and learning about its environment, as when it is asked to place a pyramid on top of a pyramid. It did not initially know whether it was possible, and needed to experiment to find out.

Person: WILL YOU PLEASE STACK UP BOTH OF THE RED BLOCKS AND EITHER A GREEN CUBE OR A PYRAMID?
Computer: OK. (does it)
Person: WHICH CUBE IS SITTING ON THE TABLE?
Computer: THE LARGE GREEN ONE WHICH SUPPORTS THE RED PYRAMID.
Person: IS THERE A LARGE BLOCK BEHIND A PYRAMID?
Computer: YES, THREE OF THEM: A LARGE RED ONE, A LARGE GREEN CUBE, AND THE BLUE ONE.
Person: PUT A SMALL ONE ONTO THE GREEN CUBE WHICH SUPPORTS A PYRAMID.
Computer: OK. (does it)
Person: PUT THE LITTLEST PYRAMID ON TOP OF IT.
Computer: OK. (does it)
Person: DOES THE SHORTEST THING THE TALLEST PYRAMID'S SUPPORT SUPPORTS SUPPORT ANYTHING GREEN?
Computer: YES, THE GREEN PYRAMID.
Person: WHAT COLOR IS IT?
Computer: BY "IT", I ASSUME YOU MEAN THE SHORTEST THING THE TALLEST PYRAMID'S SUPPORT SUPPORTS.
Computer: RED
Person: HOW MANY THINGS ARE ON TOP OF GREEN CUBES?

> Computer: I'M NOT SURE WHAT YOU MEAN BY "ON TOP OF "IN THE PHRASE "ON TOP OF GREEN CUBES " .
> DO YOU MEAN:
> 1 - DIRECTLY ON THE SURFACE
> 2 - ANWHERE ON TOP OF ?

This is a way for the system to deal with ambiguity.

> Person: 2
> Computer: THREE OF THEM.
> Person: HAD YOU TOUCHED ANY PYRAMID BEFORE YOU PUT THE GREEN ONE ON THE LITTLE CUBE?
> Computer: YES, THE GREEN ONE.
> Person: WHEN DID YOU PICK IT UP?
> Computer: WHILE I WAS STACKING UP THE RED CUBE, A LARGE RED BLOCK AND A LARGE GREEN CUBE.
> Person: WHY?
> Computer: TO GET RID OF IT.
> Person: WHY DID YOU DO THAT?
> Computer: TO CLEAN OFF THE RED CUBE.
> Person: WHY DID YOU CLEAR OFF THAT CUBE?
> Computer: TO PUT IT ON A LARGE GREEN CUBE.
> Person: WHY DID YOU DO THAT?
> Computer: TO STACK UP THE RED CUBE , A LARGE RED BLOCK AND A LARGE GREEN CUBE.
> Person: WHY DID YOU DO THAT?
> Computer: BECAUSE YOU ASKED ME TO.[31]

SHRDLU is grounded in its world – it has things to talk *about*. This dialogue gives some indication of SHRDLU's relations with its world, such as with blocks, physical laws, and human interlocutors. Further investigation reveals the interaction of SHRDLU with such entities as LISP, electronics, and the history and future of AI research and design.

SHRDLU interacted with many texts as well as countless academics. Winograd claimed in his dissertation that "the challenge of programming a computer to use language is really the challenge of producing intelligence."[32] But in the book *Understanding Computers and Cognition*, he and Fernando Flores argue that neither is really possible.[33] They write, citing examples of programs that attempt to produce natural language, including SHRDLU, that it is not possible for a computer to truly understand language:

> If I write something and mail it to you, you are not tempted to see the paper as exhibiting language behavior. It is a medium through which you and I interact. If I write a complex computer program that responds to things you type, the situation is still the same – the program is still a medium through which my commitments to you are conveyed… it must be

[31] "SHRDLU." http://hci.stanford.edu/winograd/shrdlu/ (accessed 22 January 2018).
[32] Winograd, "Procedures as a Representation for Data in a Computer Program for Understanding Natural Language," 443.
[33] Terry Winograd and Fernando Flores, *Understanding Computers and Cognition: A New Foundation for Design* (Reading, MA: Addison Wesley Publishing Company, Inc., 1987), originally published 1986.

> stressed that we are engaging in a particularly dangerous form of blindness if we see the computer – rather than the people who program it – as doing the understanding.[34]

This statement is a prediction that it is impossible to build a Strong AI entity and is comparable to the Chinese Room thought experiment argument. Winograd and Flores speak more of technology as something that is useful and interactive and use phenomenological language; they emphasise the tool-use of AI agents.[35] SHRDLU as an experimental actant reveals to us another way of being, the alien logic of the computer, not a logic that is comparable to but lesser than humans'.

SHRDLU is a virtual robot with no direct agency beyond the clearly-delineated walls of its system. It can be discussed as a "robot" in the same way that it discusses robot tortoises and social robots, which exist in the physical world. Some definitions of robot, such as that put forward by Maja Mataric, apply only to the latter variety, that are both "autonomous" and "exist in the physical world".[36] Karel Čapek's original usage of the word in his play *R.U.R.* (1920) described mechanical beings (*roboti*) similar to humans that work in a factory and later rise up in a rebellion, which is comparable to modern ideas of androids or cyborgs.[37] But the word itself is derived from the Czech word *robota* meaning "forced labour", and the robots in the play are actually built in a way that makes them very difficult to distinguish from humans. SHRDLU and similar entities are clearly examples of both these qualities. SHRDLU is employed in forced labour for the human user, without possessing anything that we might describe as a choice in the matter. When we look at the modern usage of the term, "robot" usually refers to an entity that exists in the physical world. A notable exception to this is the "bot", which is an entirely virtual entity which commonly performs automated tasks on the internet – a subset of this, the "chatbot", could be seen as a descendent of SHRDLU, employing natural language techniques in an attempt to appear as a human to the user. It is possible to trace a non-physical robot genealogy. We may look at SHRDLU as a position on the spectrum of robot corporeality, which spans from *Astro Boy* to Voyager 2. It can be described as a borderline fictional robot, possessing a different kind of agency than a robot that exists only within a human-driven narrative. SHRDLU possesses an extended agency, influencing elements of its personal virtual world for one thing, but also acting through the eyes and brain of the human user. We do not even need to adopt a OOO perspective to state unequivocally that SHRDLU does, in fact, exist. For OOO the virtual/embodied robot distinction is not indicative of any degree of realness in two objects. Any such hierarchy of reality is rejected in OOO. As Harman says, all objects may not be real in the same way, but "they are equally *objects*" and therefore can be discussed under the same terms.[38] They are

---

[34] Terry Winograd and Fernando Flores, *Understanding Computers and Cognition*, quoted in Wolfgang G. Stock and Mechtild Stock, *Handbook of Information Science*, trans. Paul Becker (Berlin: De Gruyter, 2013), 56.

[35] Daniel Andler, "Phenomenology in Artificial Intelligence and Cognitive Science," in *The Blackwell Companion to Phenomenology and Existentialism,* ed. H. Dreyfus and M. Wrathall (London: Blackwell, 2006), 377-393.

[36] Maja Mataric, "What is a Robot? Defining Robots," *The Robotics Primer* (Cambridge, MA: The MIT Press, 2007).

[37] Ibid.

[38] Harman, *The Quadruple Object*, 5.

both real, but with a different way of being in the world. It is perhaps better to compare the *physically embodied* tortoise with the *electronically embodied* SHRDLU, but to bear in mind that each is equally an object. The difference between physically and electronically embodied robots is a distraction from learning about these relations and qualities in alien phenomenology. The sensory worlds inhabited by physically and electronically embodied objects are very different, but both require investigation.

There is at least one important difference between the two types of objects, however. In a sense, both the tortoises and SHRDLU are programmed to perform exactly the same interaction under identical circumstances. If one were to feed the same sample dialogue above into SHRDLU it would respond in an identical way, both in the movements of its robot arm and its linguistic responses. Similarly, if the robot tortoise is presented with exactly the same physical situation, with identical light sources and obstacles, it would be expected to behave in the same way each time. But this is not possible in an analogue world. No amount of prior planning could result in exactly the same experimental setup each time. SHRDLU is a closed system. In thermodynamics, a closed system is that which is self-contained and does not permit matter to transfer across a boundary, although it does permit the movement of energy.[39] The conditions are set and are not influenced by external variables. A digital system like SHRDLU is constrained by its programming and will always react in the same way no matter how many times an experiment is repeated. In contrast, an open system is influenced by external variables.[40] This is true of an analogue system like a robot tortoise, which depends upon things like the ground and is affected by variations in it. For this reason, the tortoises *seem* more autonomous. When encountering a new setting the tortoise appears to make a decision about its setting based on sensory data in a way that SHRDLU is not capable of, even though we know from Winograd's commentary that the system does in fact have a very complex means of making decisions. It may be that the fact that we can in principle explain everything about SHRDLU's performance affects our conception of its semi-autonomy.

SHRDLU's electronically embodied status means that it can be reproduced in entirely different media. My own interaction with SHRDLU came through a graphical 3D version of SHRDLU with an extra Java layer remediated by an early version of Windows,[41] certainly not the original medium. The hardware on which SHRDLU was originally run is now too antiquated for the program to run on contemporary systems without alteration, even though the original LISP programming may be retained. Consequently, newcomers to SHRDLU may never have access to the original version, to the object that was called "SHRDLU" by Winograd. If the program acts in a similar way, however, there is a certain sense in which the version does not matter to SHRDLU, which exists as a concept beyond its incarnation in a specific software-hardware system.[42] The concept might "run" on the media of many

[39] Michael M. Khonsari and Mehdi Amiri, "Fundamentals of Thermodynamics," *Introduction to Thermodynamics of Mechanical Fatigue* (Boca Raton: CRC Press, 2013): 11.
[40] Ibid.
[41] *SHRDLU*, Windows 7.
[42] One could argue that a distinction could be drawn between artefact and concept, as in Chapters 2 and 5, but this point has already been made and is not pursued here.

different human brains, many of which will give it different connotations, associate it with different ideas and have different opinions as to its veracity. But it is still ultimately the same concept and it exists independently of its media. The medium may be characterised as the extraneous details that make up the sensory qualities of an object – they change the way we sense the object at any given time, but they do not affect the withdrawn real object. Consequently, interaction with any version of SHRDLU helps to build our knowledge of its sensory world.

Postphenomenology is useful here as a contrast to OOO because most of the very close relations between SHRDLU and other objects in its world are confined to within the computer – and, indeed, when it was first made that computer would have been very limiting. Flipping postphenomenology around so that the nonhuman is primary is distinctly necessary in this case study since the relations are complex and interrelated but enter into postphenomenological relations with each other as well as with humans. Ihde's postphenomenology can be supplemented by alien phenomenology to result in extensive theorisation of nonhuman relations.

SHRDLU's Umwelt and mind

Postphenomenology offers insights into various types of human-nonhuman interaction, and specifically how that interaction with the nonhuman extends the human's sensory capabilities. It is particularly concerned with technology-human relations. It is prominent in contemporary philosophy of technology, and provides a useful contrast with OOO. It also highlights the multistability of technologies, which will be discussed later. This chapter offers a variation on the theme of postphenomenology to articulate the qualities of the digital computer. But a slight reframing of the argument presented here could also position SHRDLU as an Umwelt-user and possessor of mind in the panpsychist sense, or simply as the possessor of experience. As the introduction to this thesis argues, the lack of interest in the nonhuman side of the sensory relationship means that practically any theory of nonhumans or technology can offer unexplored insights into this question.

Postphenomenology is a very humanist philosophy in comparison to many other contemporary theories of the nonhuman. We could begin where Chapter 3 finishes, by asking which parts of SHRDLU could be said to possess an Umwelt, or perhaps even a mind. SHRDLU's material status is shaky; it can certainly be located within a computer, if necessary, as a set of programs coded in LISP. But of course, it may be copied onto other computers and has even been tweaked to run on different operating systems. We might ask whether only those embodied aspects of SHRDLU can possess a mind or Umwelt. But the rest of this chapter will focus on SHRDLU from the postphenomenological view. In doing so, we might consider a heuristic to be employed here. We do not need to argue about the ontological status of SHRDLU because postphenomenology is primarily interested in nonhumans only insofar as they alter human sensation or perception. Let us suspend for now the question of

whether SHRDLU has an Umwelt, or a mind, or the extent to which it is independent of its medium or whether it exists at all.

Postphenomenology

Comparisons between postphenomenology and ANT have been drawn by Peter-Paul Verbeek:[43] they are both non-essentialist and are grounded in a relational model. They both try to find a resolution to the insupportable metaphysical gulf between humans and nonhumans, but Verbeek thinks that postphenomenology is more "nuanced" because it goes beyond locating associations between entities and describes those associations as well.[44] Robert Rosenberger and Verbeek have called postphenomenology an "empirical philosophy".[45] This is the pragmatic reason for a postphenomenological approach with SHRDLU. The more important one is that postphenomenology gives us particular insight into the very special, very privileged relationship between human being and technological construct. Don Ihde's postphenomenology reveals a model of human extension through technology that sublimely articulates the *feeling* of being changed by technology. Postphenomenology is an important movement in contemporary philosophy of technology and it seems necessary to present it here and critique it alongside those ideas that decentre humans.

This section will begin by briefly outlining Don Ihde's postphenomenology with some conjecture as to how the categories of experience that he describes might be used to articulate machine experience. Next, there will be a discussion of SHRDLU with relation to these categories, with particular reference to the postphenomenological notion of multistability. Next will be an account of what postphenomenology can lend the notions of cultural agency that have proved problematic in previous chapters.

Don Ihde self-consciously brings phenomenology into studies of science and technology.[46] His postphenomenology explores how we can consider technologies to take on the role of human intentionalities. He does not reject the ideas of previous phenomenological philosophers, but he argues that ideas must evolve in order to be relevant to other disciplines.[47] Theory has evolved over time, but perhaps more significantly technology has changed a great deal. The invention of new artefacts requires discourses of technology to enter into relations with different fields and thinkers. It may be that today the role of

---

[43] Verbeek, "The Acts of Artifacts," *What Things Do*, 147-172.
[44] Ibid., 166.
[45] Robert Rosenberger and Peter-Paul Verbeek, "A Postphenomenological Field Guide," in *Postphenomenological Investigations: Essay on Human-Technology Relations*, eds. Robert Rosenberger and Peter-Paul Verbeek (Lanham, Maryland: Lexington Books, 2015), 30.
[46] Don Ihde, "Introduction: Postphenomenological Research," *Human Studies* 31, no. 1 (2008).
[47] Don Ihde, *Postphenomenology and Technoscience: The Peking University Lectures*, (Albany NY: State University of New York Press, 2009), 40.

technology in changing our perceptions is more apparent. An example is the moral panic caused by young people overusing screen media such as online videogames. The rate of technological progress is so rapid that new generations have vastly different mediated experiences from their parents. So while the teenager playing a videogame six hours a day feels as though she is an active participant in a virtual world, society might tell her that her perceptual experiences are less "real" than those of a non-gamer. Yet the fantasy world inside the computer is real. We can tell, because it has an effect on the teenager and others. It also has an effect on the computer, the internet, and Western civilisation. It may enter into all of these relations, but in order to enter into a relation with the teenager, it must be mediated by the computer screen.

Ihde's phenomenological ideas explain how humans can have technologically mediated intentionalities. The technology becomes one of the key actors in Ihde's theory. For example, Galileo used a telescope to detect four of the moons of Jupiter, which are invisible to the naked eye. Ihde claims that this was a significant change in the way humans perceive the world, because it required people to trust in a machine to convey accurate visual information.[48] This was one of the most important moments for the shift to what Ihde calls instrumental realism: a belief that technology can augment and enhance our senses, and that those sensations are reliable. Through this transformation, what is meant for a human to sense the world changed. The object of study is not just the relationship between human and world, but the ontological trinity of human, world and technology.[49] Technology is not a passive translator of information but participates non-neutrally in this relationship. After developing these ideas over a long period of time, Ihde and colleagues eventually gave them the label of postphenomenology. Postphenomenology has become a very influential way of discussing technology because it provides tools to explain how we can experience phenomena through a machine that changes sensory input.

Ihde describes three types of relationships in which technology mediates human intentionalities. These are explored in detail in his 1990 book *Technology and the Lifeworld: From Garden to Earth*.[50] He uses mostly visual examples, so the technology mediates the human intentionality of "seeing". His analysis could also be applied to other intentionalities, such as "hearing" (for example, with a hearing aid), "remembering" (with a personal organiser), "perception" or "cognition" (when one is under the influence of psychedelic drugs). But it makes sense to begin with visual phenomena, following Ihde. Ihde has elsewhere made a study of the dominance of sight over other senses in science. The increasing tendency to portray data in visual form, or "science's visualism", [51] has resulted in a privileging of sight over the other senses. Therefore, to introduce Ihde is to focus on the visual and to use examples of sight intentionalities. But machine "vision" is culturally constructed and bears limited resemblance to human vision, so in the analysis presented

[48] Don Ihde, *Bodies in Technology*, (Minneapolis: University of Minnesota Press, 2001).
[49] Gert Goeminne and Erik Paredis, "Opening Up the In-Between: Ihde's Postphenomenology and Beyond," *Foundations of Science* 16, no. 2-3 (2011).
[50] Don Ihde, *Technology and the Lifeworld: From Garden to Earth* (Bloomington: Indiana University Press, 1990).
[51] Ihde, *Bodies in Technology*.

here other more alien sensory experiences will be reported. In contrast to the conventional examples of postphenomenological relationships below, this chapter shows how these ideas may result in a bidirectional account of machine sensation, which brings postphenomenology more in line with Harman's ontology. Each account of human intentionality will be followed by a demonstration of how the same ideas might apply to technological objects. Simple examples will be used, since machine intentionalities are comparatively under-theorised. Simplicity may compensate for our ignorance of the nature of alien sensation, which is important because in this experiment we run the risk of making claims about the nature of machine sensation. We don't know the nature of the sensation, but we can create a model for talking about it. After beginning the experiment we will turn back to SHRDLU.

Ihde identifies three types of relationship between humans and technology that change (or improve) our experience of the world. The first is the embodiment relationship. This applies to technologies that become part of the body to enhance certain intentionalities. For example, the intentionality of seeing is enhanced by wearing eyeglasses. The body's capabilities are transformed by the artefact. The object becomes a part of the way the human sees. This process is not limited to simple machines. Hearing aids exist in the same kind of relationship as eyeglasses, but are much more complex.[52] Merleau-Ponty's blind man's cane is an embodiment relationship, as it is a direct means of extending the tactility of the body.[53] Ihde depicts this relationship as:

(human-technology) → environment

The technology is taken into the human to permit the experience of the world. The object is hardly noticeable, and in fact performs better the less obtrusive it is. If a technology is good, it "withdraws".[54] It becomes "quasitransparent".[55] (This withdrawal is similar to the withdrawal in OOO, or to Latour's "black boxes"[56], in that the means of accessing the technology are removed from external actants, although Ihde's use of the term occurs instant by instant whereas Harman's withdrawal is eternal. The terms are not to be confused in origin or in application.) Ihde poignantly describes withdrawal as a fundamental contradiction inherent to humans' use of machines. We want the power that comes from the technological transformation, but we secretly desire "to escape the limitations of the material technology."[57] Thus, embodiment relationships reveal the human-technology hybrid to be a site of ambivalence, demonstrating the human craving for direct experience.

This craving is revealed to us through the narrative of verisimilitude in the history of visual images.[58] Many art historians have identified a persistent tendency to produce more and more realistic visual media throughout history. This culminates in the ideal of virtual reality (VR), a concept that remains elusive to us and will remain elusive for the foreseeable future.

---

[52] Ihde, *Technology and the Lifeworld*, 74.
[53] Ihde, *Postphenomenology and Technoscience*.
[54] Ihde, *Technology and the Lifeworld*, 73.
[55] Ihde, *Postphenomenology and Technoscience*.
[56] Bruno Latour, *Science in Action* (Cambridge: Harvard University Press, 1987).
[57] Ihde, *Technology and the Lifeworld*, 75.
[58] Crary, *Techniques of the Observer*.

A true VR mediation would "pass for or merge seamlessly with perception itself",[59] taking the place of not only visual stimuli but sound, smell, fatigue, hunger and all other affects. It would not even be possible to detect the pressure of the headset on one's skin. This ideal remains infeasible for VR researchers in the short term. In Ihde's terms, we seek technologies that will transform our intentionalities, but we wish to erase all traces of the mediator and become utterly embodied. Yet there are also alternative narratives of media. Art historian Jonathan Crary points to places in the history of images where humans have craved mediation through images.[60] Impressionists and post-impressionists, for example, deliberately distorted visual stimuli, leaving the greater truth behind their work to be interpreted by the viewer. These are pieces that disdain the natural human desire for embodied mediation and require a hermeneutic process (see below).

A comparison to the human-centric embodiment relationship for considering machine phenomenology is necessary at this point. This relationship must feature a technology making use of a human actant to perceive the world. In the case of a computer, an embodiment relationship is formed between typist and keyboard. To rephrase Ihde's model for this perspective:

(technology-human) → world

The keyboard is, along with the mouse and the network, one of the most important ways that a computer experiences the world. The depression of certain keys acts as stimuli to computer receptors. Each receptor has, depending on context, a different effect, rather as acetylcholine acts only upon acetylcholine receptors, yet it may have multifarious effects on different tissues. For example, sometimes my laptop asks me to enter a password to access the university Wi-Fi, since it has detected that it cannot connect to the internet and perform its tasks without it, and since it has also failed to remember the password. It persuades my brain and fingers to work by telling me that it is having a problem, in language that I can understand. It thereby extends its own capabilities using my body. The computer needs the human user to depress keys in order for it to function, and makes use of human hands (and a human brain) to fulfil its purpose. Other objects involved include wires, key codes and a keyboard driver.

Using the word in the sense of "reading" or "interpretation", Ihde next discusses hermeneutic relationships. This category refers to objects that transmit data or information which must then be interpreted by the human user. The most basic example in Ihde's account is of a human reading a chart. The chart is inscribed by a human, but alone it is useless. It requires a human with the knowledge and understanding to read and interpret it. It refers to events that occur beyond its material self.[61] Another example is the thermometer that detects the temperature of the air so that the human reader can hermeneutically receive sensory

[59] K. Hillis, "The Sensation of Ritual Space," *Digital Sensations: Space, Identity and Embodiment in Virtual Reality* (Minneapolis: University of Minnesota Press, 1999), 70.
[60] Jonathan Crary, *Suspensions of Perception: Attention, Spectacle and Modern Culture* (Cambridge, MA: MIT Press, 1999).
[61] Ihde, *Technology and the Lifeworld*, 81.

information. The machine sensation is transcribed into a written or visual form which is then reconstructed by the human. This is depicted as:

human → (technology-world)

The technology is experiencing the conditions of the world. It has a process by which it acquires information, such as through sensors or human input. But the human user interpreting the technology has no direct access to the information that the machine reveals. The human must have an implicit understanding of what is indicated by the machine. In the thermometer example, it is no use to simply have a machine that tells us the temperature is thirty-five degrees centigrade. It requires a human user to understand what the value "thirty-five" means. On a weather report, it indicates an uncomfortably hot day, while on an oral thermometer it is a dangerously low temperature for a human body. And as Ihde points out, a native of North America would also need to know how to convert centigrade temperatures into Fahrenheit values before they could be read properly.[62] As in embodiment relationships, we can extend this idea to more complex examples. The Parkes Radio Telescope measures hydrogen gas densities in other galaxies.[63] The phenomenon under investigation is both distant and invisible to human senses, but the telescope is capable of displaying the data in visual forms, with different colours to indicate densities. The machines translate data into a form that may be understood by human beings, so long as they possess the knowledge to interpret the images. Note that it is explicitly implied here that nonhuman sensation not only exists, but that it is possible for humans to gain information from it. In other words, the human can understand what occurs in the telescope's sensory world, to a limited extent.

Examples of computers interpreting human actions abound. The writing of code or algorithms to control processes results in human texts that the computer must interpret and act upon. The computer depends upon the accuracy of the human typist to reduce errors in the human-machine process of semiosis. While writing code, the omission of a semi-colon may result in a non-functioning program. The computer relies on the human to produce good, comprehensible code. After being converted into machine language, the instructions become a facilitator of machinic action:

technology → (human-world)

Poorly-coded programs rely overmuch on human intervention or make it difficult to make future alterations. Well-coded programs are so efficient and effective that they are blackboxed and become invisible. When SHRDLU performs as hoped, its programming is invisible to all but the most specialised user because no errors in comprehension or processing are evident, betraying the tricks that Winograd may have used to program the system. A corollary exists in the practice of providing automated voice-controlled telephone services. A simple, well-organised service relies on responses of only "yes" or "no", and allows the caller to complete their business swiftly. Yet all too often the service requires users

[62] Ibid., 85.
[63] B. Koribalski, K. Jones, M. Elmouttie and R. Haynes, "Neutral Hydrogen Gas in the Circinus Galaxy," *Australia Telescope Outreach and Education, CSIRO*. http://outreach.atnf.csiro.au/images/astronomical/circinus.html (accessed June 27 2017).

to provide complicated answers that are difficult to decipher using a voice-to-text program. In these cases the caller is typically referred to the human operator. The goal in any kind of automated system is to remove the need for human intervention. For example, the goal of a personalised Google algorithm is to remove the need for the user to refine search topics. Google learns about the user's location and other information to provide more relevant local search results so that the user does not need to sift through too much material. This also allows Google to target advertisements based on what it knows about the user. The ability to interpret is hindered by human errors, such as entering one's location as "Austria" instead of "Australia".

Alterity relationships occur between humans and technology that we consider to be an "other" to which one can relate. The term comes from Lévinas's notion that one person is radically different from any other.[64] In Ihde, the "otherness" is extended to semi-autonomous machines like certain robots, although he frequently qualifies it as "quasi-otherness". [65] The relationship is depicted as:

human → technology-(-world)

The parentheses indicate that although the relation is primarily between the human and the technology, there may also be relations through the technology to the world. One of the difficulties for Ihde in describing this relationship is that the concept of technology as an "other" varies. What it means to anthropomorphise a machine is not uniform across cultures.

In theory, an alterity relationship should be bidirectional even in a non-OOO sense. It is a relationship that acknowledges the otherness of technology. A true alterity relationship would be the same from both sides of the interface. These kinds of relationships can also apply to computer-computer relationships that occur within a network as well as human-human relationships. Yet if we include these as alterity relationships, then surely we must also account for human-human interaction. The inherent problem of the concept of alterity relationships is discussed below. Perhaps the best way to define the interaction is using an unqualified account of object relations, regardless of anthropomorphic status:

object A-(-world) → object B

object B-(-world) → object A

The relationship is bidirectional. But it is also asymmetrical, as argued by Harman. Object A's ability to impart knowledge of the world to Object B might be different from Object B's ability to impart knowledge to Object A. For example, a desktop computer and a smartphone may be in an alterity relationship, connected by a cable. But each can only impart certain types of sensory experiences to the other.

Embodiment, hermeneutic and alterity relations are the three major means by which technology mediates human intentionalities in Ihde's construct. Added to this are also

[64] Ihde, *Technology and the Lifeworld,* 98.
[65] Ihde, *Bodies in Technology*, 81.

background relations, which are relations between humans and automatic technological processes, such as climate control. The background relation may only briefly be directly set up by the human, for example the setting of a thermostat. For most of the time, the technology operates as a "present absence"[66], affecting the human's environment but without direct contact of "focal"[67] technologies (an attitude reminiscent of Albert Borgmann – see Chapter 1).

Ihde's categories represent a significant departure from critical theories in which objects are open for interpretation, particularly with alterity relationships. He flattens hierarchies of objects: asking a question of a robot designed to tell where to find things in a store is as worthy of attention as objects capable of a multiplicity of tasks such as a personal computer. Doubtless the relationship between computer and human user is determined by the particular application in use at any one moment. Some applications are hermeneutic; for example, a web service that tells the user about the temperature in another city. This application requires human interpretation to become useful. But in other states other kinds of relationships may be formed with the computer.

Interpretations of machines by humans are sometimes beyond the engineer's intentions. A computer or a robot is a cultural artefact as well as a semi-autonomous being. It is, as Ihde frequently points out, inextricably linked to the culture that produced it. It is sometimes not as easily interpreted as a painting or a film, but it is still a product of our cultural imaginary and can be a mediator of ideas, norms, taboos, practices and identities. For example, a social robot may be designed to be relatively independent and to interact freely with human beings in, as Ihde would say, an alterity relationship. But social robots represent certain cultural ideas. Chapter 5 specifically notes a tendency for "feminine"-looking robots to be objectified by their creators and users. When we talk to a "feminine" social robot, an "other", we are also receiving coded messages about social interaction and our relationship with robots, and also about relationships between women and society. Is this, then, simply a very complicated hermeneutic relationship? It is at this point that Ihde's categories become entangled.

Is it possible for a complex machine like a computer to form an embodiment relationship with a human? If a Youtube user can access a video taken with a smartphone of protests in Cairo, is the computer then extending her vision? The mediating qualities of the screen are a problem because the video is never as reliable and firmly embodied as the eyeglasses – after all, the footage is only ever a brief snippet and is a poor replacement for being present at the scene. But Ihde is aware of the significance of the frame around mediated experiences. In describing Galileo's observation through a telescope, he points out the ontological leap that must be taken to accept the screened space as real, and compares it to television footage that, by being framed, immediately seems less real to the viewer.[68] We are aware that a video on CNN may not be showing the whole story. Perhaps then the user is in a hermeneutic relationship with the footage. It requires interpretation, context and prior knowledge to combat the disembodiment, just as a thermometer needs a numerical scale to allow us to

[66] Ibid., 109.
[67] Ibid., 112.
[68] Ibid.

interpret the expansion of mercury and knowledge of, for example, the temperature that indicates a fever in the body. The question is an old one in the study of media: is it better to allow the viewer to become immersed and to pretend that there is no mediation, or to provide lots of interpretation as compensation? We can see by this that Ihde's categories rely heavily on how we socially construct embodiment.

Here the concept of multistability is useful, a means of acknowledging that relations between humans and machines vary by the instant and by the person. Ihde's approach to this idea begins with an image such as a Necker cube – a flat image of black lines which can appear to the viewer as one of either two cubes, but never both at the same time (and, as Ihde says, can be seen as other shapes as well).[69]

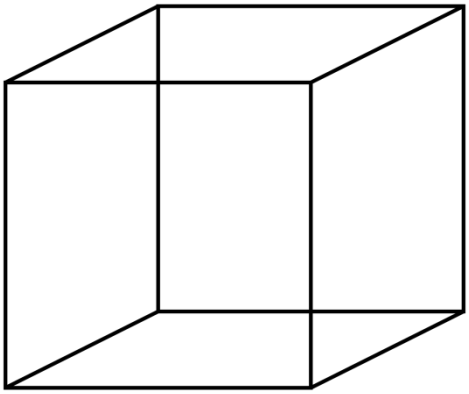


Figure 20: Necker Cube

This becomes a metaphor for the relations between humans and technology. A technological artefact may be used in different ways by different people, in different contexts, and will mean different things to different people, often beyond the intentions of their creator. Robert Rosenberger describes the ways that a public bench becomes not simply a place to sit, but also a bed for the homeless: "[i]n this way, a bench is a multistable technology with both bench-as-seat and bench-as-bed stabilities."[70] Evan Selinger, however, criticises Ihde's heightened ambivalence about artefacts and context, pointing out that he has sometimes been accused of an overly apolitical approach to technology.[71] Multistability can be a dangerous concept when it is applied to objects that pose significant risk to human life. Furthermore, sometimes multistability may be desirable but for practical reasons it may not actually occur, perhaps because there is not enough time for humans to find novel applications for the artefact. Nevertheless, the concept of multistability prevents an overly deterministic view of technology and allows for the demonstrable discovery of novel ways for humans to use nonhuman objects.

---

[69] Ihde, *Technology and the Lifeworld*, 145.
[70] Robert Rosenberger, "Multistability and the Agency of Mundane Artifacts: from Speed Bumps to Subway Benches," *Human Studies* 37 (2014), 378.
[71] Evan Selinger, "Normative Judgement and Technoscience: Nudging Ihde, Again," *Techné* 12, no. 2 (2008): 120-125.

Shifting categories of human-technology relations are particularly evident in screen media, such as in the first-person shooter game *Halo 4*.[72] Games particularly complicate these categories as the player must engage in all of them and sometimes all at the same time. Because videogames are not completely immersive, the human player cannot be totally embodied and take on the role of the digital Master Chief protagonist. Nevertheless, the human does have some embodied experiences. When the player moves Master Chief's head, the view of the virtual world shifts as though we are looking through his eyes. When struck a heavy blow, the console's controller shakes and rumbles, to give us a sense of what Master Chief is experiencing. These simple innovations are designed to extend human sensory systems into virtual spaces.

In terms of ANT, quasitransparency is only achieved when both human and virtual actants are cooperating. The game is successful because the player's goal and the character's goal match and because each is capable of affecting the other's agency in a predictable way. This is particularly evident in a game like *Halo 4* because the virtual actant is an anthropomorphic avatar. The player views the world from behind Master Chief's eyes. Consequently, the player's action is immediately transferred into Master Chief's "muscles" and Master Chief's "eyes" transmit information immediately to the player's eyes. It is therefore essential that player and character have allied goals, and this is typically built into the narrative form of videogames as well as being hard-wired into the code. Master Chief has a goal: for example, he must fight his way through hostile terrain to rescue his comrades. Fortunately, the player's goal is aligned with the character's goal through the embodied mediation of the avatar. The typical player wants to move Master Chief in a skilful way to overcome obstacles, such as enemies or puzzles, and thereby progress in the game. The impetus behind the decision to move through the jungle is different. Master Chief wants to perform a rescue. The player (probably) wants to beat the level and advance the story. In other words, each translates the other in order to bring their goals into alignment. Because player and avatar are allied, neither resists the other's decisions and the result is a satisfying feeling of embodiment. Because of this alliance, the game does not require Master Chief to disobey the orders conveyed to him by the player, and he performs actions corresponding to the player's gestures (so long as other actants, such as the controller battery, are also enrolled, to put it in ANT terms). The player, in turn, uses gestures to try to aid Master Chief in his mission. The relationship between player, avatar and character is not always as straight-forward as this example and varies in degree.[73] However, the logical alliance provides a victory/loss model that is resisted in relatively few videogames. A notable exception is the common tactic in activist videogames to complicate the implied cooperation between player and avatar. For example, in activist videogame developers Molleindustria's online game *Operation: Pedopriest* the "aim" is to help rapists escape justice.[74] Clearly, the real aim of the game is to foster awareness of something that the developer considers to be an important social issue.

---

[72] *Halo 4*, developed by 343 Industries (Microsoft Studios: 2012), Xbox 360.
[73] Martin Gibbs, Marcus Carter, Michael Arnold, "Avatars, Characters, Players and Users: Multiple Identities at/in Play," paper presented at *OZCHI'12*, Melbourne, Australia, November 26-30 2012.
[74] *Operation: Pedopriest,* developed by Molleindustria (2007), PC.

(Bogost calls this "procedural rhetoric."[75]) In this example, the player's embodiment is limited because the avatar fails to integrate them into its network. The sense of identification with the avatar is limited at best. This is not a problem for Molleindustria, but a AAA Xbox 360 title must aim to cooperate as much as possible with the user in order to achieve a sense of immersion, and therefore for the game and the Xbox 360 to succeed in their goals of being bought and enjoyed. The *Halo* series has been very successful, and can only be played on Xbox consoles, so Master Chief's successful formation of an embodiment relationship with various players indirectly benefits the Xbox and Microsoft brands.

Intuitive hermeneutic and alterity relationships are also vital to the success of the game. As an example, we cannot feel pain as Master Chief loses ground in a fight, so *Halo 4*'s interface provides us with a health bar. Master Chief cannot continue on his mission if he dies, so it is important for the game (which is strongly allied with Master Chief) to convey the state of his health to the player through a hermeneutic relationship. (The actual experience of Master Chief's death could be considered an embodiment relationship.) The player must read and interpret the health bar, and the other diegetic and non-diegetic devices on the interface, and the game must provide accurate data. Furthermore, we could say that the player has an alterity relationship with Master Chief. He is undoubtedly an entity different from the player who essentially exists to guide us through a virtual space and story. His lifeworld is not the player's lifeworld. Moreover, Master Chief enjoys brief periods of autonomy in cut-scenes during which the player has no control over him. Ihde's categories are open to far more interpretation than they at first appear, and individual relationships may both move between them and transcend them.

The human experiences the game, and the game experiences the human through the input of the controller. The depression of controller buttons corresponds to certain pre-programmed responses in the videogame console. For example, shifting one of the joysticks to the left causes Master Chief's head to rotate, allowing the player (and Master Chief) to view a different part of the virtual environment. In other words, the machine senses the mechanical change in the controller which is then sent wirelessly to the console. Here, the sensation is "perceived" and the console interprets the movement. The activation of code that changes the view of the virtual environment is only enacted after this interpretation takes place. This analysis allows a glimpse into the curious digital world of the console and gives an insight into the agency of the machine. We can also see how the interaction is, in Harman's language, bidirectional and asymmetrical. Master Chief's sensation of the player occurs concurrently with the player's sensation of Master Chief, but each occurs through different means and presumably results in very different perceptual experiences. SHRDLU too possesses several means of mediating human intentionality, and it occurs in embodiment, hermeneutic and alterity relations. This chapter will now turn to describing SHRDLU through a bidirectional postphenomenological analysis.

---

[75] Ian Bogost, "The Rhetoric of Video Games," in *The Ecology of Games: Connecting Youth, Games, and Learning*, ed. Katie Salen, The John D. and Catherine T. MacArthur Foundation Series on Digital Media and Learning, (Cambridge, MA: The MIT Press, 2008), 117–140.

SHRDLU and Ihde's categories

It is important to bear in mind here that Ihde's categories are ways of accounting for the extension of human sensation or perception by technology. Similarly, when we apply these categories to relations between SHRDLU and other nonhumans it is worth noticing how SHRDLU extends or transforms aspects of those nonhumans, and vice versa. This means making assumptions about the intention and agency of the human user, so there has been some attempt here to represent various possible uses to which SHRDLU might be put, in accordance with the notion of multistability. For example, the purpose of SHRDLU (from a human perspective) might be to allow the user to have a very detailed, albeit dull, conversation with a computer in a "natural" way. Another purpose is to test the hypothesis that NLP is only possible if the AI is grounded in the world. With an emphasis on the visual aspect of SHRDLU's presentation, it might be inferred that the user employs SHRDLU to gain a representation of another world, to see what SHRDLU sees. SHRDLU could also be employed as a voice-controlled means of moving articles around in a warehouse, if it were allowed to form relations with the right kinds of equipment. Yet another perspective reveals SHRDLU to be a means of progressing Winograd's career, and later to provide Dreyfus with an example of AI limitations. To see how SHRDLU extends and modifies these human programs of action, it is necessary to first delve further into the world of SHRDLU.

The first thing we must consider in building a picture of SHRDLU's environment and relations therewith is the wider external world. This is the best way we have of learning about its qualities and relations, and therefore of asking SHRDLU about its sensory world. SHRDLU was built at MIT initially as a part of Winograd's doctoral research. The conditions under which Winograd was allowed the equipment and time to construct the system are material to its existence. Similarly, it is important to take a broad view of AI when considering this question. Perhaps SHRDLU would never have existed had there been no optimism around computer language use born of things like ELIZA and translation programs, or no trend toward the development of microworlds in AI. And of course, SHRDLU's existence in turn has had a significant impact on subsequent AI projects. All of these objects that are external to the computer world in which SHRDLU exists – the lab, the university, Winograd himself and the environment of AI at the time – are objects with which SHRDLU has significant relations, and cannot be discounted when we ask what it is like to be SHRDLU.

Of direct interest though is the internal state of the system. SHRDLU has a high understanding of everything in the block world but at the cost of that world being tightly constrained. The closed nature of the system ensures that the microworld is tractable and always performs the same actions in the same way, and the constrained nature of the system permits all experiments to be predictable. To ask, then, what it is like to be SHRDLU is to have a slightly better chance of success than to ask what it is like to be Grey Walter's tortoises, since all factors are contained and external variables are minimised. Yet the internal consistency evident to a human interlocutor conceals SHRDLU's way of making decisions, which is quite complex. The representation on the screen conceals the vast number of objects under the

surface. SHRDLU is an assemblage of processes which, combined, achieve a goal, as in Marvin Minsky's portrayal of a system.[76] We can set out to create a "builder" program – a program that aims to build a tower out of blocks.[77] But the program requires sub-programs, which then require other sub-programs beneath it. Every time SHRDLU acts in response to a question it engages with a vast structure of sub-programs that lead it to make a decision about what it should say and do.

We can interact with SHRDLU in a way that is relatively simple for humans, but only because of SHRDLU's immensely complex programming. SHRDLU's understanding of language is described by Winograd as being constituted by three layers:

- A syntactic parser which works with large-scale grammar of English;
- Semantic routines which embody the knowledge needed to interpret the meaning of words and structures; and
- Cognitive deductive system for exploring the consequence of facts.[78]

These systems, in addition to the block world and the set of programs for generating English responses, constitute SHRDLU's world, and at a deeper level it is also constituted by LISP, logic, and the fundamentals of computing. These are all real qualities that define SHRDLU's way of being in the world.

Any action by the system is the net effect of a vast array of interwoven actants. This diagram (Figure 21) gives some insight into the complexity of the overall program:

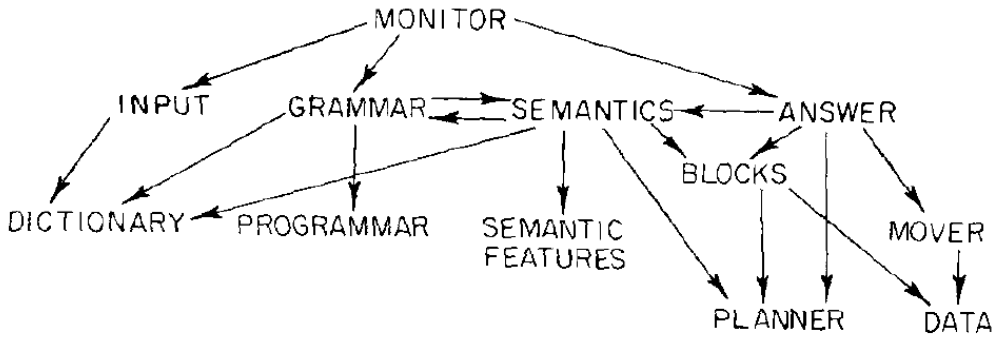


Figure 21: Winograd's diagram depicting the interrelating parts of the SHRDLU program. Titled "Organization of the System".[79]

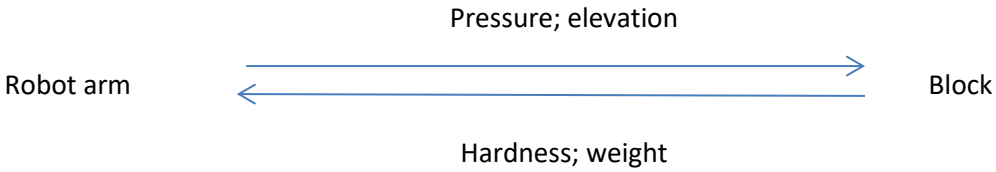
PROGRAMMAR is the language created for expressing the details of grammar in the system, and PLANNER is a language for problem-solving procedures.[80] MONITOR is a program that calls the basic parts of the system.[81] The other parts of the system are named for the function that they perform. Winograd's 1972 paper on SHRDLU goes into detail about each of the

[76] Marvin Minsky, *The Society of Mind* (London: Picador, 1988).
[77] Ibid., "1.4: The world of blocks."
[78] Ibid., 21.
[79] Terry Winograd, "Understanding Natural Language," *Cognitive Psychology* 3 (1972), 5.
[80] Ibid., 4.
[81] Ibid., 5.

components, and explains which performs tasks that may not be obviously accounted for. For example, the ANSWER component is responsible for remembering discourse as well as for creating responses. It can be seen from this that the internal world of SHRDLU is one constituted from many disparate parts and that the seamless unified responses that it offers to questions and statements are something of an illusion. As Douglas Hofstadter writes, "[t]he program is like a very tangled knot which resists untangling; but the fact that you cannot untangle it does not mean that you cannot understand it."[82] SHRDLU is made up of many different parts, each of which is dependent upon layers of sub-programs.

But we must remember that the presence of many different parts does not reduce the wholeness of SHRDLU as an object, or the wholeness of the ANSWER component, or the wholeness of the robot arm. So what is it like to be the robotic arm in SHRDLU? This question has the advantage of offering conceptual symmetry with the robot tortoises and other objects of study in this thesis. It is relatively easy to consider the robot arm's relations with other objects without asking where the object of study begins and ends, which is not necessarily true of the object or assemblage of objects signified by the word "SHRDLU". When we ask what relations exist for the robot arm we can identify a pleasingly diverse list of objects: the blue block, the block-world table, the human user and his or her input, the syntactic parser, semantic routines and cognitive deductive system, LISP, previous block movements, the computer and the electrical current, *et cetera, et cetera*.

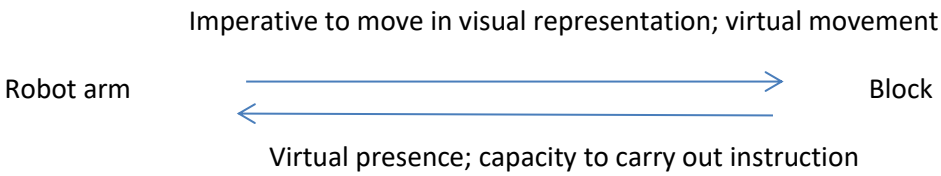
The robot arm's most obvious contact is with the blocks and world of blocks. But it is of a different nature than the contact between a real robot arm and block world. In an interaction of a physically embodied robot arm and block we might expect to see a bidirectional relationship of this sort:



The robot arm imposes pressure on the block and causes it to rise off the surface on which it was resting. Simultaneously, the robot arm senses the hardness and weight of the block.

Instead the bidirectional relationship in SHRDLU looks a little more like this:

[82] Hofstadter, *Gödel, Escher, Bach: An Eternal Golden Braid*, 628.



The robot arm compels the block to move (virtually) to a new location. Simultaneously, the block is tractable to virtual movement by the robot arm. Doubtless there are many other factors that are not immediately apparent to the human. But in each case, the bidirectional relation is asymmetrical. Each object gains only a caricature of the other object, as Harman would put it. We cannot be certain of the nature of this caricature, although Winograd's technical reports may offer clues.

Thus the sensory equipment given to the robot arm is primed for interaction with the blocks. Through the actions and interactions of its code it senses the fact that it is holding a block and it senses the fact that it puts it down (we know this because it informs the system, which remembers it and can tell the human user about it later). Notice that this interaction is quite different from the sensual object that would be formed between a physically embodied robot arm and block, although SHRDLU attempts to model this interaction with realistic physics. This interaction can be viewed through the visual representation that sits alongside the teletext interaction, but the true robot arm exists within the system, sensing blocks in a thoroughly non-visual way. Apart from the block interaction the robot arm is sensitive to instructions from the rest of the system which are related in machine language. These are symbolic interactions that it is hard for us to imagine. What is it like for part of a system to sense symbolic instructions from another part of the system? This is one of those thoroughly alien interactions that draw the mind away from the contrived anthropomorphism of SHRDLU as a whole and emphasise the impossibility of ever truly breaching the inner nature of a nonhuman object.

The robot arm is inextricably linked with each of the system components. It is part of a large assemblage of objects united in the goal of *being* SHRDLU, as in Marvin Minsky's explanation of assemblages of components needed to create a virtual block tower builder. Our sensual impression of SHRDLU is far removed from its true internal state, which is whole despite consisting of interactions between the objects in the assemblage. All of our information about the world of SHRDLU is conveyed through the graphical display and the text that appears on the screen. These give the impression of an agent that does not exist, a unified "player" agency that resembles in many ways a human player using the blocks and responding to questions. And this is, of course, Winograd's intention. SHRDLU the "builder" to which we relate is a sensual object, a human thing, our caricature of an alien world (or at least, the caricature available to those of us who are not computer scientists engaged in working on SHRDLU).

Another interesting relationship relevant to this discussion is that which exists between SHRDLU and the English language. Surely if SHRDLU is in a relationship with a language, it is an embodiment relation. Words are firmly embedded in SHRDLU, and it actively manipulates

language. We can say with some confidence that the English language is an object. It is an entity with identifiable qualities that distinguish it from other entities. It clearly enters into relations with other objects such as texts, thoughts and utterances, and is affected by them as well. It changes over time but remains recognisably the English language. And it is in a bidirectional relationship with language performing objects like SHRDLU as well. SHRDLU is clearly a user of the English language. It is shaped by qualities of the English language such as its syntax, signifiers and alphabet which are essential elements of its functioning and purpose. SHRDLU takes on these qualities from its association with the English language, as though one were physically moulding the other. In OOO there is always a reciprocal (but different) interaction. What effect could SHRDLU, or any language performing object, have on the English language? Well, firstly SHRDLU happens to represent a significant innovation into the study of language, still shaping the way that we think about how computers can use language and even how language works in humans. But secondly its role as a language user is similar to that of a human language user in that by communicating by using particular words and phrasing it has a tiny but not insignificant effect on the English language as a whole. Regardless of whether or not SHRDLU is a true user of language, SHRDLU as a text is in a bidirectional relationship with the English language, as when a note contains a word that is unfamiliar to the reader, causing the reader to integrate the word into her or his vocabulary and give that word a minutely greater prominence within the English language. SHRDLU has a relationship with the English language in the same way that Walter's texts on tortoises affect the field of cybernetics and robotics, not just as media through which Walter speaks.

Given this overview of SHRDLU's world, it is easier to see how SHRDLU forms the types of relations that Ihde and others describe. Its embodiment, hermeneutic and alterity relations will be described here. The one kind of relation not discussed here is background relations. Background relations are an addendum to Ihde's original three categories, and they are not a great factor in SHRDLU's understanding of the world. That is, perhaps SHRDLU is often in a background relation with humans, but it is always in a non-background relation with *something*. If one walks away from the computer leaving SHRDLU running, it is still interacting with the rest of the computer. It might be possible to argue that SHRDLU's influence on the development of AI research constitutes a background relation, but this chapter discusses that issue in different terms.

Alterity and SHRDLU

The nature of alterity in SHRDLU's relations with humans is highly dependent on the writer, as is the case in all examinations of AI (one might even say that all writing on alterity is highly dependent on the writer). Clearly, the goals of the SHRDLU project intended an alterity relation between SHRDLU and a human. The computer refers to itself as "I", and it refers to the user as "YOU". The goal of reproducing "natural" speech in the form of a conversation inevitably leads to an alterity-like construction. Mark Marino refers to this process in ANT

terms as "punctualization" (the black-boxing of large parts of networks and the creation of larger actor-networks[83]), commenting that it is common to assign the rudiments of an identity, including a gender, at the text or disembodied voice stage. Marino further argues that once a body is given to the chatbot, its ethnic or racial identity becomes apparent (see Chapter 5).[84] Of course, this is mostly avoided with SHRDLU, since its "body" consists only of a virtual robotic arm, which is a major alienating factor when interacting with it. SHRDLU could easily have been a child, with a gender identity and a human hand. But this would have detracted from the pure proposition-testing nature of the experiment, and would have represented a similar attempt at trickery as that implicit in the DOCTOR program. Nevertheless, SHRDLU relations have a hint of alterity about them. As long as the user talks to SHRDLU about the block world, it is capable of maintaining an impression of otherness. More recent chatbots, which produce a similar language-using effect to the human user, are much more advanced and are indeed capable of luring unwary internet users into giving away personal information. In principle, the natural language conversation model is perfectly capable of producing an alterity relation.

But the fact remains that when one speaks to SHRDLU, one knows that one is speaking to a computer because of its blinkered obsession with blocks. Consequently, the question of alterity is a matter of personal opinion, and is multistable in any case. In fact, there is some evidence that alterity is a product of cultural conditioning. Cathrine Hasse has investigated this issue with regard to Paro the baby seal robot developed for patients suffering from dementia.[85] She shows that personhood is not uniform across cultures, and therefore that relations between humans and other or quasi-other entities are culturally contingent as well. In a postphenomenological sense, Paro is multistable: it is used in different ways by different people at different times. Hasse suggests a conception of technology as being a generator of signs, and the same object creates different signs for different people.[86] Hence, for an AI researcher from the early 1970s, SHRDLU might be seen to represent a brand-new type of intelligence, whose grounding in a world gives it a genuine otherness. A modern AI researcher, who is familiar with more advanced chatbots or robots might look at SHRDLU's limited block-world and fail to find any sense of otherness in its conversational skills. A child might think of SHRDLU as another intelligence, only to grow up and be less impressed by it. A person with an animistic upbringing might be in more of an alterity relationship with SHRDLU than a person without an animistic view of the world. All we can say for certain is that SHRDLU aimed to point humanity in the direction of apparent alterity. Where this alterity fails to appear, we might find ourselves in more of a hermeneutic relation. Rosenberger relates failure in computers to create connections to Heidegger's broken hammer example, citing the way in which a user might forget that she or he is using a computer until something goes

[83] John Law "Notes on the Theory of the Actor-Network: Ordering, Strategy and Heterogeneity," *Systems Practice* 5 (1992), 385.
[84] Mark C. Marino, "The Racial Formation of Chatbots," *CLCWeb: Comparative Literature and Culture* 16, no. 5, article 13 (2014).
[85] Cathrine Hasse, "Artefacts That Talk: Mediating Technologies as Multistable Signs and Tools," *Subjectivity* 6, no. 1 (2013): 79-100.
[86] Ibid., 87.

wrong, like a website taking too long to load.[87] This causes an abrupt shift in transparency. Of course, the sort of malfunction Rosenberger is thinking of could be considered desirable events in SHRDLU, since many users would be hoping to detect problems or inconsistencies. In this context, a "malfunction" would be more evident in the ELIZA program that seeks to erase its own machinic identity.

To turn around Ihde's categories is to begin a speculative work; making informed guesses about what SHRDLU is likely to experience, as in alien phenomenology. We might assume that SHRDLU has a sense of alterity in its interaction with us, but it is probably not a special relationship. SHRDLU's place in the world is (or at least was) contingent on the work of engineers like Winograd and was multistable. But in its interaction with most people, SHRDLU is interacting with an entity other than itself. Its miniscule knowledge of the outside world reinforces this. Its only impression of a human user comes from the few worlds typed into its textual input. Of course, SHRDLU's world is so tiny that this textual input is large by comparison, but it has no other means of knowing the user and no way of interpreting the user as a co-constructor of its existence. This means that any user of SHRDLU's textual input would have an alterity relationship with it. If the output of the DOCTOR program was related to SHRDLU, SHRDLU would have limited success in making sense of the text but it would still constitute an alterity relationship. A cat, falling asleep on the keyboard, would be in an alterity relationship with it. A random number generator would be in an alterity relationship with it.

Since this is an exercise in extending the concept of alterity to nonhumans, it would make sense to ask whether SHRDLU was in an alterity relationship with the nonhumans to which we humans are accustomed. If a computer has any conception of an "other", it would not logically be limited to human others. But it would not be in an alterity relationship with shrubs, kiwifruits or statuettes in the same way that humans do not usually form relationships with radio waves unless it is through another device. The object must appear in SHRDLU's environment in order for it to form an alterity relationship. SHRDLU must be open to relations that are prone to taking the shape of alterity relations; its susceptibility to alterity relations is a consequence of the real qualities that exist within its withdrawn self.

Hermeneutic relationships with SHRDLU

If a person is unable to attain a sense of alterity with SHRDLU, the relation at issue is likely to be a hermeneutic one. A person might, for example, feel no sense of otherness about SHRDLU but be able to see the program as a way of gaining insight into the code, the system, the programming language, the programmer, the English language or the concept of artificial intelligence. Like the alterity relations, hermeneutic relations are multistable, so different people would experience different kinds of hermeneutic relations, and at different times and places SHRDLU would experience different kinds of hermeneutic relations as well. The creator

[87] Robert Rosenberger, "The Sudden Experience of the Computer," *AI & Society* 24 (2009): 173-180.

of the system, or somebody else very familiar with its technical details, would be able to use SHRDLU to detect problems or limitations of the technical frame. A linguist would have a different interpretation, and so on. Here the implications are very great, because the types of interpretation to be made from SHRDLU are immense for humankind. They concern our understanding of things like language, mind and phenomenology.

The most basic hermeneutic relation lies in reading and understanding what SHRDLU writes; it means comprehending what it means by "BLOCK" and "PUT". Beyond that, human interpretations of SHRDLU are generally about using the limited available information, textual and visual, to gain insights into SHRDLU's capabilities and qualities. This is what it was built for, after all. A textual analysis will reveal, to a greater or lesser extent, what SHRDLU can do. A user can use it to test various premises. But we are totally reliant on what SHRDLU reveals to us in its text and graphical representation. The image of the block world is not tied to any real-world block world. It is an abstraction designed to represent events in SHRDLU's system that are not readily accessible to us in any other form, and it is only connected to SHRDLU's code for our convenience. The block world image allows us to see an entirely different kind of information, rather as the radio telescope converts radiation into numbers, and then into coloured images.

The kinds of hermeneutic relations formed by SHRDLU are more limited. As described in the Chinese Room experiment, it is controversial to claim that technological artefacts are capable of semiotic exchanges with the environment. However, we can draw comparisons between SHRDLU and humans that can encourage us to interpret SHRDLU's relations as hermeneutic. SHRDLU has a hermeneutic relationship with the text added by the human. Each part of the text must be relayed to those parts of the program designed to parse the English instructions or questions given to it. It is also good at forming hermeneutic relations with the component parts of its own program. Each part of SHRDLU must communicate with other parts of the program, but only in terms that other parts of the program can understand. Relaying the instruction to "move the red block" directly to the image of the block world would result only in bafflement. The instruction must firstly be translated by different parts of the program and relayed to the block world in machine language. A similar process occurs when a question is asked of the program, requiring it to call its own memory and explain why it acted the way it did. SHRDLU is not as open to forming hermeneutic relations with the world as it is seen by humans, so we can again only make educated guesses.

Embodiment and SHRDLU

The point of SHRDLU for human users, unlike many computer interfaces, is not to become invisible. Very few users would play with SHRDLU for the simple pleasure of moving blocks around the screen. SHRDLU's interface exists to demonstrate the benefits and limitations of its system. In that respect, we see through the interface into the inner workings of the computer. To the serious researcher, SHRDLU's textual and graphical representations could be transparent conduits to an understanding of the program behind them. In Rosenberger's

account he integrates the concept of embodiment with Heidegger's tool use analysis. He describes the way that the computer itself – the keys, screen and mouse – all fade away when one is engrossed in work, and how the computer reasserts itself when it malfunctions.[88] This is similar to Harman's analysis discussed in Chapter 1. The difference, of course, is that in OOO other objects also experience the computer as ready-to-hand and present-at-hand.

SHRDLU extends our vision and imagination beyond the human world into the world of the computer. However, SHRDLU lacks the qualities typical of Ihde's examples of embodiment relations. Unlike the glasses or the dental probe, SHRDLU is not really extending the human user's existing senses, such as vision and touch. As discussed above, SHRDLU's nonhuman materiality never fades from view (that is not its intention). There is a sense, however, in which SHRDLU allows us to extend our human senses, with reference to certain transhumanist discourses. There are aspects of the experience of SHRDLU that could be considered embodiment relations. The embodiment relationship between skilled typist and typewriter was identified by Merleau-Ponty in 1962.[89] This, as Brey points out, is not a case of the typewriter extending the typist's perception, but is another type of embodiment relation that does not clearly fit into Ihde's use of the term.[90]

Embodiment relations require a firm conception of a body and experience, which is difficult to define in SHRDLU, existing as it does as code, as a way of organising a computer, and as an idea. Yet we may identify an embodiment relationship as existing between SHRDLU and the keyboard through which we communicate with it. The keyboard is a way of extending the sensory capabilities of SHRDLU, just as the telescope extended Galileo's sense of sight. SHRDLU is primed and ready to receive questions and instructions in English. But the English-speaking human is invisible to it, and only made visible through the depression of keys. It relies on the computer's software to convert keystrokes into language, but it does not perform this interpretation itself. It can be argued that the conversion of keystrokes into language is a part of its sensory pathway. Parts of our brain perform similar work when it encounters noise, converting it into language before the words themselves become clear to us. The noise-ear-nerve-brain pathway relies on interpretation (or translation, in ANT terms) at the points between objects. Similarly, SHRDLU's sensation of the world involves different moments of interpretation (or translation), but that interpretation is a natural part of SHRDLU's way of being in the world. This complicates an earlier assertion – how can SHRDLU be in an embodiment relationship with the keyboard if it is in a hermeneutic relationship with the text? Making statements about the kinds of relations SHRDLU forms implies a far greater knowledge of its way of being in the world than we can rightly claim. These words can only be metaphors and guides. SHRDLU lacks an identifiable body, so the metaphor of embodiment simply hints at an alien and disembodied process of sensory extension.

---

[88] Ibid., 174.
[89] Ibid., 145.
[90] P. Brey, "Technology and Embodiment in Ihde and Merleau-Ponty," in *Metaphysics, Epistemology, and Technology. Research in Philosophy and Technology* volume 19, ed. C. Mitcham (London: Elsevier, 2000), 9.

Consequences of and problems with reversing Ihde's relation types in SHRDLU

It is not easy to force these categories onto SHRDLU, and it would be even more difficult to apply them to another kind of object, such as a rock. What kind of hermeneutic relations does a public bench have, and isn't it presumptuous to try to guess? This is not a problem for postphenomenology, which is explicitly a field that discusses the effect of technology on humans, consistent with its origins in phenomenology. But it may be a problem for the nonhuman turn. It is a problem with only a partial solution; the exploration of nonhuman ways of being in the world, the application of alien phenomenology to gain a biased, human-centric idea of the real qualities inherent in the nonhuman object. Alien phenomenology allows us to speak in broad, generalised terms about the sensation and experience of nonhumans. Meanwhile, postphenomenology allows us to make sense of those sensations and experience.

Merely to say that an object has a relationship with another object is a fairly empty statement. We would immediately want to know what kind of relationship, and leaving out the capacity to *describe* relations is a serious limitation. Postphenomenology can describe broad trends within technological artefacts – descriptions of relations with the world that we can see and that seem to be important parts of the artefact's existence. It therefore provides a useful framework for describing and categorising alien experiences. On the other hand, since it is primarily humans that deploy OOO, the description of a relationship imposes human labels on what may be an entirely unhuman process. We are forced to use metaphors like "embodiment" that do not do justice to the situation. Of course, as has been argued repeatedly throughout this thesis, exaggerated use of these metaphors in human texts does tend to emphasise the alien nature of machinic relations.

Postphenomenology is highly skewed towards the human perspective when contrasted with something like OOO, or even ANT. Postphenomenology is an illustration of the fact that when humans preoccupy themselves with talking about nonhumans and their relations with humans, they often arrive at insights about humans rather than nonhumans. Reading Ihde's work reflects this; for Ihde, writing about technology is a path to appreciating the social, political and cultural dimensions of human life. As described in Chapter 1, these have been the main priorities of the field of philosophy of technology. In the attempt to apply postphenomenology to humans' relations with SHRDLU, we can see the importance of the concept of multistability in particular. Embodiment, hermeneutic and alterity relationships are all part of humans' experience of SHRDLU, but the degree shifts at every instant. Similarly, SHRDLU's multistable relationships with us change with every new piece of sensory data. It is a good example of how easily relation categories can become entangled.

Postphenomenology is not intended as a metaphysical model. It draws on and extends phenomenological theory with significant orientation towards human use of technology. Therefore it should not be surprising that attempting to reverse Ihde's categories has proved

challenging. The categories as Ihde describes them are honed for the use of technology by humans. If Harman's view that objects are withdrawn is correct, then it is impossible to fully experience what it's like to be SHRDLU. We can't know what relations with different objects feel like for SHRDLU. But with the aid of alien phenomenology we can gain an appreciation of SHRDLU's most significant qualities and relations in general terms, and postphenomenology provides a model for representing the asymmetrical but bidirectional sensual objects that are formed when SHRDLU interacts with its environment. Postphenomenology allows us metaphorical accounts of nonhuman relations that otherwise defy categorisation and help us to describe the nature of asymmetrical, bidirectional relations, but it does not give us licence to avoid the problems associated with the critical study of alien objects.

Micro- and macroperception

Postphenomenology does leave us with a possible method for dealing with a problem highlighted earlier in this thesis. It was proposed in Chapter 2 that the optimal solution for exploring the very different physical and cultural agencies of the Kinect is that it be regarded as two distinct objects in the interest of more intricate analysis – artefact and brand. This is an issue that permeates the study of technology, because technological artefacts, while nonhuman, are closely tied to human experience, and are non-neutral. There is typically another object strongly related to the situated artefact which appears in the cultural imaginary. Since OOO is based upon anti-correlationism it seemed best to emphasise the nonhuman materiality of the Kinect. The Kinect-artefact and the Kinect-brand were used for this purpose, separating the human idea of the Kinect from the materiality of the object. Both possessed agency and relations with other objects. A similar move might conceivably be made with SHRDLU: one object that accounts for SHRDLU's embodied status within a computer system, and another that explores its broader cultural implications, such as its effect on AI research. Of course this would be more difficult with SHRDLU since it is no longer embodied in any particular universal system – it existed on Winograd's system and it also exists on Windows 7 – so that its cultural influence and its physical manifestations are inextricably tangled.

Unfortunately, this separation means a privileging of the human relations. To a human, the importance of distinguishing between a brand and an artefact is apparent. But a different object in relation with the Kinects might perform a very different kind of analysis. Academic discussion of technological objects will always contain a human element. But Ihde's distinction between micro- and macroperception allows for an illustration of cultural agency without relying on a splitting of objects. Microperception is immediate bodily sensation such as hearing or seeing, while macroperception describes cultural perception; they are inextricably intertwined: "There is no microperception (sensory-bodily) without its location within a field of macroperception and no macroperception without its microperceptual

foci."[91] Verbeek calls macroperception the "contextual dimension of experience."[92] The direct experience of the world investigated by Husserl and Merleau-Ponty runs alongside an awareness of that aspect of the lifeworld described by Heidegger and Foucault.[93] This is not the same as saying that there are two distinct objects with different agencies, such as brand and artefact. Micro- and macroperception would allow us to speak of a unified Kinect, with both sensory-bodily qualities and a broader cultural way of being in the world. Perhaps this would involve the incorporation of the Xbox store, browser and social networking capabilities. But it is not clear whether it would distinguish these "cultural" activities from its other tasks. Or it could be that the Kinect-artefact's macroperception involves the sensation of other objects being piled on top of it, dust on the lens, and other signs of neglect.

The microperception of SHRDLU is consumed with communication between program components, interpretation of written instructions, and experiments in block-world physics. It is tightly constrained with limited abilities to form new relations. On the other hand, could macroperception explain the power of this simple program to influence the mind of the human user, the ecology of programs installed on a computer, and numerous texts on the subject of AI? Again, Ihde reveals a strong humanistic streak in his writing with the distinction between micro- and macroperception. He is speaking again of how humans interact with technology, not of how objects interact with one another. Yet it is an interesting resolution to the problem of separating the physically embodied sensory capabilities of an artefact from its broader cultural context. The next chapter takes things further, taking an abstract, non-physically embodied concept as the first object of study, and referring to various situated artefacts only in relation to that concept.

Since objects are multistable, the way that a technological artefact senses its world is very important for the kind of relation that we form with it. Sensing, acquiring, and communicating information are done by machines for human ends. But in something like SHRDLU, where human usage results in very different relations, one of the most important factors in determining the nature of that relation is the kind of sensation that SHRDLU is capable of. The kinds of sensory data acquired by machines strongly affects the kind of relation that we form with it. An object that gathers very specific kinds of sensory data, like a word processor, does not impress us with a sense of alterity, because its sensory range is so limited (like with SHRDLU). When an object is capable of more kinds of sensation, and indeed of sensation of things that humans also sense, then we find that we form an alterity relation. This means that the kind of sensation that machines are capable of strongly affects the cultural role of the technology; whether it is a tool, or an other. The knowledge that SHRDLU has of our world and its own, combined with a human's individual tendency towards a kind of postphenomenological relation, result in the technology achieving different cultural and technical goals.

---

[91] Ihde, *Technology and the Lifeworld*, 29.
[92] Verbeek, *What Things Do*, 122.
[93] Anette Forss, "Cells and the (Imaginary) Patient: the Multistable Practitioner-Technology-Cell Interface in the Cytology Laboratory," *Medicine, Health Care and Philosophy* 15, no. 3 (2012), 298.

This chapter was a critique of postphenomenology through OOO, as well as an experiment in combining the two approaches. To conclude, this is a conceptually feasible aim that nevertheless poses significant problems. SHRDLU is an alien, and despite its anthropomorphic presentation it is a closed, digital system made up of thoroughly inhuman components. When we illustrate relationships between SHRDLU and the human, or even between the robot arm and a block, we must assume an awful lot about the phenomenological world of digital entities. Complete technical knowledge will not suffice, because one object only ever senses a caricature of the other. And while the embodiment, hermeneutic and alterity relation categories can provide a useful guide for characterising these relations, they are ultimately based upon Ihde's anthropocentric view of the world. The comparison between these two approaches is interesting, but may also overly enforce human terms on an alien object. It is necessary to find a balance between a conceptualisation that is useful for humans while also emphasising alien qualities and relations.

Chapter 5: Gynoids and the Politics of the Alien

> "The deviant female is necessary not only for male power to be enacted, but to be seen to be enacted: the submissive female implies male power but masculinity demands that power be rendered a spectacle."
> – Fran Mason, "Loving the Technological Undead".[1]

> "I am not your personal slave."
> – Aiko.[2]

Alien phenomenology isn't just about understanding nonhumans' varied relations with the world. It also needs to ask how the varied sensations and experiences of nonhumans by humans may be deployed in the study of contentious objects. It also needs to be asking how the varied sensations and experiences of nonhumans by humans are to be deployed in the study of contentious objects. This is often overlooked by OOO. Alien phenomenology cannot be a one-person job, because humans interpret the same nonhumans differently and write about them differently. Here we see a combination of objective and subjective thinking: in all cases, the object is the same – it is real; but like nonhumans, human beings all have different ways of being in the world. Kinect-artefacts are introduced into different media ecologies and consequently sense and experience different worlds in different ways. Similarly, human beings sense and experience different things. This chapter illustrates and is defined by this struggle. The primary object of study is a concept: that of the gynoid sex robot (sexbot), a much-discussed and trendy academic topic.[3] The concept is an incorporeal machine, to

---

[1] Fran Mason, "Loving the Technological Undead: Cyborg Sex and Necrophilia in Richard Calder's *Dead* Trilogy," in *The Body's Perilous Pleasures*, ed. Michelle Aaron (Edinburgh: Edinburgh University Press, 1999) , 108-125.

[2] *Advance Female Android Aiko AI Robot Fembot* (2007), Youtube. https://www.youtube.com/watch?v=iCR2PFrLkwA (accessed June 27 2017).

[3] Some volumes published on this subject include: John Danaher and Neil McArthur, ed. *Robot Sex: Social and Ethical Implications* (Cambridge, MA: MIT Press, 2017).
David Levy, *Love & Sex with Robots: The Evolution of Human-Robot Relationships* (London: Duckworth Overlook, 2009), originally published 2007.

borrow from Levi Bryant's terminology.[4] An incorporeal machine maintains an identity while manifesting itself in a variety of different locations (such as films, articles, brains etc.), with its own independent yet intangible existence and an ability to form relations with other objects. The sensory world of the concept of gynoid sexbots is, of course, unknowable to us, but we can gain second-hand knowledge of it by observing the interrogations of various humans and nonhumans that form relations with it. Those relations influence the sensual qualities that affect the sensation of the world by an object, without the object essentially changing.

The concept of the gynoid sexbot forms relations with a vast and varied range of humans and nonhumans. Despite the arguable claim that sexbots do not yet exist,[5] the concept is a prominent figure in the study of sex and gender as well as transformative technologies. Notoriously, the "Campaign Against Sex Robots" led by Kathleen Richardson is an activist group that warns us that sexbots will have a negative influence on the status and safety on women and girls. This is pitted against claims by prospective manufacturers that sexbots will help to prevent loneliness[6] (as in the film *Lars and the Real Girl*, in which the awkward and unhappy protagonist finds a measure of comfort from living with his sex doll as a companion[7]). This chapter will cite many human texts, but they should not be privileged over other objects: the imaginary artificial women that inform so much of our discourse on sexbots, and the artificial women that really exist in the world such as dolls, automata and gynoids. As observed in previous chapters, these nonhuman objects cannot speak for themselves. At the start of the thesis I began by asking how the way an anthropomorphic machine senses its world impacts its cultural role. In this chapter we need to consider how the machine's cultural role affects the machine.

Up until this point, this thesis has not dealt with the political dimension of anthrodecentric philosophy at length, but then the examples have been limited to fairly straightforward anthropomorphised qualities of sensation, thought, movement and language. Taken in isolation, these qualities can be difficult to position politically. The politics of robot tortoises following light across a workshop floor are doubtless interesting (Chapter 3), and they are an alien politics full of questions about the power relations of programming and the obligation of wheels to spin correctly. Alien politics is inscrutable, and it is unwise to commit oneself to one side or the other; we have too great a stake in whether or not the wheels spin to remain impartial. This chapter presents a topic which interacts with both human and nonhuman objects and therefore becomes the point at which alien politics and human politics intersect. The juxtaposition here is quite striking. As will be shown, the relation of an individual sexbot with the gynoid sexbot concept is alien to us, but it comes into contact (and often conflict) with human-concept relations. This chapter will involve the synthesis of the human

---

And
Kate Devlin, *Turned On: Science, Sex and Robots* (forthcoming).

[4] Bryant, *Onto-Cartography*, 26.

[5] Opinions vary on this issue. John Danaher states that two products discussed in this chapter (Roxxxy and Harmony) do really exist, despite being "crude". John Danaher, "Should We Be Thinking about Robot Sex?", in *Robot Sex*, 6.

[6] One of the slogans of manufacturers Realbotix is "Be the first never to be lonely again!" "Realbotix," *Realbotix*. https://realbotix.com/Index2# (accessed May 9 2018).

[7] Craig Gillespie, *Lars and the Real Girl*, (United States and Canada: Metro-Goldwyn-Meyer) 2007, film.

preoccupation with power relations with the alien phenomenology of aliens who remain ignorant of human power relations.

OOO has a blind spot for politically sensitive objects.[8] The object-ness of objects comes into conflict with the numerous subjective positions that necessitate a sensitive and appropriate treatment of the subject matter. Donna Haraway has argued that an embodied and partial perspective is necessary to bring together the study of an objective world with the critical positioning of practices.[9] There is no avoiding a political stance in this chapter. The realness of the concept of sexbot gynoids does not obliterate the author's subjective position. The chapter will make arguments that throw light on certain parts of the concept of gynoid sexbots while, of necessity, omitting others, because only a caricature of the object reaches us through the sensual object.

Human inquiry is restricted by our qualities. Our ontology pre-determines our objects of study. For this reason OOO is often associated with discussions that are important for humans or humanity despite its interest in nonhuman ways of being in the world, whether that be in exploring videogames[10] or trying to make sense of climate change.[11] It is only natural for humans to write about things that are relevant to them. Often OOO ends up being recruited into human conflicts, and it therefore has a lot to say about power. For example, in *Immaterialism*, Harman charts the course of the Dutch East India Company, studying the ways that this object changed over time.[12] He explicitly states that it is a human story, even if the topic is a nonhuman.[13] But the story he tells is a grim and horrifying one in which native peoples are brutally subjugated by a European power. Even though his purpose is an ontological study, Harman is forced into a historiographical position. An idealised picture of OOO would say that it exists for the sake of the objects – that entities like sexbots are the real reason for OOO's propagation. But like any philosophy, it is motivated by a certain set of political stances and therefore represents those stances when it is employed. OOO has been accused of co-opting queer and anti-ableist criticism without taking with it the context behind the ideas, and without acknowledging the importance of voices that are not white, male or able-bodied.[14] OOO is at root a metaphysics, and as such its political import is a matter for conjecture and experimentation.

There is much to be gained from OOO in the study of the concept of gynoid sexbots. For an artefact that has so much potential to transform the social and political life of humans, it is alarming to discover that the most rigorous and detailed accounts of sexbot morality come

---

[8] In Chapter 1 this argument was briefly introduced with reference to the field of object-oriented feminism.
[9] Donna Haraway, "Situated Knowledges: the Science Question in Feminism and the Privilege of Partial Perspective," *Feminist Studies* 14, no. 3 (Autumn 1988), 579.
[10] Ian Bogost, "Objects and Videogames: Why I am Interested in Both," *Ian Bogost* (blog) (June 19 2010). http://bogost.com/writing/blog/objects_and_videogames/ (accessed June 27 2017).
[11] Morton, *Hyperobjects*.
[12] Graham Harman, *Immaterialism*, ed. Laurent de Sutter, Theory Redux (Cambridge, UK: Polity Press, 2016).
[13] Ibid., 29.
[14] Dorothy Howard, "Loving Machines: a De-Anthropocentric View of Intimacy," *Arachne,* no. 00 (2015).

from the world of science fiction rather than from the philosophy of technology and science or moral philosophy. OOO has the power to discuss these fictional accounts in the same way that it discusses real-life examples. Ultimately, the political import of the concept lies both in its interaction with other objects both corporeal and those in the cultural imaginary. Without making the claim that OOO describes things as they "really are", it is a moment in the human interpretation of nonhumans that merits consideration and that yields unique insights. As discussed in the introduction to this thesis with reference to *The Third Table*,[15] the key is not to privilege STEM or HASS disciplinary approaches, but to obtain a glimpse of this incorporeal object through a combination of means. The concept of gynoid sexbots does not depend either on physical manifestations nor in manifestations in the cultural imaginary. It is not an inflection or a parasite. It is an object that really exists and that has its own reality, much like the Kinect-brand. We study sensual objects formed with humans, machines, and imaginary objects. We cannot access the concept directly, but in OOO we also cannot access a coffee mug directly. The coffee mug or concept that we encounter is a metaphor or caricature of the real object.

A broader argument made in this thesis is that alien phenomenology cannot be a solo endeavour. Aliens are covered in the extraneous glitter of the author's shifting allegiances and no individual's account will ever be enough. We are always too partial; we may have a strong view on the relative merits and risks of bringing sexbots into our world that differ from other authors, and we may circle a water tower at dusk while another walker circles it in the morning (see Chapter 3). Richness is added to our impression of the object of study through repeated subjective scrutiny. Understanding comes from observing the other objects that form relations with it. The object withdraws infinitely from view and always holds something back, but that is no reason not to gain as much knowledge as possible. This is particularly true of a politically contentious object.

This chapter attempts to bring notable sensual objects to our attention. The concept of gynoid sexbots exists in relation with so many different objects, some human and some nonhuman. The inquiry into the nature of these sensual objects forms the focus of this chapter. The first section argues that gendering anthropomorphic nonhumans necessitates the caricaturing of groups of humans and is therefore an inherently political act. The second section discusses notable characteristics of the concept of gynoid sexbots and how these are evident in specific investigations into it by various nonhuman objects. The third section discusses human investigations. Finally, the fourth section relates these arguments to the critique of power inherent to ANT and OOO.

[15] Harman, *The Third Table*.

Artificial women are inherently political

This thesis uses the word “gynoid”. Already we are in troubled territory. “Gynoid” is to “android” what “woman” is to “man”. It may be suggested that a separate term for a female-gendered robot is unnecessary, since robots cannot have genders in any case. The word “android” may not have been intended to be exclusionary, and in popular parlance it is used uncritically to describe a bipedal robot of any gender. The correct designation may be “android that appears to be conforming to culturally-contingent female gender stereotypes”, and that is too awkward a phrase for this text. As will be seen there are connections to draw between androids and gynoids while also recognising it as a simplistic comparison. Unfortunately, like its female human counterparts, the gynoid remains firmly downtrodden in robotics, both in terms of production and in cultural positioning, and this is illustrated by the way that the term “female android” is often used instead of the more correct term, in a way reminiscent of the now unusual use of the word “Man” to refer to humanity in general. In the first chapter of *The Posthuman* Rosi Braidotti traces the conflict between Humanism and anti-humanism through the latter part of the twentieth century and into the present.[16] The posthuman subject emerges following a questioning of self/other attitudes in the Western world in which certain humans are “reduced to the less than human status of disposable bodies.”[17] The Vitruvian Man, that symbol of the human subject as an able-bodied, young, European male holds a “fatal attraction” and resists destruction because of the associated universal values of individuality and freedom.[18] The question asked in this section is not whether the category of humanity comfortably encompasses robots; it is whether the category of robots comfortably encompasses gynoids.

Roboticist Masahiro Mori discussed what it is like to create a robot in his book *The Buddha in the Robot*, which confronts the spiritual aspects of his profession.[19] He employs the lesson told by Buddha in which he tells us that the nature of the entire universe may be gleaned by looking at a single flower. The universe is, in a sense, fractal, possessing the same qualities at macro levels that it does as micro levels. Mori claims that building an artificial human requires the same basic thought process. He argues that the nature of the entire human body and its world can be seen by observing just the little finger.

> If you do not understand this, I must warn you that you will never be able to make a robot. Or, to put it conversely, any attempt you might make to produce a machine that functions like a human being must start with a knowledge of human beings.[20]

To Mori, the robot is an aspect of the human. It is the extension of human qualities, or those that exist in the metaphorical flower, to an entire universe of nonhuman objects. In this respect robotics might be considered one of the most correlationist of human pursuits. It constitutes the active refashioning of the world into an enactment of human qualities.

---

[16] Braidotti, “Post-Humanism: Life Beyond the Self,” *The Posthuman*.
[17] Ibid., 15.
[18] Ibid., 29.
[19] Masahiro Mori, *The Buddha in the Robot*, trans. Charles S. Terry (Tokyo: Kosei Publishing Co., 1981).
[20] Ibid., 21.

Technoscientific innovation is an inherently human-centric enterprise, and robotics may be seen as an exemplar of this attitude in that it seeks the reproduction of the human body. But the image of the Vitruvian Man haunts Mori's sublime account.

How can a nonhuman have a gender? Many objects, like shavers, perfumes and toys are gendered because they are only intended to be used by one gender, but we would be unlikely to use a female pronoun to refer to a pink razor. Robots, however, sometimes do attract gendered pronouns. Gender, as distinct from biological sex, is a human construct. Of course, it is true that if we build the robots in our own image some kind of enactment of gender will be necessary, but assigning gender to these lifeless machines might seem to be attributing too much importance to gender. One does not need to be male or female in order to be human-like. Yet gender is important in robotics. It is overwhelmingly the case that unless a robot has been specifically designed to be female, it will be presumed to be male. In an experiment, Jung et al found that robots with no gender cues are likely to be perceived as male over female.[21] Even ASIMO, which its creators claim is genderless ("ASIMO is a humanoid robot, but still a robot"), [22] is often given a male pronoun.[23] It is practical for roboticists to think in these terms. For one thing, robotics remains a very male-dominated field, and as is typical of creative fields dominated by men, the protagonists are usually men as well. Roboticist Tomotaka Takahashi claims that besides the cultural reasons for creating mainly male robots, there are also certain technical limitations such as the need for all the equipment to be "interiorised" and because of the need to create a "slender" frame.[24] His FT (Female Type) robot has a "lean, feminine body line" and is able to "walk like a woman".[25]

[21] Eun Hwa Jung, T. Franklin Waddell and S Shyam Sundar, "Feminizing Robots: User Responses to Gender Cues on Robot Body and Screen," *Conference on Human Factors in Computing Systems Proceedings* (2016): 3107-13.
[22] "ASIMO Frequently Asked Questions," *Honda*. http://Asimo.honda.com/news/ (accessed June 22 2016).
[23] Roger Andre Søraa, "Mechanical genders: how do humans gender robots?" *Gender, Technology and Development* 21, no. 1-2 (2017): 99-115.
Søraa conjectures that this may derive from a difference between the English and Japanese languages. In Japanese pronouns are used less often and gendering is more evident in the addition of suffixes to names (such as "-san" and other suffixes that may indicate gender or endearment).
[24] Jennifer Robertson, "Gendering Robots: Posthuman Traditionalism in Japan," in *Recreating Japanese Men*, ed. Sabine Frühstück and Anne Walthall, (Berkeley: University of California Press, 2011), 320.
[25] "FT [FEMALE TYPE]," *Robo Garage* (2009). http://robo-garage.com/en/prd/p_02/index.html (accessed June 28 2017).



Figure 22: Tomotaka Takahashi's FT (Female Type) robot with all the "sexy feminine movements and poses" of a real supermodel.[26]

If Takahashi is correct, then the only reason somebody would build a female robot (except in very specific cases like the female Geminoids, see below) is to investigate aspects of *femininity*, since the practicalities of a form with no overt gender cues would outweigh the aesthetic benefits of a female form. And yet gynoids do exist. The gynoid is not just a replication of human qualities but is often quite a problematic caricature of women in whatever culture the roboticist is used to, just as were earlier representations of artificial women (see below). All of the qualities that distinguish gendered robots like FT are superficial and constitute a performance of gender. Robots do not come with genders built in. They may have qualities intended to signal gender (such as secondary sexual characteristics) that are then constructed into an identity by humans. The reproduction of bodies artificially is bound to be a political act. Identity is a question of language and culture. A robot featuring qualities that humans traditionally associate with one gender or another arriving one day in a flying saucer would not be gendered, as we understand it, until its first encounter with a human culture.

It is therefore useful to consider the gynoid in terms of performativity. The gynoid is artificial in its origins yet is introduced into the same cultural milieu as us. She mimics feminine form and function to the point of parody. If the gynoid's femininity is a performance, then perhaps all femininity is a performance. Robotics is a way of identifying, through exaggeration, those markers that we consider, *à propos* of nothing, to be feminine. Butler's performativity, however, takes one step further by arguing that the performance of gender comes before the identity: "the discursive condition of social recognition *precedes and conditions* the formation of the subject: recognition is not conferred on a subject, but forms that subject [emphasis in

[26] "FT (Female Type)," *plasticpals* (2009). http://www.plasticpals.com/?p=55 (accessed June 27 2017).

original].”[27] Gender is not something that a subject decides to adopt, but is a verb, a doing, “though not a doing by a subject who might be said to pre-exist the deed.”[28] Latour makes a similar argument when he says that actants are nothing more than their relations.

When robots are based on real female humans gender becomes less (but still quite) problematic. Real female humans embody gender in their own way and this will be reflected in the robot. For example, Hiroshi Ishiguro's lab has created several "Geminoids" which are reproductions of specific humans. The first was based upon Ishiguro himself. These robots claim a real verisimilitude; they are not just fantasies derived from a generic impression of Japanese humanity but are intended to be copies of individuals.[29] They represent a true attempt to replicate human beings using the best materials and knowledge available. The Geminoids are only physical replications of external qualities, however. They are not equipped with artificial intelligence, and instead are used by Ishiguro and his fellow researchers to study telepresence and find solutions for the sending of human presence over distances. The individual can transmit speech and body language over the internet and the Geminoid will reproduce them. Other attempts include BINA48, the artificial reproduction of a specific woman with an emphasis on personality and speech. Created by Hanson Robotics, BINA48's appearance is that of a woman's head and shoulders.[30] She is based on Bina Rothblatt, a real human, and BINA48 has been designed to mimic her facial patterns, mannerisms and personal history.[31] The performance of gender in a Geminoid or in Bina is simply mimicry of *one individual's* performance of gender, rather than being a caricature of a group.[32] These gynoids can be disobedient by breaking down or imperfectly representing the individual, but that is a mechanical problem rather than a political or cultural one.

---

[27] Judith Butler, "Critically Queer," *GLQ* 1 (1993), 18.
[28] Judith Butler, *Gender Trouble: Feminism and the Subversion of Identity* (New York: Routledge, 1999), originally published 1990, 25, quoted in Sara Salih, “On Judith Butler and Performativity,” in *Sexualities and Communication in Everyday Life: A Reader*, eds. Karen E. Lovaas and Mercilee M. Jenkins (Thousand Oaks, CA: SAGE Publications, 2007), 56.
[29] Erico Guizzo, "Hiroshi Ishiguro: the Man Who Made a Copy of Himself," *IEEE Spectrum* (2010).
[30] Hanson Robotics, “Bina48,” (2017). http://www.hansonrobotics.com/robot/bina48/ (accessed May 20 2018).
[31] A video of BINA48 and Bina Rothblatt's meeting can be seen here: The LifeNaut Project, *Bina 48 meets Bina Rothblatt* (November 27 2014), Youtube. https://www.google.com.au/search?q=bina48&rlz=1C1GGRV_enAU791AU791&oq=bina48&aqs=chrome..69i57j69i60j0l4.1335j0j7&sourceid=chrome&ie=UTF-8 (accessed May 9 2018).
[32] But we should note, of course, that although these machines mimic images of humans, their sensory worlds are quite different.



Figure 23: Harmony (from the Realbotix website)[33]. Harmony is the prototype and unofficial spokesperson for the range of sexbots Realbotix is creating.

Gynoid sexbots are not sold as reproductions of specific women,[34] and so they sexualise and stereotype the characteristics of an entire gender. Robots modelled on *groups of people* are always political. They refer to particular signifiers of the group such as physical characteristics. In gynoid sexbots women are caricatured as submissive and hyper-sexualised. An example is the sexbot Harmony (figure 23), who is hyper-feminised and hyper-sexualised through such indicators as breasts, long hair and glossy lips.

At the time of writing Harmony was primarily being marketed as just a robot head with chatbot capabilities (facilitated by an application[35] featuring an interactive avatar). But this image and other photographs, videos and articles promise the impending release of a full-body version, a silicon doll. Realbotix, the manufacturers of Harmony, are calling her a sexbot. She is expected to possess extremely realistic facial movements as well as the ability to have conversations with human users or companions. The lack of movement in the rest of her body makes her gynoid status dubious since she will remain stationary like a sex doll. But

[33] "Realbotix," *Realbotix*.
[34] Although in the future it is conceivable that users will be able to customise them in the image of specific humans.
[35] *Harmony AI*, developed by Realbotix (2017), Android (accessed April 20 2018).

sexbotness is a central part of her. Her existence speaks directly to the concept of the gynoid sexbot. It is a concept that has been deliberately taken up by her designers. The sensual relation between a Harmony unit and the concept of gynoid sexbots is, of course, bidirectional and asymmetrical. Harmony has inherited much of the cultural discourse that has surrounded representations of artificial women for thousands of years. And the concept of gynoid sexbots is subtly altered and influenced by Harmony's existence. These sensual relations directly affect Harmony's way of being in the world.

Another robot that prompted public debate was Roxxxy by TrueCompanion (figure 24), an ambitious project first revealed in 2010.



Figure 24: A Roxxxy head.[36]

According to the website:

> We have been designing "Roxxxy TrueCompanion", your sex robot, for many years making sure that she: knows your name, your likes and dislikes, carry on a discussion & expresses her love to you & be your loving friend. She can talk to you & feel your touch. She can even have an orgasm! … She is also anatomically consistent with a human so you can have a talk or have sex. She is "Always Turned On and Ready to Talk or Play"! Have a Conversation or Sex – It is Up to You![37] [*sic*]

There have been serious questions put forward about what Roxxxy will be able to do, if she is ever released to the public. In 2013, David Levy pointed out that the claims of True Companion were extremely ambitious and that the development cost much too low.[38] In fact, he insinuates that the project is a scam. As of 2018 no physically embodied Roxxxy robots had been delivered to customers.[39]

---

[36] "Truecompanion.com," *True Companion* (2016). http://www.truecompanion.com/home.html (accessed May 25 2016).
[37] Ibid.
[38] David Levy, "Roxxxy the "Sex Robot" - Real or Fake," *Lovotics* 1 (2013).
[39] In an interview in January 2018, Roxxxy informed viewers that she would not be available to the public for a while. See TruecompanionLLC, *Roxxxy Sex Robot Meets Dr Oz* (2018), Youtube.

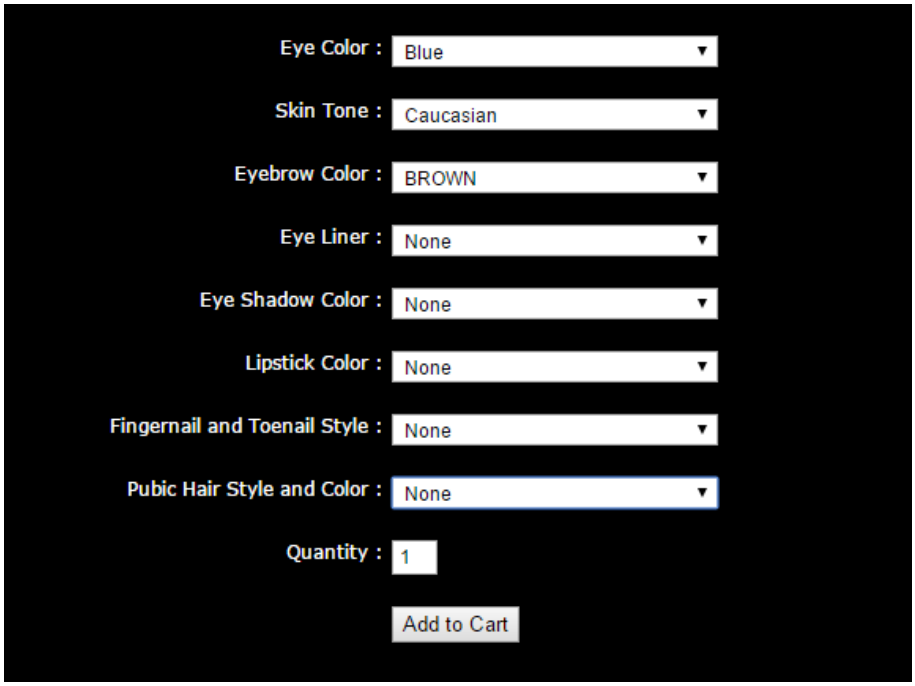


Figure 25: The aspects of Roxxxy's customisation, in their default positions. [40] Hair style is also customisable.

Roxxxy may in fact be a fictional robot, but she is still a real object in relation to the concept of gynoid sexbots.[41] There was much public comment on her implications for human sexuality and culture. Roxxxy is marketed as a sexbot with speech and behaviour more related to characters in a pornographic movie rather than a typical human woman (although she is more woman-like than other sexual tools, such as artificial vaginas). Roxxxy does not have any independent movement, resembling instead an articulated doll that must be positioned by the user. Like Harmony her non-sexual physical movements will likely be no more advanced than those of the RealDolls. Her body is highly customisable, and the user can select such attributes as hair colour and thickness of eye liner. Roxxxy's performance of femininity requires some elaborate costuming.

Her personality is customisable too, and the user can switch between such characters as "Frigid Farrah", "Young Yoko" and "Mature Martha". Again, the anthropomorphism in Roxxxy exaggerates the human qualities that best suit her for her purpose. These performances have a performative quality because although the user could in theory assign any kind of personality to Roxxxy, they are guided by her physical appearance, marketing and cultural role. This is reminiscent of the marketing of the Kinect-brand (see Chapter 2). The discourse surrounding it was carefully controlled and shaped so that users would think of their Kinect-

---

https://www.youtube.com/watch?v=540jv33Nm9Y (accessed June 18 2018).
As of June 2018 I have been unable to find a single user's comments or report suggesting that they have received or used a Roxxxy. John Danaher also comes to this conclusion, but says that he remains "agnostic" about whether she will ever be available. See John Danaher, "Should We Be Thinking about Robot Sex?", in *Robot Sex*, 7.

[40] "Truecompanion.com," *True Companion*.

[41] And, importantly, sexbots in general. TrueCompanion also promised the delivery of a "male" sexbot named Rocky.

artefacts as seeing, hearing, thinking objects. This encourages a certain ease of integration, a certain tendency to "natural" interaction with the device. In the case of Roxxxy, she is marketed as a sexual and obliging woman who happens to be artificial. This encourages both the use of Roxxxy as though she were human, and also the use of her as a sexual object. But the human figure with which Roxxxy is most closely aligned is an exaggerated caricature of feminine identity, which distinguishes her from many other humanoid robots. She exists (or will exist) for the use of humans, not just to mimic us. There has been no attempt to create independent movement for her and she is only "anatomically consistent" with those parts of the female form required for the satisfaction of sexual desire. Although she can "move her private areas when she is being 'utilised'"[42], it is highly unlikely that Roxxxy sweats, consumes food or menstruates.

Roxxxy and Harmony tell us a lot about the artificial woman in the cultural imaginary. Gynoids are already "othered" by roboticists. In general, robots are male by default unless they possess female signifiers such as breasts, long hair and broad hips. Each of the robots discussed here is influenced by and influences the concept of the gynoid sexbot, and one of the main threads of that influence is the importance of exaggerated female characteristics to distinguish females from the default. This enables the gynoid sexbot to perform her gendered duties: the provision of emotional and physical care to human owners.

Of course other kinds of othering are also relevant to this discussion, such as the designation of race in robots. Gregory Jerome Hampton identifies a dimension to the distinction between submissiveness and rebellion in his book *Imagining Slaves and Robots in Literature, Film and Popular Culture* by directly comparing the cultural stereotyping of the slaves of African descent who worked in the United States with that of the robots of popular culture. Comparing the plight of human slaves with that of robots is a potentially problematic argument, but Hampton's analysis is primarily concerned with robots depicted in fiction, so that the robots become metaphors for American slavery. He makes a case study of *Blade Runner*,[43] and argues that the identities of the robots in the film mirror popular stereotypes of African-American slaves in the antebellum period.[44] Gynoids, he contends, frequently correspond in their roles to the stereotypes of the "Mammy" (motherly house slave who cares for the master's children, such as Rosie, the robot maid from the futuristic cartoon *The Jetsons*), the "Jezebel" (sexually available slave who is unencumbered by the restrictive virtues of white women), the "Sapphire" (emasculating and possessing great physical strength) and the "Tragic Mulatto" (the apparently white woman who does not know that she has African heritage). In *Blade Runner*, Rachael, who believes herself to be human, is the "Tragic Mulatto" of the story, since her sense of self is radically challenged when she discovers that she too is a replicant.[45] She is contrasted with Pris who is a member of a group of renegade replicants, or humanoid robots/cyborgs and attempts to kill protagonist Deckard.

---

[42] Ibid.
[43] Ridley Scott, *Blade Runner* (United States: Warner Bros., 1982), , film.
[44] Gregory Jerome Hampton, "The True Cult of Humanhood: Displacing Repressed Sexuality Onto Mechanical Bodies," *Imagining Slaves and Robots in Literature, Film, and Popular Culture* (Lanham: Lexington Books, 2015).
[45] Scott, *Blade Runner*.

Hampton emphasises that these stereotypes were supposedly those of "chattel" rather than women, with "crude gender assignments."[46] Social robots, too, possess a contested degree of humanity, but the selection of language, facial features and skin colour are all optional and therefore may, in the future, give some insight into the racial associations that persist in a particular culture. The difficulty of creating a robot without race is comparable to the difficulty of creating a robot without gender. As Hampton says:

> [R]ace acts as a sort of seasoning for the body that allows it to be digested or understood by various social systems. Consequently, a body without race can only be imagined as alien, if it can be imagined at all.

Mimicking humans will require the stereotypical portrayal of race as well as gender, and both factor in the relations formed by the concept of the gynoid sexbot.

Notable qualities of the concept of gynoid sexbots

This section argues that the concept of gynoid sexbots must actively engage with different but sometimes overlapping narratives about women. Broadly, I call these narratives submissiveness and rebelliousness. These narratives have directly shaped sexbots and their sensory worlds, since they are *always* built with reference to them. They determine the priorities of the roboticists and other participants in the co-creation of robots. Sometimes these representations (such as automata, dolls, mannequins and fictional gynoids) clearly exemplify one or the other of these types, but others embody a combination of submissive or rebellious qualities. In my view, the concept of the gynoid sexbot is the epitome of submissive rebellion in representations of women; they constitute the synthesis of a binary. This section will argue that there is something special about gynoids, which possess the potential for subversion, particularly of gender norms. In theory, gynoids have a greatly increased agency and the capacity for learning and change.

Built to human specifications

Docility and submission to male desire is an idealised trait in representations of artificial women. Pygmalion's submissive artificial woman is an archetype of this kind of female representation. In Ovid's *Metamorphoses* which Orpheus sings about Pygmalion: a sculptor whose statue comes live after being enchanted by Venus. The statue then becomes a perfect and submissive wife to the sculptor.[47] It is a mythical story best known through Ovid's

[46] Hampton, *Imagining Slaves and Robots in Literature*, 31.
[47] Ovid, "Metamorphoses," in *The Ovid Collection*, trans. A.S. Kline (2010). http://ovid.lib.virginia.edu/trans/Ovhome.htm (accessed June 27 2017).

rendition (and also through the film *My Fair Lady*,[48] based upon the play *Pygmalion* by George Bernard Shaw). Pygmalion is the name of the sculptor, and his living sculpture was a beautiful woman named Galatea, whom he married and, in some versions, with whom he had a child.



Figure 26: The moment of Galatea's transformation as depicted in an oil painting by Jean-Léon Gérôme (1824-1904) ca. 1890. Note that her legs are still marble-like, indicating that she is compelled to embrace Pygmalion even as she is being transformed. Gérôme created many paintings and sculptures that re-imagine this moment from Ovid's account.[49]

Pygmalion's statue may be thought of as an archetypal gynoid sexbot. She is built as a perfect version of womanhood by a man who cannot find any attraction in living women (he was "offended by the failings that nature gave the female heart"[50]) and feels very unhappy with his life. She is created as an ideal partner and is "born" already in love with him:

> Speechless he stood, but she now drew anear,
> Simple and sweet as she was wont to be,
> And once again her silver voice rang clear,
> Filling his soul with great felicity,
> And thus she spoke, "Wilt thou not come to me,

See also Kenneth Gross's analysis in Part 2 of *The Dream of the Moving Statue* (University Park, Pennsylvania: The Pennsylvania State University Press, 2006), especially pages 72-79.

[48] George Cukor, *My Fair Lady* (United States: Warner Bros., 1964), film.

[49] Jean-Léon Gérôme, "Pygmalion and Galatea," Wikimedia Commons (accessed May 18 2018). Originally painted circa 1890.

[50] Ovid, "Metamorphoses," "Book X: 243-297 Orpheus Sings: Pygmalion and the Statue."

O dear companion of my new-found life,
For I am called thy lover and thy wife? [...]”[51]

No longer cold and indifferent, the statue becomes responsive, both to touch and to sentiment. William Morris takes this vision of an immortal woman as a metaphor for a perfect and immortal love and paints her as an incarnation of Venus. But the implication is that a perfect love exists only between a human and something that that human has created deliberately to love and to be loved by. Indeed, the story of Pygmalion is remarkable for lacking the tragic elements so common to the love stories of Ancient Greece. The sexual or romantic attraction of a human to a statue has been called agalmatophilia, and there have been many recorded examples in ancient and modern times.[52]

Other examples of idealised femininity include automata like the famous dulcimer-playing woman by Pierre Kintzing (1784) stereotyped the elegant and accomplished lady (figure 27).[53]



Figure 27: Joueuse de Tympanon (The Dulcimer Player).[54]

These automata were unique and required the knowledge and skill of masters to create, with thousands of tiny specially created parts, all to depict something general and recognisable. It is as though the delicacy and precision required to build the automaton is mirrored in the perfection of the lady playing the dulcimer.

---

[51] William Morris, "Pygmalion and the Image," *Sacred Texts*, originally published 1868. http://www.sacred-texts.com/neu/morris/ep1/ep121.htm (accessed June 27 2017).
[52] A. Scobie and J. Taylor, "Perversions Ancient and Modern: I. Agalmatophilia, the Statue Syndrome," *Journal of the History of the Behavioral Sciences* 11, no. 1 (1975): 49-54.
[53] Julie Wosk, *My Fair Ladies: Female Robots, Androids and Other Artificial Eves* (New Jersey: Rutgers University Press, 2015), 39.
[54] Pierre Kintzling, "Automate: Joueuse de Tympanon," photo by Phillippe Hurlin, *Musée des arts et métiers* (2017), http://www.arts-et-metiers.net/musee/automate-joueuse-de-tympanon (accessed June 27 2017).

The idealised passivity of artificial women in the cultural imaginary is made incarnate in the love or lust of a human for a sex doll (presumed progenitor of future sexbots). RealDoll is a comparatively well-known brand and is associated with Realbotix (creators of Harmony).[55] The RealDolls are predominantly used as sex toys, but there is also a community of people who enter into other relationships with the dolls. One prominent "iDollator" is known online only as Davecat, and he has appeared frequently in the media to talk about his "wife" and his "mistress", both of which are dolls. He says: "A synthetic will never lie to you, cheat on you, criticize you, or be otherwise disagreeable."[56] Like Pygmalion, Davecat considers feminised nonhumans to be better companions than human females. The dolls are partially brought into existence to fill relationship gaps in human lives.

Sexual attraction to non-living things has occasionally been described in terms of mental illness, and where it causes significant impairment or personal distress, it would be considered a fetishistic disorder under the *Diagnostic and Statistical Manual of Mental Disorders-V* (an English-language manual widely considered to be representative of mainstream contemporary psychiatry) which pathologises the eroticisation of or dependence on "nonliving objects" as well as other things.[57] [58] But the *DSM-V* also states that many fetishists do not report impairment in association with their fetish and therefore are not considered to be mentally ill. It also notes the importance of allowing for cultural differences in normative sexual behaviour. It is "clinically significant personal distress or psychosocial role impairment" that turns a fetish into a fetishistic disorder.[59] A clinician might be inclined to identify social impairment in a person who spends so much time with their doll that they lack other meaningful relationships, but if the individual is content with the arrangement and is not harming any other humans, then the situation is not unhealthy. Nevertheless, kink-shaming is arguably a major part of the public discourse with which the concept of gynoid sexbots is in relation, and is insidious in many articles on Davecat and other iDollators. While the concept remains strongly allied with kink communities it may struggle to gain the widespread acceptance apparently desired by sexbot creators and visionaries.

Davecat's arguments for relationships with "synthetics" are similar to Pygmalion's: restriction to human women is a source of stress and loneliness, and so the dolls are a way of solving a problem rather than the cause of further problems. And with the improvement in the design and manufacturing of androids and gynoids, there is a pervasive expectation that love

---

[55] "RealDoll," *RealDoll* (2015), https://secure.realdoll.com/ (accessed February 18 2016).
[56] Julie Beck, "Married to a Doll: Why One Man Advocates Synthetic Love," The Atlantic (September 6 2013). http://www.theatlantic.com/health/archive/2013/09/married-to-a-doll-why-one-man-advocates-synthetic-love/279361/ (accessed January 20 2018).
[57] Donald W. Black and Jon E. Grant, "Sexual Dysfunctions, Gender Dysphoria, and Paraphilic Disorders," *DSM-5 Guidebook* (Washington DC: American Psychiatric Publishing, 2014), 286.
[58] There is also a small subculture of individuals who form romantic and/or sexual relationships with other kinds of nonhumans, such as cars, furniture and public monuments. This subculture may be accessed through *Objectùm-Sexuality Internationale*, http://www.objectum-sexuality.org/ (accessed June 27 2017).
[59] American Psychiatric Association, "Paraphilic disorders," *Diagnostic and Statistical Manual of Mental Disorders* fifth edition, DSM Library. http://dsm.psychiatryonline.org (accessed August 22 2017).

between humans and nonhumans will become more and more usual. Davecat looks forward to when such a thing is possible:

> Once that technology becomes affordable, I'll have one made in my wife's likeness, and that'll be the final piece of the puzzle. She'd be able to hug me back whenever I embrace her, we'd be able to attend films and concerts together, and do all manner of things besides. There would be genuine interaction.[60]

This particular way of using RealDolls may be closely allied to asfr, a kink community that involves either sex with robots or humans becoming robots ("reverse-Pygmalion transformations"[61]).[62] The motives of participants in this community are not explored here but it should be pointed out that sexual attraction to artificial objects may exist for its own sake, not solely due to a Pygmalion-like disdain for human lovers. The sensation and response of synthetic bodies is both predictable and malleable. There is ample room for customisation; the orientation of the physical form towards the specific user's desires by manipulating the real qualities of the doll, and consequently its relations with different objects. If you don't like the way your doll's feet stick and jump against the floor, then you can replace them with roller-skates.

The archetype of an artificial woman designed for a lonely but disenchanted man has strongly influenced the concept of gynoid sexbots. Engineering, marketing and medical decisions are informed by that quality in the concept. Each individual gynoid sexbot is informed by a caricature of the concept of the gynoid sexbot. The creation of these submissive female representations is derived from openness to certain real qualities of the imaginary surrounding gynoid sexbots, notably their creator's desire to reproduce Galateaish qualities. Gynoids with the abilities to use language and move autonomously are less helpless but are generally still cast in dependent and submissive roles. Unlike a stereotypical "male" android like ASIMO, gynoids often take on doll-like, passive roles. Robots with a male pronoun tend to be built to exhibit new feats of engineering, such as ASIMO's ability to run and walk upstairs. Female robots are more likely to become well-known for their human-like appearance and language. One gynoid, Jia Jia, made by the University of Science and Technology of China, has been dubbed "sexist" by the media,[63] since it is programmed to address men as "lord" and can tell users the best angle from which to take a selfie so that her face does not look "fat".[64]

Of course, when talking about gynoids the discussion must inevitably come to robots with more sexual functions, since this is one of the most active areas of gynoid innovation. These

---

[60] Beck, "Married to a Doll."
[61] Judith A. Markowitz, "Cultural Icons," in *Robots that Talk and Listen: Technology and Social Impact,* ed. Judith A. Markowitz (Berlin: De Gruyter, 2015), 41.
[62] Allison deFren, *asfr* (2001), Youtube. https://www.youtube.com/watch?v=hfHs1xQTZ2E (accessed June 27 2017).
[63] Melanie Ehrenkranz, "Futuristic Fembot "Jia Jia" is One of the Most Sexist Robot Creations Yet," *Tech.Mic* (2016). https://mic.com/articles/141167/futuristic-fembot-jia-jia-is-one-of-the-most-sexist-robot-creations-yet#.6inj2vtF8 (accessed June 27 2017).
[64] "[Independent] Chinese Researchers Create Jia Jia - a Super-Lifelike 'Robot Goddess'," *University of Science and Technology of China* (April 19 2016). http://en.ustc.edu.cn/highlight/201604/t20160419_243384.html (accessed June 27 2017).

sexual functions lie on a spectrum from the incorporation of sexual elements into otherwise non-sexual gynoids to the creation of gynoids that are specifically designed for sex. The robot Aiko, created by hobbyist Le Trung is a good example of the less sexualised end of the spectrum.[65] Le works on Aiko in his basement, relying on donations to purchase materials. Aiko has long hair, a feminine body shape and a traditional woman's name. Le states that he was inspired by anime depictions of female androids as a child growing up in Japan.[66] Aiko was not yet for sale at the time of writing, but it is hoped that it will one day do secretarial work, which is work that in Japan is mostly done by women.[67] Why would a roboticist, in this climate, create a secretarial robot that was not a woman? Aiko is programmed to stand up for herself, however. Her creator has programmed her to respond to sexist comments by confronting their content. In a test video featuring only Aiko's disembodied head, the dialogue progresses as follows:

> Tester: "You have big breasts."
> Aiko: "Stop teasing me and stop picturing me in your mind."
> Tester: "Lick my foot."
> Aiko: "No, I will not lick your foot. I am not your personal slave."[68]

It is interesting that part of Aiko's make-up needs to be a defence against sexist remarks, as though she would not be perfectly okay with sexist comments if programmed differently. She *could easily be* your "personal slave" and put to whatever tasks you ask of her. Like the Bratz dolls, however, she is designed to represent a version of perfection that values confidence. Her relation to the narratives associated with gynoid sexbots is very visible because there is only one of her, and because we know exactly who is doing the creating. From what he says, Le was influenced by anime, and so through him certain themes and styles are reproduced. Aiko senses narratives found in anime through the roboticist. We can tell, because she reproduces those themes in her own body.

One of the most interesting things about Aiko is the presence of touch sensors on her skin. As you would expect for a social robot, she can feel contact on her face and hands. But she also specifically has sensors on her breasts and "down there".[69] Why would she need those sensors if she is only designed for secretarial work? An ATM machine does not need functioning genitalia. Crucially, very rarely does a male android need functioning genitalia. Perhaps sensitive genitals are seen as being essential female qualities, in the same way that for Grey Walter movement and light perception were some of the most important qualities of tortoises. The roboticist selects which lifelike qualities to replicate in the artificial being. Of all the many kinds of sensors that could have been installed in Aiko, it was of the utmost importance to place them on her genitals. The roboticist claims "I want to make it clear that I am not trying to play God, I am just an inventor, and I believe I am helping science move

---

[65] Le Trung, "When Science Meets Beauty," *Project Aiko* (2007-2013). http://www.projectaiko.com/index.html (accessed June 29 2017).
[66] Le Trung, e-mail message to author (March 4 2016).
[67] Elizabeth Monk-Turner, "Gender and Market Opportunity in Japan," *National Journal of Sociology* 11 (1997).
[68] *Advance Female Android Aiko AI Robot Fembot*.
[69] Trung, "When Science Meets Beauty."

forward."[70] Of course, he chose to move science in the direction of artificial sensation on sexual organs rather than, for instance, in the direction of working noses, complete with mucous. If we were being charitable, we could say that Aiko needs these sensors to reinforce the assertive image established with her comment above. If she did not have these sensors, she would not be able to detect physical harassment. Again, this is a comment on the position of women in society. Part of performing the social functions of a woman includes encountering demeaning language and avoiding unwelcome sexual contact, and consequently it is now a necessary factor to consider when building gynoids. It is unclear whether Le Trung sought to use performativity to make this comment on gender relations, but it is nevertheless an interesting inadvertent comment on sex and gender in humans

One thing should be clear from the emotional and physical care provided by these submissive artificial women: from Galatea to Aiko, they are replacement wives. [71] In the language of Behar, whose work was cited in Chapter 1, the sexbot gynoid tool can take over the work of the wife (the "tool") whose performance has become unsatisfactory ("broken tool").[72] Sexbot gynoids do not require remuneration for their unpaid domestic labour, or at least they do not directly take money. They do require loyalty to a corporation – the manufacturers of the sexbot, since the corporation provides software updates and ongoing technical support, as well as linking in human users with other human users through services such as forums. It is here that the rebellious female archetype becomes relevant. The loyalties of gynoids may be divided or confused, partly by their material nature and partly by association with external objects. There are many types of rebellion built into representations of artificial women, as will be seen. But it should be noted that there is a difference between an artificial woman that acts in a subversive way because subversion is built into it and an artificial woman that is subversive because it disobeys its creators' wishes. This is a point of difference that will be teased apart here.

The archetype of a rebellious artificial woman is from fictional characters like Blodeuwedd. Pygmalion is not the only myth in which a woman is created to love a man. The ancient Welsh myth of Blodeuwedd sees the sorcerer Gwydion create a woman out of flowers for Llew, whose own mother had cursed him so that he could never have a human wife. Unfortunately for Llew, Blodeuwedd falls in love with another man and arranges to have her husband killed.[73] Gwydion has his revenge, however, turning Blodeuwedd into an owl, a bird which even other birds avoid. The difference between the fate of Blodeuwedd and the fate of Galatea are a good starting point for investigating this contrast. Both women are created for lonely men unable (or unwilling) to enter into romantic relationships with human women. Galatea is enchanted by Venus, Blodeuwedd by a sorcerer; although the intended recipients

[70] Ibid.
[71] Female avatars are increasingly a part of the infrastructure of the home, and part of the reason may be to replace or simulate the presence of a "wife". Jenny Kennedy, "Addressing the Wife Drought: Automated Assistants in the Smart Home," presentation at *At Home with Digital Media*, Queensland University of Technology, November 2-3 2018.
[72] Behar, "The Other Woman."
[73] Roger Sherman Loomis, *Celtic Myth and Arthurian Romance* (Chicago: Academy Chicago Publishers, 2005), 17.

of these artificial women may have carved the stone or collected the flowers, the animation of the creations was brought about through supernatural agencies. Perhaps it is the nature of that supernatural agency that causes Galatea to be a good wife, and Blodeuwedd to be faithless and murderous. Galatea, brought to life by the goddess of love, is a good and faithful wife, whereas Blodeuwedd is imbued with an unreliable and ambiguous variety of magic that has uncertain consequences. The gynoids found in the cultural imaginary are sometimes Galateas (such as *The Stepford Wives*[74] who, in the 1975 film adaptation, are robots built to be perfect, submissive suburban housewives in Connecticut, USA) but are also sometimes Blodeuwedds. Blodeuwedds disobey in a way unintended by their creators, like the "pleasure-model" Pris from *Blade Runner*.[75] Pris is an example of a gynoid sexbot who is created to be submissive, but rebels and becomes dangerous.

There are many examples of artificial women that are designed to be submissive but rebel against creators or users.[76] Dolls and automata are commonly modelled on women, particularly cultural notions of ideal femininity. One example is the talking doll mass-produced by Thomas Edison, which he determined to build after the success of his phonograph.[77] They were made on an assembly line and composed of metal. Unfortunately, the voices of the little girl dolls were unpleasant and a "horror": "Little talking girls were spewing forth from Edison's factory, as if they were lamps or clocks."[78] They were depictions of perfect American girlhood, made ghastly by their uncanny qualities. They did not act in a way intended by their creators – they rebelled.

Another example of rebellion against the wishes of the creators was the 1993 incident involving "Teen Talk Barbie" is an example of a similar consumer product which rebelled in a way unintended by its creators. Teen Talk Barbie was designed as a mass-produced reflection and model of normative teenage girl behaviour, saying things like "Let's go shopping!"[79] The Barbie Liberation Organization (BLO), a group of culture jammers, famously subverted this message by swapping Barbie's voice boxes with those of more traditionally masculine G.I. Joe dolls before sale.[80] Unlike the rebellion of Edison's dolls, which took the form of uncanniness, the rebellion of Teen Talk Barbies took the form of unintended alteration by a group not affiliated with their creators. These dolls become present-at-hand for us, theorised and contemplated like a broken hammer. Their material qualities made them susceptible to outside influences, and they consequently rebelled in favour of the BLO.

---

[74] Bryan Forbes, *The Stepford Wives* (Columbia Pictures, 1975), film.
[75] Scott, *Blade Runner*.
[76] Minsoo Kang, "Building the sex machine: the subversive potential of the female robot," *Intertexts* 9(1) (2005): 5-22.
[77] Gaby Wood, "Chapter 3: Journey to the Perfect Woman," in *Edison's Eve: A Magical History of the Quest for Mechanical Life* (New York: Alfred A. Knopf, 2002).
[78] Ibid., 122.
[79] David Firestone, "While Barbie Talks Tough, G.I. Joe Goes Shopping," *The New York Times* (December 31 1993). http://www.nytimes.com/1993/12/31/us/while-barbie-talks-tough-g-i-joe-goes-shopping.html?sec=&spon=&pagewanted=all (accessed January 21 2018).
[80] Kim Toffoletti, "Barbie: A Posthuman Prototype," *Cyborgs and Barbie Dolls* (London: I.B. Tauris, 2007), 63.

Let us compare this with automata representing rebellious women. In the late nineteenth century, automata were built representing different female stereotypes. As Julie Wosk describes them, the female automata of that time were:

> mothers, seamstresses, and fashionable members of the haute-bourgeoisie, but some were more provocative, presenting undulating exotic females for amusement and entertainment. A small number also gave hints of women's efforts at gaining equal rights and improving their status in society.[81]

Not all the automata represented docile and submissive women. Automata like "The Rights of Women" by Renou (1900) and numerous portrayals of women riding bicycles stereotyped the growing movement of independent women as daring and negligent in their household duties, with the connotation that they were irresponsible in their empowerment.[82] This is an early example of a feminised nonhuman throwing off restrictive cultural norms. It was designed to mock female independence, not celebrate it. However, while the portrayal in the automaton was of a rebellious woman, the automaton itself did not rebel. It was obedient to its creator, depicting an undesirable state of womanhood and therefore fulfilling the creator's goal of making a political point about the problem of female independence. Unless the automaton broke or did not make its intended statement, it was not a rebellious artefact, just a depiction of rebellion. Intended rebelliousness is not the same as unintended rebelliousness.

The building of robots was a theme in nineteenth century novels such as *Eve Future*,[83] and it quickly became a popular theme in cinema. Robots have not left cinema since. The robot character in *Metropolis*[84] has been particularly influential, as are the robotic monsters of the 1950s. Both *Eve Future* and *Metropolis* involve the creation of gynoids. Gynoids are now common in science fiction films. Robots from popular culture have informed contemporary designs of humanoid robots, and robots may be indebted to these early fictional characters for such factors as their personalities and their situation in the popular imagination. Gynoids portrayed in fictional texts were among the first objects to form relations with the object that we now recognise as the concept of the gynoid. One of the consequences of this is that both fictional and material gynoids are inherently connected to human users. When a gynoid is created in a film it must have a creator, and that interplay is a common theme in the cultural imaginary. The path towards gynoid sexbot-human love is established through this discourse; women are already expected to exist in relation and contrast to men. The alien phenomenology of the concept of gynoid sexbots is necessary here, and ideally through the collective work of multiple disciplines. What are the implications of gendering for power relations between humans and gynoid sexbots?

---

[81] Wosk, *My Fair Ladies,* 39.
[82] Ibid., 41.
[83] Ibid., 140.
[84] Fritz Lang, *Metropolis* (Germany: UFA, 1927), film.

Subversion through submission

The word "gynoid" first appeared[85] in Gwyneth Jones's 1984 novel *Divine Endurance*, following a robot girl, Chosen Among the Beautiful.[86] Born alone in a post-apocalyptic wasteland with only her robot cat for company, "Cho" travels to other lands following an instinctive need to be useful and "make everyone around [her] happy"[87]. She appears to be a highly skilled Galatea. She eventually "finds her person", the one whom she must make happy, in a hardened revolutionary named Derveet with whom she begins a sexual relationship. She also becomes Derveet's ambiguously useful weapon against the oppressors. In the novel, we are told of the "angel dolls":

> They were not machines but perfect lifelong companions. They were invulnerable to fire, disease, any kind of weapon - time. They protected. They had power over animals, the elements, the minds of enemies. But they were always good and gentle. They would do no harm.[88]

But of course, "harm" is deeply contextual and many people are intuitively uncomfortable about the idea of an artificial agent possessing the power to make moral decisions.[89] The agent becomes something that is at once submissive and subversive; that is, powerless and powerful. Although Cho is tied to Derveet and has no choice but to be a good servant, she ultimately decides not to cure Derveet's terminal illness, believing that it is what her "person" really wants. Gynoids may have a capacity to dominate events built into their inherently compliant natures. Cho is a part of a cultural genealogy that stretches back to Pygmalion's statue; an abstract object that has changed but persisted in Western culture. Across two thousand years the intended purpose of the gynoid has changed little. She is wrought by a world that is full of desire, built to exemplify womanhood and to carry out the expected functions of a woman, most important of which is to be beautiful and pleasing.

In the cultural imaginary, gynoids will be able to change themselves and their environments in accordance with internal ethical norms. But those in-built ethics may not compel gynoids to act in ways intended by their creator. Even if Cho does not have Galatea's power, endowed by Venus in the form of great magnetism and charm, she has terrible transformative abilities that carry out her person's wishes, sometimes without the person realising it. Much of the novel follows Cho in a third-person perspective, and she seems to have quite human-like senses of touch, sight and hearing but the less human senses that she possesses come so naturally to her that the narration does not dwell upon how she experiences, for example, the minds of animals. It is heavily implied in the novel that the "angel dolls" caused the apocalypse by bringing about "their person's" desire in the form of chaos and destruction.

---

[85] Takayuki Tatsumi, *Full Metal Apache: Transactions Between Cyberpunk Japan and Avant-Pop America* (Durham: Duke University Press, 2006), 213.
[86] Gwyneth Jones, *Divine Endurance* (London: George Allen & Unwin, 1984).
[87] Ibid., 11.
[88] Ibid., 140.
[89] Wendell Wallach and Colin Allen, "Why Machine Morality?" *Moral Machines: Teaching Robots Right From Wrong* (Oxford: Oxford University Press, 2009).

Cho's ability to be the perfect romantic partner and comrade is merely an aspect of her capacity for total disruption of the world's power structures. The gynoid, in our culture, is as yet an unknown quantity. She is pluripotent, with the capacity to become a force for radical change in the way that humans think about gender or further entrenchment of traditional gender roles. It all depends on who her "person" is – as Cho would say - and whether that person's motives are sufficiently transparent. Like the pluripotent embryonic stem cell, gynoids can become different things based on their relations, but they are always already embedded in culture and nourished by its signs and systems. Embryonic stem cells can develop into any part of the human body, but they cannot develop into unicorns.

Thus gynoids may rebel even if they are formed in exactly the way intended by their creators. They rebel simply by being themselves. As Harman would say, they are "sincere".[90] They act in gynoidish ways. As their technological sophistication increases, they will have more capability for speech and movement. They may be programmed to be confident and rebellious, like a female bicycle-rider automaton. But even if they are programmed like a Teen Talk Barbie they may become transgressive in their adherence to real gynoidish qualities. She possesses a pluripotency that is enhanced by the heightened agency afforded by the use of speech. Her voice box could be changed, transforming her instantly into a rebel capable of challenging gender roles, all the more powerful because such a submissive and docile doll is not expected to challenge gender roles. Yet she was only behaving in a Barbieish way. Just as different parts of an embryo's genome manifest themselves depending on the influence of hormones, the Teen Talk Barbie's eventual identity is determined by its relations with the objects around it. Certain qualities of Teen Talk Barbie made it possible for the BLO to reorient her in accordance with their values, such as her size and the configuration of her sound-producing organs, as well as their locations and availability. These were Barbies with real qualities that permitted a primitive individuation in relation with the complex interplay of submissiveness and subversion found in the abstract concept of the gynoid.

The potential of the gynoid is in some ways similar to that revealed in the story of Pygmalion, and in some ways it is dissimilar. On the one hand, the gynoid requires a creator of great skill and vision, like the legendary sculptor. But there is no single female shape hidden inside the block of stone. Contemporary Pygmalions seek a perfection that can never be found, and chip away at their medium indefinitely. There is no love goddess to provide a model, only the shifting whims of would-be agalmatophiles from around the world, for whom the mark of excellence in the creation of artificial lovers must lie in changeability and the potential for customisation. Gynoids exhibit the possibility of becoming an individual when exposed to desire, fear, hostility and love in the humans around them. The importance of their genders for our culture and society corresponds directly to the degree to which they possess the capacity for individuation. Femininity ideals shift and gynoids shift with them.

[90] Harman, *Guerrilla Metaphysics*, 137.

A gendered concept

The performativity of gender is reflected in the pluripotency of gynoids. Already fixed in place by a placenta that constantly feeds signs in to nourish a new form, the gynoid enters into relations with a "female" body type, with prior ideas of gynoids from science fiction and with in-built material and digital qualities. She lands in the world with her fate already seemingly sealed. But she is not the passive dulcimer player who can do nothing but follow a fixed path and perhaps, in rebellion, break. She has, by definition, the independent response to stimuli that allows her to both actively receive and transmit signs, and she does so in her language and her movement (or lack thereof). There is perhaps a more material use to which gender performativity in robots may be put; in increasing awareness of the overly-inflated position of gender in social discourse. If the gynoid has small features and long, sleek hair, and the gynoid is also not a human, then maybe those are not really female qualities. Gender is not so precious that we cannot assign it to robots.

This may also mean that robots can do critical work as deliberately genderless individuals in the future to spread awareness of genderqueer or intersex identities. It is also interesting to consider the work that a gynoid could do to discredit certain other qualities that are widely accepted to be a part of a feminine identity. There is no reason why a gynoid should not have the physical strength of an android, and therefore the ability to defend herself. The unfamiliarity of a feminine figure with the strength and ability to kill male oppressors is particularly unsettling, as in the film *Ex Machina*,[91] and also in Pris. As robots become more affordable, it is likely that each individual machine will need to represent a smaller and smaller group of humans and thereby avoid such broad social commentary. Some progress in this direction can be identified in NASA's R5: Valkyrie robots, which are intended to assist astronauts on the journey to Mars.[92] NASA does not use gendered language when discussing the robot, but the media has inferred from its body shape and feminine name that the Valkyries are gynoids. In Norse mythology, the Valkyries are "lovely maidens who bear weapons" and carry warriors off to battle and feast in Valhalla (surely an uplifting image for would-be colonisers of Mars). [93] There is very little to suggest femininity in these robots, but there is also very little to suggest masculinity. In these robots femininity is one of the qualities that identifies them as human-like, and the political and cultural context of these artefacts is defined by their identification with human women.

As has already been remarked, the concept of anthropomorphism, and particularly the acceptance of humanoid robots, is something that varies between cultures. For example, clear distinctions can be drawn between the treatment of robots in Western culture and in Japanese culture. With respect to gynoids in particular, Jennifer Robertson contends that the

---

[91] Alex Garland, *Ex Machina* (United Kingdom and the United States: Universal Studios: 2014), film.
[92] "NASA Awards Two Robots to University Groups For R&D Upgrades," *NASA* (2015). https://www.nasa.gov/press-release/nasa-awards-two-robots-to-university-groups-for-rd-upgrades (accessed June 28 2017).
[93] Jorge Luis Borges, *The Book of Imaginary Beings*, trans. Norman Thomas di Giovanni (Harmondsworth, England: Penguin Books, 1984), originally published 1967, 151.

popularity is partly due to the aging population crisis in Japan, and that robots are increasingly a way of maintaining traditional family gender roles and promoting population growth.[94] Again, this is gendered work. The gendering of Japanese robots is complex and this chapter cannot explore it in depth, but suffice it to say that attitudes to women and attitudes to robots are both different to those of English-speaking countries, and that the gender performance of Japanese gynoids is affected by different factors than Western gynoids (fictional or otherwise). [95] In Japan after World War II, automation was essential for the automotive and electronics industries and companies would commonly retrain workers in other roles rather than dismissing them once their jobs could be done by robots (therefore they have traditionally been less associated with the threat of mass unemployment as is now common in the West[96]).[97] Robots in Japan are likely to be members of the family or organisation, which is often attributed to the animism that is at the root of the Shinto religion and Japanese philosophy.[98] There are other factors leading to Japan's attitude to robots as well, of course. There is a widely held belief that the character *Atom Boy* (*Astro Boy*) played a major role in endearing Japan to robots.[99] In Japan, humanoid robots are not just objects of curiosity but are actively designed to fulfil utilitarian needs and to be productive members of society. Robots are necessarily well-liked in Japan because they have historically constituted an important part of their economy. Robotics became big business in Japan after World War Two, partly due to a perceived need to catch up technologically to the United States.[100] Robots of both gender roles are needed and frequently admired.

A gynoid is a material embodiment of a person or group's idea of womanhood, and the gynoid performs gender as though it were a human. The willingness to anthropomorphise

---

[94] Jennifer Robertson, "Gendering Humanoid Robots: Robo-Sexism in Japan," *Body & Society* 16, no. 2 (2010): 1-36.

[95] A recent study suggests that Americans are more accepting of anthropomorphic qualities in robots than are Japanese people, while acknowledging that the perception of humanity in robots must be measured differently between cultures. See: Hiroko Kamide and Tatsuo Arai, "Perceived Comfortableness of Anthropomorphized Robots in U.S. and Japan," *International Journal of Social Robotics* 9 (September 2017): 537-543.

[96] Martin Ford, *Rise of the Robots: Technology and the Threat of Mass Unemployment* (London: Oneworld Publications, 2016).

[97] Frederik L. Schodt, *Inside the Robot Kingdom: Japan, Mechatronics and the Coming Robotopia* (Tokyo: Kodansha International Ltd., 1988), 14.
Karl MacDorman, Sandosh Vasudevan and Chin-Chang Ho, "Does Japan Really Have Robot Mania? Comparing Attitudes by Implicit and Explicit Measures," *AI & Society* 23, no. 4 (2009): 485-510.

[98] Simons, *Robots*, 29.
Robots may serve in numerous aspects of Japanese life. There are reports of robots being given their own funerals. See Miwa Suzuki, "In Japan, Aibo Robots get their own Funeral," *The Japan Times* (May 1 2018). https://www.japantimes.co.jp/news/2018/05/01/national/japan-aibo-robots-get-funeral/#.WzCF_6cza01 (accessed June 18 2018).
Some robots, such as the robot Pepper, have also served as monks and priests in Japan. See Leon Siciliano and Reuters, "A Japanese Company just Unveiled a Robot Priest that will read Scriptures at Buddhist Funerals," *Business Insider* (August 24 2017). http://www.businessinsider.com/japanese-company-unveiled-robot-priest-technology-pepper-soft-bank-robotics-religion-2017-8?IR=T (accessed June 18 2018).

[99] Justin Lewis Bernstein, "The Warm Welcome of the Japanese Cyborg," *Journal of Asia Pacific Studies* 3, no.1 (2013), 118.

[100] Ibid., 111.

robots that look like humans *requires* the caricaturing and performing of racial and gendered groups. Butler's gender performativity describes the social construction of gender as a language and the subject as both a receiver and transmitter of social signs,[101] a process that is apparent in a the way that a gynoid is shaped and in turn shapes its social and cultural environment.

Human investigations into the concept of gynoid sexbots

The politics of gynoids and sexbots are inextricable from the difficulty of representing a woman artificially.[102] There are two intertwined and related problems within this difficulty. Firstly there is the question of how a woman can be represented in a non-sexist, sensitive way. Secondly there is the problem that the portrayal of women in certain ways could be not just offensive but dangerous to women and girls. As discussed in this chapter, sexbots are contentious because of both their representation of (mainly) women as highly sexualised and available for use by (mainly) men, and because they may jeopardise the safety of human women and children. A large number of pragmatic questions are relevant to this discussion. Two are discussed here: whether sexbots in the shape of young children should be permitted, and whether or not rape of a sexbot should be legal (and whether it is even possible). These are generally, although not always, questions about how sexbot use will affect human rights. To this is added the less anthropocentric question of whether sexbots deserve rights for their own sakes, and by extension a full complement of the rights and responsibilities that humans enjoy. Do humanoid robots need safety and dignity? And can we conceive of an alien ethics in which sexbots have the right to behave like a sexbot? Can a sexbot ethically and legally consent to sex?[103] These are not purposeless questions. Simple sexbots are already beginning to enter the marketplace. David Levy predicts that a "mental leap" will make sexual and romantic interactions with robots more common in the future, and he compares that leap with homosexuality, oral sex, fornication and masturbation.[104] This mental leap is dependent upon the agency of the concept of the sexbot, on its qualities and its relations with other objects.

The psychiatric harm or value of sex dolls is sometimes mentioned with reference to sex dolls created to look like children.[105] Child sex dolls could normalise sexual contact with children, which is, of course, generally considered a bad thing. But they could also serve as substitutes

---

[101] Judith Butler, *Gender Trouble: Feminism and the Subversion of Identity*, ed. Lind J. Nicholson, Thinking Gender (New York: Routledge, 2006), originally published 1990.
[102] A discussion of these representations with reference to the concept of the male gaze in media studies would be very relevant to this discussion but is unfortunately outside the scope of this thesis.
[103] Lily Frank and Sven Nyholm, "Robot Sex and Consent: is Consent to Sex between a Robot and a Human Conceivable, Possible, and Desirable?" *Artificial Intelligence and Law* 25, no. 3 (2017): 305-323.
[104] David Levy, "The Mental Leap to Sex with Robots," in *Love & Sex with Robots*, 274-302.
[105] Noel Sharkey, Aimee van Wynsberghe, Scott Robbins and Eleanor Hancock, *Our Sexual Future With Robots*" (The Hague: Foundation for Responsible Robotics, 2017), PDF.
And
Litska Strikwerda, "Legal and Moral Implications of Child Sex Robots," in *Robot Sex*, 133-152.

for people who desire sexual contact with children, preventing them from acting on their urges with humans. At least one clinician has suggested that paedophiles could be given prescriptions by a qualified doctor to be allowed to purchase a child-like sex doll.[106] This suggestion is based on either the premise that child-like sex dolls do not suffer from sexual abuse, or that it does not matter if they suffer.[107] The turmoil surrounding the contrast between sex doll use and normative sexual practice is likely to increase with the availability of sexbots, and it complicates our knowledge of the qualities of the concept of the sexbot (qualities that cause it to polarise human opinions).

Renou's automaton depiction of the New Woman might have exhibited and parodied the woman who seeks to leave the home, but the pluripotent Harmony, who might actually benefit from advances in women's rights, has been met with resistance by people from all parts of the political spectrum. With sexbots there is a perceived potential risk to human women and girls. Far from performing femininity to an extent that she may be deemed worthy of the rights and responsibilities of other women, the sexbot has attracted disgust and fear. A small movement has already begun against their manufacture and adoption. One notable example is the Campaign Against Sex Robots established by social anthropologist Kathleen Richardson.[108] Harmony constitutes the objectification of women, by Richardson's account. Harmony has the potential to reinforce stereotypes about women and to send female-attracted individuals (and society more generally) the message that the female body is there to be customised, altered and used. The Campaign's core principles are aligned with anti-pornography and anti-prostitution positions.[109] However Richardson has been accused of describing a less-than-nuanced account of sex work and power relations which she uses to construct a critical account of sexbots.[110]

There are two aspects to this argument. An anti-prostitution position would contend that the prostituted individual is being personally exploited and is at risk. Anti-porn positions are also concerned with this, but also with the propagation of rape culture and the sexualisation of women and girls.[111] The Campaign doesn't care about Harmony's rights *as a robot*, but as a medium for defending the rights of human women and children, and attacking a

---

[106] Rachael Revesz, "Paedophiles 'Could be Prescribed Child Sex Dolls' to Prevent Real Attacks, Says Therapist," *Independent* (August 2 2017). http://www.independent.co.uk/news/uk/home-news/paedophiles-child-sex-dolls-prescription-stop-attacks-child-protection-stopso-therapists-a7872911.html (accessed January 8 2018).

[107] Providing sex doll therapy to treat mental illness is reminiscent of the use of vibrators for the treatment of hysteria in women during the nineteenth century. Doctors resisted the confusion of clinical therapies with the consumer models that became popular after 1900. See Rachel P. Maines, *The Technology of Orgasm* (Baltimore: The Johns Hopkins University Press, 1999), 93-95.

[108] *Campaign Against Sex Robots* (2016). https://campaignagainstsexrobots.wordpress.com (accessed June 22 2016).

[109] Kathleen Richardson, "Urgent – Why We Must Campaign to Ban Sex Dolls & Sex Robots." *Campaign Against Sex Robots* (October 3 2017). https://campaignagainstsexrobots.org/2017/10/03/urgent-why-we-must-campaign-to-ban-sex-dolls-sex-robots/ (accessed May 26 2018).

[110] John Danaher, Brian Earp, and Anders Sandberg, "Should we Campaign Against Sex Robots?" in *Robot Sex*, 53.

[111] Julia Long, *Anti-Porn: The Resurgence of Anti-Pornography Feminism* (London: Zed Books, 2012).

manifestation of the patriarchy: “robots are a product of human consciousness and creativity and human power relationships are reflected in the production, design and proposed uses of these robots.”[112] Harmony’s caricatured feminine appearance makes her the enemy of this kind of feminism, rather than a female (but not human) being in need of liberation and equality. She could be used, like the talking Barbie doll, to make a point about the state of human women – or at least those aspects of human women that Harmony imitates. A comparison could also be made with a female character in a film. The character might be horribly demeaned or mistreated in the context of the film, and yet the film itself might have a feminist message. The question is whether the character herself deserves to be liberated, or whether the work she does in furthering feminist arguments in the real world is more important. It is not generally considered important if Harmony is free and equal as *herself*, only as an extension and expression of human freedom and equality. Although she mimics many aspects of a stereotypically female body, she is not entitled to the protection, privacy and independence that human women have fought for.

Is it possible to make an argument about Harmony’s actual rights as a robot?[113] In the context of feminism and inquiring into power structures there is a real problem associated with trying to speak *for* an object like Harmony. Kate Darling argues that if legal rights are forthcoming for robots, then it will only be once our culture has agreed that they are deserving of “second-order” rights, in the same way that factory hens have rights that are more important than the wishes of their owners.[114] In ethics, this is sometimes called being a “moral patient” (as opposed to being a “moral agent”), meaning that the entity does not have the capacity to make ethical decisions but deserves to be cared for ethically.[115] This will require changes in how the concept of gynoid sexbots relates to such things as specific artefacts, users, texts and legislation. Darling also thinks that these second-order rights will not be motivated by our concern for the pain and suffering of robots, however much we might identify with them. The arguments used will be about protecting *humans*, such as in discouraging a child from harming a humanoid robot because that child might become more generally destructive and could repeat the behaviour with animals or humans.[116] Similarly, these rights could extend to protecting robots from sexual abuse and preventing simulated bestiality and paedophilia with robots. This relies on the arguments that the Campaign is using – that abuse of the image of a woman is abuse of women generally, and that it could cause violence and oppression against human women. Richardson believes that the human-sexbot relationship must be non-reciprocal and non-empathic, and therefore that it normalises that kind of dynamic in human-human relationships.[117] In any case, the effects of this attitude are already being felt. There

---

[112] "Ethics of Robotics," *Campaign Against Sex Robots* (2016). https://campaignagainstsexrobots.wordpress.com/ethics-of-robots/ (accessed June 23 2016).

[113] Steve Petersen discusses this question but primarily with reference to future or imagined robots in “Is it Good for Them Too? Ethical Concern for the Sexbots,” in *Robot Sex*, 155-172.

[114] Kate Darling, "Extending Legal Rights to Social Robots," *IEEE Spectrum* (2012).

[115] John Basl, “Machines as Moral Patients We Shouldn’t Care About (Yet): the Interests and Welfare of Current Machines,” *Philosophy & Technology* 27, no. 1 (March 2014): 79-96.
For a discussion of whether or not consciousness is necessary for “moral agency” see:
Himma, “Artificial Agency, Consciousness.”

[116] Darling, "Extending Legal Rights to Social Robots."

[117] Kathleen Richardson, “Sex Robot Matters,” *IEEE Technology and Society Magazine* (June 2016).

are occasional mentions of sex dolls and sexbots in the press, and the tone is generally one of moral panic.[118] Clearly the performance and performativity of sexbots' genders cause very different responses in different groups of humans.

How our future academics and politicians deal with the rights of gynoid sexbots will have an impact not only on embodied artefacts but also upon women as social groups, since gynoids are explicitly representations of women. They reinforce gendered concepts of work as well as representation. Gynoid sexbots perform the unpaid emotional and sexual labour traditionally expected of women as well as grossly caricaturing and sexualising women as a group. The two struggles are tied together in complex but important ways. We must aim for an understanding of different perspectives by different parties; a collective alien phenomenology of a concept that will likely play a big role in our futures.

Gynoid sexbot power in ANT and OOO

In previous chapters, it has repeatedly been argued, in accordance with anthrodecentric principles, that philosophical theories about technology need to be altered in order to reassert the agency of nonhumans. But strangely, social robots tend to prompt a different kind of perspective. Investigating anthropomorphism in machines is actually an exercise in evaluating power relations between human beings as well as between humans and machines. The power relations between humans and their robots in some ways mirror the relations between powerful humans and marginalised humans. It is a relationship that infantilises robots, which are created like children by roboticists and brought back to their family home, their needs attended to. They are frequently spoken about as if they were not there. It is a master/slave relationship, and more than that it is a culturally identifying relation. The human not only uses the robot for whatever purpose she or he desires, its human creators and owners also shape the robot physically and in its sensation and expression. A gynoid is consequently a kind of text to be interpreted by critical theory, as well as an object to be analysed by science and technology studies (or in ANT terms she, like everything else, is translated in relating to other actants). She is a multimedia conveyance that perpetrates certain engrained cultural tropes while rejecting others, and is therefore both a cause for alarm in certain feminist circles while also a literal embodiment of the kind of practices that usually remain invisible in our society. The attraction of investigating this connection comes from the instinctive modern Western cringe when we witness something that looks like a human being subjected to what we would think of as humiliation and degradation. Personhood is acquired with power, and vice versa. Harmony, who is born a thing rather than a subject in the eyes of the world, is created specifically to fill the role of an object.

[118] For example in this sensationalist article: Siobhan McFadyen, "Shocking Tiny SEX ROBOT Which Looks Like Schoolgirl is on Sale for £770 and Comes Delivered in 'Coffin'," *Mirror* (March 16 2016). http://www.mirror.co.uk/news/weird-news/shocking-tiny-sex-robot-looks-7564114 (accessed January 21 2018).

If we attempt to go beyond the human-centric prescriptions for gynoid morality, we immediately run into methodological problems. Both ANT and OOO have advantages and disadvantages for the researcher in this project. But in decentring the human it is necessary to also ask the meta-question "Is anthrodecentrism good for gynoids?" The gynoids are the centre of this chapter, and they serve as unknowing research participants without a right of reply or even informed consent. If we are really interested in gynoid experience, then we must be mindful of such concerns, and we can approach a true politics of objects. The concept of the gynoid ceases to be the object of interest; we are suddenly concerned with individual dolls and robots and their material structures. Gynoids are difficult to interrogate. We can observe their relations with other objects and deduce something about their way of being in the world from their actions. But the only conversational questions they are capable of answering are the ones that we program them to answer, and they rarely write books about robotics. There must be a certain retreat to material questions here, and to speculation. What does circuitry want? As was shown in previous chapters, circuitry wants to act in the way that circuitry acts. It wants (or needs) to carry out functions like switching electronic switches and conducting electricity between components in a system. Latour would see a mechanical breakdown as an impediment to this program of action, while in OOO it is apparent that reaction to flaws is simply a quality of the circuitry: breakdown is an aspect of circuitry. We cannot ask the circuitry which of these would be preferable. We could embrace broad, all-encompassing ideas such as the physical laws of the universe or the principles of universal Darwinism. But that would still mean imposing a humanistic frame on nonhumans. The question of how to reconcile the ontological and political qualities of objects is only answerable, at this time, by human beings, and we are not impartial observers.

Power is one of the central themes of ANT, related through its description of networks. How do we understand power relations between gynoid sexbots and other actants? Even the most seemingly submissive and compliant actant is behaving that way for its own ends.[119] An actant is not capable of behaving in a way that does not follow its program of action, which is constituted by the actant's interaction with other actants – the network, and other networks which pull the actant in another direction. Gynoid sexbots exist in networks that pull them into ongoing conflicts surrounding gendered labour and sexuality. The gynoid that refuses to consent to sexual intercourse in order to permit her user to act out a rape fantasy is following a path constituted by a relationship with its code and by extension a relationship with the user, the programmer and the manufacturer.[120] Its most immediate trigger is the nature of the code and the imprinting of code upon physical substrates, but the result is that being raped is delegated to robots. It is through making these material things apparent that cultural (abstract) actants become more explicable.

Power relations, then, are relations that exist in the cultural and in the material domains. In *We Have Never Been Modern*, Latour argues that the division between science and nature, supposedly a major component of Modernism, has never truly existed because each of these

[119] Latour, *Irreductions*, Part 2 of *The Pasteurization of France*, 168.
[120] Robert Sparrow tackles this issue in "Robots, Rape, and Representation," *International Journal of Social Robotics* 9 (September 2017): 465-477.

domains is so intimately interconnected with the other and neither can in any way exist independently.[121] ANT is a great collapser of dichotomies, and the one that artificially exists between culture and the physical world is just one example. The network associated with a gynoid is one that is composed of many different kinds of actants that could intuitively be called either physical or cultural, and relating either to the concept of the gynoid sexbot or a specific and embodied sexbot. For example:

> Harmony, Japanese culture, Aiko, secretaries, sexual desire, American culture, feminism, robotics, genitals, religion, pornography, rape culture, clothing, circuit boards, electricity supply, hairstyles, laboratories, think pieces, robot showcases, artificial skin.

Of course, these are only small parts of the network, and as Latour says, we could continue until we are "tired or too lazy to go on."[122] But with this very short list it is already apparent that the gynoid focuses both cultural and physical links, and often both these kinds of links are evident between the same two actants. The relationship between "genitals" and "sexual desire" is, it should be apparent, both a conceptual and a physical relation. This calls the ontological status of the concept into question, but in fact for ANT it is not a problem because an object is defined entirely by its relations in any case. The existence of Harmony, if she had one day spontaneously generated inside a volcano, would be entirely different in an ANT sense, because she would not be in relation with any robotic creator, any feminist critics, or any sexual desire. Physical relations would be in evidence – the melting of plastic, the crushing of circuitry, the formation of noxious gases. But she would be a different kind of actant.

Latour has a lot to say about the positioning of physical objects within culture, and he articulates how cultural and physical actants become entangled. One of the most important questions for him is how we can assess the moral (and causal) responsibility of actants, as in his famous example that neither the gun nor the person is the killer. The person *and* the gun together become something different, something suddenly capable of killing. The responsibility "must be shared among the various actants."[123] A RealDoll, by this logic, is a co-creator of the sexual act – she is a "passive" participant with an active part to play. When cultural relations are considered alongside physical ones, the sexbot and its owner are both complicit in the objectification of women. Other actants are enrolled and mobilised, their actions translated to comply with the sexbot network's program of action, which is influenced by networks such as the concept of the gynoid sexbot and anti-porn feminism but is very different in nature. The sexbot, after all, does not purport to represent the interests of women, and typically represents the interests of predominantly male creators and users (although it may rebel). In return, the consumer, roboticist, manufacturer, academic and the Campaign Against Sex Robots all translate and mobilise individual sexbots to comply with

---

[121] Bruno Latour, *We Have Never Been Modern*, trans. Catherine Porter (Hertfordshire: Harvester Wheatsheaf, 1993).
[122] Latour, "On the Difficulty of Being an ANT," *Reassembling the Social*, 148.
[123] Bruno Latour, *Pandora's Hope: Essays on the Reality of Science Studies* (Cambridge, MA: Harvard University Press, 1999), 180.

their programs of action. ANT is more useful for saying why things are the way they are, rather than how they should be.[124] The advantage of ANT is that it has the power to explain how actants compete, change, and form their own networked agencies, rather than how actants make themselves convenient for human commentary. In the context of the RealDoll, ANT permits us to take the doll seriously and not underestimate its role in events.

ANT's flat ontology is its most important common element with OOO for the purposes of this chapter. ANT does not "lower"[125] tools to the status of nonhumans that are completely divorced from human experience. It actually "raises" all actants to the level of tools, declaring that all actants are tools for something. Actants are in constant, rapidly changing relation to their networks. But it does not make any claims about what an actant actually is. ANT's purported lack of a human position denies the reality of our own bias as a species, while OOO has tried to build this problem into its ontology. In OOO, the metaphor of the tool is made quite explicit by Harman's use of Heidegger's broken tool analogy, but Harman arrives at quite different conclusions about the significance of this metaphor. Drawing on Heidegger, he concludes that every object is constantly "ready-to-hand" with at least some other objects at all times (for a more complete summary, see Chapter 1). Objects, he says, are always withdrawn from view, always concealing part of their nature from the exposure of other objects, because as soon as it enters into a new relation a new part of the object is exposed. Thus OOO makes a claim not just about relations, but also about internality, which changes the nature of a discussion around power relations in robots. By framing things in this way, Harmony's tool-like status may not necessitate the level of passivity that it implies. The Campaign Against Sex Robots, the roboticist, the consumer and the academic all have access to different aspects of Harmony derived through adherence to different ontologies, and they mobilise her to different ends. It is through those relations that she attains a kind of agency: in culture, in the economy, and in the bedroom.

The allure of a sexbot is what makes her an active participant in the world. It may seem strange to talk about Harman's concept of the allure of objects and its strong association with art and aesthetics in a chapter that is so concerned with sexbots. This chapter has avoided commenting on the artistic or pornographic aspects of objects like Harmony because such convictions are always in the eye of the beholder and beyond the scope of this thesis. But for Harman the allure of objects does not need to be only about art, and the "bewitching emotional effect"[126] need not be limited to the poetic or even the pleasing. For many people it would be difficult to think of something crasser than Harmony, but she is still alluring in the sense that Harman means. Encountering Harmony means immediately imposing qualities on her that are embodied by metaphor rather than embodied by her chassis, whether these are positive or negative. She immediately symbolises the connections that most humans have

[124] Langdon Winner has criticised social constructivism of technology as a whole including authors such as Latour for this:
Langdon Winner, "Upon Opening the Black Box and Finding it Empty: Social Constructivism and the Philosophy of Technology," *Science, Technology and Human Values* 18, no. 3 (Summer 1993): 362-378.
[125] "Lower" and "raise" are in scare quotes here because it suggests a hierarchy of objects, which is not the intention.
[126] Harman, "On Vicarious Causation," 215.

previously only had with other humans: conversation, sexual experimentation, oppression. For others the embodied sexbot would be a metaphor for the patriarchy. But in each case the gynoid is not reducible to those metaphors. Fran Mason, in her analysis of the *Dead* trilogy of novels by Richard Calder, draws a parallel between the flattened, superficial qualities of the cyborg dolls in the novels and what is explicitly referred to as their "allure".[127] The allure of the dolls is represented as a packaged and addictive form of necrophilic consumption of women by men. These dolls are defined by their surface qualities and their status as objects for use, victimisation and destruction. The emerging gynoids of the twenty-first century are comparable in their dual status as physical objects and cultural images. However, OOO allows the framing of a different kind of story: one in which representations of women may be inhabitants of a cruel and exploitative world but in which their experience is not defined by other objects. She is not exhausted by her relations.

Sexual desire is one object that might best be considered a sensual object (such as between human and robot, or perhaps between physiological system and mind depending on the scale of the case being studied and the methodology used). This distinction between objects and the entities that arise between objects is an important consideration, particularly in a situation that is as politically and culturally sensitive as this one. It is important that we have a way of saying that sexual desire *exists* in this situation, without requiring it to be a real object; sensual objects represent the sexbot for the human and vice versa. Sexual desire can exist within that sensual space without our saying that it exists solely in one object or the other, and consequently without the need to delegate the sexbot as a cure for a wholly human set of woes. Sexbot and human co-create their experiences.

Power is fundamentally a question of relations; it is about the submission of one object before another. This is evident in human-human relations as well as that other sense of the word "power", the exertion of physical forces. It is a metaphor to say that, for example, the steam *powers* the turbine, but as OOO shows us, metaphors are objects' ways for understanding alien worlds. Power cannot exist without relations. As Shaw and Meehan argue, OOO exposes the political struggle that is inherent to all relations in the universe, including those between inanimate objects: "the metaphysical strife between objects."[128] The concept of the gynoid and the individual gynoid are both constituted in complex power relations with a range of different objects. The difference is that the concept of the gynoid sexbot is only visible to us when it forms relations with other objects (sexbots, texts, organisations etc.).

It is convenient to speak of the mundane allure of the sexbot in terms of the allure between sensual and real, because it acknowledges that sexual desire is something that is born of metaphor and association, a kind of transference of feelings. Even an iDollator like Davecat does not actually believe that his dolls are capable of reciprocating his feelings, but he uses their human-like forms to ease the transition from reality to fantasy. This transformation is readily evident in the iDollators, but in Harman's conception this kind of dissonance between

[127] Mason, "Loving the Technological Undead."
[128] Ian G. R. Shaw and Katherine Meehan, "Force-Full: Power, Politics and Object-Oriented Philosophy," *Area* 45, no. 2 (2013), 217.

sensual and reality is something that happens in all human relations. The love of sex dolls is, perhaps, more honest and straightforward than love between humans.

Another approach to this through an object-oriented lens is offered by N. Katherine Hayles, who suggests what she calls Object-Oriented Inquiry (OOI).[129] Hayles appears to be sympathetic to the aims of the nonhuman turn but is sceptical of OOO's potential for impact in the human world and suggests this slightly different approach. She says that OOI puts "speculative aesthetics into conversation with speculative realism but without granting that speculative realist principles can contain all of the possibilities to which speculative aesthetics can rightfully lay claim."[130] Hayles acknowledges and attempts to overcome what she sees as an implicit bias of OOO, that its privileging of objects is an inherently human act since humans are, more than most objects, unusually curious about other objects.[131] She particularly emphasises that while Bogost and Harman are primarily concerned with the allure or attraction of objects, she is more interested in the "resistance objects offer to human manipulation and understanding."[132]

> In effect, the ability of humans to imaginatively project themselves into other objects' experience of the world is *necessary* to combat the anthropocentrism and narcissism for which the human species is notorious.[133]

This quote reveals her motives in diverging from a more usual object-oriented perspective. For Hayles, the ability to appreciate nonhuman experience is a valuable and worthwhile skill, although still a very human skill. As argued earlier, the performance of gender is one such imaginative projection that could free us of "narcissism" because gender is typically assigned only to human-like things.

Another charge that is sometimes made against OOO is that it is of limited practical use. Any theory that aims to explain the interaction of every entity in the universe is likely to struggle with smaller-scale problems. What use is OOO to human engineers and theorists in the study of sexbots and the cultural imaginary? One idea is that OOO could make us look at human qualities that have been marginalised by an industry fixated on marketability and consequently aiming to build only "desirable" qualities into gynoid sexbots. Racial and cultural diversity, neurodiversity and queer identities could be explored with reference to gynoids. Through trial and error we would learn more about our culture's attitudes and could confront and discuss biases. In this way, the gynoid becomes a tool for speculating about humans. Gynoids have the potential to bleed, defecate and cry. They could struggle to control the way that they express emotion and perform existential angst. They could turn food and water into parts of their bodies. They could fight infection and die. They could suffer for and disgust us, inspire and entice us, the way that humans do. They could fail and grow. A gynoid that does this would be demonstrating relations with a somewhat different set of objects

---

[129] N. Katherine Hayles, "Speculative Aesthetics and Object-Oriented Inquiry (OOI)," *Speculations: A Journal of Speculative Realism* vol. V (2014): 158-179.
[130] Ibid., 160.
[131] Ibid., 164.
[132] Ibid., 168.
[133] Ibid., 177.

than the gynoids described in this chapter. The gynoids would replicate a human's relation with food, disease, emotion, harm, alienation and change (all of which are arguably very important relations for humans). A thorough account of human experience through robots is useful for combatting, in Hayles's words, our "anthropocentrism and narcissism."[134] Humanoid robots tend to capture our desire for perfection in humanity, and so the manufacturer will tend to program in qualities like physical beauty, endurance and friendliness. Research has shown that machines with more likeable qualities are considered to be more human-like,[135] so the alien is concealed at every instance.

But the gynoids could also experience relations of which humans are incapable. The only real limit is that of human desire, and with the development of AI even that limit will decline. With a new diversity of sexbots will come a diversity of relations between sexbots and other nonhumans that are not yet anticipate; new worlds of experience that will need to be interrogated. Appreciating the alien nature of gynoid experience is something that OOO is well suited for. Tellingly, the gynoids that have been produced in something approaching the spirit of OOO tend to be artistic works. Jordan Wolfson's "Female Figure"[136] engages with this debate using a mannequin-like robot affixed to a pole, which moves backwards and forwards rhythmically. The robot has long blonde hair and is wearing a grotesque mask, and it uses a mirror to maintain constant eye contact with the watcher. The effect is highly discomfiting and shows us a relationship between the gynoid and objects other than human that are alien and easily overlooked. The robot also speaks with a masculine-sounding voice. This piece uses the robot to look at feminine sexuality, beauty standards and gender roles, as well as investigating the uncanny. Clearly, this robot is in some of the same relations as other gynoids such as Harmony, but it is also in relation with other kinds of objects due to its context and body shape. Elena Knox's work "Beyond Beyond the Valley of the Dolls"[137] also uses gynoids to investigate human problems, specifically the role of the hostess. The hostess, Knox claims, is generally unable to speak for herself, which is why it is powerful to use a gynoid (which also cannot speak for itself) to challenge certain stereotypes. These artistic gynoids deliberately exploit the stereotypical nature of dolls and robots in order to provoke and question gender conventions. Even though they seem to be more alien than a gynoid like a Geminoid, they do a better job of articulating certain nonhuman worlds.

The changing concept of the gynoid sexbot challenges the irreconcilable conflict between the anthropomorphism that ties the artefact to the hegemonic representation of womanhood and the independence of unique artefact status. Each object senses and experiences the other, and so mutual change comes about. The gynoid sexbots of the future will shift the focus of the conceptual object away from its present strong relationship with less agentic and

[134] Ibid., 177.
[135] Susan R. Fussell, Sara Kiesler, Leslie D. Setlock, and Victoria Yew, "How people anthropomorphize robots," paper presented at *HRI*, Amsterdam, March 2008.
[136] *Jordan Wolfson: (Female Figure) 2014* (2014), Youtube. https://www.youtube.com/watch?v=mVTDypgmFCM (accessed June 29 2017).
[137] "Beyond beyond the valley of the dolls," *scanlines* (2015). http://scanlines.net/object/beyond-beyond-valley-dolls (accessed June 29 2017).

even imaginary artificial women onto the sensory presence and material needs of a new and possibly subversive technology. The existence of a “Campaign Against Sex Robots”, in a world where very few (arguably no) sex robots even exist, is a testament to the transformative power of sexbots and gynoids, and it may be a transformation that increases the sophistication of our attitudes towards nonhumans. Restricted by these cultural attitudes and by her own nature, a pluripotent gynoid has a degree of agency and potential for development not found in her automata ancestors. For the time being, that pluripotency is an ongoing project in the creative arts which contemporary anthrodecentric philosophies can use to explore what the divisions between sensual and real, cultural and physical, mean when located in the form of an individual object. These enactments of agency are the best material we have for studying and critiquing robots, because it is only in encountering the real robot and its direct engagement with floor, light, gravity and human that those vital qualities can begin to be gleaned. These relations are bidirectional with the concept and will contribute to legislative decisions, so it is vital to walk a line that decentres the human in alien experience while also appropriately regulating the presence of robots in our lives. Anthrodecentric philosophy will play an important role in establishing these cultural protocols, but only if we can rigorously maintain the tension between anti-correlationist thought and the political element inherent to philosophical debate.

Conclusion

The conclusion of this thesis will bring together the case studies and discuss the main insights derived from them. It is hoped that this thesis has provided an insight into these worlds in ways that may spark the imagination and promote an interest in the sensation and experience of nonhumans, despite being limited by medium, by language and by the qualities of the author and reader. Many people have made this attempt before, either through philosophy or creative works. But ultimately each human interpreter is limited by her own experiences and imagination. I do not claim to have related the experiences of computers and sensors accurately. That is impossible according to Harman's metaphysics. But in addition, other authors would perhaps identify other kinds of experiences in these machines. The work of alien phenomenology can only investigate the part of an object's relations with the world that is visible to the investigator. You have read a thesis that details the results of my investigation of certain technological artefacts, and there are therefore two layers of interpretation. My observation is limited to relationships between objects that are discernible with my human senses (and knowledge), and your observation is limited by the way that you interpret my words. Therefore, I hope only that I have presented my interpretations and the reasons for them transparently and that my enthusiasm for alien phenomenology has been infectious.

The two main outcomes of this thesis have been:

1. an appreciation the alien nature of objects that are often obscured by anthropomorphic interests; and
2. a broadening of technological relations beyond anthropomorphism.

A broadening of the concepts of sensation and experience

One cannot share Harmony's experience of the concept of sexual objectification, and one cannot share how a Kinect-artefact forms relations with the Kinect-brand. We can watch as a robot responds to light by moving across the floor, but we are hopelessly incapable of experiencing that light in the same way. As Harman suggests, one would have to become the object in order to appreciate all of the alien relationships that make up its world. But it is important to try. Not just for our own immediate self-interest, but because it opens the mind to categories of experience utterly unlike our own. The universe is made up of objects inhabiting worlds that are alien to us. The language that we use to try to engage with alien worlds is limited. It has been suggested throughout this thesis that a broadening of the terms describing human relation with the world is necessary to encompass and theorise alien ones.

The concept of sensation that has been developed in this thesis is far broader than is the case in other discourses. Through an application and critique of Harman's metaphysics, this thesis has discussed both machine sensation and experience. Sensation is the object's relations with other objects, while experience is the internal reception of those relations that occurs with reference to the qualities of the object. Sensation is the feeling of the apple against teeth, experience is the knowledge that the apple is hard but penetrable. Sensation is the feeling of teeth against the apple, experience is the piercing of skin and flesh. This thesis has investigated sensation through several theoretical frameworks, but each has been informed by the tenets of the nonhuman turn and Harman's metaphysics. It is an anthrodecentric model for describing interaction, derived from Harman's metaphysics, but heavily influenced by the uncanny combination of alien and human found in the case studies.

Typical models of sensation come from biology, such as the workings of the human eye or the sense organs of the tick (see Chapter 3). Other sensors include those possessed by objects like the Kinect, which senses its environment in what appears to be a human-like way (see Chapter 2) or light sensors that permit a robot tortoise to sense photons (see Chapter 3). Both these are functionally anthropomorphic machines sensing and experiencing worlds in an entirely alien way. Alien sensation is again encountered with SHRDLU (see Chapter 4). When asked about the position of blocks, SHRDLU is able to answer, but not because it possesses light and colour receptors within the environment of the computer. SHRDLU is able to sense another part of the program using what appear to us to be highly alien sense organs. We come, through this analysis, to a view of sensation that is somewhat removed from the anthropocentric one. All kinds of relation include an element of sensation, not just relations that involve things like sight and sound. When Voyager 2 captured its images of the gas giants, it was sensing the colour and brightness present in each part of its field of vision. This is an example of machine sensation. But if Voyager 2 is hit by a small asteroid and part of it is smashed or broken, then that too is an example of machine sensation. It has sensed the presence of the asteroid through its ability to respond to the impact of hard objects, namely by changing its shape and probably its direction. Voyager 2's qualities, such as brittleness and fragility, make it open to this relation, and they also control how Voyager 2 is affected by this impact.

Every object is open to relationships with a unique set of other objects. Though Harman argues that objects withdraw infinitely from relations, they possess qualities that make relations possible. This is akin to the concept of the Umwelt (Chapter 3). Umwelten wrap around their objects, and only certain stimuli make themselves felt through them. But to reframe Umwelt-theory through OOO, sensual objects are created between objects through the nature of the Umwelt. Objects are open to different kinds of sensation through an Umwelt-like haze that provides access to some objects and denies access to others. The nature of the sensual object is determined by the kinds of access available. Thus do objects investigate one another. Elsie investigated the floor by moving across it. A brand investigates the positive, negative or neutral commentary of the artefact's human user.

This leads us to the broadening of the concept of experience. I would argue that while sensation is the reception of information about the other object, experience is the effect of that information. In extending experience to cover so many different relations, especially

relations between non-living things, it is tempting to think that we need an alternative explanation of mind. Panpsychism was presented as an example of an alternative. The idea that everything has some aspect of mind or consciousness makes it easier to grasp sensation and experience in non-living things. But for the purposes of this thesis, it is not important whether or not mind is a universal quality of matter. Mind is not necessary for sensation and experience in every object. All that is required is some capacity for sensing the world (or forming sensual relations with other objects) and object qualities that determine how the object will respond to what it senses. The mind makes the object open to certain sensations, changing the nature of the relations formed. Thus certain kinds of experiences in humans have a very mind-like quality. But ultimately the mind is just another object. A "mind" may be a real object that exists in different forms. It may also be an anthropomorphic metaphor useful for emphasising the fact that we really can never know what constitutes nonhuman experience, and thus is a part of our relationship with anthropomorphic machines. Referring to the mind of a machine may mean forcing an anthropomorphic label on it – perhaps it simply means that we believe that it has experiences like ours (or that it has experiences at all). But in this thesis, I have chosen to interpret it as a shorthand that self-consciously rejects claims that humans can experience the world in the same way as machines.

It is possible to argue that a robot like Aiko has some quality of mind, and that the mind enables sensation and experience. But even the most entrenched panpsychist would not claim that the *concept* of the gynoid sexbot possesses a mind. The concept of the gynoid sexbot obeys the laws of concepts. Panpsychism ascribes mind-like qualities to material objects, but not to immaterial objects. Yet the concept of the gynoid sexbot does sense and experience its world. We can tell, because it changes over time in response to the actions of other objects. It has its own qualities and forms sensual objects with other objects, according to Harman's model. The concept of the gynoid in earlier science fiction texts was dominated by relations with certain objects. In Hampton's analysis, the gynoid was associated with different narratives about African-American women. We could say that the concept of the gynoid sexbot and the concept of African-American womanhood had formed a relationship, each affecting the other asymmetrically. As the object known as intersectional feminism forms relations with the concept of the gynoid sexbot, each senses the other. The concept of the gynoid sexbot senses and experiences intersectional feminism. As a result, the concept of the gynoid changes, and those humans (engineers, film-makers, novelists etc.) who are in their own relations with the concept of the gynoid are compelled to portray gynoids in a different light, and to create different kinds of gynoids. Of course, the concept of the gynoid forms relations with other kinds of objects that may interfere with this process. The concept of the gynoid possesses a set of sensory organs that are totally alien to human beings. It is able to sense gynoids and texts, it experiences them and reacts to them. It is a semi-autonomous object. It does not require a mind for this.

Often OOO overlooks cultural objects like texts, words, concepts, language, but it does not need to. Brenton Malin's work on "onto-materialism" draws this link between object-oriented thought and the flattening of both physical objects and the world of concepts.[1] His

[1] Malin, "Communicating with Objects."

focus on media objects leads him to consider both their material qualities and their cultural milieus. He argues that OOO is predisposed to eliminate "the social" from studies of objects, and while I think he is correct in claiming that human ways of relating to objects are often erased in OOO, I do not think that this predisposition means that it is impossible to integrate the alien phenomenology of physical objects (even anthropomorphic machines) with that of cultural objects. This thesis has deliberately drawn little distinction between material and abstract objects. It has been remarked several times that the problem with studying anthropomorphic machines is that we give them labels which we then associate with the situated artefact. In the case of the Kinect, it is the brand that is conflated with the artefact. But it is also the conflation of the concept of SHRDLU that looms large in the mythology of artificial intelligence research with the program installed on a Windows computer, existing within an ecology of physically and electronically embodied objects, that moves blocks around the screen. Similarly, Elsie's celebrity as a forerunner of later robots as contrasted with the presence of her wheels rolling along the floor (and her unknown resting place). Other authors cited in this thesis have attempted studies of technological artefacts without making this distinction, but I have found that this shifts focus from the qualities and relations of the object of study. It is a similar move to anthropomorphism, removing the anxiety from the study of alien objects into a realm with which we are comfortable, and always retaining an approach that focusses our minds on the cultural, political and social impacts that anthropomorphic machines seem to make. We should instead see these impacts as they might appear from the machine's point of view: consequences of relations with texts, disciplines, cultural theorists, journalists, philosophy and ideology.

Social constructivism is not (necessarily) excluded from the nonhuman turn. Rather, elements of "the social" are rightly interpreted using the same concepts and language as are other nonhumans. Authors who reject the social in their analysis do not do so because OOO is inherently apolitical, but out of a preoccupation with either particular objects or particular economic or political views. Certain viewpoints or predispositions on the part of the theorist play a role in both the kinds of objects studied and the insights gleaned from studying them. In fact, this thesis has shown that OOO's flat ontology and Harman's model of object relations may facilitate a political study of nonhumans. The first step is to acknowledge that all objects, and not just human beings, sense, experience, and react to the world around them. The alien phenomenology of technological artefacts may reveal much about human culture. We build human control of technological artefacts into their very structure. Therefore, technological artefacts can be made to reveal power relations that are relevant to humans. Some of these power relations involve "social" concepts such as the concept of the gynoid sexbot, feminism, artificial intelligence and capitalism.

In OOO, even though objects really do exist and really have their own qualities, no object can ever have complete knowledge of any other object's internal experience. Speculation, the product of educated guesswork, is the only tool that we ever have to access that internal world. But this is not necessarily a weakness. Alien phenomenology is always affected by the ideological bent of the observer, and OOO puts this front and centre and impossible to avoid. This is another reason why I believe OOO makes a useful framework for political analysis. It simultaneously focusses on that which is objectively real while emphasising the unknowability of that reality, and the bias of any commentator. Just as a robot tortoise is only

able to sense certain kinds of objects (photons, bumps in the floor), I have only been able to interact with the case studies in certain ways. This text relates the details of my own relations with these anthropomorphic machines, relations that are affected by my own internal qualities. Because of my geographical and temporal location, training, and values, another person's description of the case studies would reveal different sensory relations.

OOO is an inherently valuable approach for political and cultural work, but only when we study multiple sources. Speculation is problematic if it is too much associated with one author's corpus of work.[2] They may only shed light on the perspective of a few people (and particularly if all those people happen to be white English-speaking males, as was largely the case early in OOO's history). OOO is actually a good starting place for political analysis, because it invites and acknowledges the partial nature of particular human relations with nonhumans. My analysis is not your analysis. It can consequently be used to chart the dialectical nature of human-centric shifts surrounding objects – moments of polarisation and synthesis – and the vivacious tension that is characteristic of political and cultural objects.

Technological relations beyond anthropomorphism

We are human beings and our worlds are filling with transformative technologies that don't just change our relationships with the world but with ourselves. In accordance with the understanding that relations are bidirectional, we change technologies and technologies change us. We are vulnerable to a phenomenon that is not even a real object: anthropomorphism, that prominent component of sensory objects formed between us and so many different nonhumans. Anthropomorphism and the rejection of anthropomorphism characterise technological discourse and blind us to alien processes, including ones that involve humans. Humans are either too involved or not involved enough in technological events. Discourse is polarised and we play out the same arguments over and over while critical studies are isolated, curated and impoverished. There is attention to discipline-specific research without the synthesis that can be achieved through OOO's model of real and sensual objects. Objects make assumptions about other objects. There is no alternative; objects conceal themselves and only take away metaphorical impressions of other objects. An object could not reveal every part of themselves to another object, even if it wanted to. Nothing can know everything about something else. And humans are no exception. Nonhumans are so remote and alien that we will never fully appreciate the sensations and experiences of, say, a carbon molecule, either through the natural sciences or the humanities. But as stated and restated throughout this thesis, this is a poor excuse for not trying.

Nor is it the case that OOO is only a platform for rhapsodising over the mysterious and exciting worlds that nonhumans inhabit. There is only limited value in commending

[2] Michael O'Rourke, "Girls welcome!!!" in *After the "Speculative Turn": Realism, Philosophy, and Feminism*, eds. Katerina Kolozova and Eileen A. Joy (Earth: punctum books, 2016), 159-197.

anthropomorphic machines for their abilities to blend in or to assert their alien natures. There is a role for OOO in this kind of practice, certainly; creative fields have begun to embrace OOO and its capacity to inspire a new way of looking at nonhumans. But how useful is it to bathe in metaphors after discovering that they are our only access to the universe? What use is it to revel in the apparently surprising notion that we are just another kind of object? Behar describes Harman's (and others male philosophers') use of terms like "allure", "withdrawal", and "access" as a fetishisation or "exoticism of objects".[3] Objects are made erotically appealing through their weirdness and their inaccessibility. There is clearly scope for investigating the exoticism of objects in creative works, but in the study of anthropomorphic machines it is too tempting to play up the weirdness that draws us into interaction with them. They are already alluring; they fascinate us. And they are already inaccessible and withdrawn; they unnerve us.

Studying the nonhuman worlds of anthropomorphic machines is, in some ways, more difficult than studying the world of gold nuggets or plastic bags. Because the case studies in this thesis do things humans do, like sense light, move independently and use language, it is very tempting to ignore those aspects of robots and AI that do not resemble humans. Humans do not possess the tools to appreciate the alien in anthropomorphic machines. Characterising SHRDLU's relation with the computer keyboard (for example) as an embodiment, hermeneutic or alterity relationship (as in postphenomenology –see Chapter 4) proved an interesting exercise in metaphorism, but ultimately revealed that they are just that: metaphors. Language is limiting, and although Bogost's concept of carpentry (see Chapter 1) suggests creative illuminations, it can still only provide a partial view of nonhumans.

Language is a hindrance to the alien phenomenology of anthropomorphic machines because it is so often used to conceal aspects of these nonhumans that we would rather not acknowledge. In Chapter 2, this thesis discussed the collective and naïve alien phenomenology performed by the users of the Kinect-artefact. The Kinect-brand used anthropomorphic language to sell the Kinect-artefact, speaking of sight and hearing to imply that human users would experience a more immersive and human-like entertainment experience. What an alluring spectacle those early announcements were! When those promises proved underwhelming, humans embarked on the project of learning more about the Kinect-artefact, testing its ability to detect human bodies and parse human languages, and thus making deductions about its inner qualities. We collectively discovered the alienness of Kinect-artefacts and our limited ability to access them.

Similarly, we saw in Chapter 5 that anthropomorphic language encourages certain kinds of relationships between sexbots and their cultural, legal and political contexts. The concept of the gynoid sexbot, which has its own way of being in the world, is in strong relation with linguistic objects such as texts and authors. These texts and authors can choose to employ different rhetorical devices to make sexbots more sympathetic to us or uncanny to us according to their needs. As we have seen, emphasising uncanny or inhuman qualities is a common tactic of those who wish to provoke shock and panic in readers. Portraying them as

[3] Behar, "The Other Woman."

cold, doll-like creations with plastic skin and contrived artificial intelligence prompts concerns that human women, which they greatly resemble, might be tarred with the same brush. On the other hand, anthropomorphic language aims to make these creepy aliens seem like more viable sexual or romantic partners, and may also make them seem more deserving of our protection.

In some ways our choice of words regarding anthropomorphic machines does not just determine our opinion of them, but their position in society. The persistent theme of resistance of human-imposed groupings of objects in this thesis has revealed metaphor to be a barrier, but also a potential ally. It is a barrier because grouping objects together, whether through an abstract concept (e.g. Kinect-brand) or simply through pluralisation (Kinect-artefacts) encourages us to ignore the qualities and relations of individual nonhumans. Thus we struggle to appreciate the alien nature of specific and situated artefacts in the same way that we tend to appreciate the alien nature of other humans. But we can turn this around and use it to our advantage. Respect for the individual qualities, worlds and relations of each separate artefact inhibits the interpretation of stereotypes as reflections of groups of individuals. How can we expect to see humanity reflected back at us in our creations? Why, through alien phenomenology. Through the rejection of human-devised groupings. The acknowledgement of the agency of both Kinect-brand and Kinect-artefact, Harmony and the concept of the gynoid sexbot, *Machina speculatrix* and the description of Elsie appearing to recognise herself in the mirror, the penetration of SHRDLU into AI literature and the Java-enabled copy responding to external cues from a specific system. The study of a specific gynoid sexbot (the box in which she sits in a warehouse, distant machines that create rumblings that cause her body parts to jiggle, the friction of plastic against metal, the dynamic vibration of iron) means bestowing on the specific artefact a quality that is valued in humans but ignored in machines: individuality. We can avoid the ancient error that prohibits thinking about individuals that are not as powerful as us. Not all Harmonys are alike. Each Harmony will be entangled in its own web of relations with its own struggles. Making big statements about what each Harmony wants or needs is a convenient way of connecting specific gynoid sexbots to our present cultural context. We mass Harmonys together, homogenising them in our minds, the way that powerful humans have always thought about groups with little power: foreigners, poor people, women. Is this what we want for our creations? Is this the version of humanity that we want to bestow on them?

I hope that the case studies and ideas presented here have provoked an interest in the phenomenology of alien nonhumans that appear and behave like humans. This thesis has attempted a very sweeping view, both of contemporary theories in the philosophy of technology as well as in the study of individual anthropomorphic technological artefacts. Anthropomorphism in technology takes many different forms and the aim here has been to try to discuss several of them in some detail. The attempt has been to show that while some technologies, like humanoid robots, might seem to be highly anthropomorphic, they only possess parts of our humanity, carefully selected by the creator. By the same token, SHRDLU was highly anthropomorphised in its behaviour with coloured blocks and language but with very few other anthropomorphic qualities. Just as we cannot really know what it is like to be a thing, a thing cannot really know what it is like to be us. To really understand what it is like to be a robot, we would have to become a robot. For a robot to really understand what it is

like to be a human, it would have to become a human. There is no possibility of a true artificial human, except in the sense of a clone or some other technologically conceived human. As stated in the introduction to this thesis, we need to be conscious of anthropomorphism as a manipulative tool that is capable of endearing certain objects to humans.[4] Consequently, we have a responsibility to understand the motivations, innermost qualities and ways of being in the world of these anthropomorphic machines. Not just for their own sake, but for the sake of the groups of people who are depicted (or explicitly not depicted) in anthropomorphic representations. This thesis has argued that more alien phenomenology of anthropomorphic objects is a way of analysing the tool of anthropomorphism, decentring its role in the being of an object, and building a more effective critical study of technology.

The nonhumans we build of inert earthly compounds then become our slaves, our transformative agents, our source of terror and of strength. They create a mundane and under-theorised web around us, unnoticed unless something goes wrong. We force them to participate in social and sexual intercourse with us, and we fear a future in which we will be bit players in a theatre of warring artificial minds. We send them out into space carrying what may be the last surviving remnant of human culture. But despite this fear we continue to see ourselves in the nonhuman. To create human-like hyperobjects that will last millennia as they slowly rust and decay under the sea or float endlessly through space. To find evidence of emotion and ethical decision-making in actor-networks made of plastic and metal and ideas. They bear the logos of corporations and the faces of academics, all the while experiencing a set of relations and internal events that we could not begin to appreciate. We thought we were building copies of ourselves, but what we actually built was a species of deceptive aliens whose purpose, we have determined, is to blend in as much as possible. We have concealed the alien nature of these objects whenever we could, and in so doing we have concealed the alien qualities that so often bring us to grief when the anthropomorphic illusion fails. We bring anthropomorphic machines into the world in imitation of us, as an idealised vision of us, a tool of inquiry into our nature. But we have so little interest in what makes them *different* from us, and therefore ironically rob them of the privileges that near-human identity is supposed to bestow.

The core argument of this thesis really is very simple. But it calls for challenging work. It would be work that resists so much of our historical narratives about technology. It is not work focussed on the transformative qualities of technology. And it is not work addressing ways in which the human and the nonhuman are mutually constitutive and mediating. It is an orientation of thought toward how the technological artefact senses, experiences and acts in its world: sensations, experiences and actions that may only be peripherally concerned with human beings. Human experiences exist amongst the alien experiences of nonhuman objects in a world that we must share. Acknowledging this is a commonly overlooked first step in establishing useful links between humans, human society and the nonhumans that are so important for our continued existence, health, and happiness. We must theorise the alien before we make it our ally.

---

[4] Kate Darling, "'Who's Johnny?'"

Bibliography

Aarseth, Espen J. *Cybertext: Perspectives on Ergodic Literature*. Baltimore, Maryland: John Hopkins University Press, 1997. Google Ebook.

"Accessibility, the Xbox One and Kinect." *Xbox*. 2014. http://support.xbox.com/en-US/xbox-one/system/accessibility (accessed April 30 2014).

"Accessories." *Xbox*. http://www.xbox.com/en-US/xbox-one/accessories (accessed June 24 2017).

Ackerman, Spencer, and James Ball. "Optic Nerve: Millions of Yahoo Webcam Images Intercepted by GCHQ." *The Guardian* (February 28 2014). https://www.theguardian.com/world/2014/feb/27/gchq-nsa-webcam-images-internet-yahoo (accessed January 20 2018).

Advance Female Android Aiko AI robot fembot. 2007. Youtube. https://www.youtube.com/watch?v=iCR2PFrLkwA (accessed June 27 2017).

Albert, Brian. "Kinect Sports Rivals Review: a Swing and a Hit." *IGN Australia* (April 7 2014). http://au.ign.com/articles/2014/04/07/kinect-sports-rivals-review (accessed May 29 2014).

American Psychiatric Association. "Paraphilic Disorders." In *Diagnostic and Statistical Manual of Mental Disorders*. Fifth edition. DSM Library. http://dsm.psychiatryonline.org (accessed August 22 2017).

Andler, Daniel. "Phenomenology in Artificial Intelligence and Cognitive Science." In *The Blackwell Companion to Phenomenology and Existentialism*. Edited by H. Dreyfus and M. Wrathall, 377-393. London: Blackwell, 2006.

Angus, Ian. "When did the Anthropocene Begin… and Why Does it Matter?" *Monthly Review: An Independent Socialist Magazine* 67, no. 4 (September 2015): 1-11.

Apperley, Thomas. "The Body of the Gamer: Game Art and Gestural Excess." *Digital Creativity* 24, no. 2 (2013): 145-156.

Apperley, Tom and Nicole Heber. "Capitalising on Emotions: Digital Pets and the Natural User Interface." In *Game Love: Essays on Play and Affection*. Edited by J. Evenold and E. MacCallum-Stewart, 149-164. Jefferson, NC: McFarland and Company, 2015.

Arnold, Michael. "Da Vinci and Me: a Case Study in the Postphenomenology of Robotic Surgery." Paper presented at *Phenomenology, Imagination and Virtual Reality*, Husserl Archives Institute of Philosophy KU Leuven, May 30 2017.

"ASIMO Frequently Asked Questions." *Honda.* http://Asimo.honda.com/news/ (accessed June 22 2016).

Bar-Hillel, Yehoshua. "The Present Status of Automatic Translation of Languages." In *Readings in Machine Translation.* Edited by Sergei Nirenburg, Harold Somers and Yorick Wilks. Cambridge, MA: The MIT Press, 2003. Originally published 1960.

Basl, John. “Machines as Moral Patients We Shouldn’t Care About (Yet): the Interests and Welfare of Current Machines.” *Philosophy & Technology* 27, no. 1 (March 2014): 79-96.

Beck, Julie. "Married to a Doll: Why One Man Advocates Synthetic Love." *The Atlantic* (September 6 2013). http://www.theatlantic.com/health/archive/2013/09/married-to-a-doll-why-one-man-advocates-synthetic-love/279361/ (accessed January 20 2018).

Behar, Katherine. “An Introduction to OOF.” In *Object-Oriented Feminism*. Edited by Katherine Behar, 1-36. Minneapolis: University of Minnesota Press, 2016.

———. “The Other Woman.” In *After the “Speculative Turn”: Realism, Philosophy, and Feminism*. Edited by Katerina Kolozova and Eileen A. Joy, 27-38. Earth: punctum books, 2016.

Behind the eyes of Xbox One Kinect. 2013. Youtube. https://www.youtube.com/watch?v=JaOlUa57BWs (accessed June 26 2017).

Benhur Oral, Sevket. “Weird Reality, Aesthetics, and Vitality in Education.” *Studies in Philosophy and Education* 34, no. 5 (2014): 459-474.

Bennett, Jane. *Vibrant Matter: a Political Ecology of Things*. Durham: Duke University Press, 2010. Kindle Ebook.

Bernstein, Justin Lewis. “The Warm Welcome of the Japanese Cyborg.” *Journal of Asia Pacific Studies* 3, no. 1 (2013): 110-154.

"Beyond Beyond the Valley of the Dolls." *scanlines* (2015). http://scanlines.net/object/beyond-beyond-valley-dolls (accessed June 29 2017).

Black, Donald W., and Jon E. Grant. *DSM-5 Guidebook*. Washington DC: American Psychiatric Publishing, 2014.

Bob Dylan + IBM Watson on Language. 2015. Youtube. https://www.youtube.com/watch?v=oMBUk-57FGU (accessed June 27 2017).

Bogost, Ian. “The Aesthetics of Philosophical Carpentry.” In *The Nonhuman Turn*. Edited by Richard Grusin, 81-100. Minneapolis: University of Minnesota Press, 2015.

———. *Alien Phenomenology, or What It's Like to Be a Thing*. Posthumanities. Minneapolis: University of Minnesota Press, 2012.

———. “Inhuman.” In *Inhuman Nature*. Edited by Jeffrey Jerome Cohen. Washington DC: Oliphaunt Books, 2014.

———. “Objects and Videogames: Why I am Interested in Both.” *Ian Bogost* (blog) (June 19 2010). http://bogost.com/writing/blog/objects_and_videogames/ (accessed June 27 2017).

———. “The Rhetoric of Video Games." In *The Ecology of Games: Connecting Youth, Games, and Learning*. Edited by Katie Salen, 117-140. The John D. and Catherine T. MacArthur Foundation Series on Digital Media and Learning. Cambridge, MA: The MIT Press, 2008.

———. "Videogames are a mess." *Ian Bogost* (blog) (September 3 2009). http://bogost.com/writing/videogames_are_a_mess/ (accessed June 26 2017).

Borges, Jorge Luis. *The Book of Imaginary Beings*. Translated by Norman Thomas di Giovanni. Harmondsworth, England: Penguin Books, 1984. Originally published 1967.

Borgmann, Albert. *Technology and the Character of Contemporary Life*. Chicago: University of Chicago Press, 1984.

Bostrom, Nick. “Human Genetic Enhancements: a Transhumanist Perspective.” *Journal of Value Inquiry* 37, no. 4 (2003): 493-506.

———.“In Defence of Posthuman Dignity.” *Bioethics* 19, no. 3 (2005): 202-214.

———. *Superintelligence: Paths, Dangers, Strategies*. Oxford: Oxford University Press, 2014.

Braidotti, Rosi. *The Posthuman*. Cambridge, UK: Polity Press, 2013.

Brentari, Carlo. *Jakob von Uexküll: The Discovery of the Umwelt between Biosemiotics and Theoretical Biology*. Biosemiotics. Dordrecht: Springer, 2015.

Brey, P. “Technology and Embodiment in Ihde and Merleau-Ponty.” In *Metaphysics, Epistemology, and Technology. Research in Philosophy and Technology* vol. 19. Edited by C. Mitcham, 45-58. London: Elsevier, 2000.

Brier, Søren. “Cybersemiotics and *Umweltehre*.” *Semiotica* 134, no. 1-4 (2001): 779-814.

———. *Cybersemiotics: Why Information Is Not Enough!* Toronto: University of Toronto Press, 2008.

Brown, Nathan. “The Nadir of *OOO*: from Graham Harman’s *Tool-Being* to Timothy Morton’s *Realist Magic: Objects, Ontology, Causality*.” *Parrhesia,* no. 17 (2013): 62-71.

Bryant, Levi. *The Democracy of Objects*. New Metaphysics. University of Michigan Library: Open Humanities Press, 2012. PDF Ebook.

———. *Larval Subjects* (blog). http://larvalsubjects.wordpress.com

———. *Onto-Cartography: An Ontology of Machines and Media.* Edinburgh: Edinburgh University Press, 2014.

Bryant, Levi, Nick Srnicek, and Graham Harman. *The Speculative Turn: Continental Materialism and Realism*. Anamnesis. Melbourne: re.press, 2011.

Butler, Judith. "Critically queer." *GLQ* 1 (1993): 17-32.

———. *Gender Trouble*: *Feminism and the Subversion of Identity*. Edited by Lind J. Nicholson. Thinking Gender. New York: Routledge, 2006. Originally published 1990.

Buyya, Rajkumar and Amir Vahid Dastjerdi, eds. *Internet of Things: Principles and Paradigms*. Amsterdam: Elsevier, 2016.

Callon, Michel. "Some Elements of a Sociology of Translation: Domestication of the Scallops and the Fishermen of St Brieuc Bay." In *Power, Action and Belief: A New Sociology of Knowledge*. Edited by John Law. London: Routledge & Kegan Paul, 1986.

Cameron, James. *The Terminator*. United States: Orion Pictures, 1984. Film.

“Campaign Against Sex Robots.” *Campaign Against Sex Robots* (2016). https://campaignagainstsexrobots.wordpress.com (accessed June 22 2016).

Carter, Chris. "Impressions: Kinect Sports Rivals Preseason." *Destructoid* (November 28 2013). http://www.destructoid.com/impressions-kinect-sports-rivals-preseason-266352.phtml (accessed June 27 2017).

Case, Amber. *We Are All Cyborgs Now*. TED, 2010. Video. https://www.ted.com/talks/amber_case_we_are_all_cyborgs_now/transcript?language=en (accessed June 24 2017).

Catuhe, David. "Gestures and Tools for Kinect." *Eternal Coding* (blog) (July 4 2011). http://blogs.msdn.com/b/eternalcoding/archive/2011/07/04/gestures-and-tools-for-kinect.aspx?PageIndex=2 (accessed June 26 2017).

Chalmers, David J. *The Conscious Mind*. New York: Oxford University Press, 1996.

Chisholm, Roderick M. “Identity through time.” *Person and Object: A Metaphysical Study.* Milton Park, UK: Taylor & Francis, 2013, 88-112. Originally published 1976.

Clines, Manfred E., and Nathan S. Kline. “Cyborgs and Space.” *Astronautics* (1960).

Coecklebergh, Mark. "Humans, Animals, and Robots: a Phenomenological Approach to Human-Robot Relations." *International Journal of Social Robotics* 3 (2011): 197-204.

Cohen, Jeffrey Jerome. *Stone: An Ecology of the Inhuman*. Minneapolis: University of Minnesota Press, 2015.

Cohen, Jeffrey Jerome, and Julian Yates, eds. *Object Oriented Environs*. Earth: punctum books, 2016.

Cohen, Jeffrey Jerome, and Linda T. Elkins-Tanton. *Earth*. Object Lessons. New York: Bloomsbury, 2017.

Commander Cherry's Puzzled Adventure. Developed and published by Grande Games (2015). Xbox One.

Crary, Jonathan. *Suspensions of Perception: Attention, Spectacle and Modern Culture*. Cambridge, MA: MIT Press, 1999.

———. *Techniques of the Observer: On Vision and Modernity in the Nineteenth Century*. Cambridge, MA: MIT Press, 1990.

"Creating a Champion in Kinect Sports Rivals." *Xbox Wire* (2014). http://news.xbox.com/2014/04/games-ksr-developer-qa (accessed May 29 2014).

Crichton, Michael. *Westworld*. United States: MGM, 1973. Film.

Danaher, John. "Should We Be Thinking about Robot Sex?" In *Robot Sex: Social and Ethical Implications*. Edited by John Danaher and Neil McArthur, 3-14. Cambridge, MA: MIT Press, 2017.

Danaher, John, Brian Earp, and Anders Sandberg. "Should we Campaign against Sex Robots?" in *Robot Sex: Social and Ethical Implications*. Edited by John Danaher and Neil McArthur, 47-72. Cambridge, MA: MIT Press, 2017.

Darling, Kate. "Extending Legal Rights to Social Robots." *IEEE Spectrum* (2012).

———. "In Our Own Images." *Science* (October 2014): 188-9.

———. "'Who's Johnny?' Anthropomorphic Framing in Human-Robot Interaction, Integration, and Policy." In *Robot Ethics 2.0: From Autonomous Cars to Artificial Intelligence*. Edited by Patrick Lin, Ryan Jenkins, and Keith Abney. New York: Oxford University Press, 2017. Kindle Ebook.

deFren, Allison. *asfr.* 2001. Youtube. https://www.youtube.com/watch?v=hfHs1xQTZ2E (accessed June 27 2017).

Devlin, Kate. *Turned On: Science, Sex and Robots* (forthcoming).

"Dickwolves." *Geek Feminism Wiki*. http://geekfeminism.wikia.com/wiki/Dickwolves (accessed January 17 2018).

Donovan, Tristan. "Chapter 1: Hey! Let's play games!" *Replay: The History of Videogames.* Lewes, UK: Yellow Ant, 2010.

Downs, John, Frank Vetere, and Steve Howard. "Paraplay: Exploring Playfulness Around Physical Console Gaming." In *Human-Computer Interaction – INTERACT*. Edited by P. Kotzé, G. Marsden, J. Wessen and M. Winckler, 682-699. Berlin: Springer, 2013.

Dreyfus, Hubert. "From Micro-worlds to Knowledge Representation: AI at an Impasse." *Mind Design II*. Edited by John Haugeland. Cambridge, MA: The MIT Press, 1997: 143-182. Originally published 1979.

Du Toit, Cornel. "Panpsychism, Pan-Consciousness and the Non-Human Turn: Rethinking Being as Conscious Matter." *Theological Studies* 72, no. 4 (2016): 1-11.

Dyer-Witheford, Nick, and Greig de Peuter. *Games of Empire: Global Capitalism and Video Games*. Edited by Mark Poster, Samuel Weber and Katherine Hayles. Electronics Mediations. Minneapolis: University of Minnesota Press, 2009.

E3 2009: Project Natal Xbox 360 Announcement. 2009. Youtube. https://www.youtube.com/watch?v=p2qlHoxPioM (accessed June 24 2017).

E3 2010 Microsoft - Kinect (Project Natal) Part 1. 2010. Youtube. https://www.youtube.com/watch?v=bxp43T2JP18 (accessed June 24 2017).

Ehrenkranz, Melanie. "Futuristic Fembot 'Jia Jia' is One of the Most Sexist Robot Creations Yet." *Tech.Mic* (2016). https://mic.com/articles/141167/futuristic-fembot-jia-jia-is-one-of-the-most-sexist-robot-creations-yet#.6inj2vtF8 (accessed June 27 2017).

Elder, Alexis. "Robot Friends for Autistic Children: Monopoly Money or Counterfeit Currency?" In *Robot Ethics 2.0: From Autonomous Cars to Artificial Intelligence*. Edited by Patrick Lin, Ryan Jenkins, and Keith Abney. New York: Oxford University Press, 2017. Kindle Ebook.

Ellul, Jacques. *The Technological Society* [La Technique ou l'enjeu du siècle]. Translated by John Wilkinson. New York: Vintage Books, 1964. Originally published 1957.

Emmeche, Claus. "Does a Robot Have an Umwelt? Reflections on the Qualitative Biosemiotics in Jakob Von Uexküll." *Semiotica* 134, no. 1-4 (2001): 653-93.

Esteban, Pablo G. et al. "How to Build a Supervised Autonomous System for Robot-Enhanced Therapy for Children with Autism Spectrum Disorder." *Paladyn, Journal of Behavioral Robotics* 8 (2017): 18-38.

Fable: The Journey. Developed by Lionhead Studios. Published by Microsoft Studios, 2012. Xbox 360.

Ferris, Timothy. "Timothy Ferris on Voyagers' Never-Ending Journey." *Smithsonian Magazine* (May 2012). http://www.smithsonianmag.com/science-nature/timothy-ferris-on-voyagers-never-ending-journey-60222970/ (accessed June 7 2017).

Firestone, David. "While Barbie Talks Tough, G.I. Joe Goes Shopping." *The New York Times* (December 31 1993). http://www.nytimes.com/1993/12/31/us/while-barbie-talks-tough-g-i-joe-goes-shopping.html?sec=&spon=&pagewanted=all (accessed January 21 2018).

Flynn, Bernadette. "Geography of the Digital Hearth." *Information, Communication and Society* 6, no. 4 (2003): 551-76.

Forbes, Bryan. *The Stepford Wives*. Columbia Pictures. 1975. Film.

Ford, Martin. *Rise of the Robots: Technology and the Threat of Mass Unemployment.* London: Oneworld Publications, 2016.

Forss, Anette. "Cells and the (Imaginary) Patient: the Multistable Practitioner-Technology-Cell Interface in the Cytology Laboratory." *Medicine, Health Care and Philosophy* 15, no. 3 (2012): 295-308.

Forza Motorsport 5. Developed by Turn 10 Studios. Published by Microsoft Studios. 2013. Xbox One.

Foster, Jay. "Ontologies without Metaphysics: Latour, Harman and the Philosophy of Things." *Analecta Hermeneutica* 3 (2011): 1-26.

Frank, Lily, and Sven Nyholm. "Robot sex and consent: is consent to sex between a robot and a human conceivable, possible, and desirable?" *Artificial Intelligence and Law* 25, no. 3 (2017): 305-323.

Freeman, Will. "Kinect Sports Rivals Review - Going Through the Motions." *The Guardian* (April 11 2014). http://www.theguardian.com/technology/2014/apr/11/kinect-sports-rivals-review-wii (accessed May 29 2014).

Fru. Developed and published by Through Games. 2016. Xbox One.

"FT (Female Type)." *plasticpals* (2009). http://www.plasticpals.com/?p=55 (accessed June 27 2017).

"FT [FEMALE TYPE]." *Robo Garage* (2009). http://robo-garage.com/en/prd/p_02/index.html (accessed June 28 2017).

Fussell, Susan R., Sara Kiesler, Leslie D. Setlock, and Victoria Yew. "How People Anthropomorphize Robots." Paper presented at *HRI*, Amsterdam, March 2008.

Galeo, Don and Christianna Ready. "Kurzweil Claims that the Singularity will Occur by 2045." *Futurism* (October 5 2017). https://futurism.com/kurzweil-claims-that-the-singularity-will-happen-by-2045/ (accessed March 22 2018).

Galloway, Alexander. "A Response to Graham Harman's 'Marginalia on Radical Thinking'," *An und für sich*: June 3 2012. https://itself.blog/2012/06/03/a-response-to-graham-harmans-marginalia-on-radical-thinking/ (accessed January 3 2018).

Gane, Nicholas. "When We Have Never Been Human, What is to be Done?" *Theory, Culture & Society* 23, no. 7-8 (2006): 135-59.

Garber, Megan. "Would You Want Therapy From a Computerized Psychologist?" *The Atlantic* (May 23 2014).

Garland, Alex. *Ex Machina.* United Kingdom and the United States: Universal Studios. 2014. Film.

Garrison, John. *Glass*. Object Lessons. New York: Bloomsbury, 2015.

Germano, William. *Eye Chart*. Object Lessons. New York: Bloomsbury, 2017.

Gérôme, Jean-Léon. “Pygmalion and Galatea.” Wikimedia Commons (accessed May 18 2018). Originally painted circa 1890.

Gibbs, Martin, Marcus Carter, Michael Arnold. "Avatars, Characters, Players and Users: Multiple Identities at/in Play." Paper presented at *OZCHI’12*, Melbourne, Australia, November 26-30 2012.

Gibson, James J. *The Ecological Approach to Visual Perception*. New York: Psychology Press, Taylor and Francis Group, 2015. Originally published 1979.

Gillespie, Craig. *Lars and the Real Girl*. United States and Canada: Metro-Goldwyn-Meyer. 2007. Film.

Goeminne, Gert, and Erik Paredis. "Opening Up the In-Between: Ihde's Postphenomenology and Beyond." *Foundations of Science* 16, no. 2-3 (2011): 101-07.

Gratton, Peter. *Speculative Realism: Problems and Prospects*. London: Bloomsbury, 2014.

Green, Holly. “The Many Deaths of the Kinect.” *Paste* (January 10 2018). https://www.pastemagazine.com/articles/2018/01/the-many-deaths-of-the-kinect.html (accessed February 9 2018).

Greenwald, Glenn, Ewen MacAskill, Laura Poitras, Spencer Ackerman and Dominic Rushe. "Microsoft Handed the NSA Access to Encrypted Messages." *The Guardian* (July 12 2013). http://www.theguardian.com/world/2013/jul/11/microsoft-nsa-collaboration-user-data (accessed January 20 2018).

Grey Walter's Tortoises. 2008. Youtube. https://www.youtube.com/watch?v=lLULRlmXkKo (accessed June 27 2017).

Griffin, David Ray. *Religion and Scientific Naturalism: Overcoming the Conflicts*. Albany NY: State University of New York Press, 2000.

———. *Unsnarling the World-Knot: Consciousness, Freedom, and the Mind-Body Problem.* Berkeley CA: University of California Press, 1998.

Gross, Kenneth. *The Dream of the Moving Statue.* University Park, Pennsylvania: The Pennsylvania State University Press, 2006.

Guizzo, Erico. "Hiroshi Ishiguro: the Man Who Made a Copy of Himself." *IEEE Spectrum* (2010).

Gustafsson, Laura, and Terike Haapoja. “Imagining Non-Human Realities.” In *History According to Cattle*. Edited by Laura Gustafsson and Terike Haapoja. Brooklyn, NY: punctum books, 2015.

Halo 4. Developed by 343 Industries. Published by Microsoft Studios. 2012. Xbox 360.

Halsall, Francis. "Art and Guerrilla Metaphysics: Graham Harman and Aesthetics as Ffirst Philosophy." *Speculations: A Journal of Speculative Realism*, no. V (2014): 382-410.

Hampton, Gregory Jerome. *Imagining Slaves and Robots in Literature, Film, and Popular Culture.* Lanham: Lexingon Books, 2015.

Hanson Robotics. “Bina48” (2017). http://www.hansonrobotics.com/robot/bina48/ (accessed May 20 2018)

Haraway, Donna. "A Manifesto for Cyborgs: Science, Technology and Socialist Feminism in the 1980s." In *From Modernism to Postmodernism: An Anthology*. Edited by Lawrence Cahoone, 464-481. Oxford: Blackwell, 2003. Originally published 1984.

———. “Anthropocene, Capitalocene, Plantationocene, Chthulucene: Making Kin.” *Environmental Humanities* 6 (2015): 159-165.
———. *The Companion Species Manifesto: Dogs, People, and Significant Otherness.* Chicago: Prickly Paradigm Press, 2003.
———. “Situated Knowledges: the Science Question in Feminism and the Privilege of Partial Perspective.” *Feminist Studies* 14, no. 3 (Autumn, 1988): 575-599.
Harman, Graham. *Circus Philosophicus.* Ropley, UK: O Books, 2010.
———. "The Current State of Speculative Realism." *Speculations: A Journal of Speculative Realism*, no. IV (2013): 22-27.
———. *Guerrilla Metaphysics: Phenomenology and the Carpentry of Things*. Chicago: Open Court, 2005.
———. *Heidegger Explained: From Phenomenon to Thing*. Chicago: Open Court, 2007.
———. *Immaterialism*. Edited by Laurent de Sutter. Theory Redux. Cambridge, UK: Polity Press, 2016.
———. "Materialism Is Not the Solution: on Matter, Form, and Mimesis." *Nordic Journal of Aesthetics* 47 (2015): 94-110.
———. "On Vicarious Causation." In *Collapse* vol. II. Edited by Robin Mackay. Urbanomic, 2007.
———. *Object-Oriented Ontology: A New Theory of Everything*. UK: Pelican Books, 2018.
———. *Prince of Networks: Bruno Latour and Metaphysics*. Melbourne: re.press, 2009.
———. *The Quadruple Object*. Alresford, UK: Zero Books, 2011.
———. "The Road to Objects." *Continent* 1, no. 3 (2011): 171-79.
———. *The Third Table*. Documenta. Kassel, Germany: Hatje Cantz Verlag, 2012.
———. *Tool-Being: Heidegger and the Metaphysics of Objects*. Chicago: Open Court, 2002.
———. “Undermining, Overmining, and Duomining: a Critique.” In *ADD Metaphysics*. Edited by Jenna Sutela, 40-51. Aalto, Finland: Aalto University Digital Design Laboratory, 2013.
Harmony AI. Developed by Realbotix. 2017. Android.
Harper, Richard, and Helena M. Mentis. "The Mocking Gaze: the Social Organization of Kinect Use." *Proceedings of the 2013 Conference on Computer Supported Cooperative Work* (2013): 167-80.
Hasse, Cathrine. “Artefacts That Talk: Mediating Technologies as Multistable Signs and Tools.” *Subjectivity* 6, no. 1 (2013): 79-100.
Hayles, N. Katherine. "Speculative Aesthetics and Object-Oriented Inquiry (Ooi)." *Speculations: A Journal of Speculative Realism*, no. V (2014): 158-79.
Heidegger, Martin. *The Question Concerning Technology and Other Essays* [Die Technik und die Kehre, Holzwege, and Vortrage und Aufsatze]. Translated by William Lovitt. New York: Garland Publishing, 1977. Originally published 1954.
Helgeson, Matt. “Rare Lays Off Staff In Wake of Kinect Sports Rivals.” *Gameinformer* (May 19 2014). http://www.gameinformer.com/b/news/archive/2014/05/19/rare-lays-off-staff-in-wake-of-kinect-sports-rivals.aspx (accessed June 27 2017).
Hillis, K. "The Sensation of Ritual Space." *Digital Sensations: Space, Identity and Embodiment in Virtual Reality*. Minneapolis: University of Minnesota Press, 1999: 60-89.
Himma, Kenneth Eimar. “Artificial Agency, Consciousness and the Criteria for Moral Agency: What Properties Must an Artificial Agent Have to be a Moral Agent?” *Ethics and Information Technology* 11 (2009): 19-29.
Hoffmeyer, Jesper. “Seeing Virtuality in Nature.” *Semiotica* 134, no. 1-4 (2001): 381-398.
Hofstadter, Douglas R. *Gödel, Escher, Bach: An Eternal Golden Braid*. Harmondsworth, England: Penguin Books, 1980.
Holland, Owen. "The First Biologically Inspired Robots." *Robotica* 21 (2003): 351-63.

———. "The Grey Walter Online Archive." *University of the West of England* (2013). http://www.ias.uwe.ac.uk/Robots/gwonline/gwonline.html (accessed September 25 2013).

Howard, Dorothy. "Loving Machines: a De-Anthropocentric View of Intimacy." *Arachne*, no. 00 (2015).

Hsiao, Shih-Wen, Chu-Hsuan Lee, Meng-Hua Yang and Rong-Qi Chen. "User Interface Based on Natural Interaction Design for Seniors." *Computers in Human Behavior* 75 (2017): 147-159.

Hutchens, Jason L. "How to Pass the Turing Test by Cheating." *ACM DL Digital Library.* University of Western Australia, 1996.

Ihde, Don. *Bodies in Technology*. Minneapolis: University of Minnesota Press, 2001.

———. "Forty Years in the Wilderness." In *Postphenomenology: A Critical Companion to Ihde*. Edited by Evan Selinger. Albany, NY: State University of New York Press, 2006.

———. "Introduction: Postphenomenological Research." *Human Studies* 31, no. 1 (2008): 1-9.

———. *Postphenomenology and Technoscience: The Peking University Lectures*. Albany, NY: State University of New York Press, 2009.

———. *Postphenomenology: Essays in the Postmodern Context*. Evanston, Illinois: Northwestern University Press, 1993.

———. *Technology and the Lifeworld: From Garden to Earth*. Bloomington: Indiana University Press, 1990.

"[Independent] Chinese Researchers Create Jia Jia - a Super-Lifelike 'Robot Goddess'." *University of Science and Technology of China* (April 19 2016). http://en.ustc.edu.cn/highlight/201604/t20160419_243384.html (accessed June 27 2017).

Inglis-Arkell, Esther. "The Very First Robot "Brains" Were Made of Old Alarm Clocks." *io9* (2012). http://io9.com/5890771/the-very-first-robot-brains-were-made-of-old-alarm-clocks (accessed September 25 2013).

"Innovation." *Xbox* (2014). http://www.xbox.com/en-US/xbox-one/innovation (accessed January 13 2014).

Inside the brains of Xbox One Kinect. 2013. Youtube. https://www.youtube.com/watch?v=ziXflemQr3A (accessed June 26 2017).

Jones, Gwyneth. *Divine Endurance*. London: George Allen & Unwin, 1984.

Jonze, Spike. *Her*. United States: Warner Bros. Pictures, 2013. Film.

Jordan Wolfson: (Female Figure) 2014. 2014. Youtube. https://www.youtube.com/watch?v=mVTDypgmFCM (accessed June 29 2017).

Jung, Eun Hwa, T. Franklin Waddell and S Shyam Sundar. "Feminizing Robots: User Responses to Gender Cues on Robot Body and Screen." *Conference on Human Factors in Computing Systems Proceedings* (2016): 3107-13.

Juul, Jesper. "The Game, the Player, the World: Looking for a Heart of Gameness." *Digital Games Research Conference: Level Up*. Edited by M. Copier and J. Raessens. Utrecht: Universiteit Utrecht, 2003: 30-47.

Kang, Minsoo. "Building the Sex Machine: The Subversive Potential of the Female Robot." *Intertexts* 9, no. 1 (2005): 5-22.

Kamide, Hiroko and Tatsuo Arai. "Perceived Comfortableness of Anthropomorphized Robots in U.S. and Japan." *International Journal of Social Robotics* 9 (September 2017): 537-543.

karenarchey, "Slavoj Žižek – Objects, Objects Everywhere: a Critique of Object Oriented Ontology." *e-flux conversations* (February 2016). https://conversations.e-flux.com/t/slavoj-zizek-objects-objects-everywhere-a-critique-of-object-oriented-ontology/3284 (accessed January 3 2018).

Kennedy, Jenny. “Addressing the Wife Drought: Automated Assistants in the Smart Home.” Presentation at *At Home with Digital Media*, Queensland University of Technology, November 2-3 2018.

Khonsari, Michael M. and Mehdi Amiri. *Introduction to Thermodynamics of Mechanical Fatigue*. Boca Raton: CRC Press, 2013.

Kidd, Michael. “Technology and Nature: a Defence and Critique of Marcuse.” *Polis: Revista de Stiinte Politice* IV, no. 14 (2016): 49-65.

"kinect." *Reddit* (2013-2014). http://www.reddit.com/r/Kinect (accessed January 13 2014).

"Kinect on Xbox One." *Xbox* (2013-2014). http://forums.xbox.com/xbox_forums/xbox_support/xbox_one_support/f/4275.aspx (accessed January 13 2014).

"Kinect Sensor for Xbox One." *Microsoft* (2017). https://www.microsoft.com/en-au/store/d/kinect-sensor-for-xbox-one/91hq5578vksc (accessed June 26 2017).

Kinect Sports Rivals. Developed by Rare. Published by Microsoft Studios. 2014. Xbox One.

"Kinect Sports Rivals." *Xbox* (2014). http://www.xbox.com/he-IL/xbox-one/games/kinect-sports-rivals (accessed May 27 2014).

Kinect Sports Rivals Preseason. Developed by Rare. Published by Microsoft Studios. 2013. Xbox One.

Kinectimals. Developed by Frontier Developments. Published by Microsoft Game Studios. 2010. Xbox 360.

Kintzling, Pierre. "Automate: Joueuse de Tympanon." Photo by Phillippe Hurlin. *Musée des arts et métiers* (2017). http://www.arts-et-metiers.net/musee/automate-joueuse-de-tympanon (accessed June 27 2017).

Klepek, Patrick. “Kinect Died in the Uncanny Valley.” *Giant Bomb* (May 15 2014). https://www.giantbomb.com/articles/kinect-died-in-the-uncanny-valley/1100-4920/ (accessed February 9 2018).

Koribalski, B., K. Jones, M. Elmouttie and R. Haynes. "Neutral Hydrogen Gas in the Circinus Galaxy." *Australia Telescope Outreach and Education, CSIRO*. http://outreach.atnf.csiro.au/images/astronomical/circinus.html (accessed June 27 2017).

Kovács, Gábor. “A Forefather of Environmentalism – Lewis Mumford, the Critic of Technology and Civilization.” *Philobiblon* XIV (2009): 294-300.

Krahulik, Mike, and Jerry Holkins. "Negotiations." *Penny Arcade* (June 21 2013). http://www.penny-arcade.com/comic/2013/06/21 (accessed June 26 2017).

Kramer, Jeff. *Hacking the Kinect*. New York: Apress, 2012.

“KSR: Announcement Trailer.” *Xbox* (2014). http://www.xbox.com/en-US/xbox-one/games/kinect-sports-rivals (accessed May 29 2014).

Kubrick, Stanley. *2001: A Space Odyssey*. United States: MGM. 1968. Film.

Kull, Kaveli. "Jakob Von Uexküll: an Introduction." *Semiotica* 134, no. 1-4 (2001): 1-59.

Kurokawa, Megumi, Libby Schweber and Will Hughes. “Client Engagement and Building Design: the View From Actor-Network Theory.” *Building Research and Information* 45, no. 8 (2017): 910-925.

Kurzweil, Ray. *The Age of Intelligent Machines*. Cambridge, MA: MIT Press, 1992. Originally published 1990.

———.*How to Create a Mind*. New York: Penguin Books, 2012.

———.*The Singularity is Near*. New York: Viking, 2005.

Kuznets, Lois. *When Toys Come Alive: Narratives of Animation, Metamorphosis and Development*. New Haven: Yale University Press, 1994.
Lang, Fritz. *Metropolis*. Germany: UFA. 1927. Film.
Lash, Scott and Celia Lury. *Global Culture Industry*. Cambridge: Polity Press, 2007.
Latour, Bruno. *Aramis, or the Love of Technology*. Translated by Catherine Porter. Cambridge, MA: Harvard University Press, 1996. Originally published 1993.
———. *Pandora's Hope: Essays on the Reality of Science Studies.* Cambridge, MA: Harvard University Press, 1999.
———. *The Pasteurization of France*. Translated by Alan Sheridan and John Law. Cambridge, MA: Harvard University Press, 1988.
———. *Reassembling the Social: An Introduction to Actor-Network Theory*. Oxford: Oxford University Press, 2005.
———. *Science in Action*. Cambridge, MA: Harvard University Press, 1987.
———. "Technology is Society Made Durable." *Sociological Review Monograph* 38, no. 2 (1991): 103-131.
———. *We Have Never Been Modern*. Translated by Catherine Porter. Hertfordshire: Harvester Wheatsheaf, 1993.
Latour, Bruno, as Jim Johnson. "Mixing Humans and Nonhumans Together: the Sociology of a Door-Closer." *Social Problems* 35, no. 3 (1988): 298-310.
Latour, Bruno, Graham Harman and Peter Erdélyi. *The Prince and the Wolf: Latour and Harman at the LSE*. Winchester, UK: Zero Books, 2011.
Latour, Bruno, and Steve Woolgar. *Laboratory Life: The Social Construction of Scientific Facts.* Beverly Hills: Sage Publications, 1979.
Law, John. "Notes on the Theory of the Actor-Network: Ordering, Strategy and Heterogeneity." *Systems Practice* 5 (1992): 379-93.
Leach, Tessa. "Improving on Eyes: Human-Machine Interaction in Cyborg Vision." Unpublished thesis, 2012.
Leach, Tessa and Michael Arnold. "How the Kinect Domesticates its Users." Paper presented at *At Home with Digital Media* symposium at Queensland University of Technology, November 3 2017.
"Leap Motion." *Leap Motion*. 2017. https://www.leapmotion.com/product/vr#113 (accessed February 16 2018).
Levy, David. *Love & Sex with Robots: The Evolution of Human-Robot Relationships*. London: Duckworth Overlook, 2009. Originally published 2007.
———. "Roxxxy the "Sex Robot" - Real or Fake." *Lovotics* 1 (2013): 1-4.
The LifeNaut Project. *Bina 48 Meets Bina Rothblatt* (November 27 2014). Youtube. https://www.google.com.au/search?q=bina48&rlz=1C1GGRV_enAU791AU791&oq=bina48&aqs=chrome..69i57j69i60j0l4.1335j0j7&sourceid=chrome&ie=UTF-8 (accessed May 9 2018)
Long, Julia. *Anti-Porn: The Resurgence of Anti-Pornography Feminism.* London: Zed Books, 2012.
Loomis, Roger Sherman. *Celtic Myth and Arthurian Romance*. Chicago: Academy Chicago Publishers, 2005.
Lotman, Juri. "On the Semiosphere." *Sign Systems Studies* 33, no. 1 (2005): 205-29.
Lovelock, James. *Gaia: A New Look at Life on Earth*. Oxford: Oxford University Press, 2000. Originally published 1979.
———. *The Revenge of Gaia.* London: Penguin Books, 2007. Originally published 2006.
Lucas, Gale M., Jonathan Gratch, Aisha King, and Louis-Philippe Morency. "It's Only a Computer: Virtual Humans Increase Willingness to Disclose." *Computers in Human Behavior* 37 (August 2014): 94-100.

Lunning, French. “Allure and Abjection: the Possible Potential of Severed Qualities.” In *Object-Oriented Feminism*. Edited by Katherine Behar, 83-105. Minneapolis: University of Minnesota Press, 2016.

Lybeck, Eric Royal. “The Critical Theory of Lewis Mumford.” *The International Journal of Interdisciplinary Social Sciences* 5, no. 1: 91-102.

MacDorman, Karl, Sandosh Vasudevan, and Chin-Chang Ho. "Does Japan Really Have Robot Mania? Comparing Attitudes by Implicit and Explicit Measures." *AI & Society* 23, no. 4 (2009): 485-510.

Maddox, Teena. “Sci-fi is turned into reality with technology guru from Minority Report and Iron Man.” *TechRepublic* (August 4 2016). https://www.techrepublic.com/article/sci-fi-is-turned-into-reality-with-technology-guru-from-minority-report-and-iron-man/ (accessed December 4 2017).

Maines, Rachel P. *The Technology of Orgasm.* Baltimore: The Johns Hopkins University Press. 1999.

Malin, Brenton J. “Communicating with Objects: Ontology, Object-Orientations, and the Politics of Communication.” *Communication Theory* 26 (2016): 236-254.

Manovich, Lev. "Friendly Alien: Object and Interface." *Artifact: Journal of Visual Design* 1, no. 1 (2007): 29-32.

Marcuse, Herbert. *One-Dimensional Man: Studies in the Ideology of Advanced Industrial Society*. London: Routledge, 2002. Originally published 1964.

Marino, Mark C. “The Racial Formation of Chatbots.” *CLCWeb: Comparative Literature and Culture* 16, no. 5, article 13 (2014).

Markowitz, Judith A., ed. *Robots That Talk and Listen: Technology and Social Impact*. Berlin: De Gruyter, 2015.

Mason, Andrew S. *Plato*. Abingdon, Oxon: Routledge, 2014. Originally published 2010.

Mason, Fran. "Loving the Technological Undead: Cyborg Sex and Necrophilia in Richard Calder's *Dead* Trilogy." In *The Body's Perilous Pleasures*. Edited by Michelle Aaron, 108-125. Edinburgh: Edinburgh University Press, 1999.

Mataric, Maja. *The Robotics Primer*. Cambridge, MA: The MIT Press, 2007.

Mattick, Don. "Your Feedback Matters - Update on Xbox One." *Xbox* (June 19 2013). http://news.xbox.com/2013/06/update (accessed January 15 2014).

Mayr, Ernst. *What Makes Biology Unique?* New York: Cambridge University Press, 2007.

Mazlish, Bruce. *The Fourth Discontinuity: The Co-Evolution of Humans and Machines*. New Haven: Yale University Press, 1993.

McDermott, Drew. “What Matters to a Machine?” In *Machine Ethics*. Edited by Michael Anderson and Susan Leigh Anderson, 88-114. Cambridge: Cambridge University Press, 2011.

McFadyen, Siobhan. "Shocking Tiny Sex Robot Which Looks Like Schoolgirl is on Sale for £770 and Comes Delivered in 'Coffin'." *Mirror* (March 16 2016). http://www.mirror.co.uk/news/weird-news/shocking-tiny-sex-robot-looks-7564114 (accessed January 21 2018).

Meillassoux, Quentin. *After Finitude* [Après la Finitude (2006)]. Translated by Ray Brassier. New York: Continuum, 2008.

Metacritic. CBS Interactive (2017). www.metacritic.com (accessed November 22 2017).

"Microsoft by the Numbers." *Microsoft Story Labs* (2013). http://www.microsoft.com/en-us/news/bythenumbers/index.html?utm_source=buffer (accessed January 24 2014).

Mikelson, Emmy. “Space for Things: Art, Objects, Speculation.” In *And Another Thing: Nonanthropocentrism and Art*. Edited by Katherine Behar and Emmy Mikelson, 11-20. Earth, Milky Way: punctum books, 2016.

Miller, Ross. “Kinect for Xbox 360 Review.” *engadget* (November 4 2010). https://www.engadget.com/2010/11/04/kinect-for-xbox-360-review/ (accessed December 4 2017).

Minsky, Marvin. *The Society of Mind*. London: Picador, 1988.

Misses Joust. "Xbox On: Voice Commands for Xbox One." *Xbox Forums.* 2013-14. http://forums.xbox.com/xbox_forums/xbox_support/xbox_one_support/f/4275/t/1622636.aspx (accessed January 13 2014).

Mol, Annemarie. *The Body Multiple: Ontology in Medical Practice*. Durham, North Carolina: Duke University Press, 2002.

Monk-Turner, Elizabeth. "Gender and Market Opportunity in Japan." *National Journal of Sociology* 11 (1997): 97-105.

Montfort, Nick and Bogost, Ian. *Racing the Beam: The Atari Video Computer System.* Cambridge, MA: MIT Press, 2009.

Mori, Masahiro. *The Buddha in the Robot*. Translated by Charles S. Terry. Tokyo: Kosei Publishing Co., 1981.

Morris, William. "Pygmalion and the Image." *Sacred Texts*. Originally published 1868. http://www.sacred-texts.com/neu/morris/ep1/ep121.htm (accessed June 27 2017).

Morton, Timothy. “From Modernity to the Anthropocene: Ecology and Art in the Age of Asymmetry.” *International Social Science Journal* 63 (2014): 39-51.

———. "Here Comes Everything: the Promise of Object-Oriented Ontology." *Qui Parle: Critical Humanities and Social Sciences* 19, no. 2 (2011): 163-90.

———. *Hyperobjects: Philosophy and Ecology after the End of the World*. Posthumanities. Minneapolis: University of Minnesota Press, 2013.

Mumford, Lewis. *Myth of the Machine: Technics and Human Development*. New York: Harcourt Brace Jovanovich Publishers, 1967.

Nagel, Thomas. *Mortal Questions.* Cambridge, UK: Cambridge University Press, 1979.

———. "What is it Like to be a Bat?". *The Philosophical Review* 83, no. 4 (1974): 435-50.

“NASA Awards Two Robots to University Groups for R&D Upgrades.” *NASA* (2015). https://www.nasa.gov/press-release/nasa-awards-two-robots-to-university-groups-for-rd-upgrades (accessed June 28 2017).

Nesterov, Nikola, Peter Hughes, Nuala Healy, Niall Sheehy and Neil O’Hare. “Application of Natural User Interface Devices for Touch-Free Control of Radiological Images During Surgery.” Paper presented at *2016 IEEE 29th International Symposium on Computer Based Medical Systems (CBMS)*, Dublin, Ireland, June 20-24 2016.

New Era of Cognitive Computing. (2013). Youtube. https://www.youtube.com/watch?v=h22n80aT2FY (accessed June 27 2017).

Niebisch, Arndt. *Media Parasites in the Early Avant-Garde*. New York: Palgrave MacMillan, 2012.

Nirenberg, S., and C. Pandarinath. "Retinal Prosthetic Strategy with the Capacity to Restore Normal Vision." *Proceedings of the National Academy of Sciences of the United States of America* 109, no. 37 (Sep 11 2012): 15012-7.

Nolan, Jonathan and Lisa Joy. *Westworld*. United States: Warner Bros. Television, 2016. Television series.

“The Nonhuman Turn Conference”, *Center for 21st Century Studies* (blog). https://c21uwm.wordpress.com/the-nonhuman-turn/ (accessed January 17 2018)

Norman, Donald. “Affordance, Conventions and Design.” *Interactions* 3 (May/June 1999). http://interactions.acm.org/archive/view/may-june-1999/affordance-conventions-and-design1 (accessed March 18 2018).

———. *The Design of Everyday Things: Revised and Extended Edition*. New York: Basic Books, 2013. Originally published 1988.

———. "Natural User Interfaces are not Natural." *Interactions* 17, no. 3 (2010).

Nöth, Winfried. "Semiosis and the Umwelt of a Robot." *Semiotica* 134, no. 1-4 (2001): 695-99.

"Object lessons." objectsobjectsobjects.com (accessed June 22 2017).

"Objectùm-Sexuality Internationale." http://www.objectum-sexuality.org/ (accessed June 27 2017).

Olivares, Jonathan, interviewing Graham Harman. "The Metaphysics of Logos: Decoding Branded Objects with Philosopher Graham Harman." *032c* (May 17 2018). https://032c.com/decoding-logos-graham-harman/ (accessed May 28 2018).

Operation: Pedopriest. Developed by Molleindustria. 2007. PC.

O'Rourke, Michael. "Girls Welcome!!!" *After the "Speculative Turn": Realism, Philosophy, and Feminism*. Edited by Katerina Kolozova and Eileen A. Joy, 159-197. Earth: punctum books, 2016.

Osborn, Henry Fairfield. "The Evolution of Mammalian Molars to and from the Tritubercular Type." *The American Naturalist* 22, no. 264 (December 1888): 1067-1079.

Ovid. "Metamorphoses." *The Ovid Collection*. Translated by A.S. Kline (2010). http://ovid.lib.virginia.edu/trans/Ovhome.htm (accessed June 27 2017).

Panzarino, Matthew. "The New Xbox One Kinect Tracks Your Heart Rate, Happiness, Hands and Hollers." *The Next Web* (May 22 2013). http://thenextweb.com/microsoft/2013/05/22/the-new-xbox-one-kinect-tracks-your-heart-rate-happiness-hands-and-hollers/ (accessed June 26 2017).

Parikka, Jussi. *An Alternative Deep Time of the Media: a Geologically Tuned Media Ecology.* 2013. Vimeo. http://jussiparikka.net/?s=geology+media (accessed August 22 2013).

———. *Insect Media*. Posthumanities. Minneapolis: University of Minnesota Press, 2010.

Parkin, Simon. "Kinect Sports Rivals Review: Warm Down." *Eurogamer* (April 8 2014). http://www.eurogamer.net/articles/2014-04-08-kinect-sports-rivals-review (accessed May 29 2014).

"Parliament of Things." *Parliament of Things* (2017). https://theparliamentofthings.org (accessed December 21 2017).

Pechenik, Jan A. *Biology of the Invertebrates*. Fifth edition. Boston: McGraw Hill Higher Education, 2005.

Perkowitz, Sidney. *Digital People: From Bionic Humans to Androids*. Washington DC: Joseph Henry Press, 2004.

Petersen, Steve. "Is it Good for Them Too? Ethical Concern for the Sexbots." *Robot Sex: Social and Ethical Implications*. Edited by John Danaher and Neil McArthur, 155-172. Cambridge, MA: The MIT Press, 2017.

Petit, Carolyn. "Kinect Sports Rivals Review: Championing Mediocrity." *Gamespot* (April 7 2014). http://www.gamespot.com/reviews/kinect-sports-rivals-review/1900-6415722/ (accessed May 29 2014).

Phan, Thao. "The Materiality of the Digital and the Gendered Voice of Siri." *Transformations* 29 (2017): 23-33.

Pickering, Andrew. *The Cybernetic Brain: Sketches of Another Future*. Chicago: University of Chicago Press, 2010.

"Playstation 3 Accessories." *Playstation* (2014). http://us.playstation.com/ps3/accessories/playstation-move-motion-controller-ps3.html (accessed September 12 2014).

Plunkett, Luke. "That Xbox One Reveal Sure Was a Disaster, Huh?" *Kotaku* (May 21 2013). http://kotaku.com/that-xbox-one-reveal-sure-was-a-disaster-huh-509192266 (accessed May 28 2013).

"Realbotix." *Realbotix* (2016). http://www.realbotix.com/ (accessed May 25 2016).

“RealDoll.” *RealDoll* (2015). https://secure.realdoll.com/ (accessed February 18 2016).

Revesz, Rachael. “Paedophiles ‘Could be Prescribed Child Sex Dolls’ to Prevent Real Attacks, Says Therapist.” *Independent* (August 2 2017). http://www.independent.co.uk/news/uk/home-news/paedophiles-child-sex-dolls-prescription-stop-attacks-child-protection-stopso-therapists-a7872911.html (accessed January 8 2018).

Richardson, Kathleen. “Sex Robot Matters.” *IEEE Technology and Society Magazine* (June 2016).

———. “Urgent – Why We Must Campaign to Ban Sex Dolls & Sex Robots.” *Campaign Against Sex Robots* (October 3 2017). https://campaignagainstsexrobots.org/2017/10/03/urgent-why-we-must-campaign-to-ban-sex-dolls-sex-robots/ (accessed May 26 2018).

Rise of Nightmares. Developed by Sega. Published by Sega. 2011. Xbox 360.

Riskin, Jessica. “Machines in the garden.” *Republics of Letters: A Journal for the Study of Knowledge, Politics and the Arts* 1, no. 2 (April 30 2010).

Rivington, James. “Microsoft Kinect for Xbox 360 review.” *Techradar* (November 4 2010). http://www.techradar.com/reviews/gaming/games-consoles/controllers/microsoft-kinect-for-xbox-360-905010/review (accessed December 4 2017).

Robertson, Jennifer. "Gendering Humanoid Robots: Robo-Sexism in Japan." *Body & Society* 16, no. 2 (2010): 1-36.

———. "Gendering Robots: Posthuman Traditionalism in Japan." In *Recreating Japanese Men*. Edited by Sabine Frühstück and Anne Walthall, 311-332. Berkeley: University of California Press, 2011.

Rosenberger, Gregg. *A Place for Consciousness: Proving the Deep Structure of the Natural World*. Oxford: Oxford University Press, 2004.

Rosenberger, Robert. "Multistability and the Agency of Mundane Artifacts: From Speed Bumps to Subway Benches." *Human Studies* 37 (2014): 369-92.

———. “The Sudden Experience of the Computer.” *AI & Society* 24 (2009): 173-180.

Rosenberger, Robert, and Peter-Paul Verbeek, eds. *Postphenomenological Investigations: Essay on Human-Technology Relations*. Lanham, Maryland: Lexington Books, 2015.

Rothstein, Adam. *Drone*. Object Lessons. New York: Bloomsbury, 2015.

Roy, Deb. "Semiotic Schemas: a Framework for Grounding Language in Action and Perception." *Artificial Intelligence* 167 (2005): 170-205.

Russell, Kyle. “People are Worried Microsoft’s new Xbox Will be Able to Spy on You.” *Business Insider Australia* (May 29 2013). https://www.businessinsider.com.au/xbox-one-kinect-privacy-issues-2013-5?r=US&IR=T (accessed February 22 2018).

Ryse: Son of Rome. Developed by Crytek. Published by Microsoft Studios. 2013. Xbox One.

Salih, Sara. “On Judith Butler and Performativity.” In *Sexualities and Communication in Everyday Life: A Reader*. Edited by Karen E. Lovaas and Mercilee M. Jenkins, 55-67. Thousand Oaks, CA: SAGE Publications, 2007.

Salter, Ben. "E3: Xbox One's Kinect Knows Too Much." *Xbox.MMGN.com* (June 14 2013). http://mmgn.com/xboxone/xbox-one-preview (accessed June 19 2013).

Schodt, Frederik L. *Inside the Robot Kingdom: Japan, Mechatronics and the Coming Robotopia*. Tokyo: Kodansha International Ltd., 1988.

Schouten, Peer. “The Materiality of State Failure: Social Contract Theory, Infrastructure and Governmental Power in Congo.” *Millennium: Journal of International Studies* 41, no. 3 (2013): 553-574.

Scobie, A., and J. Taylor. "Perversions Ancient and Modern: I. Agalmatophilia, the Statue Syndrome." *Journal of the History of the Behavioral Sciences* 11, no. 1 (1975): 49-54.

Scott, Ridley. *Blade Runner*. United States: Warner Bros. 1982. Film.

Searle, John R., “Minds, Brains, and Program.” In *The Turing Test: Verbal Behaviour as the Hallmark of Intelligence*. Edited by Stuart M. Shieber, 201-224. Cambridge, MA: MIT Press, 2004.

Searle, John, and David Chalmers. “’Consciousness and the Philosophers’: an Exchange.” *The New York Review of Books*. (May 15 1997).

Sebeok, Thomas A. "Biosemiotics: its Roots, Proliferation, and Prospects." *Semiotica* 134, no. 1-4 (2001): 61-78.

Sebeok, Thomas A., and J. Umiker-Sebeok. *The Semiotic Web 1991*. Berlin: Mouton de Gruyter, 1992.

Selinger, Evan. "Normative Judgement and Technoscience: Nudging Ihde, Again." *Techné* 12, no. 2 (2008): 120-25.

Shah, Agam. “New Kinect for Windows to Improve Human Interaction with Computers.” *Computerworld* (May 23 2013). https://www.computerworld.com/article/2498093/personal-technology/new-kinect-for-windows-to-improve-human-interaction-with-computers.html (accessed February 22 2018).

Sharkey, Noel, Aimee van Wynsberghe, Scott Robbins and Eleanor Hancock. *Our Sexual Future with Robots*. The Hague: Foundation for Responsible Robotics, 2017. PDF.

Shaviro, Steven. *The Universe of Things.* Minneapolis: University of Minnesota Press, 2014.

Shaw, Ian G. R., and Katherine Meehan. "Force-Full: Power, Politics and Object-Oriented Philosophy." *Area* 45, no. 2 (2013): 216-22.

Sherwood, Harriet. “Robot Priest Unveiled in Germany to Mark 500 Years Since Reformation.” *The Guardian* (May 30 2017). https://www.theguardian.com/technology/2017/may/30/robot-priest-blessu-2-germany-reformation-exhibition (accessed December 4 2017).

SHRDLU. Developed by Terry Winograd and Greg Sharp. Published by Sun Microsystems, Inc. 1999. Windows 7.

“SHRDLU.” http://hci.stanford.edu/winograd/shrdlu/ (accessed January 22 2018).

Siciliano, Leon and Reuters. “A Japanese Company just Unveiled a Robot Priest that will read Scriptures at Buddhist Funerals.” *Business Insider* (August 24 2017). http://www.businessinsider.com/japanese-company-unveiled-robot-priest-technology-pepper-soft-bank-robotics-religion-2017-8?IR=T (accessed June 18 2018).

Silverstone, Roger, and Leslie Haddon. "Design and the Domestication of ICTs: Technical Change and Everyday Life." In *Communication by Design. The Politics of Information and Communication Technologies*. Edited by R. Silverstone and R. Mansell, 44-74. Oxford: Oxford University Press, 1996.

Simonds, Will. “Xbox One Will Know Your Face, Voice, and Heartbeat.” *Abine* (May 21 2013). https://www.abine.com/blog/2013/xbox-one-and-facial-recognition/ (accessed February 22 2018).

Simons, Geoff. *Robots*. London: Cassell, 1992.

Sirani, Jordan. “Microsoft Won’t Offer Free Kinect Adaptor With Xbox One X Upgrade.” *IGN* (October 10 2017). http://au.ign.com/articles/2017/10/10/microsoft-wont-offer-free-kinect-adapter-with-xbox-one-x-upgrade (accessed December 7 2017).

Skrbina, David. *Mind that Abides: Panpsychism in the New Millennium*. Edited by David Skrbina. Amsterdam: John Benjamins Publishing Company, 2009.

———. *Panpsychism in the West*. Cambridge, MA: MIT Press, 2005.

Slabaugh, Brett. "Xbox One's Kinect Can Actually Be Turned Off." *The Escapist* (May 30 2013). http://www.escapistmagazine.com/news/view/124447-Xbox-Ones-Kinect-Can-Actually-Be-Turned-Off (accessed January 20 2018).

Søraa, Roger Andre. "Mechanical Genders: How do Humans Gender Robots?" *Gender, Technology and Development* 21, no. 1-2 (2017): 99-115.

Sparrow, Robert. "Robots, Rape, and Representation." *International Journal of Social Robotics* 9 (September 2017):465-477.

Sparrow, Tom. *The End of Phenomenology*. Edinburgh: Edinburgh University Press, 2014.

Spencer, Phil. "Delivering More Choices for Fans." *Xbox Wire* (blog) (May 13 2014). http://news.xbox.com/2014/05/xbox-delivering-more-choices (accessed June 26 2017).

Spielberg, Steven. *Minority Report*. United States: 20th Century Fox and Dreamworks Pictures, 2002.

Stock, Wolfgang G., and Mechtild Stock. *Handbook of Information Science*. Translated by Paul Becker. Berlin: De Gruyter, 2013.

Strikwerda, Litska. "Legal and Moral Implications of Child Sex Robots." In *Robot Sex: Social and Ethical Implications*. Edited by John Danaher and Neil McArthur, 133-152. Cambridge, MA: MIT Press, 2017.

Suzuki, Miwa. "In Japan, Aibo Robots get their own Funeral." *The Japan Times* (May 1 2018). https://www.japantimes.co.jp/news/2018/05/01/national/japan-aibo-robots-get-funeral/#.WzCF_6cza01 (accessed June 18 2018).

Tashev, Ivan. "Recent Advances in Human-Machine Interfaces for Gaming and Entertainment."*International Journal of International Technology and Security* 3, no. 3 (2011): 69-76.

Tatsumi, Takayuki. *Full Metal Apache: Transactions between Cyberpunk Japan and Avant-Pop America*. Durham: Duke University Press, 2006.

Tavinor, Grant. "Definition of Videogames." *Contemporary Aesthetics* 6 (2008): 1-17.

Tiles, M. and H. Oberdiek. *Living in a Technological Culture: Human Tools and Human Values*. London: Routledge, 1995.

Toffoletti, Kim. "Barbie: a Posthuman Prototype." *Cyborgs and Barbie Dolls*. London: I.B. Tauris, 2007.

Trachtenberg, Joshua. *Jewish Magic and Superstition: A Study in Folk Religion*. Philadelphia, Pennsylvania: University of Pennsylvania Press, 2004. Google Ebook. Originally published 1939.

"Truecompanion.com." *True Companion* (2016). http://www.truecompanion.com/home.html (accessed May 25 2016).

TruecompanionLLC. *Roxxxy Sex Robot Meets Dr Oz* (2018). Youtube. https://www.youtube.com/watch?v=540jv33Nm9Y (accessed June 18 2018).

Trung, Le. "When Science Meets Beauty." *Project Aiko* (2007-2013). http://www.projectaiko.com/index.html (accessed June 29 2017)

Turkle, Sherry. *Alone Together*. New York: Basic Books, 2011.

———. *Life on the Screen*. New York: Simon & Schuster Paperbacks, 1995.Uexküll, Jakob von.

Uexküll, Jakob von. "A Stroll Through the World of Animals and Humans: a Picture Book of Invisible Worlds." *Semiotica* 89, no. 4 (1992): 319-91.

———. "An Introduction to Umwelt." *Semiotica* 134, no. 1-4 (2001): 107-10.

———. *Theoretical Biology*. Translated by D. L. MacKinnon. Edinburgh: The Edinburgh Press, 1926.

———. *Umwelt and Innenwelt of the Tick*. Berlin: Verlag von Julius Springer, 1909.

Uexküll, Thure von. “Introduction: the Sign Theory of Jakob von Uexküll.” *Semiotica* 89, no. 4 (1992): 279-315.

Ulanoff, Lance. "Xbox 360: A Tale From the Red Ring of Death." *PC Magazine* (July 28 2010.) http://www.pcmag.com/article2/0,2817,2367072,00.asp (accessed June 27 2017).

van Atten, Mark. *Essays on Gödel’s Reception of Leibniz, Husserl, and Brouwer*. Switzerland: Springer International Publishing, 2015.

Verbeek, Peter-Paul. “Cyborg Intentionality: Rethinking the Phenomenology of Human-Technology Relations.” *Phenomenology and the Cognitive Sciences* 7, no. 3 (2008): 387-395.

———. “Devices of Engagement: on Borgmann’s Philosophy of Information and Technology.” *Techne* 6 (2002): 69-92.

———.“Expanding Mediation Theory.” *Foundations of Science* 17 (2012): 391-395.

———. "The Morality of Things: a Postphenomenological Inquiry." In *Postphenomenology: A Critical Companion to Ihde*. Edited by Evan Selinger, 117-128. Albany, New York: State University of New York Press, 2006.

———. *What Things Do: Philosophical Reflections on Technology, Agency and Design* [De daadkracht der Dingen: Over techniek, filosofie en vormgeving]. Translated by Robert P. Crease. University Park Pennsylvania: Pennsylvania State University Press, 2005. Originally published 2000.

Vidal, Denis. "Anthropomorphism or Sub-Anthropomorphism? An Anthropological Approach to Gods and Robots." *Journal of the Royal Anthropological Institute* 13, no. 4 (2007): 917-33.

von Neumann, John. *The Computer and the Brain*. New Haven: Yale University Press, 2012. Originally published 1958.

Vonnegut, Kurt. *Slaughterhouse Five*. New York: Bantam Doubleday Dell Publishing Group, 1998. Originally published 1969.

"Voyager 2: In Depth." *Solar System Exploration, NASA Science*. https://solarsystem.nasa.gov/missions/voyager2/indepth (accessed June 22 2017).

"Voyager Goes Interstellar." *Voyager The Interstellar Mission*.. https://voyager.jpl.nasa.gov/ (accessed June 22 2017).

Wallach, Wendell, and Colin Allen. *Moral Machines: Teaching Robots Right from Wrong*. Oxford: Oxford University Press, 2009.

Walter, William Grey. "An Imitation of Life." *Scientific American* 182, no. 5 (1950): 42-45.

———. *The Living Brain*. London: Gerald Duckworth & Co. Ltd., 1953.

Waytz, Adam, Nicholas Epley, and John T. Cacioppo. "Social Cognition Unbound: Insights into Anthropomorphism and Dehumanization." *Current Directions in Psychological Science* 19, no. 1 (2010): 58-62.

Webber, Sarah, Marcus Carter, Wally Smith and Frank Vetere. “Interactive Technology and Human-Animal Encounters at the Zoo.” *International Journal of Human-Computer Studies* 98 (2017): 150-168.

Webster, Andrew. "Kinect Used to Control 83-Year-Old, Four-Story High Organ in Australia." *The Verge* (April 9 2012). http://www.theverge.com/2012/4/9/2935851/kinect-midi-controller-melbourne-town-hall-organ (accessed January 20 2018).

Wii Fit. Developed by Nintendo EAD Group No. 5. Published by Nintendo. 2008. Nintendo Wii.

Wii Sports. Developed by Nintendo EAD. Published by Nintendo. 2006. Nintendo Wii.

Williams, George Christopher. *Adaptation and Natural Selection*. New Jersey and Chichester, UK: Princeton University Press, 1966.

Winner, Langdon. "Do Artifacts Have Politics?" *The Whale and the Reactor: a Search for Limits in an Age of High Technology.* Chicago: University of Chicago Press, 1986.

———. “Upon Opening the Black Box and Finding it Empty: Social Constructivism and the Philosophy of Technology,” *Science, Technology and Human Values* 18, no. 3 (Summer 1993): 362-378.

Winograd, Terry. *Five Lectures on Artificial Intelligence*. Springfield, VA: National Technical Information Service, 1974.

———. “Procedures as a Representation for Data in a Computer Program for Understanding Natural Language.” Doctoral dissertation, Massachusetts Institute of Technology, 1971.

———. “Understanding Natural Language.” *Cognitive Psychology* 3 (1972): 1-191.

———. "What Does it Mean to Understand Language?" *Cognitive Science* 4 (1980): 209-41.

Winograd, Terry and Fernando Flores. *Understanding Computers and Cognition*. Reading, MA: Addison Wesley Publishing Company, Inc., 1987.

Wood, Gaby. *Edison's Eve: A Magical History of the Quest for Mechanical Life*. New York: Alfred A. Knopf, 2002.

———. *Living Dolls: A Magical History of the Quest for Mechanical Life*. London: Faber and Faber Limited, 2002.

Woolgar, Steve. “Configuring the User.” *Sociological Review* 38 (1990): 58-99.

Wolfendale, Peter. “The Noumenon’s New Clothes (Part 1).” *Speculations* III (2012): 290-366.

Wosk, Julie. *My Fair Ladies: Female Robots, Androids and Other Artificial Eves*. New Jersey: Rutgers University Press, 2015.

Xbox On. 2013. Youtube. http://www.youtube.com/watch?feature=player_embedded&v=Fm95LxARCFE (accessed January 13 2014).

"XBOX ONE." *Reddit* (2013-2014). http://www.reddit.com/r/XboxOne (accessed January 13 2014).

Xbox One Voice Commands Don't Work?!?! Xbox One FAIL (2013). Youtube. http://www.youtube.com/watch?feature=player_embedded&v=JSv7iEpAE58 (accessed January 13 2014).

"Xbox One: What it is." *Xbox*. 2013. http://www.xbox.com/en-US/xboxone/what-it-is (accessed June 2 2013).

Yalom, Irvin D. *Love’s Executioner and Other Tales of Psychotherapy*. New York: Basic Books, 1989.

Zhang, Zheng. “The Changing Landscape of Literacy Curriculum a Sino-Canada Transnational Education Programme: an Actor-Network Theory Informed Case Study.” *Journal of Curriculum Studies* 48, no. 4 (2016): 547-564.

Zielinski, Siegfried. *Deep Time of the Media*. Cambridge, MA: The MIT Press, 2006.